

SIEMENS

SINUMERIK

SINUMERIK ONE simDrive Create MyVirtual Machine simDrive parameters/alarms

List Manual

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Valid for:
MACHINUM
Create MyVirtual Machine V1.4
SINUMERIK ONE
SINUMERIK CNC software V6.23

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Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER

indicates that death or severe personal injury will result if proper precautions are not taken.
--

WARNING
--

indicates that death or severe personal injury may result if proper precautions are not taken.

CAUTION
--

indicates that minor personal injury can result if proper precautions are not taken.
--

NOTICE

indicates that property damage can result if proper precautions are not taken.
--

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

WARNING
--

Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.
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Trademarks

All names identified by ® are registered trademarks of Siemens Aktiengesellschaft. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Introduction

1.1 About SINUMERIK

From simple, standardized CNC machines to premium modular machine designs – the SINUMERIK CNCs offer the right solution for all machine concepts. Whether for individual parts or mass production, simple or complex workpieces – SINUMERIK is the highly dynamic automation solution, integrated for all areas of production. From prototype construction and tool design to mold making, all the way to large-scale series production.

Visit our website for more information SINUMERIK (<https://www.siemens.com/sinumerik>).

1.2 About this documentation

Target group

This publication is intended for:

- CNC configuration and commissioning personnel
- CNC programming personnel for job planning

Purpose

The intended target group can use the List Manual to test and commission the system or the plant professionally and safely.

Benefits

The List Manual provides information on parameters and their effects on the system.

Standard scope

This documentation only describes the functionality of the standard version. This may differ from the scope of the functionality of the system that is actually supplied. Please refer to the ordering documentation only for the functionality of the supplied drive system.

It may be possible to execute other functions in the system which are not described in this documentation. This does not, however, represent an obligation to supply such functions with a new control or when servicing.

For reasons of clarity, this documentation cannot include all of the detailed information on all product types. Further, this documentation cannot take into consideration every conceivable type of installation, operation and service/maintenance.

The machine manufacturer must document any additions or modifications they make to the product themselves.

Websites of third-party companies

This document may contain hyperlinks to third-party websites. Siemens is not responsible for and shall not be liable for these websites and their content. Siemens has no control over the information which appears on these websites and is not responsible for the content and information provided there. The user bears the risk for their use.

1.3 Documentation on the internet

1.3.1 Documentation overview SINUMERIK ONE

Comprehensive documentation about the functions provided in SINUMERIK ONE Version 6.13 and higher is provided in the Documentation overview SINUMERIK ONE (<https://support.industry.siemens.com/cs/ww/en/view/109768483>).



You can display documents or download them in PDF and HTML5 format.

The documentation is divided into the following categories:

- User: Operating
- User: Programming
- Manufacturer/Service: Functions
- Manufacturer/Service: Hardware
- Manufacturer/Service: Configuration/Setup
- Manufacturer/Service: Safety Integrated
- Information and training
- Manufacturer/Service: SINAMICS

1.3.2 Documentation overview SINUMERIK operator components

Comprehensive documentation about the SINUMERIK operator components is provided in the Documentation overview SINUMERIK operator components (<https://support.industry.siemens.com/cs/document/109783841/technische-dokumentation-zu-sinumerik-bedienkomponenten?dti=0&lc=en-WW>).

You can display documents or download them in PDF and HTML5 format.

The documentation is divided into the following categories:

- Operator Panels
- Machine control panels
- Machine Pushbutton Panel
- Handheld Unit/Mini handheld devices
- Further operator components

An overview of the most important documents, entries and links to SINUMERIK is provided at SINUMERIK Overview - Topic Page (<https://support.industry.siemens.com/cs/document/109766201/sinumerik-an-overview-of-the-most-important-documents-and-links?dti=0&lc=en-WW>).

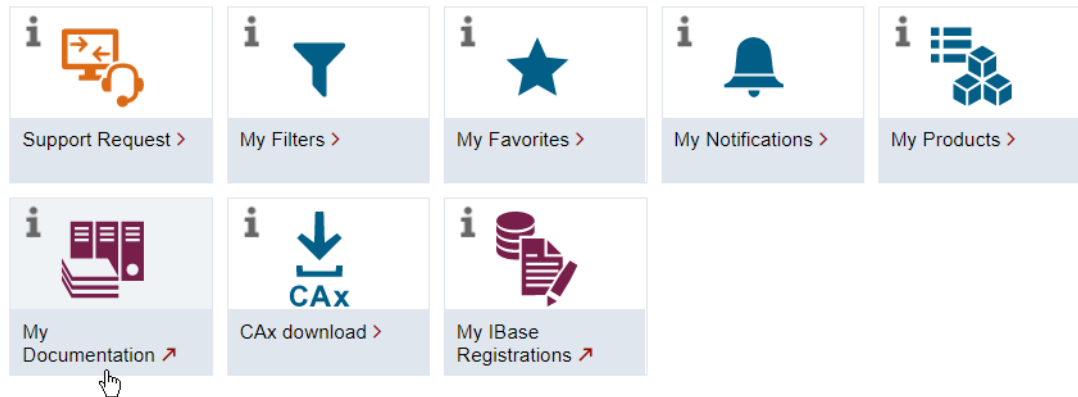
1.4 Feedback on the technical documentation

If you have any questions, suggestions, or corrections regarding the technical documentation published in the Siemens Industry Online Support, use the link "Give feedback" link which appears at the end of the entry.

1.5 mySupport documentation

With the "mySupport documentation" web-based system you can compile your own individual documentation based on Siemens content, and adapt it for your own machine documentation.

To start the application, click on the "My Documentation" tile on the "mySupport links and tools" (<https://support.industry.siemens.com/cs/ww/en/my>) portal page:

mySupport Links and Tools

The configured manual can be exported in RTF, PDF or XML format.

Note

Siemens content that supports the mySupport documentation application can be identified by the presence of the "Configure" link.

1.6 Service and Support

Product support

You can find more information about products on the internet:

Product support (<https://support.industry.siemens.com/cs/ww/en/>)

The following is provided at this address:

- Up-to-date product information (product announcements)
- FAQs (frequently asked questions)
- Manuals
- Downloads
- Newsletters with the latest information about your products
- Global forum for information and best practice sharing between users and specialists
- Local contact persons via our Contacts at Siemens database (→ "Contact")
- Information about field services, repairs, spare parts, and much more (→ "Field Service")

Technical support

Country-specific telephone numbers for technical support are provided on the internet at address (<https://support.industry.siemens.com/cs/ww/en/sc/4868>) in the "Contact" area.

If you have any technical questions, please use the online form in the "Support Request" area.

Training

You can find information on SITRAIN at the following address (<https://www.siemens.com/sitrain>).

SITRAIN offers training courses for automation and drives products, systems and solutions from Siemens.

Siemens support on the go



With the award-winning "Industry Online Support" app, you can access more than 300,000 documents for Siemens Industry products – any time and from anywhere. The app can support you in areas including:

- Resolving problems when implementing a project
- Troubleshooting when faults develop
- Expanding a system or planning a new system

Furthermore, you have access to the Technical Forum and other articles from our experts:

- FAQs
- Application examples
- Manuals
- Certificates
- Product announcements and much more

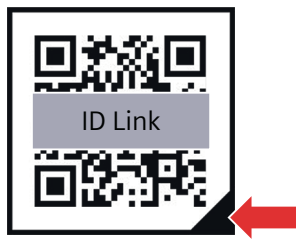
The "Industry Online Support" app is available for Apple iOS and Android.

Digital product information as QR code

The ID link according to IEC 61406 provides you with digital information about your product.

The ID link is a globally unique identifier and appears as a QR code on the product.

You can recognize the ID link by the frame with a black corner at the bottom right.



In addition to the digital nameplate you will find the following product information:

- Technical specifications
- FAQs
- Manuals
- Certificates
- Product announcements
- Application examples

1.7 Using OpenSSL

This product can contain the following software:

- Software developed by the OpenSSL project for use in the OpenSSL toolkit
- Cryptographic software created by Eric Young.
- Software developed by Eric Young

You can find more information on the internet:

- OpenSSL (<https://www.openssl.org>)
- Cryptsoft (<https://www.cryptsoft.com>)

1.8 Compliance with the General Data Protection Regulation

Siemens observes standard data protection principles, in particular the data minimization rules (privacy by design).

For this product, this means:

The product does not process or store any personal data, only technical function data (e.g. time stamps). If the user links this data with other data (e.g. shift plans) or if he/she stores person-related data on the same data medium (e.g. hard disk), thus personalizing this data, he/she must ensure compliance with the applicable data protection stipulations.

Fundamental safety instructions

2.1 General safety instructions

 WARNING
Danger to life if the safety instructions and residual risks are not observed If the safety instructions and residual risks in the associated hardware documentation are not observed, accidents involving severe injuries or death can occur. • Observe the safety instructions given in the hardware documentation. • Consider the residual risks for the risk evaluation.

 WARNING
Malfunctions of the machine as a result of incorrect or changed parameter settings As a result of incorrect or changed parameterization, machines can malfunction, which in turn can lead to injuries or death. • Protect the parameterization against unauthorized access. • Handle possible malfunctions by taking suitable measures, e.g. emergency stop or emergency off.

2.2 Warranty and liability for application examples

Application examples are not binding and do not claim to be complete regarding configuration, equipment or any eventuality which may arise. Application examples do not represent specific customer solutions, but are only intended to provide support for typical tasks.

As the user you yourself are responsible for ensuring that the products described are operated correctly. Application examples do not relieve you of your responsibility for safe handling when using, installing, operating and maintaining the equipment.

2.3 Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

2.3 Cybersecurity information

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit
<https://www.siemens.com/cybersecurity-industry>.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed under
<https://new.siemens.com/cert>.

Further information is provided on the Internet:

Industrial Security Configuration Manual (<https://support.industry.siemens.com/cs/ww/en/view/108862708>)



WARNING

Unsafe operating states resulting from software manipulation

Software manipulations, e.g. viruses, Trojans, or worms, can cause unsafe operating states in your system that may lead to death, serious injury, and property damage.

- Keep the software up to date.
- Incorporate the automation and drive components into a state-of-the-art, integrated industrial cybersecurity concept for the installation or machine.
- Make sure that you include all installed products in the integrated industrial cybersecurity concept.
- Protect files stored on exchangeable storage media from malicious software by with suitable protection measures, e.g. virus scanners.
- Carefully check all cybersecurity-related settings once commissioning has been completed.

Explanation of the parameters

3.1 Structure of the data tables

Basic structure of the parameter description

The tables of SINAMICS parameters all have the following standard structure:

Parameter number	BICO: Full parameter name / short parameter name		
Drive object (function module)	Can be changed:	Calculated:	Access level:
	Data type:	Dynamic index:	Function diagram:
	P group:	Unit group:	Unit selection:
	Not for motor type:	Normalizing	Expert list:
	Min.	Max.	Factory setting

Example

r0025[0...3]	CO: Input voltage, smoothed		
A_INF, S_INF	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dynamic index:	Function diagram: 8850, 8950
	P group: Display, signals	Unit group: -	Unit selection: -
	Not for motor type: -	Normalization p2001	Expert list: 1
	Min - [Vrms]	Max - [Vrms]	Factory setting - [Vrms]

References

You can find a detailed description of SINAMICS parameters in the following document:
SINAMICS S120/S150 List Manual

3.2 Meaning of table fields

Parameter number pxxxx[0...n]

The "parameter number" field contains the number of the parameter.

The number consists of a leading "p" or "r", followed by the parameter number and the index (optional).

3.2 Meaning of table fields

Below are some examples of how the number is represented in the parameter list:

p... Adjustable parameter (read and write parameter)
r... Display parameter (read-only)
p0918 Adjustable parameter 918
p0608[0..3] Adjustable parameter 608, indices 0 to 3
p0610[0..n] Adjustable parameter 6010, indices 0 to n (n = configurable)
r0944 Display parameter 944
r2129.0..15 Display parameter 2129 with bit array from bit 0 (lowest bit) to bit 15 (highest bit)

BICO: Full parameter name / short parameter name

The following abbreviations can appear in front of the parameter name:

BI	Binector Input This parameter selects the source of a digital signal.
BO	Binector Output This parameter is available as an digital signal for interconnection with other parameters.
CI	Connector Input This parameter selects the source of an "analog" signal.
CO	Connector Output This parameter is available as an "analog" signal for interconnection with other parameters.
CO/BO	Connector/Binector Output This parameter is available as both an "analog" and also as digital signals for interconnection with other parameters.

Drive object (function module)

A drive object (DO) is an independent, "self-contained" functional unit with its own parameters, faults messages, and alarms.

The parameter list specifies the associated drive object and function module for each individual parameter.

Additional information is available in the following documentation: /FH1/ SINAMICS S120 Function Manual Drive Functions

Can be changed

The "-" sign indicates that the parameter can be changed in any object state and that the change will be activated immediately.

The information C1(x), C2(x), T, U ((x): optional) means that the parameter can be changed only in the specified drive object state and that the change will not be activated until the object switches to another state. This can be one or more states.

The following states are available:

C1(x)	Device commissioning Device is in the process of being commissioned (p0009 > 0). The pulses cannot be enabled. The parameter can only be changed in the following device commissioning settings (p0009 > 0):	C1: Commissioning 1
	• C1: can be changed for all settings p0009 > 0. • C1(x): can only be changed for the settings p0009 = x. A modified parameter value will not be activated until the device commissioning mode is exited with p0009 = 0.	
C2(x)	Drive object commissioning The drive is in the process of being commissioned (p0009 = 0 and p0010 > 0). The pulses cannot be enabled. The parameter can only be changed in the following device commissioning settings (p0010 > 0):	C2: Commissioning 2
	• C2: can be changed for all settings p0010 > 0. • C2(x): can be changed for the settings p0010 = x. A modified parameter value will not be activated until drive commissioning mode is exited with p0010 = 0	
U	Operation The pulses are enabled	U: Run
T	Ready The pulses are not enabled Status "C1(x)" or "C2(x)" is not active.	T: Ready to run

Calculated

Specifies whether the parameter is influenced by automatic calculations.

The calculation attribute defines which activities influence the parameter.

Access level

Specifies the access level required to be able to display and change this parameter. The required access level can be set using p0003.

The system uses the following access levels

1. Standard
2. Extended
3. Expert
4. Service

Parameters with this access level are protected by a password.

Data type

The information on the data type can consist of the following two items (separated by a slash).

- First information
Data type of the parameter.
- Second information (for binector or connector input only)
Data type of the signal source to be interconnected (binector/connector output).

Parameters can have the following data types:

I8	Integer8	8-bit integer number
I16	Integer16	16-bit integer number
I32	Integer32	32-bit integer number
U8	Unsigned8	8 bits without sign
U16	Unsigned16	16 bits without sign
U32	Unsigned32	32 bits without sign
Float	FloatingPoint32	32-bit floating point number

Dynamic index

For parameters with a dynamic index [0 to n], the following information is specified:

- Data set (if available).
- Parameter for the number of indices ($n = \text{number} - 1$).

Function diagram

The parameter is included in this function diagram. The structure of the parameter function and its relationship with other parameters are shown in the function diagram.

The function diagrams are available in the following documentation:

/LH1/ SINAMICS S120/S150 List Manual

P group

Specifies the functional group to which this parameter belongs. The required parameter group can be set via p0004.

Unit, unit group, unit selection

The standard unit of a parameter is specified in square brackets after the values for "Min", "Max", and "Factory setting".

For parameters where the unit can be changed, "Unit Group" and "Unit Selection" specify which group this parameter belongs and which parameter can be used to change the unit.

Not for motor type

Specifies for which motor type this parameter has no significance

ASM	Induction motor
FEM	Separately excited synchronous motor
PEM	Permanent-magnet synchronous motor
REL	Reluctance motor/SIEMOSYN motor

Normalizing

Specification of the reference variable with which a signal value is automatically converted for a BICO interconnection.

The following reference variables are possible:

p2000 ... p2007: Reference speed, reference voltage, etc.

TEMP: 100°C = 100%

PERCENT: 1.0 = 100 %

4000H: 4000 hex = 100 %

Expert list

Specifies whether this parameter is available in the expert list of the specified drive objects in the commissioning software.

1: Parameter does exist in the expert list.

0: Parameter does not exist in the expert list

3.3 Number ranges of the parameters

Number ranges

Parameters are grouped into the following number ranges:

Range		Description
From	to	
0000	0099	Display and operation
0100	0199	Commissioning
0200	0299	Power unit
0300	0399	Motor
0400	0499	Encoder
0500	0599	Technology and units, motor-specific data, probes
0600	0699	Thermal monitoring, maximum current, operating hours, motor data, central probe
0700	0799	Control Unit terminals, measuring sockets

3.3 Number ranges of the parameters

Range		Description
From	to	
0800	0839	CDS, DDS data sets, motor changeover
0840	0879	Sequence control (e.g. signal source for ON/OFF1)
0880	0899	ESR, parking, control and status words
0900	0999	PROFIBUS/PROFIdrive
1000	1199	Setpoint channel (e.g. ramp-function generator)
1200	1299	Functions (e.g. motor holding brake)
1300	1399	U/f control
1400	1799	Closed-loop control
1800	1899	Gating unit
1900	1999	Power unit and motor identification
2000	2009	Reference values
2010	2099	Communication (fieldbus)
2100	2139	Faults and alarms
2140	2199	Signals and monitoring
2200	2359	Technology controller
2360	2399	Staging, hibernation
2500	2699	Position control (LR) and basic positioning (EPOS)
2700	2719	Reference values, display
2720	2729	Load gearbox
2800	2819	Logic operations
2900	2930	Fixed values (e.g. percent, torque)
3000	3099	Motor identification results
3100	3109	Real-time clock (RTC)
3110	3199	Faults and alarms
3200	3299	Signals and monitoring
3400	3659	Infeed closed-loop control
3660	3699	Voltage Sensing Module (VSM), Braking Module internal
3700	3779	Advanced Positioning Control (APC)
3780	3819	Synchronization
3820	3849	Friction characteristic
3850	3899	Functions (e.g. long stator)
3900	3999	Management
4000	4599	Terminal Board, Terminal Module (e.g. TB30, TM31)
4600	4699	Sensor Module
4700	4799	Trace
4800	4849	Function Generator
4950	4999	OA application
5400	5499	Line droop control (e.g. shaft generator)
5500	5599	Dynamic grid support (solar)
5600	5913	PROFenergy
5900	6999	SINAMICS GM/SM/GL/SL
7000	7499	Parallel connection of power units

3.3 Number ranges of the parameters

Range		Description
From	to	
7500	7599	SINAMICS SM120
7700	7729	External messages
7770	7789	NVRAM, system parameters
7800	7839	EEPROM read/write parameters
7840	8399	Internal system parameters
8400	8449	Real-time clock (RTC)
8500	8599	Data and macro management
8600	8799	CAN bus
8800	8899	Communication Board Ethernet (CBE), PROFIdrive
8900	8999	Industrial Ethernet, PROFINET, CBE20
9000	9299	Topology
9300	9399	Safety Integrated
9400	9499	Parameter consistency and storage
9500	9899	Safety Integrated
9900	9949	Topology
9950	9999	Diagnostics, internal
10000	10199	Safety Integrated
11000	11299	Free technology controller 0, 1, 2
20000	20999	Free function blocks (FBLOCKS)
21000	25999	Drive Control Chart (DCC)
50000	53999	SINAMICS DC MASTER (DC closed-loop control)
61000	61001	PROFINET

3.3 Number ranges of the parameters

simDrive parameters

Version: 05207300

All objects: A_INF_840_VD, CU_I_840_VD, CU_NX_840_VD, SERVO_DBSI_VD

r0002	Control Unit operating display / CU op_display		
CU_I_840_VD, CU_NX_840_VD	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	117	-
Description:	Operating display for the Control Unit (CU).		
Value:	0: Operation 10: Ready 20: Wait for run-up 25: Wait for automatic FW update of DRIVE-CLiQ components 31: Commissioning tool download active 33: Remove/acknowledge topology error 34: Exit commissioning mode 35: Carry out first commissioning 70: Initialization 80: Reset active 99: Internal software error 101: Specify topology 111: Insert drive object 112: Delete drive object 113: Change drive object number 114: Change component number 115: Run parameter download 117: Delete component		

NOTICE

For several missing enable signals, the corresponding value with the highest number is displayed.

r0002	Drive operating display / Drv op_display		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 1
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	250	-
Description:	Operating display for the drive.		
Value:	0: Operation - everything enabled 10: Operation - set "enable setpoint" = "1" (p1142, p1152) 11: Operation - set "enable speed controller" = "1" (p0856)		

- 12: Operation - RFG frozen, set "RFG start" = "1" (p1141)
- 13: Operation - set "enable RFG" = "1" (p1140)
- 14: Oper. - MotID, excit. running and/or brake opens, SS2, STOP C
- 15: Operation - open brake (p1215)
- 16: Operation - withdraw braking with OFF1 using "ON/OFF1" = "1"
- 17: Operation - braking with OFF3 can only be interrupted with OFF2
- 18: Operation - brake on fault, remove fault, acknowledge
- 19: Operation - armature short-circ./DC brake act. (p1230, p1231)
- 21: Ready for operation - set "Enable operation" = "1" (p0852)
- 22: Ready for operation - de-magnetizing running (p0347)
- 23: Ready for operation - set "Infeed operation" = "1" (p0864)
- 31: Ready for switching on - set "ON/OFF1" = "0/1" (p0840)
- 35: Switching on inhibited - carry out first commissioning (p0010)
- 41: Switching on inhibited - set "ON/OFF1" = "0" (p0840)
- 42: Switching on inhibited - set "OC/OFF2" = "1" (p0844, p0845)
- 43: Switching on inhibited - set "OC/OFF3" = "1" (p0848, p0849)
- 44: Switching on inhibited - supply STO terminal w/ 24 V (hardware)
- 45: Switching on inhibited - rectify fault, acknowledge fault, STO
- 46: Switching on inhibited - exit commissioning mode (p0009, p0010)
- 60: Drive object deactivated/not operational
- 70: Initialization
- 200: Wait for booting/partial booting
- 250: Device signals a topology error

Dependency:

See also: r0046

NOTICE

For several missing enable signals, the corresponding value with the highest number is displayed.

Note

OC: Operating condition
EP: Enable Pulses (pulse enable)
RFG: Ramp-function generator
COMM: Commissioning
MotID: Motor data identification
SS2: Safe Stop 2
STO: Safe Torque Off

p0009CU_I_840_VD,
CU_NX_840_VD**Device commissioning parameter filter / Dev com par_filt****Can be changed:** C1, T**Data type:** Integer16**P group:** All groups**Not for motor type:** -**Min:**

0

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** -**Max:**

1

Access level: 1**Function diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting:**

0

Description:

Sets the device and basic drive commissioning.

By appropriately setting this parameter, those parameters are filtered that can be written into in the various commissioning steps.

Value:

0: Ready

1: Device configuration

NOTICE

For p0009 = 0:

The check and the calculation can take a longer period of time, so that in the worst case scenario, communication with the commissioning tool is interrupted. As a consequence, alternatively p0009 = 10000 can be used.

For p0009 = 10000:

After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.

Note

The drives can only be switched on outside the device commissioning (the inverter enabled). In this case, p0009 must be 0 (Ready) and the individual drive objects must have already gone into operation (p0010).

p0009 = 0: ready

When changing into this state, the device configuration is checked and commissioning completed (a new start can be initiated).

p0009 = 1: Device configuration

At the first commissioning of the device, after booting, the device is in the "device configuration" state. To start the internal automatic first commissioning of the drive unit, p0009 should be set to 0 (Ready) after the ID for the actual topology (r0098) was transferred into the ID for the target topology (p0099). To do this, it is sufficient to set a single index value of p0099[x] the same as r0098[x]. Before the device has been completely commissioned, no other parameter can be changed. After the first commissioning was carried out, in this state, when required, other basic device configuration parameters can be adapted (e.g. the basic sampling time in p0110).

p0009 = 2: Defines the drive type / function module

In this state, the drive object types and/or the function modules can be changed or selected for the individual drive objects. To do this, the drive object type can be set using p0107[0...15] and the function can be set using p0108[0...15] (refer to p0101[0...15]).

p0009 = 3: Drive basic configuration

In this state, after the device has been commissioned for the first time, basic changes can be made for the individual drive objects (e.g. sampling times in p0111, p0112, p0115 and the number of data sets in p0120, p0130, p0140, p0170, p0180).

p0009 = 4: Data set basic configuration

In this state, after the device has been commissioned for the first time, for the individual drive objects changes can be made regarding the assignment of the components (p0121, p0131, p0141, p0151, p0161) to the individual data sets and the assignment of the power unit, motor and encoder to the drive data sets (p0185, ...).

p0009 = 29: Device download

In the case of a download via the commissioning tool, the device is automatically brought into this state. After the download has been completed, p0009 is automatically set to 0 (ready). It is not possible to manually set p0009 to this value.

p0009 = 30: Parameter reset

In order to bring the complete unit into the "first commissioning" state or to load the parameters saved using p0977, to start, p0009 must be set to this value. p0976 can then be changed to the required value.

p0009 = 50: configure Technology Extension

After the device has been commissioned for the first time, in this state, Technology Extensions can be activated/deactivated for the individual drive objects (p4956).

p0009 = 55: install Technology Extension

Technology Extensions can be installed/uninstalled in this state.

p0009 = 101: Topology input

In this state, the DRIVE-CLiQ target topology can be entered using p9902 and p9903.

p0009 = 111: Insert drive object

This state allows a new drive object to be inserted using p9911.

p0009 = 112: Delete drive object

This state allows existing drive objects to be deleted using p9912 after the device has been commissioned for the first time.

p0009 = 113: Change drive object number

This state allows the drive object number of existing drive objects to be changed using p9913 after the device has been commissioned for the first time.

p0009 = 114: Change component number

This state allows the component number of existing components to be changed using p9914 after the device has been commissioned for the first time.

p0009 = 115: Parameter download

This state allows the complete device and drive commissioning using the parameter services.

p0009 = 117: Delete component

This state allows components to be deleted using p9917 after the device has been commissioned for the first time.

p0009 = 10000: ready (asynchronous)

When changing into this state, internally p0009 is set = 0. Additional calculations and checks are carried out in the background (asynchronously).

p0010	Drive commissioning parameter filter / Drv com par_filt		
SERVO_DBSI_VD	Can be changed: C2(1), T	Calculated: -	Access level: 1
	Data type: Integer16	Dyn. index: -	Function diagram: 2800, 2818
	P group: All groups	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	10000	1
Description:	Sets the parameter filter to commission a drive. Setting this parameter filters out the parameters that can be written into in the various commissioning steps.		
Value:	0: Ready 1: Quick commissioning 2: Power unit commissioning 3: Motor commissioning 4: Encoder commissioning 5: Technological application/units 15: Data sets 17: Basic positioner commissioning 25: Position control commissioning 29: Only Siemens internal 30: Parameter reset 95: Safety Integrated commissioning 10000: Ready with immediate feedback signal		
NOTICE			
For p0010 = 10000 the following applies: After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.			
Note			
The drive can only be switched on outside the drive commissioning (inverter enable). To realize this, this parameter must be set to 0. By setting p3900 to a value other than 0, the quick commissioning is completed, and this parameter is automatically reset to 0. Procedure for "Reset parameter": Set p0010 to 30 and p0970 to 1. p0010 = 10000 corresponds to p0010 = 0. Unlike with p0010 = 0, the parameter modification is applied immediately and the calculations are made in the background. Further parameter modifications cannot be made while the calculations are being performed. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

r0021	CO: Actual speed smoothed / n_act smth		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Displays, signals Not for motor type: - Min: - [rpm]	Calculated: - Dyn. index: - Unit group: 3_1 Scaling: p2000 Max: - [rpm]	Access level: 2 Function diagram: 4700, 4710 Unit selection: p0505 Expert list: 1 Factory setting: - [rpm]
Description:	Display and connector output for the smoothed actual value of the motor speed.		
Dependency:	See also: r0022, r0063		

Note

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The speed actual value is available smoothed (r0021, r0022) and unsmoothed (r0063).

r0021

SERVO_DBSI_VD (Lin)

CO: Actual velocity smoothed / v_act smth

Can be changed: -

Data type: FloatingPoint32

P group: Displays, signals

Not for motor type: -

Min:

- [m/min]

Calculated: -

Dyn. index: -

Unit group: 4_1

Scaling: p2000

Max:

- [m/min]

Access level: 2

Function diagram: 4700, 4710

Unit selection: p0505

Expert list: 1

Factory setting:

- [m/min]

Description:

Display and connector output for the smoothed actual value of the motor velocity.

Dependency:

See also: r0022, r0063

Note

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The velocity actual value is available smoothed (r0021, r0022) and unsmoothed (r0063).

r0022

SERVO_DBSI_VD

Speed actual value rpm smoothed / n_ActV rpm smth

Can be changed: -

Data type: FloatingPoint32

P group: Displays, signals

Not for motor type: -

Min:

- [rpm]

Calculated: -

Dyn. index: -

Unit group: -

Scaling: p2000

Max:

- [rpm]

Access level: 2

Function diagram: 4700, 4710

Unit selection: -

Expert list: 1

Factory setting:

- [rpm]

Description:

Displays the smoothed actual value of the motor speed.

r0022 is identical to r0021, however, it always has units of rpm and contrary to r0021 cannot be changed over.

Dependency:

See also: r0021, r0063

Note

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The speed actual value is available smoothed (r0021, r0022) and unsmoothed (r0063).

r0022

SERVO_DBSI_VD (Lin)

Actual velocity smoothed / v_act smth

Can be changed: -

Data type: FloatingPoint32

P group: Displays, signals

Not for motor type: -

Min:

- [m/min]

Calculated: -

Dyn. index: -

Unit group: -

Scaling: p2000

Max:

- [m/min]

Access level: 2

Function diagram: 4700, 4710

Unit selection: -

Expert list: 1

Factory setting:

- [m/min]

Description:

Displays the smoothed actual value of the motor velocity.

Dependency:

See also: r0021, r0063

Note

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The velocity actual value is available smoothed (r0021, r0022) and unsmoothed (r0063).

r0026	CO: DC link voltage smoothed / Vdc smth		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Displays, signals Not for motor type: - Min: - [V]	Calculated: - Dyn. index: - Unit group: - Scaling: p2001 Max: - [V]	Access level: 2 Function diagram: 5730, 8750, 8850, 8950 Unit selection: - Expert list: 1 Factory setting: - [V]
Description:	Displays the smoothed actual value of the DC link voltage. NOTICE For SINAMICS S120 AC Drive (AC/AC) the following applies: When measuring a DC link voltage < 200 V, for the Power Module (e.g. PM340) a valid measured value is not supplied. In this case, when an external 24 V power supply is connected, a value of approx. 24 V is displayed in the display parameter. Note SERVO, VECTOR: Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The DC link voltage is available smoothed (r0026) and unsmoothed (r0070).		
r0033	Torque utilization smoothed / M_util smth		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Displays, signals Not for motor type: - Min: - [%]	Calculated: - Dyn. index: - Unit group: - Scaling: PERCENT Max: - [%]	Access level: 3 Function diagram: 8012 Unit selection: - Expert list: 1 Factory setting: - [%]
Description:	Displays the smoothed torque utilization as a percentage. The torque utilization is obtained from the required smoothed torque in reference to the torque limit, scaled using p2196. Note Smoothing time constant = 100 ms The signal is not suitable as a process quantity and may only be used as a display quantity. The torque utilization is available smoothed (r0033) and unsmoothed (r0081). For M_set total (r0079) > M_max offset (p1532), the following applies: - demanded torque = M_set total - M_max offset - actual torque limit = M_max upper effective (r1538) - M_max offset For M_set total (r0079) <= M_max offset (p1532), the following applies: - demanded torque = M_max offset - M_set total - actual torque limit = M_max offset - M_max lower effective (r1539) For the actual torque limit = 0, the following applies: r0033 = 100 % For the actual torque limit < 0, the following applies: r0033 = 0 %		
r0033	Force utilization smoothed / F_util smth		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Displays, signals Not for motor type: - Min: - [%]	Calculated: - Dyn. index: - Unit group: - Scaling: PERCENT Max: - [%]	Access level: 3 Function diagram: 8012 Unit selection: - Expert list: 1 Factory setting: - [%]
Description:	Displays the smoothed force utilization as a percentage. The force utilization is obtained from the required smoothed force referred to the force limit.		

Note

Smoothing time constant = 100 ms

The signal is not suitable as a process quantity and may only be used as a display quantity.

The force utilization is available smoothed (r0033) and unsmoothed (r0081).

For $F_{set\ total} (r0079) > F_{max\ offset} (p1532)$, the following applies:

- demanded torque = $F_{set\ total} - F_{max\ offset}$

- actual force limit = $F_{max\ upper\ effective} (r1538) - F_{max\ offset}$

For $F_{set\ total} (r0079) \leq F_{max\ offset} (p1532)$, the following applies:

- demanded force = $F_{max\ offset} - F_{set\ total}$

- actual force limit = $F_{max\ offset} - F_{max\ lower\ effective} (r1539)$

For the actual force limit = 0, the following applies: r0033 = 100 %

For the actual force limit < 0, the following applies: r0033 = 0 %

r0035

SERVO_DBSI_VD

CO: Motor temperature / Mot temp**Can be changed:** -**Data type:** FloatingPoint32**P group:** Displays, signals**Not for motor type:** -**Min:**

- [°C]

Calculated: -**Dyn. index:** -**Unit group:** 21_1**Scaling:** p2006**Max:**

- [°C]

Access level: 2**Function diagram:** 8016, 8017**Unit selection:** p0505**Expert list:** 1**Factory setting:**

- [°C]

Description:

Display and connector output for the actual temperature in the motor.

Note

For r0035 not equal to -200.0 °C, the following applies:

- this temperature display is valid.

- a KTY/PT1000 temperature sensor is connected.

- the thermal model for the induction motor is activated (p0612 bit 1 = 1 and temperature sensor deactivated: p0600 = 0 or p0601 = 0).

For r0035 equal to -200.0 °C, the following applies:

- this temperature display is not valid (temperature sensor error).

- a PTC sensor or bimetallic NC contact is connected.

- the temperature sensor of the synchronous motor is deactivated (p0600 = 0 or p0601 = 0).

r0044

SERVO_DBSI_VD

Thermal converter utilization / Conv_util therm**Can be changed:** -**Data type:** FloatingPoint32**P group:** Displays, signals**Not for motor type:** -**Min:**

- [%]

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** PERCENT**Max:**

- [%]

Access level: 2**Function diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting:**

- [%]

Description:

Displays the thermal converter utilization as a percentage.

With this value, various thermal monitoring functions are taken into account.

Note

The thermal motor utilization is displayed in parameter r0034.

p0045	Display values smoothing time constant / Disp_val Tc_smth		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 4715, 5610, 5730, 6714, 8012
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 10000.00 [ms]	Factory setting: 1.00 [ms]
Description:	Sets the smoothing time constant for the following display values: SERVO: r0078[1], r0079[1], r0081 (calculated from the quantities smoothed with p0045), r0082[1]. VECTOR: r0063[1], r0068[1], r0080[1], r0082[1].		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

r0046.0...31	CO/BO: Missing enable signal / Missing enable sig				
SERVO_DBSI_VD	Can be changed: -		Calculated: -		Access level: 1
	Data type: Unsigned32		Dyn. index: -		Function diagram: 2634
	P group: Displays, signals		Unit group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min:		Max:		Factory setting:
	-		-		-
Description:	Display and BICO output for missing enable signals that are preventing the closed-loop drive control from being commissioned.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	OFF1 enable missing	Yes	No	7954
	01	OFF2 enable missing	Yes	No	-
	02	OFF3 enable missing	Yes	No	-
	03	Operation enable missing	Yes	No	-
	04	Armature short-circuit / DC braking enable missing	Yes	No	7014, 7016
	05	STOP2 enable missing	Yes	No	-
	08	Safety enable missing	Yes	No	-
	09	Infeed enable missing	Yes	No	-
	10	Ramp-function generator enable missing	Yes	No	-
	11	Ramp-function generator start missing	Yes	No	-
	12	Setpoint enable missing	Yes	No	-
	16	OFF1 enable internal missing	Yes	No	-
	17	OFF2 enable internal missing	Yes	No	-
	18	OFF3 enable internal missing	Yes	No	-
	19	Pulse enable internal missing	Yes	No	-
	20	Armature short-circuit/DC braking internal enable missing	Yes	No	7014, 7016
	21	STOP2 enable internal missing	Yes	No	-
	25	Function bypass active	Yes	No	-
	26	Drive inactive or not operational	Yes	No	-
	27	De-magnetizing not completed	Yes	No	-
	28	Brake open missing	Yes	No	-
	29	Cooling unit ready signal missing	Yes	No	-
	30	Speed controller inhibited	Yes	No	-

	31	Jog setpoint active	Yes	No	-
Dependency:	See also: r0002				

Note

The value r0046 = 0 indicates that all enable signals for this drive are present.

Bit 00 = 1 (enable signal missing), if:

- the signal source in p0840 is a 0 signal.
- there is a "switching on inhibited".

Bit 01 = 1 (enable signal missing), if:

- the signal source in p0844 or p0845 is a 0 signal.

Bit 02 = 1 (enable signal missing), if:

- the signal source in p0848 or p0849 is a 0 signal.

Bit 03 = 1 (enable signal missing), if:

- the signal source in p0852 is a 0 signal.

Bit 04 = 1 (armature short-circuit active), if:

- the signal source in p1230 has a 1 signal

Bit 05, Bit 06: Being prepared

Bit 08 = 1 (enable signal missing), if:

- safety functions have been enabled and STO is active.
- a safety-relevant signal is present with a STOP A response.

STO enabled via terminals:

- the pulse enable via terminal EP is missing (booksize: X21, chassis: X41), or the signal source in p9620 is for a 0 signal.

STO enabled via PROFIsafe or TM54F:

- STO is selected via PROFIsafe or TM54F.

Bit 09 = 1 (enable signal missing), if:

- the signal source in p0864 is a 0 signal.

Bit 10 = 1 (enable signal missing), if:

- the signal source in p1140 is a 0 signal.

Bit 11 = 1 (enable signal missing) if the speed setpoint is frozen, because:

- the signal source in p1141 is a 0 signal.
- the speed setpoint is entered from jogging and the two signal sources for jogging, bit 0 (p1055) and bit 1 (p1056) have a 1 signal.

Bit 12 = 1 (enable signal missing), if:

- the signal source in p1142 is a 0 signal.
- When activating the function module "basic positioner" (r0108.4 = 1), the signal source in p1142 is set to a 0 signal.

Bit 16 = 1 (enable signal missing), if:

- there is an OFF1 fault response. The system is only enabled if the fault is removed and was acknowledged and the "switching on inhibited" withdrawn with OFF1 = 0.

Bit 17 = 1 (enable signal missing), if:

- commissioning mode is selected (p0009 > 0 or p0010 > 0).
- there is an OFF2 fault response.
- the drive is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).

Bit 18 = 1 (enable signal missing), if:

- OFF3 has still not been completed or an OFF3 fault response is present.

Bit 19 = 1 (internal pulse enable missing), if:

- synchronization is running between the basic clock cycle, DRIVE-CLiQ clock cycle and application clock cycle.

Bit 20 = 1 (internal armature short-circuit active), if:

- the drive is not in the state "S4: Operation" or "S5x" (refer to function diagram 2610).
- the internal pulse enable is missing (r0046.19 = 0).

Bit 21 = 1 (enable signal missing), if:

The pulses have been enabled and the speed setpoint has still not been enabled, because:

- the holding brake opening time (p1216) has still not expired.
- the motor has still not been magnetized (induction motor).
- the encoder has not been calibrated (U/f vector and synchronous motor)

Bit 22: Being prepared

Bit 26 = 1 (enable signal missing), if:

- the drive is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).
- the function "parking axis" is selected (BI: p0897 = 1 signal)..
- all power units of a parallel connection are deactivated (p0125, p0895).
- the drive device is in the "PROFInergy energy-saving mode" (r5600, CU-specific).

Bit 27 = 1 (enable signal missing), if:

- de-magnetizing has still not been completed (only for vector).

Bit 28 = 1 (enable signal missing), if:

- the holding brake is closed or has still not been opened.

Bit 29 = 1 (enable signal missing), if:

- the cooling unit ready signal via binector input p0266[1] missing.

Bit 30 = 1 (speed controller inhibited), if one of the following reasons is present:

- a 0 signal is available via binector input p0856.
- the function generator with current input is active.
- the measuring function "current controller reference frequency characteristic" is active.
- the pole position identification is active.
- motor data identification is active (only certain steps).

Bit 31 = 1 (enable signal missing), if:

- the speed setpoint from jog 1 or 2 is entered.

r0046.0...31		CO/BO: Missing enable signal / Missing enable sig			
SERVO_DBSI_VD (Lin)	Can be changed: -		Calculated: -		Access level: 1
	Data type: Unsigned32		Dyn. index: -		Function diagram: 2634
	P group: Displays, signals		Unit group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min:		Max:		Factory setting:
	-		-		-
Description:	Display and BICO output for missing enable signals that are preventing the closed-loop drive control from being commissioned.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	OFF1 enable missing	Yes	No	7954
	01	OFF2 enable missing	Yes	No	-
	02	OFF3 enable missing	Yes	No	-
	03	Operation enable missing	Yes	No	-
	04	Armature short-circuit / DC braking enable missing	Yes	No	7014, 7016
	05	STOP2 enable missing	Yes	No	-
	08	Safety enable missing	Yes	No	-
	09	Infeed enable missing	Yes	No	-
	10	Ramp-function generator enable missing	Yes	No	-
	11	Ramp-function generator start missing	Yes	No	-
	12	Setpoint enable missing	Yes	No	-
	16	OFF1 enable internal missing	Yes	No	-
	17	OFF2 enable internal missing	Yes	No	-
	18	OFF3 enable internal missing	Yes	No	-
	19	Pulse enable internal missing	Yes	No	-
	20	Armature short-circuit/DC braking internal enable missing	Yes	No	7014, 7016
	21	STOP2 enable internal missing	Yes	No	-
	25	Function bypass active	Yes	No	-
	26	Drive inactive or not operational	Yes	No	-
	27	De-magnetizing not completed	Yes	No	-
	28	Brake open missing	Yes	No	-
	29	Cooling unit ready signal missing	Yes	No	-
	30	Velocity controller inhibited	Yes	No	-
	31	Jog setpoint active	Yes	No	-

Dependency: See also: r0002

Note

The value r0046 = 0 indicates that all enable signals for this drive are present.

Bit 00 = 1 (enable signal missing), if:

- the signal source in p0840 is a 0 signal.
- there is a "switching on inhibited".

Bit 01 = 1 (enable signal missing), if:

- the signal source in p0844 or p0845 is a 0 signal.

Bit 02 = 1 (enable signal missing), if:

- the signal source in p0848 or p0849 is a 0 signal.

Bit 03 = 1 (enable signal missing), if:

- the signal source in p0852 is a 0 signal.

Bit 04 = 1 (armature short-circuit active), if:

- the signal source in p1230 has a 1 signal.

Bit 05, Bit 06: Being prepared

Bit 08 = 1 (enable signal missing), if:

- safety functions have been enabled and STO is active.
- a safety-relevant signal is present with a STOP A response.

STO enabled via terminals:

- the pulse enable via terminal EP is missing (booksize: X21, chassis: X41), or the signal source in p9620 is for a 0 signal.

STO enabled via PROFIsafe or TM54F:

- STO is selected via PROFIsafe or TM54F.

Bit 09 = 1 (enable signal missing), if:

- the signal source in p0864 is a 0 signal.

Bit 10 = 1 (enable signal missing), if:

- the signal source in p1140 is a 0 signal.

Bit 11 = 1 (enable signal missing) if the velocity setpoint is frozen, because:

- the signal source in p1141 is a 0 signal.
- the velocity setpoint is entered from jogging and the two signal sources for jogging, bit 0 (p1055) and bit 1 (p1056) have a 1 signal.

Bit 12 = 1 (enable signal missing), if:

- the signal source in p1142 is a 0 signal.
- When activating the function module "basic positioner" (r0108.4 = 1), the signal source in p1142 is set to a 0 signal.

Bit 16 = 1 (enable signal missing), if:

- there is an OFF1 fault response. The system is only enabled if the fault is removed and was acknowledged and the "switching on inhibited" withdrawn with OFF1 = 0.

Bit 17 = 1 (enable signal missing), if:

- commissioning mode is selected (p0009 > 0 or p0010 > 0).
- there is an OFF2 fault response.
- the drive is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).

Bit 18 = 1 (enable signal missing), if:

- OFF3 has still not been completed or an OFF3 fault response is present.

Bit 19 = 1 (internal pulse enable missing), if:

- synchronization is running between the basic clock cycle, DRIVE-CLiQ clock cycle and application clock cycle.

Bit 20 = 1 (internal armature short-circuit active), if:

- the drive is not in the state "S4: Operation" or "S5x" (refer to function diagram 2610).
- the internal pulse enable is missing (r0046.19 = 0).

Bit 21 = 1 (enable signal missing), if:

The pulses have been enabled and the velocity setpoint has still not been enabled, because:

- the holding brake opening time (p1216) has still not expired.
- the motor has still not been magnetized (induction motor).

Bit 22: Being prepared

Bit 26 = 1 (enable signal missing), if:

- the drive is inactive (p0105 = 0) or is not operational (r7850[DO-Index]=0).
- the function "parking axis" is selected (BI: p0897 = 1 signal)..
- all power units of a parallel connection are deactivated (p0125, p0895).
- the drive device is in the "PROFInergy energy-saving mode" (r5600, CU-specific).

Bit 27 = 1 (enable signal missing), if:

- de-magnetizing has still not been completed (only for vector).
- Bit 28 = 1 (enable signal missing), if:
 - the holding brake is closed or has still not been opened.
- Bit 29 = 1 (enable signal missing), if:
 - the cooling unit ready signal via binector input p0266[1] missing.
- Bit 30 = 1 (velocity controller inhibited), if one of the following reasons is present:
 - a 0 signal is available via binector input p0856.
 - the function generator with current input is active.
 - the measuring function "current controller reference frequency characteristic" is active.
 - the pole position identification is active.
 - motor data identification is active (only certain steps).
- Bit 31 = 1 (enable signal missing), if:
 - the velocity setpoint from jog 1 or 2 is entered.

r0049[0...3] Motor data set/encoder data set effective / MDS/EDS effective

SERVO_DBSI_VD	Can be changed: - Data type: Unsigned8 P group: Displays, signals Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 2 Function diagram: 8565 Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the effective Motor Data Set (MDS) and the effective Encoder Data Sets (EDS).		
Index:	[0] = Motor Data Set MDS effective [1] = Encoder 1 Encoder Data Set EDS effective [2] = Encoder 2 Encoder Data Set EDS effective [3] = Encoder 3 Encoder Data Set EDS effective		
Dependency:	See also: p0187, p0188, p0189, r0838		
Note Value 99 means the following: No encoder assigned (not configured).			

r0050.0...3 CO/BO: Command Data Set CDS effective / CDS effective

SERVO_DBSI_VD	Can be changed: - Data type: Unsigned8 P group: Displays, signals Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 2 Function diagram: 8560 Unit selection: - Expert list: 1 Factory setting: -																									
Description:	Displays the effective Command Data Set (CDS).																											
Bit array:	<table><tr><th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FD</th></tr><tr><td>00</td><td>CDS effective bit 0</td><td>ON</td><td>OFF</td><td>-</td></tr><tr><td>01</td><td>CDS effective bit 1</td><td>ON</td><td>OFF</td><td>-</td></tr><tr><td>02</td><td>CDS effective bit 2</td><td>ON</td><td>OFF</td><td>-</td></tr><tr><td>03</td><td>CDS effective bit 3</td><td>ON</td><td>OFF</td><td>-</td></tr></table>	Bit	Signal name	1 signal	0 signal	FD	00	CDS effective bit 0	ON	OFF	-	01	CDS effective bit 1	ON	OFF	-	02	CDS effective bit 2	ON	OFF	-	03	CDS effective bit 3	ON	OFF	-		
Bit	Signal name	1 signal	0 signal	FD																								
00	CDS effective bit 0	ON	OFF	-																								
01	CDS effective bit 1	ON	OFF	-																								
02	CDS effective bit 2	ON	OFF	-																								
03	CDS effective bit 3	ON	OFF	-																								
Dependency:	See also: r0836																											
	Note The Command Data Set selected using a binector input (e.g. p0810) is displayed using r0836.																											

r0051.0...4 CO/BO: Drive Data Set DDS effective / DDS effective

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned8**Dyn. index:** -**Function diagram:** 8565**P group:** Displays, signals**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the effective Drive Data Set (DDS).

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	DDS effective bit 0	ON	OFF	-
01	DDS effective bit 1	ON	OFF	-
02	DDS effective bit 2	ON	OFF	-
03	DDS effective bit 3	ON	OFF	-
04	DDS effective bit 4	ON	OFF	-

Dependency:

See also: r0837

Note

The drive data set changeover is suppressed when selecting the motor identification, during the rotating measurement, the encoder calibration and the friction characteristic record.

r0060

SERVO_DBSI_VD

CO: Speed setpoint before the setpoint filter / n_set before filt**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 2701, 2704, 5020, 6030, 6799**P group:** Displays, signals**Unit group:** 3_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** p2000**Expert list:** 1**Min:****Max:****Factory setting:**

- [rpm]

- [rpm]

- [rpm]

Description:

Displays the actual speed setpoint at the input of the speed controller or U/f characteristic (after the interpolator).

Note

The speed setpoint is available smoothed (r0020) and unsmoothed (r0060).

r0060

SERVO_DBSI_VD (Lin)

CO: Velocity setpoint before the setpoint filter / v_set before filt**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 2701, 2704, 5020, 6030, 6799**P group:** Displays, signals**Unit group:** 4_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** p2000**Expert list:** 1**Min:****Max:****Factory setting:**

- [m/min]

- [m/min]

- [m/min]

Description:

Displays the actual velocity setpoint at the input of the velocity controller or U/f characteristic (after the interpolator).

Note

The velocity setpoint is available smoothed (r0020) and unsmoothed (r0060).

r0061[0...1] SERVO_DBSI_VD	CO: Actual speed unsmoothed / n_act unsmoothed		
	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 4700, 4710, 4715
	P group: Displays, signals	Unit group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min: - [rpm]	Max: - [rpm]	Factory setting: - [rpm]
Description:	Displays the unsmoothed actual speed values sensed by the encoders.		
Index:	[0] = Encoder 1 [1] = Encoder 2		
r0061[0...1] SERVO_DBSI_VD (Lin)	CO: Actual velocity unsmoothed / v_act unsmoothed		
	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 4700, 4710, 4715
	P group: Displays, signals	Unit group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min: - [m/min]	Max: - [m/min]	Factory setting: - [m/min]
Description:	Displays the unsmoothed actual velocity values sensed by the encoders.		
Index:	[0] = Encoder 1 [1] = Encoder 2		
r0062 SERVO_DBSI_VD	CO: Speed setpoint after the filter / n_set after filter		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5019, 5020, 5030, 5042, 5210
	P group: Displays, signals	Unit group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min: - [rpm]	Max: - [rpm]	Factory setting: - [rpm]
Description:	Display and connector output for the speed setpoint after the setpoint filters.		
r0062 SERVO_DBSI_VD (Lin)	CO: Velocity setpoint after the filter / v_set after filter		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5019, 5020, 5030, 5042, 5210
	P group: Displays, signals	Unit group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min: - [m/min]	Max: - [m/min]	Factory setting: - [m/min]
Description:	Display and connector output for the velocity setpoint after the setpoint filters.		

r0063 SERVO_DBSI_VD	CO: Actual speed smoothed / n_act smth		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 4700, 4710, 5019, 5300, 8019
	P group: Displays, signals	Unit group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min: - [rpm]	Max: - [rpm]	Factory setting: - [rpm]
Description:	Display and connector output for the smoothed speed actual value.		
Dependency:	See also: r0021, r0022, r0061, p1441		
	Note In encoderless operation, the speed actual value is calculated and can be smoothed using p1451. For operation with encoder, r0063 is smoothed with p1441. The speed actual value is available smoothed (r0021, r0022) and unsmoothed (r0061).		
r0063 SERVO_DBSI_VD (Lin)	CO: Actual velocity smoothed / v_act smth		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 4700, 4710, 5019, 5300, 8019
	P group: Displays, signals	Unit group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min: - [m/min]	Max: - [m/min]	Factory setting: - [m/min]
Description:	Display and connector output for the smoothed velocity actual value.		
Dependency:	See also: r0021, r0022, r0061, p1441		
	Note In encoderless operation, the velocity actual value is calculated and can be smoothed using p1451. For operation with encoder, r0063 is smoothed with p1441. The velocity actual value is available smoothed (r0021, r0022) and unsmoothed (r0061).		
r0064 SERVO_DBSI_VD	CO: Speed controller system deviation / n_ctrl SysDev		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5040, 6040
	P group: Displays, signals	Unit group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min: - [rpm]	Max: - [rpm]	Factory setting: - [rpm]
Description:	Displays the actual system deviation of the speed controller.		
	Note In servo control mode with active reference model, the system deviation to the P component of the speed controller is displayed.		
r0064 SERVO_DBSI_VD (Lin)	CO: Velocity controller system deviation / v_ctrl SysDev		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5040, 6040
	P group: Displays, signals	Unit group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min: - [m/min]	Max: - [m/min]	Factory setting: - [m/min]

Description:	Displays the actual system deviation of the velocity controller.		
	Note In servo control mode with active reference model, the system deviation to the P component of the velocity controller is displayed.		
r0077	CO: Current setpoint torque-generating / Iq_set		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Displays, signals Not for motor type: REL Min: - [Arms]	Calculated: - Dyn. index: - Unit group: 6_2 Scaling: p2002 Max: - [Arms]	Access level: 3 Function diagram: 5700, 5714, 5722 Unit selection: p0505 Expert list: 1 Factory setting: - [Arms]
Description:	Display and connector output for the torque/force-generating current setpoint.		
	Note This value is irrelevant for the U/f control mode.		
r0077	CO: Current setpoint force-generating / Iq_set		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Displays, signals Not for motor type: REL Min: - [Arms]	Calculated: - Dyn. index: - Unit group: 6_2 Scaling: p2002 Max: - [Arms]	Access level: 3 Function diagram: 5700, 5714, 5722 Unit selection: p0505 Expert list: 1 Factory setting: - [Arms]
Description:	Display and connector output for the force-generating current setpoint.		
	Note This value is irrelevant for the U/f control mode.		
r0078[0...1]	CO: Current actual value torque-generating / Iq_act		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Displays, signals Not for motor type: - Min: - [Arms]	Calculated: - Dyn. index: - Unit group: 6_2 Scaling: p2002 Max: - [Arms]	Access level: 3 Function diagram: 5700, 5714, 5730 Unit selection: p0505 Expert list: 1 Factory setting: - [Arms]
Description:	Display and connector output for the torque-generating current actual value (Iq_act).		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		
Dependency:	See also: p0045		
	Note These values are irrelevant for the U/f control mode. The torque-generating current actual value is available smoothed (r0030 with 100 ms, r0078[1] with p0045) and unsmoothed (r0078[0]).		

r0078[0...1]	CO: Current actual value force-generating / Iq_act		
SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5700, 5714, 5730
	P group: Displays, signals	Unit group: 6_2	Unit selection: p0505
	Not for motor type: -	Scaling: p2002	Expert list: 1
	Min:	Max:	Factory setting:
	- [Arms]	- [Arms]	- [Arms]
Description:	Display and connector output for the force-generating current actual value (Iq_act).		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		
Dependency:	See also: p0045		
	Note		
	These values are irrelevant for the U/f control mode.		
	The force-generating current actual value is available smoothed (r0030 with 100 ms, r0078[1] with p0045) and unsmoothed (r0078[0]).		

r0079[0...1]	CO: Torque setpoint total / M_set total		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5610
	P group: Displays, signals	Unit group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min:	Max:	Factory setting:
	- [Nm]	- [Nm]	- [Nm]
Description:	Display and connector output for the torque setpoint at the output of the speed controller (before clock cycle interpolation).		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		

r0079[0...1]	CO: Total force setpoint / F_set total		
SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5610
	P group: Displays, signals	Unit group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min:	Max:	Factory setting:
	- [N]	- [N]	- [N]
Description:	Display and connector output for the force setpoint at the output of the velocity controller (before clock cycle interpolation).		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045		

r0080	CO: Torque actual value / M_ActV		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5730
	P group: Displays, signals	Unit group: 7_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2003	Expert list: 1
	Min:	Max:	Factory setting:
	- [Nm]	- [Nm]	- [Nm]
Description:	Display and connector output for actual torque value.		

Note

The value is available smoothed (r0031) and unsmoothed (r0080).

r0080**CO: Force actual value / F_ActV**

SERVO_DBSI_VD (Lin)

Can be changed: -**Data type:** FloatingPoint32**P group:** Displays, signals**Not for motor type:** -**Min:**

- [N]

Calculated: -**Dyn. index:** -**Unit group:** 8_1**Scaling:** p2003**Max:**

- [N]

Access level: 3**Function diagram:** 5730**Unit selection:** p0505**Expert list:** 1**Factory setting:**

- [N]

Description:

Display and connector output for actual force value.

Note

The force actual value is available smoothed (r0031) and unsmoothed (r0080).

r0081**CO: Torque utilization / M_utilization**

SERVO_DBSI_VD

Can be changed: -**Data type:** FloatingPoint32**P group:** Displays, signals**Not for motor type:** -**Min:**

- [%]

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** PERCENT**Max:**

- [%]

Access level: 3**Function diagram:** 8012**Unit selection:** -**Expert list:** 1**Factory setting:**

- [%]

Description:

Displays the torque utilization as a percentage.

The torque utilization is obtained from the required smoothed torque referred to the torque limit.

Dependency:

See also: r0033

Note

The torque utilization is available smoothed (r0033) and unsmoothed (r0081).

The torque utilization is obtained from the required torque referred to the torque limit as follows:

- Positive torque: $r0081 = ((r0079 + p1532) / (r1538 - p1532)) * 100 \%$

- Negative torque: $r0081 = ((-r0079 + p1532) / (-r1539 + p1532)) * 100 \%$

The calculation of the torque utilization depends on the selected smoothing time constant (p0045).

r0081**CO: Force utilization / F_utilization**

SERVO_DBSI_VD (Lin)

Can be changed: -**Data type:** FloatingPoint32**P group:** Displays, signals**Not for motor type:** -**Min:**

- [%]

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** PERCENT**Max:**

- [%]

Access level: 3**Function diagram:** 8012**Unit selection:** -**Expert list:** 1**Factory setting:**

- [%]

Description:

Displays the force utilization as a percentage.

The force utilization is obtained from the required smoothed force referred to the force limit.

Dependency:

See also: r0033

Note

The force utilization is available smoothed (r0033) and unsmoothed (r0081).

The calculation of the force utilization depends on the selected smoothing time constant (p0045).

r0082[0...3] SERVO_DBSI_VD	CO: Active power actual value / P_act		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5730
	P group: Displays, signals	Unit group: 14_5	Unit selection: p0505
	Not for motor type: -	Scaling: r2004	Expert list: 1
	Min: - [kW]	Max: - [kW]	Factory setting: - [kW]
Description:	Displays the instantaneous active power.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045 [2] = Power drawn [3] = Power drawn smoothed		
	Note The mechanical active power is available smoothed (r0032 with 100 ms, r0082[1] with p0045) and unsmoothed (r0082[0]). For index [3]: Smoothing time constant = 4 ms		

r0082[0...3]	CO: Active power actual value / P_act		
SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5730
	P group: Displays, signals	Unit group: 14_8	Unit selection: p0505
	Not for motor type: -	Scaling: r2004	Expert list: 1
	Min: - [kW]	Max: - [kW]	Factory setting: - [kW]
Description:	Displays the instantaneous active power.		
Index:	[0] = Unsmoothed [1] = Smoothed with p0045 [2] = Power drawn [3] = Power drawn smoothed		
	Note The mechanical active power is available smoothed (r0032 with 100 ms, r0082[1] with p0045) and unsmoothed (r0082[0]). For index [3]: Smoothing time constant = 4 ms		

p0101[0...n] CU_I_840_VD, CU_NX_840_VD	Drive object numbers / DO numbers		
	Can be changed: C1(1)	Calculated: -	Access level: 2
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Topology	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 62	Factory setting: 0
Description:	The parameter contains the object number via which every drive object can be addressed. The number of an existing drive object is entered into each index. Value = 0: No drive object is defined.		

Note

The numbers are automatically allocated.

For the commissioning tool, this object number cannot be entered using the expert list, but is automatically assigned when inserting an object.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r0102[0...1]

CU_I_840_VD,
CU_NX_840_VD

Number of drive objects / DO count

Can be changed: -

Data type: Unsigned16

P group: Topology

Not for motor type: -

Min:

-

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

-

Access level: 2

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

-

Description:

Displays the number of existing or existing and prepared drive objects.

Index:

[0] = Existing drive objects

[1] = Existing and prepared drive objects

Dependency:

See also: p0101

Note

The numbers of the drive objects are in p0101.

For index [0]:

Displays the number of drive objects that have already been set up.

For index [1]:

Displays the number of drive objects that have already been set up and, in addition, the drive objects that still have to be set up.

p0103[0...n]

CU_I_840_VD,
CU_NX_840_VD

Application-specific view / Appl_spec view

Can be changed: C1(2)

Data type: Unsigned16

P group: -

Not for motor type: -

Min:

0

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

999

Access level: 2

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

0

Description:

The application-specific view of an existing drive object is entered into each index.

The parameter cannot be changed.

Dependency:

See also: p0107

Note

In the non-volatile memory, the application-specific views are defined in files with the following structure:

PDxxxyyy.ACX

xxx: Application-specific view (p0103)

yyy: Type of drive object (p0107)

Example:

PD052011.ACX

--> "011" stands for the drive object, type SERVO

--> "052" is the number of the view for this drive object

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r0103	Application-specific view / Appl_spec view		
A_INF_840_VD, SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	65535	-
Description:	Displays the application-specific view of the individual drive object.		
Dependency:	See also: r0107		

p0105	Activate/deactivate drive object / DO act/deact		
SERVO_DBSI_VD	Can be changed: T	Calculated: -	Access level: 2
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	2	1
Description:	Setting to activate/deactivate a drive object.		
Value:	0: Deactivate drive object 1: Activate drive object 2: Drive object deactivate and not present		
Dependency:	When activating drive objects with the safety functions enabled, the following applies: After reactivating, a warm restart (p0009 = 30, p0976 = 2, 3) or POWER ON should be carried out. See also: A01314		

NOTICE

The following applies when activating:

If components are inserted for the first time and the appropriate drive object is activated, then the drive system is automatically booted. To do this, the pulses of all of the drive objects must be suppressed.

Note

For value = 0, 2:

When a drive object is deactivated it no longer outputs any errors.

If value = 0:

All components of the drive object were completely commissioned and are deactivated using this value. They can be removed from the DRIVE-CLiQ without any error.

If a component has been deactivated, only the component with the correct serial number may be inserted, or none at all.

If value = 1:

All components of the drive object must be available for error-free operation.

If value = 2:

Components of a drive object in a project generated offline and set to this value must never be inserted in the actual topology from the very start. This means that the components are marked to be bypassed in the DRIVE-CLiQ line.

For components that comprise several individual components (e.g. Double Motor Modules), it is not permissible to set just one subset to this value.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0107[0...n] CU_I_840_VD, CU_NX_840_VD	Drive object type / DO type		
	Can be changed: C1(2) Data type: Integer16 P group: - Not for motor type: - Min: 0	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 11	Access level: 2 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0
Description:	The type of an existing drive object is entered into each index.		
Value:	0: - 3: SINAMICS I 4: SINAMICS NX/CX32 10: ACTIVE INFEED CONTROL 11: SERVO		
Dependency:	See also: p0103		
 CAUTION If you change this parameter and exit the device commissioning mode, then the complete software will be set up again and all of the previous drive parameter settings are deleted.			
Note The number (p0101) and the associated drive object type are in the same index. For SINAMICS S a drive object type can only be changed between SERVO and VECTOR. If you change the parameter and exit drive commissioning (p0009 from 2 to 0) the drive parameters are set up again. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

r0107 SERVO_DBSI_VD	Drive object type / DO type		
	Can be changed: - Data type: Integer16 P group: Closed-loop control Not for motor type: - Min: 11	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 11	Access level: 2 Function diagram: - Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the type of each drive object.		
Value:	11: SERVO		
Dependency:	See also: r0103		
Note DO: Drive Object			

p0108[0...n] CU_I_840_VD, CU_NX_840_VD	Drive objects function module / DO fct_mod		
	Can be changed: C1(2) Data type: Unsigned32 P group: - Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 2 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0000 0000 0000 0000 0000 0000 0000 0000 bin

Description: The function module of an existing drive object is entered into each index (see p0101, p0107).

The following bits are available for the Control Unit (Index 0):

Bit 18: Free function blocks

Bit 29: CAN

Bit 30: COMM BOARD

Bit 31: PROFINET

For all other drive objects (Index > 0), the significance of the bits should be taken from the display parameters r0108 of the drive object.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Bit 0	ON	OFF	-
	01	Bit 1	ON	OFF	-
	02	Bit 2	ON	OFF	-
	03	Bit 3	ON	OFF	-
	04	Bit 4	ON	OFF	-
	05	Bit 5	ON	OFF	-
	06	Bit 6	ON	OFF	-
	07	Bit 7	ON	OFF	-
	08	Bit 8	ON	OFF	-
	09	Bit 9	ON	OFF	-
	10	Bit 10	ON	OFF	-
	11	Bit 11	ON	OFF	-
	12	Bit 12	ON	OFF	-
	13	Bit 13	ON	OFF	-
	14	Bit 14	ON	OFF	-
	15	Bit 15	ON	OFF	-
	16	Bit 16	ON	OFF	-
	17	Bit 17	ON	OFF	-
	18	Bit 18	ON	OFF	-
	19	Bit 19	ON	OFF	-
	20	Bit 20	ON	OFF	-
	21	Bit 21	ON	OFF	-
	22	Bit 22	ON	OFF	-
	23	Bit 23	ON	OFF	-
	24	Bit 24	ON	OFF	-
	25	Bit 25	ON	OFF	-
	26	Bit 26	ON	OFF	-
	27	Bit 27	ON	OFF	-
	28	Bit 28	ON	OFF	-
	29	Bit 29	ON	OFF	-
	30	Bit 30	ON	OFF	-
	31	Bit 31	ON	OFF	-

Dependency: Active messages can prevent or influence activating a function module.

See also: A06860, A07089, F13010

Note

A "function module" is a functional expansion of a drive object that can be activated when commissioning.

DO: Drive Object

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r0108	Drive objects function module / DO fct_mod																																																																																			
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned32 P group: Closed-loop control Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 2 Function diagram: - Unit selection: - Expert list: 1 Factory setting: -																																																																																	
Description:	Displays the activated function module for the particular drive object.																																																																																			
Bit array:	<table> <tr> <th>Bit</th><th>Signal name</th><th>1 signal</th><th>0 signal</th><th>FD</th></tr> <tr><td>01</td><td>Extended torque control / Ext M_ctrl</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>02</td><td>Speed/torque control / n/M</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>03</td><td>Position control / Pos ctrl</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>04</td><td>Basic positioner / EPOS</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>06</td><td>DSC with spline / DSC spline</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>07</td><td>Advanced Positioning Control (APC) / APC</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>08</td><td>Extended setpoint channel / Ext setp</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>09</td><td>Extended Stopping and Retraction / ESR</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>10</td><td>Moment of inertia estimator / OBT / J_estimator / OBT</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>11</td><td>Spindle diagnostics / Spin_diag</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>12</td><td>Linear motor / Lin</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>13</td><td>Safety rotary axis / Safety rot</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>14</td><td>Extended brake control / Ext brake</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>17</td><td>Extended messages/monitoring / Ext msg</td><td>Activated</td><td>Not activated</td><td>-</td></tr> <tr><td>21</td><td>Extended current setpoint filter / Ext I_setp_filt</td><td>Activated</td><td>Not activated</td><td>-</td></tr> </table>	Bit	Signal name	1 signal	0 signal	FD	01	Extended torque control / Ext M_ctrl	Activated	Not activated	-	02	Speed/torque control / n/M	Activated	Not activated	-	03	Position control / Pos ctrl	Activated	Not activated	-	04	Basic positioner / EPOS	Activated	Not activated	-	06	DSC with spline / DSC spline	Activated	Not activated	-	07	Advanced Positioning Control (APC) / APC	Activated	Not activated	-	08	Extended setpoint channel / Ext setp	Activated	Not activated	-	09	Extended Stopping and Retraction / ESR	Activated	Not activated	-	10	Moment of inertia estimator / OBT / J_estimator / OBT	Activated	Not activated	-	11	Spindle diagnostics / Spin_diag	Activated	Not activated	-	12	Linear motor / Lin	Activated	Not activated	-	13	Safety rotary axis / Safety rot	Activated	Not activated	-	14	Extended brake control / Ext brake	Activated	Not activated	-	17	Extended messages/monitoring / Ext msg	Activated	Not activated	-	21	Extended current setpoint filter / Ext I_setp_filt	Activated	Not activated	-			
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p0115[0...6]	Sampling times for internal control loops / t_samp int ctrl			
SERVO_DBSI_VD	Can be changed: C1(3) Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: - Min: 0.00 [μs]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 16000.00 [μs]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: [0] 125.00 [μs] [1] 125.00 [μs] [2] 125.00 [μs] [3] 4000.00 [μs] [4] 1000.00 [μs] [5] 4000.00 [μs] [6] 4000.00 [μs]	
Description:	Sets the sampling times for the control loops. The default setting is made using p0112 and can only be individually changed for p0112 = 0 (expert).			
Recommendation:	When changing the sampling times of the current and speed controllers (also refer to p0115), then we recommend that after exiting commissioning (p0009 = 0) the controller settings are re-calculated using p0340 = 4.			

Index:

- [0] = Current controller
- [1] = Speed controller
- [2] = Flux controller
- [3] = Setpoint channel
- [4] = Position controller
- [5] = Positioning
- [6] = Technology controller

Dependency: The sampling times can only be separately set if p0112 is 0 (expert). If a sampling time is modified in the expert mode, then all of the sampling times with higher indices are automatically changed in the same ratio as the sampling time itself was changed. Slower time slices are only taken if the calculated sampling time is also permitted. Upper limit is 8 ms.

Higher-level controls must be calculated in integral ratios to lower-level controls (e.g. $p0115[1] = N * p0115[0]$; where N is an integer number). The sampling time of the speed controller ($p0115[1]$) can have as a maximum a value of 800% of the current controller sampling time ($p0115[0]$).

For servo drives, the maximum sampling time of the current controller is 250 μ s and for vector drives, 500 μ s.

The sampling times for setpoint channel ($p0115[3]$), position controller ($p0115[4]$), positioning ($p0115[5]$) and technology controller ($p0115[6]$) must have at least 2x the value of the current controller sampling time ($p0115[0]$).

Note

For function modules that can be activated (e.g. technology controller), the parameters values are pre-assigned.

For the Active Line Module (ALM) and Smart Line Module (SLM), the current and DC link voltage controllers operate with the same sampling time. For ALM/SLM the maximum current controller sampling time is 400 μ s.

For the Basic Line Module (BLM), the DC link voltage measurement operates in the current controller sampling time. For BLM booksize, only the current controller sampling time of 250 μ s is permitted. For BLM chassis, only the current controller sampling time of 2000 μ s is permitted.

For power unit type PM340 (r0203), only current controller sampling times of 62.5 μ s, 125 μ s, 250 μ s and 500 μ s can be set. The maximum current controller sampling time for servo drives and the minimum current controller sampling time for vector drives is 250 μ s.

If sampling times in p0115 are individually changed for p0112 = 0 (expert) then it must always be observed that the selected sampling times of the setpoint channel ($p0115[3]$), position controller ($p0115[4]$), positioning ($p0115[5]$) and technology controller ($p0115[6]$) are always greater than or equal to twice the current controller sampling time ($p0115[0]$).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0115[0...6] Sampling times for internal control loops / t_samp int ctrl

SERVO_DBSI_VD (Lin)	Can be changed: C1(3)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [μ s]	16000.00 [μ s]	[0] 125.00 [μ s]
			[1] 125.00 [μ s]
			[2] 125.00 [μ s]
			[3] 4000.00 [μ s]
			[4] 1000.00 [μ s]
			[5] 4000.00 [μ s]
			[6] 4000.00 [μ s]

Description: Sets the sampling times for the control loops.

The default setting is made using p0112 and can only be individually changed for p0112 = 0 (expert).

Recommendation: When changing the sampling times of the current and velocity controllers (p0115), then we recommend that after exiting commissioning (p0009 = 0) the controller settings are re-calculated using p0340 = 4.

Index:	[0] = Current controller [1] = Velocity controller [2] = Flux controller [3] = Setpoint channel [4] = Position controller [5] = Positioning [6] = Technology controller
Dependency:	The sampling times can only be separately set if p0112 is 0 (expert). If a sampling time is modified in the expert mode, then all of the sampling times with higher indices are automatically changed in the same ratio as the sampling time itself was changed. Slower time slices are only taken if the calculated sampling time is also permitted. Upper limit is 8 ms. Higher-level controls must be calculated in integral ratios to lower-level controls (e.g. $p0115[1] = N * p0115[0]$; where N is an integer number). The sampling time of the velocity controller (p0115[1]) can have as a maximum a value of 800% of the current controller sampling time (p0115[0]). For servo drives, the maximum sampling time of the current controller is 250 μs and for vector drives, 500 μs.
Note For function modules that can be activated (e.g. technology controller), the parameters values are pre-assigned. For the Active Line Module (ALM) and Smart Line Module (SLM), the current and DC link voltage controllers operate with the same sampling time. For ALM/SLM the maximum current controller sampling time is 400 μ s. For the Basic Line Module (BLM), the DC link voltage measurement operates in the current controller sampling time. For BLM booksize, only the current controller sampling time of 250 μ s is permitted. For BLM chassis, only the current controller sampling time of 2000 μ s is permitted. For power unit type PM340 (r0203), only current controller sampling times of 62.5 μ s, 125 μ s, 250 μ s and 500 μ s can be set. The maximum current controller sampling time for servo drives and the minimum current controller sampling time for vector drives is 250 μ s. If sampling times in p0115 are individually changed for p0112 = 0 (expert) then it must always be observed that the selected sampling times of the setpoint channel (p0115[3]), position controller (p0115[4]), positioning (p0115[5]) and technology controller (p0115[6]) are always greater than or equal to twice the current controller sampling time (p0115[0]). This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.	

p0120	Number of Power unit Data Sets (PDS) / PDS count		
SERVO_DBSI_VD	Can be changed: C1(3), C2(15)	Calculated: -	Access level: 2
	Data type: Unsigned8	Dyn. index: -	Function diagram: -
	P group: Data sets	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	1	8	1
Description:	Sets the number of Power unit Data Sets (PDS).		
	The value corresponds to the number of power units connected together for a parallel circuit configuration.		
Dependency:	See also: r0107		
	Note		
	This parameter is only significant for drive objects A_INF and VECTOR with a parallel circuit configuration.		
	This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p0121[0...n]	Power unit component number / PU comp_no		
SERVO_DBSI_VD	Can be changed: C1(4), C2(15)	Calculated: -	Access level: 3
	Data type: Unsigned8	Dyn. index: PDS, p0120	Function diagram: -
	P group: Data sets	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 199	Factory setting: 0
Description:	The power unit data set is assigned to a power unit using this parameter. This unique component number is assigned when parameterizing the topology. Only component numbers can be entered into this parameter that correspond to a power unit.		
Dependency:	See also: r0107		
	Note For parallel circuit configurations, the parameter index is assigned to a power unit. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p0121[0...n]	Power unit component number / PU comp_no		
A_INF_840_VD	Can be changed: C1(4), C2(15)	Calculated: -	Access level: 3
	Data type: Unsigned8	Dyn. index: PDS	Function diagram: -
	P group: Data sets	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 199	Factory setting: 0
Description:	The power unit data set is assigned to a power unit using this parameter. This unique component number is assigned when parameterizing the topology. Only component numbers can be entered into this parameter that correspond to a power unit.		
Dependency:	See also: p0107		
	Note For parallel circuit configurations, the parameter index is assigned to a power unit.		

p0130	Number of Motor Data Sets (MDS) / MDS count		
SERVO_DBSI_VD	Can be changed: C1(3), C2(15)	Calculated: -	Access level: 2
	Data type: Unsigned8	Dyn. index: -	Function diagram: 8575
	P group: Data sets	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 1	Max: 16	Factory setting: 1
Description:	Sets the number of Motor Data Sets (MDS).		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p0140 SERVO_DBSI_VD	Number of Encoder Data Sets (EDS) / EDS count		
	Can be changed: C1(3), C2(15)	Calculated: -	Access level: 2
	Data type: Unsigned8	Dyn. index: -	Function diagram: 8570
	P group: Data sets	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 1	Max: 16	Factory setting: 1

Description: Sets the number of Encoder Data Sets (EDS).

Note

When parameterizing the drive with "no encoder" there must be at least one encoder data set (p0140 >= 1). This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0170 Number of Command Data Sets (CDS) / CDS count

SERVO_DBSI_VD	Can be changed: C1(3)	Calculated: -	Access level: 2
	Data type: Unsigned8	Dyn. index: -	Function diagram: -
	P group: Commands	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	1	2	1

Description: Sets the number of Command Data Sets (CDS).

Note

It is possible to toggle between command parameters (BICO parameters) using this data set changeover. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0180 Number of Drive Data Sets (DDS) / DDS count

SERVO_DBSI_VD	Can be changed: C1(3), C2(15)	Calculated: -	Access level: 2
	Data type: Unsigned8	Dyn. index: -	Function diagram: 8565
	P group: Data sets	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	1	32	1

Description: Sets the number of Drive Data Sets (DDS).

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0187[0...n] Encoder 1 encoder data set number / Enc 1 EDS number

SERVO_DBSI_VD	Can be changed: C1(4), C2(15)	Calculated: -	Access level: 3
	Data type: Unsigned8	Dyn. index: DDS, p0180	Function diagram: 4700, 8570
	P group: Data sets	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	99	99

Description: Assign a drive data set (= index) the corresponding encoder data set (EDS) for encoder 1.

The value corresponds to the number of the assigned encoder data set.

Example:

Encoder 1 in drive data set 2 should be assigned encoder data set 0.

--> p0187[2] = 0

NOTICE

Writing to p0187 is rejected if the pole position identification is selected (p1982 = 1) and additional data sets with the same MDS data set (p0186) are available, which however have a different encoder data set number in p0187.

If all data sets with this MDS p0187 are to be changed, then the pole position identification of the data sets involved should be temporarily deselected (p1982 = 0), p0187 changed for all MDS data sets and then the pole position identification reselected (p1982 = 1).

If a motor with pole position identification is to be operated with two different encoders, then for this motor, two motor data sets should be introduced.

Note

A value of 99 means that no encoder has been assigned to this drive data set (not configured).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0188[0...n]

SERVO_DBSI_VD

Encoder 2 encoder data set number / Enc 2 EDS number**Can be changed:** C1(4), C2(15)**Calculated:** -**Access level:** 3**Data type:** Unsigned8**Dyn. index:** DDS, p0180**Function diagram:** 4700, 8570**P group:** Data sets**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

99

99

Description:

Assign a drive data set (= index) the corresponding encoder data set (EDS) for encoder 2.

The value corresponds to the number of the assigned encoder data set.

Example:

Encoder 2 in drive data set 2 should be assigned to encoder data set 1.

--> p0188[2] = 1

Note

A value of 99 means that no encoder has been assigned to this drive data set (not configured).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0189[0...n]

SERVO_DBSI_VD

Encoder 3 encoder data set number / Enc 3 EDS number**Can be changed:** C1(4), C2(15)**Calculated:** -**Access level:** 3**Data type:** Unsigned8**Dyn. index:** DDS, p0180**Function diagram:** 4700, 8570**P group:** Data sets**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

99

99

Description:

Assign a drive data set (= index) the corresponding encoder data set (EDS) for encoder 3.

The value corresponds to the number of the assigned encoder data set.

Note

A value of 99 means that no encoder has been assigned to this drive data set (not configured).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0199[0...24]	Drive object name / DO name		
All objects	Can be changed: C1	Calculated: -	Access level: 2
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 65535	Factory setting: 0
Description:	Freely assignable name for a drive object. For the commissioning tool, this name cannot be entered using the expert list, but is specified in the configuration wizards. The object name can be subsequently modified in the Project Navigator using standard Windows resources.		
	Note The parameter is not influenced by setting the factory setting.		
r0200[0...n]	Power unit code number actual / PU code no act		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: PDS, p0120	Function diagram: -
	P group: Converter	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Displays the unique code number of the power unit.		
	Note r0200 = p0201: No power unit found For parallel circuit configurations, the parameter index is assigned to a power unit.		
p0201[0...n]	Power unit code number / PU code no		
SERVO_DBSI_VD	Can be changed: C2(2)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: PDS, p0120	Function diagram: -
	P group: Converter	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 65535	Factory setting: 0
Description:	Sets the actual code number from r0200 to acknowledge the power unit being used. When commissioned for the first time, the code number is automatically transferred from r0200 into p0201.		
Dependency:	See also: F07815		
	NOTICE When p0201 = 10000, the rated power unit data is reloaded and dependent parameters are set (e.g. p0205, p0210, p0230, p0857, p1800). p0201 is then automatically assigned the value of r0200 if the code number of the power unit could be read. A warm start must be performed after this procedure (automatically if necessary).		
	Note The parameter is used to identify when the drive is being commissioned for the first time. The power unit commissioning can only be exited (p0201 = r0200), if the actual and acknowledged code numbers are identical (p0010 = 2). However, if the comparator in p9906 or p9908 is at 2 (low) or 3 (minimum), the power unit commissioning is automatically set to p0201 = r0200 upon exiting. When the code number is changed, the connection voltage (p0210) is checked and, if necessary, adjusted. For parallel circuit configurations, the parameter index is assigned to a power unit. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p0300[0...n]	Motor type selection / Mot type sel		
SERVO_DBSI_VD	Can be changed: C2(1, 3) Data type: Integer16 P group: Motor Not for motor type: - Min: 0	Calculated: - Dyn. index: MDS, p0130 Unit group: - Scaling: - Max: 10100	Access level: 1 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0
Description:	Selects the motor type or start to read in the motor parameters for a motor with DRIVE-CLiQ (p0300 = 10000). For p0300 < 10000 the following applies: The first digit of the parameter value always defines the general motor type and corresponds to the third-party motor belonging to a motor list: 1 = rotating induction motor 2 = rotating synchronous motor 3 = linear induction motor (reserved) 4 = linear synchronous motor The type information must be entered to filter motor-specific parameters and to optimize the operating characteristics and behavior. For example, for synchronous motors, power factor (p0308) is neither used nor displayed (in the BOP/AOP). The following applies for values < 100: Motor data must be manually entered. The following applies for values >= 100: Motor data are automatically loaded from an internal list.		
Value:	0: No motor 1: Induction motor 2: Synchronous motor 4: Synchronous motor linear 102: 1PH2 induction motor 103: 1PH3 induction motor 104: 1PH4 induction motor 107: 1PH7 induction motor 108: 1PH8 induction motor 111: xxxx induction motor OEM 132: 1FE2 induction motor 133: 2KJ3 induction motor 134: 1PM4 induction motor 136: 1PM6 induction motor 166: 1PL6 induction motor 191: 2SP1 induction motor 200: 1PH8 synchronous motor 203: 1FT2 synchronous motor 206: 1FT6 synchronous motor 207: 1FT7 synchronous motor 222: xxxx synchronous motor OEM 231: 1FG1 synchronous geared motor 232: 1FE2 synchronous motor 236: 1FK6 synchronous motor 237: 1FK7 synchronous motor 261: 1FE1 synchronous motor 272: 1FK2 synchronous motor 276: 1FS6 synchronous motor		

283:	1FW3 synchronous motor
286:	1FW6 synchronous motor
291:	2SP1 synchronous motor
401:	1FN1 synchronous motor linear
403:	1FN3 synchronous motor linear
406:	1FN6 synchronous motor linear
444:	xxxx synchronous motor linear OEM
10000:	Motor with DRIVE-CLiQ
10001:	Motor with DRIVE-CLiQ 2nd data set
10100:	Motor with DRIVE-CLiQ (only read in motor data)

Dependency:

When the motor type is changed, the code number in p0301 may be reset to 0.

If p0300 is changed during quick commissioning (p0010 = 1), then the matching technological application (p0500) is automatically pre-assigned. This does not occur when commissioning the motor (p0010 = 3). If p0300 = 10000 is written for a parameter download, p0500 is pre-assigned with DRIVE-CLiQ corresponding to the motor type.

NOTICE

If a catalog motor is selected (p0300 ≥ 100) and an associated motor code number (p0301), then the parameters that are associated with this list cannot be changed (write protection). The write protection is canceled if the motor type p0300 is set to a non-Siemens motor that matches p0301 (e.g. p0300 = 2 for p0301 = 2xxxx). Write protection is automatically canceled when the results of motor data identification are copied to the motor parameters.

The motor type of a catalog motor corresponds to the upper three digits of the code number or the following assignment (if the particular motor type is listed):

Type/code number ranges

102 / 102xx, 122xx

103 / 103xx

104 / 104xx, 114xx, 124xx

107 / 107xx, 117xx, 127xx

108 / 108xx, 118xx, 128xx, 138xx, 148xx, 158xx

132 / 132xx, 142xx

134 / 134xx, 144xx, 154xx

136 / 136xx, 146xx, 156xx

166 / 166xx, 176xx, 186xx

191 / 191xx

200 / 200xx, 210xx, 220xx

203 / 203xx, 213xx, 233xx, 243xx, 253xx, 273xx

206 / 206xx, 216xx, 226xx

207 / 207xx, 217xx, 227xx

231 / 231xx, 241xx, 251xx

232 / 232xx, 242xx, 252xx

235 / 235xx, 245xx, 255xx

237 / 237xx, 247xx, 257xx

261 / 261xx, 262xx, 263xx

272 / 272xx, 282xx, 292xx

283 / 283xx, 293xx

286 / 286xx, 296xx

403 / 403xx, 413xx

406 / 406xx, 416xx, 426xx

For OEM motors:

111 / 111xx, 112xx, 113xx

222 / 222xx, 223xx, 224xx

444 / 444xx, 445xx, 446xx

Note

With p0300 = 10000, for a motor with DRIVE-CLiQ, the motor parameters are automatically downloaded, with p0300 = 10001, the motor parameters of a second data set (if available).

If a motor type has not been selected (p0300 = 0), then the drive commissioning routine cannot be exited.

A motor type with a value above p0300 >= 100 describes motors for which a motor parameter list exists.

Motor types with a value below p0300 < 100 correspond to the selection of a third-party motor. When appropriately selected, this means that the motor parameters are pre-assigned the settings for a third-party motor.

This also applies for parameters for a motor with DRIVE-CLiQ. In this case p0300 can only be set to p0300 = 10000 or 10001 (read motor parameters) or to the corresponding non-Siemens motor (first digit of the motor code number) in order to be able to cancel the write protection.

With p0300 = 10100, when the system powers up, for a motor with DRIVE-CLiQ, the motor data are loaded, without subsequently newly calculating the control parameters. This means that control parameters that are already optimized are kept. To load the data, motor code number p0301 must match the code number of the connected encoder r0302.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0305[0...n]

SERVO_DBSI_VD

Rated motor current / Mot I_{rated}**Can be changed:** C2(1, 3)**Calculated:** -**Access level:** 1**Data type:** FloatingPoint32**Dyn. index:** MDS, p0130**Function diagram:** -**P group:** Motor**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.00 [Arms]

10000.00 [Arms]

0.00 [Arms]

Description:

Sets the rated motor current (rating plate).

NOTICE

When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.

If p0305 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).

Note

When the parameter value is entered the connection type of the motor (star-delta) must be taken into account.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0311[0...n]

SERVO_DBSI_VD

Rated motor speed / Mot n_{rated}**Can be changed:** C2(1, 3)**Calculated:** -**Access level:** 1**Data type:** FloatingPoint32**Dyn. index:** MDS, p0130**Function diagram:** -**P group:** Motor**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.0 [rpm]

210000.0 [rpm]

0.0 [rpm]

Description:

Sets the rated motor speed (rating plate).

Dependency:

If p0311 is changed and for p0314 = 0, the pole pair (r0313) is re-calculated automatically.

NOTICE

When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.

If p0311 is changed during quick commissioning (p0010 = 1), the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0311[0...n]			
SERVO_DBSI_VD (Lin)	Rated motor velocity / Mot v_{rated} Can be changed: C2(1, 3) Data type: FloatingPoint32 P group: Motor Not for motor type: - Min: 0.0 [m/min]	Calculated: - Dyn. index: MDS, p0130 Unit group: - Scaling: - Max: 6000.0 [m/min]	Access level: 1 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0.0 [m/min]
Description:	Sets the rated motor velocity (rating plate).		
Dependency:	The pole pair width is set in p0315.		
NOTICE When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection. If p0311 is changed during quick commissioning (p0010 = 1), the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).			
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p0315[0...n]			
SERVO_DBSI_VD (Lin)	Motor pole pair width / Mot PolePair width Can be changed: C2(1, 3) Data type: FloatingPoint32 P group: Motor Not for motor type: - Min: 1.00 [mm]	Calculated: - Dyn. index: MDS, p0130 Unit group: - Scaling: - Max: 1000.00 [mm]	Access level: 2 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 30.00 [mm]
Description:	Sets the pole pair width of the linear motor.		
NOTICE When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.			

p0316[0...n]			
SERVO_DBSI_VD	Motor torque constant / Mot kT Can be changed: C2(1), T, U Data type: FloatingPoint32 P group: Motor Not for motor type: ASM, SESM, REL, RESM Min: 0.000 [Nm/A]	Calculated: - Dyn. index: MDS, p0130 Unit group: 28_1 Scaling: - Max: 4000.000 [Nm/A]	Access level: 1 Function diagram: - Unit selection: p0100 Expert list: 1 Factory setting: 0.000 [Nm/A]
Description:	Sets the torque constant of the synchronous motor. p0316 = 0: The torque constant is calculated from the motor data. p0316 > 0: The selected value is used as torque constant.		
NOTICE When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.			
Note This parameter is not used for induction motors (p0300 = 1xx). This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This quarantees the functionality of the simplified drive model.			

p0316[0...n] Motor force constant / Mot kT

SERVO_DBSI_VD (Lin)

Can be changed: C2(1), T, U**Data type:** FloatingPoint32**P group:** Motor**Not for motor type:** ASM, SESM, REL, RESM**Min:**

0.000 [N/Arms]

Calculated: -**Dyn. index:** MDS, p0130**Unit group:** 29_1**Scaling:** -**Max:**

150000.000 [N/Arms]

Access level: 1**Function diagram:** -**Unit selection:** p0100**Expert list:** 1**Factory setting:**

0.000 [N/Arms]

Description:

Sets the force constant of the synchronous motor.

p0316 = 0:

The force constant is calculated from the motor data.

p0316 > 0:

The selected value is used as force constant.

NOTICE

When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.

p0322[0...n] Maximum motor speed / Mot n_max

SERVO_DBSI_VD

Can be changed: C2(1, 3)**Data type:** FloatingPoint32**P group:** Motor**Not for motor type:** -**Min:**

0.0 [rpm]

Calculated: -**Dyn. index:** MDS, p0130**Unit group:** -**Scaling:** -**Max:**

210000.0 [rpm]

Access level: 1**Function diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting:**

0.0 [rpm]

Description:

Sets the maximum motor speed.

Dependency:

See also: p1082

NOTICE

When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.

If p0322 is changed during quick commissioning (p0010 = 1), the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).

Note

The parameter has no significance for a value of p0322 = 0.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0322[0...n] Motor maximum velocity / Mot v_max

SERVO_DBSI_VD (Lin, Spin_diag)

Can be changed: C2(1, 3)**Data type:** FloatingPoint32**P group:** Motor**Not for motor type:** -**Min:**

0.0 [m/min]

Calculated: -**Dyn. index:** MDS, p0130**Unit group:** -**Scaling:** -**Max:**

2000.0 [m/min]

Access level: 1**Function diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting:**

0.0 [m/min]

Description:

Sets the maximum motor velocity.

Dependency:

See also: p1082

NOTICE

When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.

If p0322 is changed during quick commissioning (p0010 = 1), the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).

Note

The parameter has no significance for a value of p0322 = 0.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0322[0...n]

SERVO_DBSI_VD
(Spin_diag)

Maximum motor speed / Mot n_max

Can be changed: C2(1, 3)

Data type: FloatingPoint32

P group: Motor

Not for motor type: -

Min:

0.0 [rpm]

Calculated: -

Dyn. index: MDS, p0130

Unit group: -

Scaling: -

Max:

260000.0 [rpm]

Access level: 1

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

0.0 [rpm]

Description:

Sets the maximum motor speed.

Dependency:

See also: p1082

NOTICE

When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.

If p0322 is changed during quick commissioning (p0010 = 1), the maximum speed p1082, which is also associated with quick commissioning, is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).

Note

The parameter has no significance for a value of p0322 = 0.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0323[0...n]

SERVO_DBSI_VD

Maximum motor current / Mot I_max

Can be changed: C2(1, 3)

Data type: FloatingPoint32

P group: Motor

Not for motor type: ASM, SESM, RESM

Min:

0.00 [Arms]

Calculated: -

Dyn. index: MDS, p0130

Unit group: -

Scaling: -

Max:

20000.00 [Arms]

Access level: 1

Function diagram: 5722

Unit selection: -

Expert list: 1

Factory setting:

0.00 [Arms]

Description:

Sets the maximum permissible motor current (e.g. de-magnetizing current for synchronous motors).

NOTICE

When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.

If p0323 is changed during quick commissioning (p0010 = 1), then the maximum current p0640 is pre-assigned accordingly. This is not the case when commissioning the motor (p0010 = 3).

Note

The parameter has no effect for induction motors.

For synchronous motors, a value must always be entered for the maximum motor current.

p0323 is a motor data. The user-selectable current limit is entered into p0640.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0341[0...n]			
SERVO_DBSI_VD	Motor moment of inertia / Mot m_inert		
	Can be changed: T, U	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dyn. index: MDS, p0130	Function diagram: 5042, 5210, 6020, 6030, 6031
	P group: Motor	Unit group: 25_1	Unit selection: p0100
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: - [kgm²]	Max: - [kgm²]	Factory setting: - [kgm²]
Description:	Sets the motor moment of inertia (without load).		
Dependency:	IEC drives (p0100 = 0): unit kg m^2 NEMA drives (p0100 = 1): unit lb ft^2 The parameter value is included, together with p0342, in the rated starting time of the motor.		
NOTICE			
When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.			
Note			
SERVO: p0341 * p0342 + p1498 influence the speed/torque precontrol in encoderless operation.			
VECTOR: The product of p0341 * p0342 is used when the speed controller (p0340 = 4) is calculated automatically. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p0341[0...n]			
SERVO_DBSI_VD (Lin)	Motor weight / Mot weight		
	Can be changed: T, U	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dyn. index: MDS, p0130	Function diagram: 5042, 5210
	P group: Motor	Unit group: 27_1	Unit selection: p0100
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: - [kg]	Max: - [kg]	Factory setting: - [kg]
Description:	Sets the high moments of inertia (without load).		
Dependency:	IEC drives (p0100 = 0): unit kg m^2 NEMA drives (p0100 = 1): unit lb ft^2 The parameter value is included, together with p0342, in the rated starting time of the motor.		
NOTICE			
When selecting a catalog motor (p0301), this parameter is automatically pre-assigned and is write protected. Information in p0300 should be carefully observed when removing write protection.			
Note			
p0341 * p0342 + p1498 influence the speed/torque precontrol in encoderless operation.			

p0404[0...n]			
SERVO_DBSI_VD	Encoder configuration effective / Enc_config eff		
	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: EDS, p0140	Function diagram: 4010, 4704
	P group: Encoder	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: 0000 0000 0000 0000 0000 0000 0000 0000 bin
Description:	Settings for the basic encoder properties.		

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Linear encoder	Yes	No	-
	01	Absolute encoder	Yes	No	-
	02	Multiturn encoder	Yes	No	-
	03	Track A/B square-wave	Yes	No	-
	04	Track A/B sine	Yes	No	-
	05	Track C/D	Yes	No	-
	06	Hall sensor	Yes	No	-
	08	EnDat encoder	Yes	No	-
	09	SSI encoder	Yes	No	-
	10	DRIVE-CLiQ encoder	Yes	No	-
	11	Digital encoder	Yes	No	-
	12	Equidistant zero mark	Yes	No	-
	13	Irregular zero mark	Yes	No	-
	14	Distance-coded zero mark	Yes	No	-
	15	Commutation with zero mark (not ASM)	Yes	No	-
	16	Acceleration	Yes	No	-
	17	Track A/B analog	Yes	No	-
	20	Voltage level 5 V	Yes	No	-
	21	Voltage level 24 V	Yes	No	-
	22	Remote sense (only SMC30)	Yes	No	-
	23	Resolver excitation	Yes	No	-

NOTICE

This parameter is automatically preassigned for encoders from the encoder list and for identify encoder (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

Note

ZM: Zero mark

SMC: Sensor Module Cabinet

If a technique to determine the commutation information/data has not been selected (e.g. track C/D, Hall sensor), and the encoder pulse number is an integer multiple of the pole number, then the following applies:

The track A/B is adjusted to match the magnetic position of the motor.

For bit 01, 02 (absolute encoder, multiturn encoder):

These bits can only be selected for EnDat encoders, SSI encoders or DRIVE-CLiQ encoders.

For bit 10 (DRIVE-CLiQ encoder):

This bit is only used for the large-scale integrated DRIVE-CLiQ encoders that provide their encoder data directly in DRIVE-CLiQ format without converting this data. This bit is not, therefore, set for first-generation DRIVE-CLiQ encoders.

For bit 12 (equidistant zero mark):

The zero marks occur at regular intervals (e.g. rotary encoder with 1 zero mark per revolution or linear encoder with constant zero mark distance).

The bit activates monitoring of the zero mark distance (p0424/p0425, linear/rotary) or in the case of the linear encoder with 1 zero mark and p0424 = 0 zero mark monitoring is activated.

For bit 13 (irregular zero mark):

The zero marks occur at irregular intervals (e.g. a linear scale with only 1 zero mark in the traversing range). The zero mark distance is not monitored.

For bit 14 (distance-coded zero mark):

The distance (clearance) between two or several consecutive zero marks allows the absolute position to be calculated.

For bit 15 (commutation with zero mark):

Only applicable for synchronous motors.

The function can be deselected by priority via p0430.23.

For distance-coded zero marks, the following applies:

The phase sequence of the C/D track (if available) must be the same as the phase sequence of the encoder (A/B track).

The phase sequence of the Hall signal (if available) must be the same as the phase sequence of the motor. Further, the position of the Hall sensor must be mechanically adjusted to the motor EMF.

The fine synchronization is only started after two zero marks have been passed.

p0407[0...n]

SERVO_DBSI_VD

Linear encoder grid division / Enc grid div

Can be changed: C2(4)

Calculated: -

Access level: 3

Data type: Unsigned32

Dyn. index: EDS, p0140

Function diagram: 4010, 4704

P group: Encoder

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0 [nm]

250000000 [nm]

16000 [nm]

Description:

Sets the grid division for a linear encoder.

In conjunction with the values in p0418/p0419, the grid division defines the transfer format for position actual values Gn_XIST1 (r0482) and Gn_XIST2 (r0483).

NOTICE

This parameter is automatically preassigned for encoders from the encoder list and for identify encoder (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

Note

The lowest permissible value is 250 nm.

This value does not always correspond to the grid division of the measuring device. For a DRIVE-CLiQ encoder, a value is entered here that facilitates optimum transfer of the resolution (p0422).

p0408[0...n]	Rotary encoder pulse number / Rot enc pulse no		
SERVO_DBSI_VD	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: EDS, p0140	Function diagram: 4010, 4704
	P group: Encoder	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	16777215	2048

Description: Sets the number of pulses for a rotary encoder.
In conjunction with the values in p0418/p0419, the pulse number defines the transfer format for position actual values Gn_XIST1 (r0482) and Gn_XIST2 (r0483).

NOTICE

This parameter is automatically preassigned for encoders from the encoder list and for identify encoder (p0400).
When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

Note

The smallest permissible value is 1 pulse.

The number of pole pairs for a resolver is entered here.

This value does not always correspond to the pulse number of the measuring device. For a DRIVE-CLiQ encoder, a value is entered here that facilitates optimum transfer of the resolution (p0423).

p0410[0...n]	Encoder inversion actual value / Enc inv ActV		
SERVO_DBSI_VD	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: EDS, p0140	Function diagram: 4704, 4710, 4711, 4715
	P group: Encoder	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0000 bin

Description: Setting to invert actual values.

Bit	Signal name	1 signal	0 signal	FD
00	Invert speed actual value	Yes	No	4710, 4711, 4715
01	Invert position actual value	Yes	No	4704

Note

The inversion influences the following parameters:

Bit 00: r0061, r0063 (exception: encoderless control), r0094

Bit 01: r0482, r0483

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0410[0...n]	Encoder inversion actual value / Enc inv ActV		
SERVO_DBSI_VD (Lin)	Can be changed: C2(4)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: EDS, p0140	Function diagram: 4704, 4710, 4711, 4715
	P group: Encoder	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0000 bin

Description: Setting to invert actual values.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Invert velocity actual value	Yes	No	4710, 4711, 4715
	01	Invert position actual value	Yes	No	4704

Note

The inversion influences the following parameters:

Bit 00: r0061, r0063 (exception: encoderless control), r0094

Bit 01: r0482, r0483

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0418[0...n] Fine resolution Gx_XIST1 (in bits) / Enc fine Gx_XIST1

SERVO_DBSI_VD

Can be changed: C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned8**Dyn. index:** EDS, p0140**Function diagram:** 4010, 4704**P group:** Encoder**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

18

11

Description:

Sets the fine resolution in bits of the incremental position actual values.

Note

The parameter applies for the following process data:

- Gx_XIST1

- Gx_XIST2 for reference mark or flying measurement

The fine resolution specifies the fraction between encoder pulses. Depending on the physical measurement principle, an encoder pulse can be broken down into a different number of fractions (e.g. square-wave encoder: 2 bit = resolution 4, sin/cos encoder: Typical 11 bit = resolution 2048).

For a square-wave encoder, with the factory setting, the least significant bits have the value zero, i.e. they do not supply any useful information.

For especially high quality measuring systems, the fine resolution must be increased corresponding to the available accuracy.

p0419[0...n] Fine resolution absolute value Gx_XIST2 (in bits) / Enc fine Gx_XIST2

SERVO_DBSI_VD

Can be changed: C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned8**Dyn. index:** EDS, p0140**Function diagram:** 4704, 4710**P group:** Encoder**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

18

9

Description:

Sets the fine resolution in bits of the absolute position actual values.

Dependency:

See also: p0418

Note

This parameter applies to process data Gx_XIST2 when reading the absolute value.

p0421[0...n] Absolute encoder rotary multiturn resolution / Enc abs multiturn

SERVO_DBSI_VD

Can be changed: C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** EDS, p0140**Function diagram:** 4704**P group:** Encoder**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

4294967295

4096

Description: Sets the number of rotations that can be resolved for a rotary absolute encoder.

NOTICE

This parameter is automatically preassigned for encoders from the encoder list and for identify encoder (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

p0424[0...n]**Encoder linear zero mark distance / Enc lin ZM_dist**

SERVO_DBSI_VD

Can be changed: C2(4)**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** EDS, p0140**Function diagram:** -**P group:** Encoder**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0 [mm]

65535 [mm]

20 [mm]

Description:

Sets the distance between two zero marks for a linear encoder.

This information is used for zero mark monitoring.

NOTICE

This parameter is automatically preassigned for encoders from the encoder list and for identify encoder (p0400). When selecting a catalog encoder, this parameter cannot be changed (write protection). Information in p0400 should be carefully observed when removing write protection.

Note

For distance-coded zero marks this parameter signifies the basic distance.

r0456[0...2]**Encoder configuration supported / Enc_config sup**

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Encoder**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the encoder configuration supported by the Sensor Module.

Index:

[0] = Encoder 1

[1] = Encoder 2

[2] = Encoder 3

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Linear encoder	Yes	No	-
01	Absolute encoder	Yes	No	-
02	Multiturn encoder	Yes	No	-
03	Track A/B square-wave	Yes	No	-
04	Track A/B sine	Yes	No	-
05	Track C/D	Yes	No	-
06	Hall sensor	Yes	No	-
08	EnDat encoder	Yes	No	-
09	SSI encoder	Yes	No	-
10	DRIVE-CLiQ encoder	Yes	No	-
11	Digital encoder	Yes	No	-
12	Equidistant zero mark	Yes	No	-
13	Irregular zero mark	Yes	No	-
14	Distance-coded zero mark	Yes	No	-
15	Commutation with zero mark (not ASM)	Yes	No	-

16	Acceleration	Yes	No	-
17	Track A/B analog	Yes	No	-
20	Voltage level 5 V	Yes	No	-
21	Voltage level 24 V	Yes	No	-
22	Remote sense (only SMC30)	Yes	No	-
23	Resolver excitation	Yes	No	-

Dependency: See also: p0404

Note

ZM: Zero mark

This parameter is only used for diagnostics.

A value of zero is displayed if an encoder is not present.

r0463[0...2] Encoder serial number part 4 / Enc ser_no 4

SERVO_DBSI_VD

Can be changed: -

Calculated: -

Access level: 3

Data type: Unsigned32

Dyn. index: -

Function diagram: -

P group: Encoder

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description: Displays the actual serial number part 4 of the appropriate encoder.

Index:
[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Dependency: See also: r0464

r0464[0...2] Encoder serial number part 5 / Enc ser_no 5

SERVO_DBSI_VD

Can be changed: -

Calculated: -

Access level: 3

Data type: Unsigned32

Dyn. index: -

Function diagram: -

P group: Encoder

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description: Displays the actual serial number part 5 of the appropriate encoder.

Index:
[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

Dependency: See also: r0463

r0469[0...2] Absolute encoder linear measuring step / Enc lin meas step

SERVO_DBSI_VD

Can be changed: -

Calculated: -

Access level: 3

Data type: Unsigned32

Dyn. index: -

Function diagram: -

P group: Encoder

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

- [nm]

- [nm]

- [nm]

Description: Displays the resolution of the absolute position for a linear absolute encoder.

Index:
[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

r0470[0...2]	Redundant coarse position value valid bits / Valid bits		
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned16 P group: Encoder Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the valid bits of the redundant coarse position value.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	See also: p9323, p9523		
r0471[0...2]	Redundant coarse position value fine resolution bits / Fine bit		
SERVO_DBSI_VD	Can be changed: - Data type: Integer16 P group: Encoder Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the number of valid bits for the fine resolution of the redundant coarse position value.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Dependency:	See also: p9324, p9524		
r0472[0...2]	Redundant coarse position value relevant bits / Relevant bits		
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned16 P group: Encoder Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the number of relevant bits for the redundant coarse position value.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
r0473[0...2]	Non safety-relevant measuring steps position value pos1 / nsrPos1		
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned32 P group: Encoder Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the non safety-relevant measuring steps of POS1.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		

r0474[0...2] Redundant coarse position value configuration / Red pos config

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Encoder**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the encoder configuration for the redundant coarse position value.

Index:

[0] = Encoder 1

[1] = Encoder 2

[2] = Encoder 3

Bit array:**Bit** **Signal name****1 signal****0 signal****FD**

00 Incrementer

Yes

No

-

01 Encoder CRC least significant byte first

Yes

No

-

02 Redundant coarse position val. most significant bit left-aligned

Yes

No

-

04 Binary comparison not possible

Yes

No

-

05 Single-channel encoder

Yes

No

-

Dependency:

See also: p9315, p9515

r0475[0...2] Gx_XIST1 coarse position safety most significant bit / Gx_XIST1 safe MSB

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Encoder**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the bit number for the safe most significant bit (MSB) of the Gx_XIST1 coarse position.

Index:

[0] = Encoder 1

[1] = Encoder 2

[2] = Encoder 3

Note

MSB: Most Significant Bit

r0479[0...2] CO: Diagnostics encoder position actual value Gn_XIST1 / Diag Gn_XIST1

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Integer32**Dyn. index:** -**Function diagram:** 4704**P group:** Encoder**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and connector output for the encoder actual position value Gn_XIST1 according to PROFIdrive for diagnostics. In contrast to r0482, the value is updated in each DRIVE-CLiQ basic clock cycle and displayed with sign.

Index:

[0] = Encoder 1

[1] = Encoder 2

[2] = Encoder 3

⚠ CAUTION

Following ramping-up or after a data set changeover, the new value is present at connector inputs which are interconnected to connector output r0479 and under certain circumstances take 100 ms to become available.

Reason:

These interconnections are updated in the background, unlike interconnections involving other connector outputs (e.g. CO: r0482).

The value is immediately available when non-cyclically reading r0479 (e.g. via the expert list).

r0482[0...2]	CO: Encoder actual position value Gn_XIST1 / Enc Gn_XIST1		
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned32 P group: Encoder Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 4700, 4702, 4704, 4735, 4740, 4750 Unit selection: - Expert list: 1 Factory setting: -
Description:	Display and connector output for the encoder actual position value Gn_XIST1 according to PROFIdrive.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
Note			
- this value is reset if necessary when the "parking encoder" (r0481.14) function is deselected.			
- in this value, the measuring gear (p0432, p0433) is only taken into account when the position tracking is activated (p0411.0 = 1).			
- the update time for the position control (EPOS) corresponds to the position controller clock cycle (p0115[4]).			
- the update time in isochronous operation corresponds to the bus cycle time r2064[1].			
- the update time in isochronous operation and with position control (EPOS) corresponds to the position controller sampling time (p0115[4]).			
- the update time in non-isochronous operation or without position control (EPOS) must be determined from the default bus cycle time and the minimum cycle time:			
The default bus cycle time is the lowest common multiple (LCM) of all current controller sampling times (p0115[0]) in the drive group (infeed + drives).			
The minimum cycle time is four times the maximum of all current controller sampling times (p0115[0]) in the drive group (infeed + drives).			
If the minimum cycle time is greater than the default bus cycle time, then the update time corresponds to the minimum cycle time; otherwise, the update time corresponds to the default bus cycle time.			
The minimum update time is 1 ms.			
Example 1: infeed, servo			
Default bus cycle time = KGV(250 µs, 125 µs) = 250 µs			
Minimum cycle time = 4 * MAX(250 µs, 125 µs) = 4 * 250 µs = 1 ms			
-> update time = 1 ms			
Example 2: infeed, servo, vector			
Default bus cycle time = KGV(250 µs, 125 µs, 400 µs) = 2 ms			
Minimum cycle time = 4 * MAX(250 µs, 125 µs, 400 µs) = 4 * 400 µs = 1.6 ms			
-> update time = 2 ms			

r0483[0...2]	CO: Encoder actual position value Gn_XIST2 / Enc Gn_XIST2		
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned32 P group: Encoder Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 4704, 4750 Unit selection: - Expert list: 1 Factory setting: -

Description: Displays the encoder actual position value Gn_XIST2 according to PROFIdrive.

Index:
[0] = Encoder 1
[1] = Encoder 2
[2] = Encoder 3

NOTICE

The encoder position actual value must be requested using the encoder control word Gn_STW.13.

Note

- in this value, the measuring gear (p0432, p0433) is only taken into account when the position tracking is activated (p0411.0 = 1).
- if GxZSW.15 = 1 (r0481), then an error code with the following significance is located in Gx_XIST2 (r0483):
1: Encoder fault.
2: Possible position shift in Gx_XIST1.
3: Encoder parking not possible.
4: Cancellation, reference block search (e.g. reference mark not available or input terminal for external zero mark not set). Zero mark is requested, however according to p0404.12/13/14 there is no zero mark (alarm A07565).
5: Cancellation, fetch reference value (e.g. illegal change from reference mark search to flying measurement).
6: cancellation, flying measurement (e.g. input terminal for probe not set).
7: Cancellation, fetch measured value (e.g. illegal change from flying measurement to reference mark search).
8: Abort, absolute value transfer.
3841: Function not supported.
4097: Abort, reference mark search due to an initialization error. Possible cause: defective Control Unit hardware.
4098: Abort, flying measurement due to an initialization error. Possible cause: defective Control Unit hardware.
4099: Abort, reference mark search due to a measuring error. Possible cause: too many measuring pulses have occurred.
4100: Abort, flying measurement due to a measuring error. Possible cause: too many measuring pulses have occurred.

r0836.0...3 CO/BO: Command Data Set CDS selected / CDS selected

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned8**Dyn. index:** -**Function diagram:** 8560**P group:** Displays, signals**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description: Displays the command data set (CDS) selected via the binector input.

Bit	Signal name	1 signal	0 signal	FD
00	CDS selection bit 0	ON	OFF	-
01	CDS selection bit 1	ON	OFF	-
02	CDS selection bit 2	ON	OFF	-
03	CDS selection bit 3	ON	OFF	-

Dependency: See also: r0050

Note

Command data sets are selected via binector input p0810 and following.
The currently effective command data set is displayed in r0050.

r0837.0...4 CO/BO: Drive Data Set DDS selected / DDS selected

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned8**Dyn. index:** -**Function diagram:** 8565**P group:** Displays, signals**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description: Displays the drive data set (DDS) selected via the binector input.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	DDS selection bit 0	ON	OFF	-
	01	DDS selection bit 1	ON	OFF	-
	02	DDS selection bit 2	ON	OFF	-
	03	DDS selection bit 3	ON	OFF	-
	04	DDS selection bit 4	ON	OFF	-
Dependency:	See also: r0051				

Note

Drive data sets are selected via binector input p0820 and following.

The currently effective drive data set is displayed in r0051.

If there is only one data set, then a value of 0 is displayed in this parameter and not the selection via binector inputs.

r0838[0...3]	Motor/Encoder Data Set selected / MDS/EDS selected		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned8	Dyn. index: -	Function diagram: 8565
	P group: Displays, signals	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Displays the selected Motor Data Set (MDS) and the selected Encoder Data Sets (EDS).		
Index:	[0] = Motor Data Set MDS selected [1] = Encoder 1 Encoder Data Set EDS selected [2] = Encoder 2 Encoder Data Set EDS selected [3] = Encoder 3 Encoder Data Set EDS selected		
Dependency:	See also: r0049, p0187, p0188, p0189		
	Note Value 99 means the following: No encoder assigned (not configured).		

r0863.0...2	CO/BO: Drive coupling status word/control word / CoupleZSW/STW				
A_INF_840_VD, SERVO_DBSI_VD	Can be changed: -		Calculated: -		Access level: 2
	Data type: Unsigned16		Dyn. index: -		Function diagram: -
	P group: Commands		Unit group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min:		Max:		Factory setting:
	-		-		-
Description:	Display and BICO output for the status word and control word of the drive coupling.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Closed-loop control operation	Yes	No	2610, 8710, 8810, 8910
	01	Energize contactor	Yes	No	2610, 2634, 7990, 8734, 8834, 8934
	02	Infeed line supply failure	Yes	No	-
Dependency:	See also: p0864				

Note

For bit 00:

Bit 0 signals that the infeed is ready.

When the operating signal is transferred via binector output r0863.0 this allows several drives to start (run-up) staggered over time when they are simultaneously switched on.

To realize this, the following connections/interconnections are required:

Drive 1: Interconnect binector input p0864 with binector output r0863.0 of the infeed

Drive 2: Interconnect binector input p0864 with binector output r0863.0 of drive 1

Drive 3: Interconnect binector input p0864 with binector output r0863.0 of drive 2 etc.

The first drive only transfers the operating signal to the next drive after it has reached its ready condition.

For bit 01:

Bit 1 is used to control an external line contactor.

For bit 02:

This bit only signals line supply failure for Active Infeed (A_INF) and Smart Infeed (S_INF).

p0864

SERVO_DBSI_VD

BI: Infeed operation / INF operation**Can be changed:** T**Calculated:** -**Access level:** 2**Data type:** Unsigned32 / Binary**Dyn. index:** -**Function diagram:** 2610, 8710, 8910**P group:** Commands**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

0

Description:

Sets the signal source for the operating signal of the infeed (e.g. BO: r0863.0).

Dependency:

See also: r0863

Note

The sequence control of a servo/vector drive requires this signal.

The following applies for an infeed without DRIVE-CLiQ:

For these infeeds, the "ready" message is available via an output terminal. This signal must be connected to a digital input. The drives supplied from this infeed must use this signal as ready signal (BI: p0864 = digital input).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r0887.0...13

SERVO_DBSI_VD (ESR)

BO: ESR status word / ESR ZSW**Can be changed:** -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** -**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the status word for the "ESR" function.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	ESR monitoring activated	Yes	No	-
01	Profile "Extended stopping (integrated in the drive)" active	Yes	No	-
02	Profile "Extended retraction (integrated in the drive)" active	Yes	No	-
03	Profile "Generator operation (Vdc controller)" active	Yes	No	-
09	ESR enabled (p0889)	Yes	No	-
10	ESR trigger active (p0890)	Yes	No	-
11	ESR OFF ramp OFF1/OFF3 (p0891)	OFF1	OFF3	-
12	ESR response initiated	Yes	No	3082
13	ESR response presently running	Yes	No	-

Dependency: See also: p0888

Note

ESR: Extended Stop and Retract

p0888 ESR configuration / ESR configuration

SERVO_DBSI_VD (ESR)	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: Integer16	Dyn. index: -	Function diagram: 3082
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	4	0

Description: Sets the configuration for the "ESR" function.

Value:	0: No function
	1: Extended stopping (integrated in the drive), n_set
	2: Extended retraction (function integrated in the drive)
	3: Generator operation (Vdc controller)
	4: Extended stopping (integrated in the drive), n_act

Dependency: See also: p0892, p0893, p1240

CAUTION

For p0888 = 3 generator operation (Vdc controller) must be correspondingly configured using p1240.

Note

ESR: Extended Stop and Retract

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0892 ESR timer / ESR timer

SERVO_DBSI_VD (ESR)	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [s]	20.00 [s]	0.50 [s]

Description: Sets the timer for the brake delay for the "ESR" function.

Dependency: The following applies when Safety Integrated is enabled:
p0892 < p9580 (SI motion, pulse cancellation delay bus failure)
If the setting for the times is not observed, then ESR OFF ramp will not be able to be fully completed.
See also: p0888

Note

ESR: Extended Stop and Retract

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0893 ESR velocity / ESR velocity

SERVO_DBSI_VD (ESR, Lin)	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 3082
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-340.28235E36 [m/min]	340.28235E36 [m/min]	0 [m/min]

Description: Sets the speed/velocity, which, when triggered, is approached with an OFF3 ramp.
The timer in p0892 is the total time that elapses for the approach and constant velocity travel. After this, depending on the setting in p0891, an OFF1 ramp or OFF3 ramp is realized.

Dependency: See also: p0888, p0892

Note

This parameter is only of significance for the "Retract" profile (p0888 = 2).

ESR: Extended Stop and Retract

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0893 ESR speed / ESR speed

SERVO_DBSI_VD (ESR)	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 3082
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-340.28235E36 [rpm]	340.28235E36 [rpm]	0 [rpm]

Description: Sets the speed/velocity, which, when triggered, is approached with an OFF3 ramp.
The timer in p0892 is the total time that elapses for the approach and constant velocity travel. After this, depending on the setting in p0891, an OFF1 ramp or OFF3 ramp is realized.

Dependency: See also: p0888, p0892

Note

This parameter is only of significance for the "Retract" profile (p0888 = 2).

ESR: Extended Stop and Retract

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r0896.0 BO: Parking axis, status word / Parking axis, ZSW

SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned8	Dyn. index: -	Function diagram: -
	P group: Displays, signals	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-

Description: Displays the status word for the "parking axis" function.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Parking axis active	Yes	No	-

r0898.0...14 CO/BO: Control word sequence control / STW seq_ctrl

SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dyn. index: -	Function diagram: 2501
	P group: Displays, signals	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-

Description: Display and connector output for the control word of the sequence control.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	ON/OFF1	Yes	No	-
	01	OC / OFF2	Yes	No	-
	02	OC / OFF3	Yes	No	-

03	Enable operation	Yes	No	-
04	Enable ramp-function generator	Yes	No	-
05	Continue ramp-function generator	Yes	No	-
06	Enable speed setpoint	Yes	No	-
07	Command open brake	Yes	No	-
08	Jog 1	Yes	No	3001
09	Jog 2	Yes	No	3001
10	Master control by PLC	Yes	No	-
12	Speed controller enable	Yes	No	-
14	Command close brake	Yes	No	-

Note

OC: Operating condition

r0898.0...14 CO/BO: Control word sequence control / STW seq_ctrl

SERVO_DBSI_VD (Lin)

Can be changed: -**Data type:** Unsigned16**P group:** Displays, signals**Not for motor type:** -**Min:**

-

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** -**Max:**

-

Access level: 2**Function diagram:** 2501**Unit selection:** -**Expert list:** 1**Factory setting:**

-

Description:

Display and connector output for the control word of the sequence control.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	ON/OFF1	Yes	No	-
01	OC / OFF2	Yes	No	-
02	OC / OFF3	Yes	No	-
03	Enable operation	Yes	No	-
04	Enable ramp-function generator	Yes	No	-
05	Continue ramp-function generator	Yes	No	-
06	Enable velocity setpoint	Yes	No	-
07	Command open brake	Yes	No	-
08	Jog 1	Yes	No	3001
09	Jog 2	Yes	No	3001
10	Master control by PLC	Yes	No	-
12	Velocity controller enable	Yes	No	-
14	Command close brake	Yes	No	-

Note

OC: Operating condition

r0898.0...10 CO/BO: Control word sequence control infeed / STW seq_ctrl INF

A_INF_840_VD

Can be changed: -**Data type:** Unsigned16**P group:** Displays, signals**Not for motor type:** -**Min:**

-

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** -**Max:**

-

Access level: 2**Function diagram:** 8820, 8920**Unit selection:** -**Expert list:** 1**Factory setting:**

-

Description:

Display and connector output for the control word of the sequence control for the infeed.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	ON/OFF1	Yes	No	-

01	OC / OFF2	Yes	No	-
03	Enable operation	Yes	No	-
05	Inhibit motoring operation	Yes	No	-
06	Inhibit regenerative	Yes	No	-
10	Master control by PLC	Yes	No	-

Note

OC: Operating condition

r0899.0...15

SERVO_DBSI_VD

CO/BO: Status word sequence control / ZSW seq_ctrl**Can be changed:** -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 2503**P group:** Displays, signals**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and BICO output for the status word of the sequence control.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Ready for switching on	Yes	No	-
01	Ready	Yes	No	-
02	Operation enabled	Yes	No	-
03	Jog active	Yes	No	-
04	No coasting active	OFF2 inactive	OFF2 active	-
05	No Quick Stop active	OFF3 inactive	OFF3 active	-
06	Switching on inhibited active	Yes	No	-
07	Drive ready	Yes	No	-
08	Controller enable	Yes	No	-
09	Control request	Yes	No	-
11	Pulses enabled	Yes	No	-
12	Open holding brake	Yes	No	-
13	Command close holding brake	Yes	No	-
14	Pulse enable from the brake control	Yes	No	-
15	Setpoint enable from the brake control	Yes	No	-

Note

For bits 00, 01, 02, 04, 05, 06, 09:

For PROFIdrive, these signals are used for status word 1.

For bit 13:

When the "Safe Brake Control" (SBC) is activated and selected, the brake is no longer controlled using this signal.

For bit 14, 15:

These signals are only of significance when the "extended brake control" function module is activated (r0108.14 = 1).

r0899.0...12

A_INF_840_VD

CO/BO: Status word sequence control infeed / ZSW seq_ctrl INF**Can be changed:** -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 8826, 8926**P group:** Displays, signals**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and BICO output for the status word of the sequence control of the infeed unit.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
-----	-------------	----------	----------	----

00	Ready for switching on	Yes	No	-
01	Ready	Yes	No	-
02	Operation enabled	Yes	No	-
04	No OFF2 active	OFF2 inactive	OFF2 active	-
06	Switching on inhibited	Yes	No	-
08	Switching-on operation active	Yes	No	-
09	Control request	Yes	No	-
11	Precharging complete	Yes	No	-
12	Line contactor closed	Yes	No	8934

Note

For bit 12:

The feedback signal of a line contactor (auxiliary contact) can be interconnected via BI: p0860.

p0922CU_I_840_VD,
CU_NX_840_VD**IF1 PROFIdrive PZD telegram selection / IF1 PZD tel****Can be changed:** C2(1), T**Data type:** Unsigned16**Calculated:** -**Dyn. index:** -**Access level:** 1**Function diagram:** 2401, 2420,
2423, 2481, 2483**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

390

999

391

Description:

Sets the send and receive telegram.

Value:

390: SIEMENS telegram 390, PZD-2/2

391: SIEMENS telegram 391, PZD-3/7

999: Free telegram configuration with BICO

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0922

SERVO_DBSI_VD

IF1 PROFIdrive PZD telegram selection / IF1 PZD tel**Can be changed:** C2(1), T**Data type:** Unsigned16**Calculated:** -**Dyn. index:** -**Access level:** 1**Function diagram:** 2401, 2415,
2416, 2419, 2420, 2421, 2422,
2423**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

136

136

136

Description:

Sets the send and receive telegram.

Value:

136: SIEMENS telegram 136, PZD-15/19

Note

If a value is not equal to 999, and therefore a telegram is set, then interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0922	IF1 PROFIdrive PZD telegram selection / IF1 PZD tel		
SERVO_DBSI_VD (EPOS, EPOS, Pos ctrl, Pos ctrl, Spin_diag)	Can be changed: C2(1), T Data type: Unsigned16	Calculated: - Dyn. index: -	Access level: 1 Function diagram: 2401, 2415, 2416, 2419, 2420, 2421, 2422, 2423
	P group: Communications Not for motor type: - Min: 7	Unit group: - Scaling: - Max: 999	Unit selection: - Expert list: 1 Factory setting: 999
Description:	Sets the send and receive telegram.		
Value:	7: Standard telegram 7, PZD-2/2 9: Standard telegram 9, PZD-10/5 110: SIEMENS telegram 110, PZD-12/7 111: SIEMENS telegram 111, PZD-12/12 999: Free telegram configuration with BICO		
	Note If a value is not equal to 999, and therefore a telegram is set, then interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p0922		IF1 PROFIdrive PZD telegram selection / IF1 PZD tel	
SERVO_DBSI_VD (Pos ctrl, Pos ctrl, Spin_diag)	Can be changed: C2(1), T	Calculated: -	Access level: 1
	Data type: Unsigned16	Dyn. index: -	Function diagram: 2401, 2415, 2416, 2419, 2420, 2421, 2422, 2423
	P group: Communications	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 999	Max: 999	Factory setting: 999
Description:	Sets the send and receive telegram.		
Value:	999: Free telegram configuration with BICO		
Note			
If a value is not equal to 999, and therefore a telegram is set, then interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p0922	IF1 PROFIdrive PZD telegram selection / IF1 PZD tel		
SERVO_DBSI_VD (Spin_diag)	Can be changed: C2(1), T Data type: Unsigned16	Calculated: - Dyn. index: -	Access level: 1 Function diagram: 2401, 2415, 2416, 2419, 2420, 2421, 2422, 2423
	P group: Communications Not for motor type: - Min: 1	Unit group: - Scaling: - Max: 999	Unit selection: - Expert list: 1 Factory setting: 999
Description:	Sets the send and receive telegram.		
Value:	1: Standard telegram 1, PZD-2/2 2: Standard telegram 2, PZD-4/4 3: Standard telegram 3, PZD-5/9		

4:	Standard telegram 4, PZD-6/14
5:	Standard telegram 5, PZD-9/9
6:	Standard telegram 6, PZD-10/14
102:	SIEMENS telegram 102, PZD-6/10
103:	SIEMENS telegram 103, PZD-7/15
105:	SIEMENS telegram 105, PZD-10/10
106:	SIEMENS telegram 106, PZD-11/15
116:	SIEMENS telegram 116, PZD-11/19
118:	SIEMENS telegram 118, PZD-11/19
125:	SIEMENS telegram 125, PZD-14/10
126:	SIEMENS telegram 126, PZD-15/15
136:	SIEMENS telegram 136, PZD-15/19
138:	SIEMENS telegram 138, PZD-15/19
139:	SIEMENS telegram 139, PZD-15/19
146:	SIEMENS telegram 146, PZD-19/20
148:	SIEMENS telegram 148, PZD-19/20
149:	SIEMENS telegram 149, PZD-19/20
220:	SIEMENS telegram 220, PZD-10/10
999:	Free telegram configuration with BICO

Note

If a value is not equal to 999, and therefore a telegram is set, then interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p0922

A_INF_840_VD

IF1 PROFIdrive PZD telegram selection / IF1 PZD tel**Can be changed:** C2(1), T**Data type:** Unsigned16**Calculated:** -**Dyn. index:** -**Access level:** 1**Function diagram:** 2401, 2420, 2423, 2447, 2457, 2481, 2483**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

370

999

999

Description:

Sets the send and receive telegram.

Value:

370: SIEMENS telegram 370, PZD-1/1

371: SIEMENS telegram 371, PZD-5/8

999: Free telegram configuration with BICO

Note

If a value is not equal to 999, a telegram is set and the automatically set interconnections in the telegram are inhibited. The inhibited interconnections can only be changed again after setting value 999.

r0944

All objects

CO: Counter for fault buffer changes / Fault buff change**Can be changed:** -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 8060**P group:** Messages**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description: Display and connector output for the counter for changes of the fault buffer.
This counter is incremented every time the fault buffer changes.

Recommendation: Used to check whether the fault buffer has been read out consistently.

Dependency: See also: r0945, r0947, r0948, r0949, r2109

r0945[0...63]

CU_I_840_VD,
CU_NX_840_VD,
SERVO_DBSI_VD

Fault code / Fault code

Can be changed: -
Data type: Unsigned16
P group: Messages
Not for motor type: -
Min:
-

Calculated: -
Dyn. index: -
Unit group: -
Scaling: -
Max:
-

Access level: 2
Function diagram: 8050, 8060
Unit selection: -
Expert list: 1
Factory setting:
-

Description: Displays the numbers of faults that have occurred.

Dependency: See also: r0947, r0948, r0949, r2109, r2130, r2136, r3120, r3122

NOTICE

The properties of the fault buffer should be taken from the corresponding product documentation.
--

Note

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).
Fault buffer structure (general principle):
r0945[0], r0949[0], r0948[0], r2109[0], r3115[0] --> actual fault case, fault 1
...
r0945[7], r0949[7], r0948[7], r2109[7], r3115[7] --> actual fault case, fault 8
r0945[8], r0949[8], r0948[8], r2109[8], r3115[8] --> 1st acknowledged fault case, fault 1
...
r0945[15], r0949[15], r0948[15], r2109[15], r3115[15] --> 1st acknowledged fault case, fault 8
...
r0945[56], r0949[56], r0948[56], r2109[56], r3115[56] --> 7th acknowledged fault case, fault 1
...
r0945[63], r0949[63], r0948[63], r2109[63], r3115[63] --> 7th acknowledged fault case, fault 8

r0946[0...65534] Fault code list / Fault code list

CU_I_840_VD,
CU_NX_840_VD

Can be changed: -
Data type: Unsigned16
P group: Messages
Not for motor type: -
Min:
-

Calculated: -
Dyn. index: -
Unit group: -
Scaling: -
Max:
-

Access level: 3
Function diagram: -
Unit selection: -
Expert list: 0
Factory setting:
-

Description: Lists the fault codes stored in the drive unit.
The indices can only be accessed with a valid fault code.
Example:
r0946[0...999] = 0 --> fault code 0 ... 999 is not available
r0946[1000] = 1000 --> fault code 1000 is available
r0946[1001] = 1001 --> fault code 1001 is available
...
r0946[1008] = 0 --> fault code 1008 is not available
...

Dependency: The parameter assigned to the fault code is entered in r0951 under the same index.

r0947[0...63]		Fault number / Fault number	
All objects	Can be changed: - Data type: Unsigned16 P group: Messages Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 8050, 8060 Unit selection: - Expert list: 1 Factory setting: -
Description:		This parameter is identical to r0945.	

r0948[0...63]		Fault time received in milliseconds / t_fault rcv ms			
All objects	Can be changed: - Data type: Unsigned32 P group: Messages Not for motor type: - Min: - [ms]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: - [ms]	Access level: 3 Function diagram: 8050, 8060 Unit selection: - Expert list: 1 Factory setting: - [ms]		
Description:		Displays the system runtime in milliseconds when the fault occurred.			
Dependency:		See also: r0945, r0947, r0949, r2109, r2130, r2136, r3120, r3122			
<table><tr><td>NOTICE</td></tr><tr><td>The time comprises r2130 (days) and r0948 (milliseconds).</td></tr></table>				NOTICE	The time comprises r2130 (days) and r0948 (milliseconds).
NOTICE					
The time comprises r2130 (days) and r0948 (milliseconds).					
Note The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the fault buffer and the assignment of the indices is shown in r0945. When the parameter is read via PROFIdrive, the TimeDifference data type applies.					

r0949[0...63]		Fault value / Fault value	
All objects	Can be changed: - Data type: Integer32 P group: Messages Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 8050, 8060 Unit selection: - Expert list: 1 Factory setting: -
Description:		Displays additional information about the fault that occurred (as integer number).	
Dependency:		See also: r0945, r0947, r0948, r2109, r2130, r2136, r3120, r3122	
Note The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the fault buffer and the assignment of the indices is shown in r0945.			

p0971		Save drive object parameters / DO par save	
SERVO_DBSI_VD	Can be changed: T, U Data type: Unsigned16 P group: Factory settings Not for motor type: - Min: 0	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 1	Access level: 1 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0
Description:		Setting to save the parameter of the particular drive object in the non-volatile memory. When saving, only the adjustable parameters intended to be saved are taken into account.	
Value:		0: Inactive	

1: Save drive object

Dependency: See also: r3996

**WARNING**

If the Control Unit power supply is switched off while data is being saved, then the backup of all adjustable parameters can be lost, and the Control Unit must be recommissioned.

NOTICE

The Control Unit power supply may only be switched off after data has been saved (i.e. after data save has been started, wait until the parameter again has the value 0).
Writing to parameters is inhibited while saving.
The progress while saving is displayed in r3996.

Note

Starting from the particular drive object, the following parameters are saved:

CU3xx: Device-specific parameters and PROFIBUS device parameters.

Other objects: Parameters of the actual object and PROFIBUS device parameters.

Prerequisite:

Before saving with p0971, all parameters (topology, all drive objects) must have been saved at least once using p0977 = 1.

p0978[0...n]

CU_I_840_VD,
CU_NX_840_VD

List of drive objects / List of the DO

Can be changed: C1(1)

Data type: Unsigned8

P group: Topology

Not for motor type: -

Min:

0

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

255

Access level: 2

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

[0] 1

[1...24] 0

Description:

This parameter is an image of p0101 in conformance with PROFIdrive.

Parameters p0101 and p0978 contain the following information:

1) The same number of drive objects

2) The same drive objects

In this sense, they are consistent.

Difference between p0101 and p0978:

p0978 can be re-sorted and a zero inserted in order to identify those drive objects that participate in the process data exchange and to define their sequence in the process data exchange. Drive objects that are listed after the first zero, are excluded from the process data exchange.

For p0978, in addition, the value 255 can be inserted a multiple number of times.

p0978[n] = 255 means: The drive object is visible for the PROFIBUS master and is empty (without any actual process data exchange). This allows cyclic communications of a PROFIBUS master with unchanged configuring to the drive units with a lower number of drive objects.

Dependency:

See also: p0101, p0971

Note

p0978 cannot be changed when the drive system is commissioned for the first time. The reason for this is that at this time the actual topology has still not been acknowledged (p0099 is still not equal to r0098 and p0009 is set to 0).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r0979[0...30]	PROFIdrive encoder format / PD encoder format		
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned32 P group: Encoder Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 4704 Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the actual position encoder used according to PROFIdrive.		
Index:	[0] = Header [1] = Type encoder 1 [2] = Resolution encoder 1 [3] = Shift factor G1_XIST1 [4] = Shift factor G1_XIST2 [5] = Distinguishable revolutions encoder 1 [6...10] = Reserved [11] = Type encoder 2 [12] = Resolution encoder 2 [13] = Shift factor G2_XIST1 [14] = Shift factor G2_XIST2 [15] = Distinguishable revolutions encoder 2 [16...20] = Reserved [21] = Type encoder 3 [22] = Resolution encoder 3 [23] = Shift factor G3_XIST1 [24] = Shift factor G3_XIST2 [25] = Distinguishable revolutions encoder 3 [26...30] = Reserved		
	Note Information about the individual indices can be taken from the following literature: PROFIdrive Profile Drive Technology		

r0979[0...30]	PROFIdrive encoder format / PD encoder format		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: Unsigned32 P group: Encoder Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 4704 Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the actual position encoder used according to PROFIdrive.		

Index:	[0] = Header
	[1] = Type encoder 1
	[2] = Resolution encoder 1
	[3] = Shift factor G1_XIST1
	[4] = Shift factor G1_XIST2
	[5] = Distinguishable distance encoder 1
	[6...10] = Reserved
	[11] = Type encoder 2
	[12] = Resolution encoder 2
	[13] = Shift factor G2_XIST1
	[14] = Shift factor G2_XIST2
	[15] = Distinguishable distance encoder 2
	[16...20] = Reserved
	[21] = Type encoder 3
	[22] = Resolution encoder 3
	[23] = Shift factor G3_XIST1
	[24] = Shift factor G3_XIST2
	[25] = Distinguishable distance encoder 3
	[26...30] = Reserved

Note

Information about the individual indices can be taken from the following literature:
PROFIdrive Profile Drive Technology

p1081

SERVO_DBSI_VD

Maximum speed scaling / n_max scal
Can be changed: T, U

Data type: FloatingPoint32

P group: Setpoints

Not for motor type: -

Min:

100.00 [%]

Calculated: -

Dyn. index: -

Unit group: -

Scaling: PERCENT

Max:

105.00 [%]

Access level: 2

Function diagram: 3050, 3095

Unit selection: -

Expert list: 1

Factory setting:

100.00 [%]

Description:

Sets the scaling for the maximum speed (p1082).

For a higher-level speed control, this scaling allows the maximum speed to be briefly exceeded.

Dependency:

See also: p1082

NOTICE

Continuous operation above a scaling of 100 % is not permitted.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1081

SERVO_DBSI_VD (Lin)

Maximum velocity scaling / v_max scal
Can be changed: T, U

Data type: FloatingPoint32

P group: Setpoints

Not for motor type: -

Min:

100.00 [%]

Calculated: -

Dyn. index: -

Unit group: -

Scaling: PERCENT

Max:

105.00 [%]

Access level: 2

Function diagram: 3050, 3095

Unit selection: -

Expert list: 1

Factory setting:

100.00 [%]

Description:

Sets the scaling for the maximum velocity (p1082).

For a higher-level velocity control, this scaling allows the maximum velocity to be briefly exceeded.

Dependency:

See also: p1082

NOTICE

Continuous operation above a scaling of 100 % is not permitted.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1082[0...n]

SERVO_DBSI_VD

Maximum speed / n_max**Can be changed:** C2(1), T**Data type:** FloatingPoint32**P group:** Setpoints**Not for motor type:** -**Min:**

0.000 [rpm]

Calculated: CALC_MOD_ALL**Dyn. index:** DDS, p0180**Unit group:** 3_1**Scaling:** -**Max:**

210000.000 [rpm]

Access level: 1**Function diagram:** 2820, 3020, 3050, 3060, 3070, 3095**Unit selection:** p0505**Expert list:** 1**Factory setting:**

1500.000 [rpm]

Description:

Sets the highest possible speed.

Dependency:

See also: p0115, p0322

NOTICE

After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.

Note

The parameter applies for both motor directions.

The parameter has a limiting effect and is the reference quantity for all ramp-up and ramp-down times (e.g. down ramps, ramp-function generator, motor potentiometer).

Since the parameter is part of quick commissioning (p0010 = 1), it is defined appropriately when p0310, p0311, p0322, p0324, p0530, p0531, and p0532 are changed.

$$p1082 \leq \min(p0324, p0532) \text{ if } p0324 > 0 \text{ and } p0532 > 0$$

$$p1082 \leq p0322, \text{ if } p0324 = 0 \text{ or } p0532 = 0 \text{ and } p0322 > 0$$

$$p1082 \leq 60 / (10.0 * p0115[0] * r0313)$$

$$p1082 \leq 60 * \text{Maximum power unit pulse frequency} / (5.0 * r0313)$$

For the automatic calculation (p0340 = 1) the value of the parameter is pre-assigned the maximum motor speed (p0322). If p0322 = 0, the rated motor speed (p0311) is used as default (pre-assignment) value. For induction motors that are not catalog motors (p0301 = 0), the synchronous no-load speed is used as default (pre-assignment) value (p0310 * 60 / r0313).

For synchronous motors, the following additionally applies:

In the automatic calculation (p0340 = 1), p1082 is, on one hand, limited to speeds for which the rated power unit current (S1 continuous duty r0207[3]) is not sufficient as field current:

$$p1082 < p0348 / (1 - r0207 / r0331), \text{ applicable for } r0207[3] < r0331$$

On the other hand, an additional limit is effective, which prevents the EMF from exceeding the maximum DC link voltage (see p0643 and p1231).

The effective assignment of the motor data set parameter (e.g. p0311) to the drive data set parameter p1082 when pre-assigning should be taken from p0186.

p1082 is also available in the quick commissioning (p0010 = 1); this means that when exiting via p3900 > 0, the value is not changed.

p1082[0...n]

SERVO_DBSI_VD (Lin)

Maximum velocity / v_max**Can be changed:** C2(1), T**Data type:** FloatingPoint32**P group:** Setpoints**Not for motor type:** -**Min:**

0.000 [m/min]

Calculated: CALC_MOD_ALL**Dyn. index:** DDS, p0180**Unit group:** 4_1**Scaling:** -**Max:**

1300.000 [m/min]

Access level: 1**Function diagram:** 2820, 3020, 3050, 3060, 3070, 3095**Unit selection:** p0505**Expert list:** 1**Factory setting:**

1000.000 [m/min]

Description: Sets the highest possible velocity.

Dependency: See also: p0115, p0322

NOTICE

After the value has been modified, no further parameter modifications can be made and the status is shown in r3996. Modifications can be made again when r3996 = 0.

Note

The parameter applies for both motor directions.

The parameter has a limiting effect and is the reference quantity for all ramp-up and ramp-down times (e.g. down ramps, ramp-function generator, motor potentiometer).

Since the parameter is part of quick commissioning (p0010 = 1), it is defined appropriately when p0310, p0311, p0322, p0324, p0530, p0531, and p0532 are changed.

$p1082 \leq \min(p0324, p0532)$ if $p0324 > 0$ and $p0532 > 0$

$p1082 \leq p0322$, if $p0324 = 0$ or $p0532 = 0$ and $p0322 > 0$

$p1082 \leq 60 / (10.0 * p0115[0] * r0313)$

$p1082 \leq 60 * \text{Maximum power unit pulse frequency} / (5.0 * r0313)$

For the automatic calculation (p0340 = 1) the value of the parameter is pre-assigned the maximum motor speed (p0322). If p0322 = 0, the rated motor speed (p0311) is used as default (pre-assignment) value. For induction motors that are not catalog motors (p0301 = 0), the synchronous no-load speed is used as default (pre-assignment) value ($p0310 * 60 / r0313$).

For synchronous motors, the following additionally applies:

In the automatic calculation (p0340 = 1), p1082 is, on one hand, limited to speeds for which the rated power unit current (S1 continuous duty r0207[3]) is not sufficient as field current:

$p1082 < p0348 / (1 - r0207 / r0331)$, applicable for $r0207[3] < r0331$

On the other hand, an additional limit is effective, which prevents the EMF from exceeding the maximum DC link voltage (see p0643 and p1231).

The effective assignment of the motor data set parameter (e.g. p0311) to the drive data set parameter p1082 when pre-assigning should be taken from p0186.

p1082 is also available in the quick commissioning (p0010 = 1); this means that when exiting via p3900 > 0, the value is not changed.

p1083[0...n]

SERVO_DBSI_VD

CO: Speed limit in positive direction of rotation / n_limit pos

Can be changed: T, U

Calculated: -

Access level: 2

Data type: FloatingPoint32

Dyn. index: DDS, p0180

Function diagram: 3050, 3095

P group: Setpoints

Unit group: 3_1

Unit selection: p0505

Not for motor type: -

Scaling: p2000

Expert list: 1

Min:

Max:

Factory setting:

0.000 [rpm]

210000.000 [rpm]

210000.000 [rpm]

Description:

Sets the maximum speed for the positive direction.

NOTICE

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

p1083[0...n]

SERVO_DBSI_VD (Lin)

CO: Velocity limit positive direction / v_limit pos

Can be changed: T, U

Calculated: -

Access level: 2

Data type: FloatingPoint32

Dyn. index: DDS, p0180

Function diagram: 3050, 3095

P group: Setpoints

Unit group: 4_1

Unit selection: p0505

Not for motor type: -

Scaling: p2000

Expert list: 1

Min:

Max:

Factory setting:

0.000 [m/min]

1000.000 [m/min]

1000.000 [m/min]

Description:

Sets the maximum velocity for the positive direction.

NOTICE

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

r1084	CO: Speed limit positive effective / n_limit pos eff		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Setpoints Not for motor type: - Min: - [rpm]	Calculated: - Dyn. index: - Unit group: 3_1 Scaling: p2000 Max: - [rpm]	Access level: 3 Function diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting: - [rpm]
Description:	Display and connector output for the active positive speed limit.		
Dependency:	See also: p1082, p1083		

r1084	CO: Velocity limit positive effective / v_limit pos eff		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Setpoints Not for motor type: - Min: - [m/min]	Calculated: - Dyn. index: - Unit group: 4_1 Scaling: p2000 Max: - [m/min]	Access level: 3 Function diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting: - [m/min]
Description:	Display and connector output for the active positive velocity limit.		
Dependency:	See also: p1082, p1083		

p1086[0...n]	CO: Speed limit in negative direction of rotation / n_limit neg		
SERVO_DBSI_VD	Can be changed: T, U Data type: FloatingPoint32 P group: Setpoints Not for motor type: - Min: -210000.000 [rpm]	Calculated: - Dyn. index: DDS, p0180 Unit group: 3_1 Scaling: p2000 Max: 0.000 [rpm]	Access level: 2 Function diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting: -210000.000 [rpm]
Description:	Sets the speed limit for the negative direction.		
NOTICE			
A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			

p1086[0...n]	CO: Velocity limit negative direction / v_limit neg		
SERVO_DBSI_VD (Lin)	Can be changed: T, U Data type: FloatingPoint32 P group: Setpoints Not for motor type: - Min: -1000.000 [m/min]	Calculated: - Dyn. index: DDS, p0180 Unit group: 4_1 Scaling: p2000 Max: 0.000 [m/min]	Access level: 2 Function diagram: 3050, 3095 Unit selection: p0505 Expert list: 1 Factory setting: -1000.000 [m/min]
Description:	Sets the velocity limit for the negative direction.		
NOTICE			
A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			

r1087	CO: Speed limit negative effective / n_limit neg eff		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 3050, 3095
	P group: Setpoints	Unit group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min:	Max:	Factory setting:
	- [rpm]	- [rpm]	- [rpm]
Description:	Display and connector output for the active negative speed limit.		
Dependency:	See also: p1082, p1086		

r1087	CO: Velocity limit negative effective / v_limit neg eff		
SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 3050, 3095
	P group: Setpoints	Unit group: 4_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2000	Expert list: 1
	Min:	Max:	Factory setting:
	- [m/min]	- [m/min]	- [m/min]
Description:	Display and connector output for the active negative velocity limit.		
Dependency:	See also: p1082, p1086		

p1135[0...n]	OFF3 ramp-down time / OFF3 t_RD		
SERVO_DBSI_VD	Can be changed: C2(1), T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 3060, 3070
	P group: Setpoints	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.000 [s]	600.000 [s]	0.000 [s]
Description:	Sets the ramp-down time from the maximum speed down to zero speed for the OFF3 command.		
	Note		
	This time can be exceeded if the DC link voltage reaches its maximum value.		

p1135[0...n]	OFF3 ramp-down time / OFF3 t_RD		
SERVO_DBSI_VD (Lin)	Can be changed: C2(1), T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 3060, 3070
	P group: Setpoints	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.000 [s]	600.000 [s]	0.000 [s]
Description:	Sets the ramp-down time from the maximum velocity down to zero speed for the OFF3 command.		
	Note		
	This time can be exceeded if the DC link voltage reaches its maximum value.		

p1192[0...n] SERVO_DBSI_VD	DSC encoder selection / DSC enc selection		
Description: Value:	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: DDS, p0180	Function diagram: 3090
	P group: Setpoints	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 1	Max: 3	Factory setting: 1
Sets the number of the encoder used for DSC.			
1: Encoder 1 (motor encoder)			
2: Encoder 2			
3: Encoder 3			
Note DSC: Dynamic Servo Control Value 1 corresponds to encoder 1 (motor encoder); the encoder data set is assigned via p0187. Value 2 corresponds to encoder 2; the encoder data set is assigned via p0188. Value 3 corresponds to encoder 3; the encoder data set is assigned using p0189. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1193[0...n] SERVO_DBSI_VD	DSC encoder adaptation factor / DSC enc adpt fact		
Description: Dependency:	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 3090
	P group: Setpoints	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -1000000.00000	Max: 1000000.00000	Factory setting: 1.00000
Sets the factor to adapt the encoder when using either encoder 2 or 3 for DSC.			
The factor sets the ratio of the pulse difference between the motor encoder (encoder 1) and the selected encoder for the same distance moved through.			
This factor takes into account e.g. gear ratios, differences in the number of encoder pulses.			
See also: p1192			
Note DSC: Dynamic Servo Control Example: Encoder 1: Motor encoder with 2048 pulses/revolution, ballscrew with 10 mm/revolution pitch Encoder 2: Linear scale with 20 µm grid division as direct measuring system p1193 = number of pulses, encoder 1 per motor revolution / number of pulses, encoder 2 per motor revolution p1193 = 2048 / (10 mm / 20 µm) = 4.096 This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1215 SERVO_DBSI_VD	Motor holding brake configuration / Brake config		
Description: Value:	Can be changed: T	Calculated: -	Access level: 2
	Data type: Integer16	Dyn. index: -	Function diagram: 2701, 2707, 2711
	P group: Functions	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 3	Factory setting: 0
Sets the holding brake configuration.			
0: No motor holding brake available			

- 1: Motor holding brake acc. to sequence control
- 2: Motor holding brake always open
- 3: Motor holding brake like sequence control connection via BICO

Dependency: See also: p1216, p1217, p1226, p1227, p1228

**CAUTION**

For the setting p1215 = 0, if a brake is used, it remains closed. If the motor moves, this will destroy the brake.

NOTICE

If p1215 was set to 1 or if p1215 was set to 3, then when the pulses are suppressed, the brake is closed even if the motor is still rotating. Pulse suppression can either be caused by a 0 signal at p0844, p0845 or p0852 or as a result of a fault with OFF2 response. If this is not desirable (e.g. for a flying restart), then the brake can be kept open using a 1 signal at p0855.

Note

If the configuration is set to "no holding brake present" when powering up, then the motor holding brake will be automatically identified. If a motor holding brake is detected, the configuration is set to "motor holding brake as for sequence control".

If a motor holding brake is used via the brake connection of the Motor Module integrated in the drive, then it is not permissible that p1215 is set to 3.

If an external motor holding brake is being used, then p1215 should be set to 3 and r0899.12 should be interconnected as control signal.

When the function module "extended brake control" is activated (r0108.14 = 1), r1229.1 should be interconnected as control signal.

The parameter can only be set to zero when the pulses are inhibited.

The parameterization "no motor holding brake available" and "Safe Brake Control" enabled (p1215 = 0, p9602 = 1, p9802 = 1) is not practical if there is no motor holding brake.

The parameterization "motor holding brake the same as sequence control, connection via BICO" and "Safe Brake Control" enabled (p1215 = 3, p9602 = 1, p9802 = 1) is not practical.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1216

SERVO_DBSI_VD

Motor holding brake opening time / Brake t_{open}

Can be changed: T, U

Calculated: -

Access level: 2

Data type: FloatingPoint32

Dyn. index: -

Function diagram: 2701, 2711

P group: Functions

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0 [ms]

10000 [ms]

100 [ms]

Description: Sets the time to open the motor holding brake.

After controlling the holding brake (opens), the speed/velocity setpoint remains at zero for this time. After this, the speed/velocity setpoint is enabled.

Recommendation: This time should be set longer than the actual opening time of the brake. This ensures that the drive cannot accelerate when the brake is applied.

Dependency: See also: p1215, p1217

Note

For a motor with DRIVE-CLiQ and integrated brake, for p0300 = 10000, this time is pre-assigned the value saved in the motor.

For p1216 = 0 ms, the monitoring and the message A07931 "Brake does not open" are deactivated.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1217	Motor holding brake closing time / Brake t_close		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2701, 2711
	P group: Functions	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0 [ms]	Max: 10000 [ms]	Factory setting: 100 [ms]
Description:	Sets the time to apply the motor holding brake. After OFF1 or OFF3 and the holding brake is controlled (the brake closes), then the drive remains closed-loop controlled for this time stationary with a speed setpoint/velocity setpoint of zero. The pulses are suppressed when the time expires.		
Recommendation:	This time should be set longer than the actual closing time of the brake. This ensures that the pulses are only suppressed after the brake has closed.		
Dependency:	See also: p1215, p1216		
	NOTICE If the selected closing time is too short with respect to the actual closing time of the brake, then the load can sag. If the closing time is selected to be too long with respect to the actual closing time of the brake, the control works against the brake and therefore reduces its lifetime.		
	Note For a motor with DRIVE-CLiQ and integrated brake, for p0300 = 10000, this time is pre-assigned the value saved in the motor. For p1217 = 0 ms, the monitoring and the message A07932 "Brake does not close" are deactivated. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1221	Open motor holding brake threshold / Open brake thr		
SERVO_DBSI_VD (Ext brake)	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2707
	P group: Functions	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [%]	Max: 200.00 [%]	Factory setting: 0.00 [%]
Description:	Sets the threshold value for the command "open brake".		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1226[0...n]	Threshold for zero speed detection / n_standst n_thr		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 2701, 2704
	P group: Functions	Unit group: 3_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [rpm]	Max: 210000.00 [rpm]	Factory setting: 20.00 [rpm]

Description:	Sets the speed threshold for the standstill identification. Acts on the actual value and setpoint monitoring. When braking with OFF1 or OFF3, when the threshold is undershot, standstill is identified. The following applies when the brake control is activated: When the threshold is undershot, the brake control is started and the system waits for the brake closing time in p1217. The pulses are then suppressed. if the brake control is not activated, the following applies: When the threshold is undershot, the pulses are suppressed and the drive coasts down.
Dependency:	See also: p1215, p1216, p1217, p1227

NOTICE

For reasons relating to the compatibility to earlier firmware versions, a parameter value of zero in indices 1 to 31 is overwritten with the parameter value in index 0 when the Control Unit boots.

Note

Standstill is identified in the following cases:

- the speed actual value falls below the speed threshold in p1226 and the time started after this in p1228 has expired.
 - the speed setpoint falls below the speed threshold in p1226 and the time started after this in p1227 has expired.
- The actual value sensing is subject to measuring noise. For this reason, standstill cannot be detected if the speed threshold is too low.

p1226[0...n]

SERVO_DBSI_VD (Lin)

Standstill detection velocity threshold / v_standst v_thr**Can be changed:** T, U**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dyn. index:** DDS, p0180**Function diagram:** 2701, 2704**P group:** Functions**Unit group:** 4_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.00 [m/min]

1000.00 [m/min]

0.20 [m/min]

Description:	Sets the velocity threshold for the standstill identification. Acts on the actual value and setpoint monitoring. When braking with OFF1 or OFF3, when the threshold is undershot, standstill is identified. The following applies when the brake control is activated: When the threshold is undershot, the brake control is started and the system waits for the brake closing time in p1217. The pulses are then suppressed. if the brake control is not activated, the following applies: When the threshold is undershot, the pulses are suppressed and the drive coasts down.
Dependency:	See also: p1215, p1216, p1217, p1227

NOTICE

For reasons relating to the compatibility to earlier firmware versions, a parameter value of zero in indices 1 to 31 is overwritten with the parameter value in index 0 when the Control Unit boots.

Note

Standstill is identified in the following cases:

- the velocity actual value falls below the velocity threshold in p1226 and the time started after this in p1228 has expired.
 - the velocity setpoint falls below the velocity threshold in p1226 and the time started after this in p1227 has expired.
- The actual value sensing is subject to measuring noise. For this reason, standstill cannot be detected if the velocity threshold is too low.

p1227	Zero speed detection monitoring time / n_standst t_mon		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2701, 2704
	P group: Functions	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.000 [s]	Max: 300.000 [s]	Factory setting: 4.000 [s]
Description:	Sets the monitoring time for the standstill identification. When braking with OFF1 or OFF3, standstill is identified after this time has expired, after the setpoint speed has fallen below p1226 (also refer to p1145). After this, the brake control is started, the system waits for the closing time in p1217 and then the pulses are suppressed.		
Dependency:	See also: p1215, p1216, p1217, p1226		
NOTICE			
For p1145 > 0.0 (RFG tracking) the setpoint is not equal to zero dependent on the selected value. This can therefore cause the monitoring time in p1227 to be exceeded. In this case, for a driven motor, the pulses are not suppressed.			
Note			
Standstill is identified in the following cases:			
- the speed actual value falls below the speed threshold in p1226 and the time started after this in p1228 has expired.			
- the speed setpoint falls below the speed threshold in p1226 and the time started after this in p1227 has expired.			
For p1227 = 300.000 s the following applies:			
Monitoring is deactivated.			
For p1227 = 0.000 s, the following applies:			
With OFF1 or OFF3 and a ramp-down time = 0, the pulses are immediately suppressed and the motor "coasts" down.			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1228	Pulse suppression delay time / Pulse suppr t_del		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2701, 2704
	P group: Functions	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.000 [s]	Max: 299.000 [s]	Factory setting: 0.000 [s]
Description:	Sets the delay time for pulse suppression. After OFF1 or OFF3, the pulses are canceled, if at least one of the following conditions is fulfilled:		
	- the speed actual value falls below the threshold in p1226 and the time started after this in p1228 has expired.		
	- the speed setpoint falls below the threshold in p1226 and the time started after this in p1227 has expired.		
Dependency:	See also: p1226, p1227		
NOTICE			
When the motor holding brake is activated, pulse cancellation is additionally delayed by the brake closing time (p1217).			

p1240[0...n]	Vdc controller or Vdc monitoring configuration / Vdc ctrl config		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: DDS, p0180	Function diagram: 3082, 5650
	P group: Functions	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0	Max: 9	Factory setting: 0
Description:	Sets the configuration of the controller or monitoring for the DC link voltage (Vdc).		
Value:	0: Inhibit Vdc ctrl		

- 1: Enable Vdc_max controller
- 2: Enable Vdc_min controller (kinetic buffering)
- 3: Enable Vdc_min controller and Vdc_max controller
- 4: Activate Vdc_max monitoring
- 5: Activate Vdc_min monitoring
- 6: Activate Vdc_min monitoring and Vdc_max monitoring
- 7: Enable Vdc_max controller without accelerating
- 8: Enable Vdc_min controller without braking
- 9: Enable Vdc_min and Vdc_max controller w/o braking/accelerating

Dependency:

See also: p1244, p1248, p1532

NOTICE

During a few steps of the rotating measurement (p1960 = 1) the Vdc_min controller and/or Vdc_max controller is disabled.
--

Note

p1240 = 1, 3:

When the upper DC link voltage threshold is reached (p1244), then the following applies:

- the Vdc_max controller limits the regenerative energy in order that the DC link voltage is kept below the maximum DC link voltage when braking.

- when other drives regenerate into the DC link, then the Vdc_max controller causes the motor to accelerate.

p1240 = 2, 3:

When the lower DC link voltage threshold is reached (p1248), the following applies:

- the Vdc_min controller limits the energy taken from the DC link in order to keep the DC link voltage above the minimum DC link voltage when accelerating.

- the motor is braked in order to use its kinetic energy to buffer the DC link.

p1240 = 4, 5, 6:

When the threshold in p1244 or p1248 is reached, the DC link voltage monitoring initiates a fault with a response and therefore reduces additional negative effects on the DC link voltage.

p1240 = 7, 9:

as for p1240 = 1, 3. However, the motor is prevented from accelerating as a result of the regenerative feedback (energy recovery) of the other drives. The effective lower torque limit cannot exceed the offset of the torque limit (p1532).

p1240 = 8, 9:

as for p1240 = 2, 3. However, the motor is prevented from braking due to the fact that the DC link voltage has been lowered. The effective upper torque limit cannot be less than the offset of the torque limit (p1532).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1244[0...n]

SERVO_DBSI_VD

DC link voltage threshold upper / Vdc upper thr

Can be changed: T, U

Data type: FloatingPoint32

P group: Functions

Not for motor type: REL

Min:

165 [V]

Calculated: CALC_MOD_CON

Dyn. index: DDS, p0180

Unit group: -

Scaling: -

Max:

1200 [V]

Access level: 3

Function diagram: 5650

Unit selection: -

Expert list: 1

Factory setting:

750 [V]

Description:

Sets the upper threshold for the DC link voltage.

For p1240 = 1, 3, 7, 9, this threshold is used as limit setpoint for the Vdc_max controller.

For p1240 = 4, 6, for DC link voltages above this threshold, an appropriate fault is output.

Dependency:

See also: p1240, p1248

Note

For $p1244 < 1.07 * \text{"parameterized DC link voltage"}$, the input of values is rejected.

For $p0204.0 = 1$, the following applies:

"Parameterized DC link voltage" = $p0210$

For $p0204.0 = 0$, the following applies:

"Parameterized DC link voltage" = $p0210 * 1.4142$

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1248[0...n]

SERVO_DBSI_VD

DC link voltage threshold lower / Vdc lower thr

Can be changed: T, U

Data type: FloatingPoint32

P group: Functions

Not for motor type: REL

Min:

50 [V]

Calculated: CALC_MOD_CON

Dyn. index: DDS, p0180

Unit group: -

Scaling: -

Max:

1000 [V]

Access level: 3

Function diagram: 5650

Unit selection: -

Expert list: 1

Factory setting:

285 [V]

Description:

Sets the lower threshold for the DC link voltage.

For $p1240 = 2, 3, 8, 9$, this threshold is used as limit setpoint for the Vdc_min controller.

For $p1240 = 5, 6$, for DC link voltages below this threshold, an appropriate fault is output.

Dependency:

See also: p1240, p1244

Note

For $p1248 > 0.93 * \text{"parameterized DC link voltage"}$, the input of values is rejected.

For $p0204.0 = 1$, the following applies:

"Parameterized DC link voltage" = $p0210$

For $p0204.0 = 0$, the following applies:

"Parameterized DC link voltage" = $p0210 * 1.4142$

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1300[0...n]

SERVO_DBSI_VD

Open-loop/closed-loop control operating mode / Op/cl-lp ctrl_mode

Can be changed: C2(1), T

Data type: Integer16

P group: V/f open-loop control

Not for motor type: -

Min:

20

Calculated: -

Dyn. index: DDS, p0180

Unit group: -

Scaling: -

Max:

23

Access level: 2

Function diagram: 5060, 8012

Unit selection: -

Expert list: 1

Factory setting:

21

Description:

Sets the open and closed-loop control mode of a drive.

Value:

20: Speed control (encoderless)

21: Speed control (with encoder)

23: Torque control (with encoder)

Dependency:

Closed-loop speed or torque control (with encoder) cannot be selected if the encoder type is not entered (p0400).

See also: r0108, p0300, p0311

NOTICE

General conditions for encoderless operation can be found in the following literature:

SINAMICS S120 Function Manual Drive Functions

Note

The closed-loop torque control can only be changed over in operation (p1300 = 20, 21) by selecting the closed-loop speed control (p1501). At the changeover, the setting of p1300 does not change. In this case, the actual state is displayed in r1407, bit 2 and bit 3.

For encoderless operation (p1404 = 0 or p1300 = 20), the following applies:

- the following condition must be fulfilled: $p1800 \geq 1 / (4 * p0115[0])$
- for motors with a low power rating (< 300 W), we recommend to set $p1800 \geq 1 / p0115[0]$.
- although pulse frequencies $p1800 = 1 / (n * p0115[0])$ with $n = 3$ or 4 are possible, for $p0115[0] > 62.5 \mu s$, they result in unsteady closed-loop control and should be avoided.
- for a blocksize unit with a current controller sampling time $p0115[0] < 80 \mu s$ and a pulse frequency of $p1800 = 0.5 / p0115[0]$, then it may be necessary to increase the switchover speed of the model p1755.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1300[0...n]

SERVO_DBSI_VD (Lin)

Open-loop/closed-loop control operating mode / Op/cl-lp ctrl_mode**Can be changed:** C2(1), T**Data type:** Integer16**P group:** V/f open-loop control**Not for motor type:** -**Min:**

20

Calculated: -**Dyn. index:** DDS, p0180**Unit group:** -**Scaling:** -**Max:**

23

Access level: 2**Function diagram:** 5060, 8012**Unit selection:** -**Expert list:** 1**Factory setting:**

21

Description:

Sets the open and closed-loop control mode of a drive.

Value:

20: Velocity control (encoderless)

21: Velocity control (with encoder)

23: Force control (with encoder)

Dependency:

Closed-loop velocity or force control (with encoder) cannot be selected if the encoder type is not entered (p0400).

See also: r0108, p0300, p0311

NOTICE

General conditions for encoderless operation can be found in the following literature:
SINAMICS S120 Function Manual Drive Functions

Note

It is only possible to change over to force control during operation (p1501) if velocity control is selected (p1300 = 20, 21). At the changeover, the setting of p1300 does not change. In this case, the actual state is displayed in r1407, bit 2 and bit 3.

For encoderless operation (p1404 = 0 or p1300 = 20), the following applies:

- the following condition must be fulfilled: $p1800 \geq 1 / (4 * p0115[0])$
- for motors with a low power rating (< 300 W), we recommend to set $p1800 \geq 1 / p0115[0]$.
- although pulse frequencies $p1800 = 1 / (n * p0115[0])$ with $n = 3$ or 4 are possible, for $p0115[0] > 62.5 \mu s$, they result in unsteady closed-loop control and should be avoided.

p1400[0...n]

SERVO_DBSI_VD

Speed control configuration / n_ctrl config**Can be changed:** T, U**Data type:** Unsigned32**P group:** Closed-loop control**Not for motor type:** REL**Min:**

-

Calculated: -**Dyn. index:** DDS, p0180**Unit group:** -**Scaling:** -**Max:**

-

Access level: 2**Function diagram:** 5019, 5490**Unit selection:** -**Expert list:** 1**Factory setting:**

0000 0100 0000 0000 0000

0011 1010 0000 bin

Description:

Sets the configuration for the closed-loop speed control.

Bit array:**Bit** **Signal name**

03 Reference model speed setpoint I component

1 signal

ON

0 signal

OFF

FD

5030

04	Torque limiting active in motoring/regenerative mode	Yes	No	-
05	Kp/Tn adaptation active	Yes	No	-
07	Interpolation speed precontrol active	Yes	No	-
08	Interpolation torque setpoint active	Yes	No	-
10	Speed precontrol	For symmetrizing	For setp_filter 2	-

Note

For bit 07:

The interpolator is only active for isochronous PROFIBUS operation, and with the sign-of-life received from the master (STW2.12 ... STW2.15). Further, for active Dynamic Servo Control (DSC) an additional dead time of one speed controller sampling time occurs.

For bit 10:

The precontrol signal via connector input p1430 only becomes effective at p1402.4 = 1 (torque-speed precontrol with encoder) at p1400.10 = 0 (for setp_filter 2).

p1400[0...n]		Velocity control configuration / v_ctrl config			
SERVO_DBSI_VD (Lin)	Can be changed: T, U		Calculated: -		Access level: 2
	Data type: Unsigned32		Dyn. index: DDS, p0180		Function diagram: 5019, 5490
	P group: Closed-loop control		Unit group: -		Unit selection: -
	Not for motor type: REL		Scaling: -		Expert list: 1
	Min: -		Max: -		Factory setting: 0000 0100 0000 0000 0000 0011 1010 0000 bin
Description:		Sets the configuration for the closed-loop velocity control.			
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	03	Reference model velocity setpoint I component	ON	OFF	5030
	04	Force limiting active in motoring/regenerative mode	Yes	No	-
	05	Kp/Tn adaptation active	Yes	No	-
	07	Interpolation velocity controller precontrol active	Yes	No	-
	08	Interpolation force setpoint active	Yes	No	-
	10	Velocity precontrol	For symmetrizing	For setp_filter 2	-

Note

For bit 07:

The interpolator is only active for isochronous PROFIBUS operation, and with the sign-of-life received from the master (STW2.12 ... STW2.15). Further, for active Dynamic Servo Control (DSC) an additional dead time of one velocity controller sampling time occurs.

For bit 10:

The precontrol signal via connector input p1430 only becomes effective at p1402.4 = 1 (force-velocity precontrol with encoder) at p1400.10 = 0 (for setp_filter 2).

For bit 11:

If the motor rotates when the pulses are enabled, then we recommend p1400.11 = 1 (starting value = setpoint) with the matching sign.

If the motor remains stationary (zero speed) when the pulses are enabled, then we recommend p1400.11 = 0 (starting value = 0.0).

For bit 12:

If a changeover is made from operation with encoder to encoderless operation while accelerating (with the threshold from p1404), then we recommend p1400.12 = 0.

If the changeover is made from operation with an encoder to encoderless at constant velocity (e.g. with a DDS changeover or if there is an encoder fault via p0491), then we recommend p1400.12 = 1.

For bit 17:

In order to avoid limit cycles (e.g. as a result of disturbing forces) for DSC with a high Kv factor, the position controller output can be limited using a root function corresponding to the currently available deceleration capability of the drive. In this case, the total mass (m_tot) must be parameterized precisely (if necessary, determine the mass p0341, p0342 and p1498 using the motor data identification). If the limiting function responds, then this is indicated in r1407.19.

As a result of the absolute value limiting above $v[m/min] = 5.7 \times F_{max}[N] / (K_v[1000/min] \times m_{tot}[kg])$, the dynamic response of the position controller is no longer linear (F_{max} , see r1538, r1539). This is the reason that velocity precontrol is recommended.

For bit 18:

Only active when the "moment of inertia estimator" function module is active (r0108.10 = 1).

The result of the moment of inertia estimator is displayed in r1493 when the function is activated.

The function assumes that the velocity changes without the load changing. If the velocity must be changed with associated load change, then during this time, the estimated mass should be frozen using binector input p1502.

r1407.0...28

SERVO_DBSI_VD

CO/BO: Status word speed controller / ZSW n_ctrl

Can be changed: -

Data type: Unsigned32

P group: Closed-loop control

Not for motor type: REL

Min:

-

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

-

Access level: 3

Function diagram: 2522

Unit selection: -

Expert list: 1

Factory setting:

-

Description:

Display and BICO output for the status word of the speed controller.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	U/f control active	Yes	No	-
01	Encoderless operation active	Yes	No	-
02	Torque control active	Yes	No	8010
04	Speed setpoint from DSC	Yes	No	2522
05	Speed controller I component frozen	Yes	No	-
06	Speed controller I component set	Yes	No	-
07	Torque limit reached	Yes	No	5610
08	Upper torque limit active	Yes	No	5610
09	Lower torque limit active	Yes	No	5610
11	Speed setpoint limited	Yes	No	-
13	Encoderless operation due to a fault	Yes	No	-
19	DSC position controller limited	Yes	No	3090
20	DSC with spline on	Yes	No	-

21	Speed precontrol for DSC with spline on	Yes	No	-
22	Torque precontrol for DSC with spline on	Yes	No	-
23	Torque-speed precontrol with encoder on	Yes	No	-
24	Moment of inertia estimator active	Yes	No	-
25	Load estimate active	Yes	No	-
26	Moment of inertia estimator stabilized	Yes	No	-
28				-

Note

For bit 01, 13:

If, after a fault, the encoder still provides a valid commutation position ($p1992.10 = 1$), then a switch is not immediately made into encoderless operation. Both bits remain at 0 for this time.

For bit 04:

The following conditions must be fulfilled to set to 1:

- connector input p1190 and p1191 must be interconnected with a signal source that is not equal to zero.

- OFF1, OFF3 or STOP2 must not be active.

- it is not permissible that the motor data identification is active.

- Master control must not be active.

The following conditions can mean that the DSC function is not active in spite of the fact that the bit is set:

- isochronous operation is not selected (r2054 not equal to 4).

- the PROFIBUS is not isochronous (r2064[0] not equal to 1).

- DSC is not activated on the control side, therefore $KPC = 0$ is transferred as value to connector input p1191.

r1407.0...28**CO/BO: Status word velocity controller / ZSW v_ctrl**

SERVO_DBSI_VD (Lin)

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** 2522**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and BICO output for the status word of the velocity controller.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	U/f control active	Yes	No	-
01	Encoderless operation active	Yes	No	-
02	Force control active	Yes	No	8010
04	Velocity setpoint from DSC	Yes	No	2522
05	Velocity controller I component frozen	Yes	No	-
06	Velocity controller I component set	Yes	No	-
07	Force limit reached	Yes	No	5610
08	Upper force limit active	Yes	No	5610
09	Lower force limit active	Yes	No	5610
11	Velocity setpoint limited	Yes	No	-
13	Encoderless operation due to a fault	Yes	No	-
19	DSC position controller limited	Yes	No	3090
20	DSC with spline on	Yes	No	-
21	Velocity precontrol for DSC with spline on	Yes	No	-
22	Force precontrol for DSC with spline on	Yes	No	-
23	Torque-speed precontrol with encoder on	Yes	No	-
24	Mass estimate active	Yes	No	-
25	Load estimate active	Yes	No	-
26	Moment of inertia estimator stabilized	Yes	No	-

Note

For bit 01, 13:

If, after a fault, the encoder still provides a valid commutation position ($p1992.10 = 1$), then a switch is not immediately made into encoderless operation. Both bits remain at 0 for this time.

For bit 04:

The following conditions must be fulfilled to set to 1:

- connector input p1190 and p1191 must be interconnected with a signal source that is not equal to zero.
- OFF1, OFF3 or STOP2 must not be active.
- it is not permissible that the motor data identification is active.
- Master control must not be active.

The following conditions can mean that the DSC function is not active in spite of the fact that the bit is set:

- isochronous operation is not selected ($r2054$ not equal to 4).
- the PROFIBUS is not isochronous ($r2064[0]$ not equal to 1).
- DSC is not activated on the control side, therefore $KPC = 0$ is transferred as value to connector input p1191.

p1409[0...n] SERVO_DBSI_VD	Speed control extended configuration / n_ctrl ext config				
	Can be changed: C1(3)		Calculated: -	Access level: 2	
	Data type: Unsigned32		Dyn. index: DDS, p0180		Function diagram: -
	P group: -		Unit group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min: -		Max: -		Factory setting: 0000 bin
Description:	Sets the extended configuration for the closed-loop speed control.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Interpolation supplementary torque active	Yes	No	5060

p1409[0...n]		Velocity control extended configuration / v_ctrl ext config			
SERVO_DBSI_VD (Lin)	Can be changed: C1(3)	Calculated: -		Access level: 2	
	Data type: Unsigned32	Dyn. index: DDS, p0180		Function diagram: -	
	P group: -	Unit group: -		Unit selection: -	
	Not for motor type: -	Scaling: -		Expert list: 1	
	Min: -	Max: -		Factory setting: 0000 bin	
Description:		Sets the extended configuration for the closed-loop velocity control.			
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Interpolation supplementary force active	Yes	No	5060

p1414[0...n]		Speed setpoint filter activation / n_set_filt act			
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3		
	Data type: Unsigned16	Dyn. index: DDS, p0180	Function diagram: 5020		
	P group: Closed-loop control	Unit group: -	Unit selection: -		
	Not for motor type: REL	Scaling: -	Expert list: 1		
	Min: -	Max: -	Factory setting: 0000 bin		
Description:		Setting for activating/deactivating the speed setpoint filter.			
Recommendation:		If only one filter is required, filter 1 should be activated and filter 2 deactivated, to avoid excessive processing time.			
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Activate filter 1	Yes	No	-
	01	Activate filter 2	Yes	No	-

Dependency: The individual speed setpoint filters are parameterized as of p1415.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1414[0...n] Velocity setpoint filter activation / v_set_filt act

SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0000 bin

Description: Setting for activating/deactivating the velocity setpoint filter.

Recommendation: If only one filter is required, filter 1 should be activated and filter 2 deactivated, to avoid excessive processing time.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Activate filter 1	Yes	No	-
	01	Activate filter 2	Yes	No	-

Dependency: The individual velocity setpoint filters are parameterized as of p1415.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1415[0...n] Speed setpoint filter 1 type / n_set_filt 1 type

SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	2	0

Description: Sets the type for speed setpoint filter 1.

Value:	0:	Low pass: PT1
	1:	Low pass: PT2
	2:	General 2nd order filter

Dependency: PT1 low pass: p1416
PT2 low pass: p1417, p1418
General filter: p1417 ... p1420

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1415[0...n] Velocity setpoint filter 1 type / v_set_filt 1 type

SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	2	0

Description: Sets the type for speed setpoint filter 1.

Value:	0:	Low pass: PT1
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	1: Low pass: PT2
	2: General 2nd order filter
Dependency:	PT1 low pass: p1416 PT2 low pass: p1417, p1418 General filter: p1417 ... p1420
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.	

p1416[0...n]	Speed setpoint filter 1 time constant / n_set_filt 1 Tc		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 5000.00 [ms]	Factory setting: 0.00 [ms]
Description:	Sets the time constant for the speed setpoint filter 1 (PT1).		
Dependency:	See also: p1414, p1415		
Note			
This parameter is only effective if the filter is set as a PT1 low pass.			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1416[0...n]	Velocity setpoint filter 1 time constant / v_set_filt 1 Tc		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 5000.00 [ms]	Factory setting: 0.00 [ms]
Description:	Sets the time constant for the velocity setpoint filter 1 (PT1).		
Dependency:	See also: p1414, p1415		
	Note This parameter is only effective if the filter is set as a PT1 low pass. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1417[0...n]	Speed setpoint filter 1 denominator natural frequency / n_set_filt1 fn_den		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 2000.0 [Hz]
Description:	Sets the denominator natural frequency for speed setpoint filter 1 (PT2, general filter).		
Dependency:	See also: p1414, p1415		

Note

This parameter is only effective if the speed filter is parameterized as a PT2 low pass or as general filter.

The filter is only effective if the natural frequency is less than half of the sampling frequency.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1417[0...n]	Velocity setpoint filter 1 denominator natural frequency / v_set_filt1 fn_den		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 2000.0 [Hz]
Description:	Sets the denominator natural frequency for velocity setpoint filter 1 (PT2, general filter).		
Dependency:	See also: p1414, p1415		

Note

This parameter is only effective if the velocity filter is parameterized as a PT2 low pass or as general filter.

The filter is only effective if the natural frequency is less than half of the sampling frequency.

p1418[0...n]	Speed setpoint filter 1 denominator damping / n_set_filt 1 D_den		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.001	Max: 10.000	Factory setting: 0.700
Description:	Sets the denominator damping for speed setpoint filter 1 (PT2, general filter).		
Dependency:	See also: p1414, p1415		

Note

This parameter is only effective if the speed filter is parameterized as a PT2 low pass or as general filter.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1418[0...n]	Velocity setpoint filter 1 denominator damping / v_set_filt 1 D_den		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.001	Max: 10.000	Factory setting: 0.700
Description:	Sets the denominator damping for velocity setpoint filter 1 (PT2, general filter).		
Dependency:	See also: p1414, p1415		

Note

This parameter is only effective if the velocity filter is parameterized as a PT2 low pass or as general filter.

p1419[0...n]	Speed setpoint filter 1 numerator natural frequency / n_set_filt1 fn_num		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 2000.0 [Hz]
Description:	Sets the numerator natural frequency for speed setpoint filter 1 (general filter).		
Dependency:	See also: p1414, p1415		
Note			
This parameter is only effective if the speed filter is set as a general filter.			
The filter is only effective if the natural frequency is less than half of the sampling frequency.			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1419[0...n]	Velocity setpoint filter 1 numerator natural frequency / v_set_filt1 fn_num		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 2000.0 [Hz]
Description:	Sets the numerator natural frequency for velocity setpoint filter 1 (general filter).		
Dependency:	See also: p1414, p1415		
	Note		
	This parameter is only effective if the velocity filter is set as a general filter.		
	The filter is only effective if the natural frequency is less than half of the sampling frequency.		

p1420[0...n]	Speed setpoint filter 1 numerator damping / n_set_filt 1 D_num		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.000	Max: 10.000	Factory setting: 0.700
Description:	Sets the numerator damping for speed setpoint filter 1 (general filter).		
Dependency:	See also: p1414, p1415		
	Note This parameter is only effective if the speed filter is set as a general filter. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1420[0...n] SERVO_DBSI_VD (Lin)	Velocity setpoint filter 1 numerator damping / v_set_filt 1 D_num		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.000	Max: 10.000	Factory setting: 0.700

Description: Sets the numerator damping for velocity setpoint filter 1 (general filter).

Dependency: See also: p1414, p1415

Note

This parameter is only effective if the velocity filter is set as a general filter.

p1421[0...n]

SERVO_DBSI_VD

Speed setpoint filter 2 type / n_set_filt 2 type

Can be changed: T, U

Data type: Integer16

P group: Closed-loop control

Not for motor type: REL

Min:

0

Calculated: -

Dyn. index: DDS, p0180

Unit group: -

Scaling: -

Max:

2

Access level: 3

Function diagram: 5020

Unit selection: -

Expert list: 1

Factory setting:

0

Description: Sets the type for speed setpoint filter 2.

Value:

0: Low pass: PT1

1: Low pass: PT2

2: General 2nd order filter

Dependency:

PT1 low pass: p1422

PT2 low pass: p1423, p1424

General filter: p1423 ... p1426

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1421[0...n]

SERVO_DBSI_VD (Lin)

Velocity setpoint filter 2 type / v_set_filt 2 type

Can be changed: T, U

Data type: Integer16

P group: Closed-loop control

Not for motor type: REL

Min:

0

Calculated: -

Dyn. index: DDS, p0180

Unit group: -

Scaling: -

Max:

2

Access level: 3

Function diagram: 5020

Unit selection: -

Expert list: 1

Factory setting:

0

Description: Sets the type for velocity setpoint filter 2.

Value:

0: Low pass: PT1

1: Low pass: PT2

2: General 2nd order filter

Dependency:

PT1 low pass: p1422

PT2 low pass: p1423, p1424

General filter: p1423 ... p1426

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1422[0...n]

SERVO_DBSI_VD

Speed setpoint filter 2 time constant / n_set_filt 2 Tc

Can be changed: T, U

Data type: FloatingPoint32

P group: Closed-loop control

Not for motor type: REL

Min:

0.00 [ms]

Calculated: -

Dyn. index: DDS, p0180

Unit group: -

Scaling: -

Max:

5000.00 [ms]

Access level: 3

Function diagram: 5020

Unit selection: -

Expert list: 1

Factory setting:

0.00 [ms]

Description: Sets the time constant for the speed setpoint filter 2 (PT1).

Dependency: See also: p1414, p1421

Note

This parameter is only effective if the speed filter is set as a PT1 low pass.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1422[0...n] Velocity setpoint filter 2 time constant / v_set_filt 2 Tc

SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 5000.00 [ms]	Factory setting: 0.00 [ms]

Description: Sets the time constant for the velocity setpoint filter 2 (PT1).

Dependency: See also: p1414, p1421

Note

This parameter is only effective if the velocity filter is set as a PT1 low pass.

p1423[0...n] Speed setpoint filter 2 denominator natural frequency / n_set_filt2 fn_den

SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 2000.0 [Hz]

Description: Sets the denominator natural frequency for speed setpoint filter 2 (PT2, general filter).

Dependency: See also: p1414, p1421

Note

This parameter is only effective if the speed filter is parameterized as a PT2 low pass or as general filter.

The filter is only effective if the natural frequency is less than half of the sampling frequency.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1423[0...n] Velocity setpoint filter 2 denominator natural frequency / v_set_filt2 fn_den

SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 2000.0 [Hz]

Description: Sets the denominator natural frequency for velocity setpoint filter 2 (PT2, general filter).

Dependency: See also: p1414, p1421

Note

This parameter is only effective if the velocity filter is parameterized as a PT2 low pass or as general filter.

The filter is only effective if the natural frequency is less than half of the sampling frequency.

p1424[0...n] SERVO_DBSI_VD	Speed setpoint filter 2 denominator damping / n_set_filt 2 D_den		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.001	Max: 10.000	Factory setting: 0.700
Description:	Sets the denominator damping for speed setpoint filter 2 (PT2, general filter).		
Dependency:	See also: p1414, p1421		
	Note This parameter is only effective if the speed filter is parameterized as a PT2 low pass or as general filter. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
p1424[0...n] SERVO_DBSI_VD (Lin)	Velocity setpoint filter 2 denominator damping / v_set_filt 2 D_den		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.001	Max: 10.000	Factory setting: 0.700
Description:	Sets the denominator damping for velocity setpoint filter 2 (PT2, general filter).		
Dependency:	See also: p1414, p1421		
	Note This parameter is only effective if the velocity filter is parameterized as a PT2 low pass or as general filter.		
p1425[0...n] SERVO_DBSI_VD	Speed setpoint filter 2 numerator natural frequency / n_set_filt2 fn_num		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 2000.0 [Hz]
Description:	Sets the numerator natural frequency for speed setpoint filter 2 (general filter).		
Dependency:	See also: p1414, p1421		
	Note This parameter is only effective if the speed filter is set as a general filter. The filter is only effective if the natural frequency is less than half of the sampling frequency. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
p1425[0...n] SERVO_DBSI_VD (Lin)	Velocity setpoint filter 2 numerator natural frequency / v_set_filt2 fn_num		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5020
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 2000.0 [Hz]
Description:	Sets the numerator natural frequency for velocity setpoint filter 2 (general filter).		

Dependency: See also: p1414, p1421

Note

This parameter is only effective if the velocity filter is set as a general filter.
The filter is only effective if the natural frequency is less than half of the sampling frequency.

p1426[0...n]

SERVO_DBSI_VD

Speed setpoint filter 2 numerator damping / n_set_filt 2 D_num

Can be changed: T, U

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dyn. index: DDS, p0180

Function diagram: 5020

P group: Closed-loop control

Unit group: -

Unit selection: -

Not for motor type: REL

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0.000

10.000

0.700

Description: Sets the numerator damping for speed setpoint filter 2 (general filter).

Dependency: See also: p1414, p1421

Note

This parameter is only effective if the speed filter is set as a general filter.
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1426[0...n]

SERVO_DBSI_VD (Lin)

Velocity setpoint filter 2 numerator damping / v_set_filt 2 D_num

Can be changed: T, U

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dyn. index: DDS, p0180

Function diagram: 5020

P group: Closed-loop control

Unit group: -

Unit selection: -

Not for motor type: REL

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0.000

10.000

0.700

Description: Sets the numerator damping for velocity setpoint filter 2 (general filter).

Dependency: See also: p1414, p1421

Note

This parameter is only effective if the velocity filter is set as a general filter.

p1428[0...n]

SERVO_DBSI_VD

Speed precontrol symmetrizing dead time / n_prectrSym t_dead

Can be changed: T, U

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dyn. index: DDS, p0180

Function diagram: 5030, 5042, 5210

P group: Closed-loop control

Unit group: -

Unit selection: -

Not for motor type: REL

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0.0

2.0

0.0

Description: Sets the dead time to symmetrize the speed setpoint for active torque precontrol.

The selected multiplier refers to the sampling time of the speed controller (dead time= p1428 * p0115[1]).

Dependency: In conjunction with p1429, this parameter can emulate the characteristics of how the torque is established (dynamic response of closed current control loop).

See also: p1429

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1428[0...n]	Velocity precontrol symmetrizing dead time / v_prectrSym t_dead		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5030, 5042, 5210
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.0	Max: 2.0	Factory setting: 0.0
Description:	Sets the dead time to symmetrize the velocity setpoint for active force precontrol. The selected multiplier refers to the sampling time of the velocity controller (dead time= p1428 * p0115[1]).		
Dependency:	In conjunction with p1429, this parameter can emulate the characteristics of how the force is established (dynamic response of closed current control loop). See also: p1429		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1429[0...n]	Speed precontrol symmetrizing time constant / n_prectrl sym Tc		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5030, 5042, 5210, 6031
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 10000.00 [ms]	Factory setting: 0.00 [ms]
Description:	Sets the time constant (PT1) for symmetrizing the speed setpoint for active torque precontrol.		
Dependency:	In conjunction with p1428, this parameter can emulate the characteristics of how torque is established (dynamic response of the closed current control loop). For VECTOR (r0107) the following applies: The parameter is only effective if the acceleration model is supplied using external acceleration signals (p1400.2 = 1). For p1400.2 = 0, time constant p1442 (or p1452 for sensorless vector control) is used. See also: p1428		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1429[0...n]	Velocity precontrol symmetrizing time constant / v_prectrl sym Tc		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5030, 5042, 5210
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 10000.00 [ms]	Factory setting: 0.00 [ms]
Description:	Sets the time constant (PT1) for symmetrizing the velocity setpoint for active force precontrol.		
Dependency:	In conjunction with p1428, this parameter can emulate the characteristics of how the force is established (dynamic response of closed current control loop). See also: p1428		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

r1432[0...1] SERVO_DBSI_VD	CO: Speed precontrol / n_prectrl		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5020, 5030
	P group: Closed-loop control	Unit group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2000	Expert list: 1
	Min: - [rpm]	Max: - [rpm]	Factory setting: - [rpm]
Description:	Display and connector output for the speed precontrol. For index [0]: Displays the speed precontrol value after symmetrizing for the torque build-up (emulates the closed current control loop). For index [1]: Displays the speed precontrol value before the switch p1400.10. When the "DSC with spline" function module is activated (r0108.6 = 1, signal source of p1194.0 = 1), then this is the precontrol value generated by the spline. Otherwise, this is the value from the signal source of p1430 (possibly after linear interpolation). The precontrol value generated by the spline is zero if the speed precontrol for the spline is deactivated (signal source of p1194.4 = 0).		
Index:	[0] = After symmetrization [1] = Before switch p1400.10		
Dependency:	Symmetrizing can be parameterized with p1428 and/or p1429.		

r1432[0...1] SERVO_DBSI_VD (Lin)	CO: Velocity precontrol / v_precontrol		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5020, 5030
	P group: Closed-loop control	Unit group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2000	Expert list: 1
	Min: - [m/min]	Max: - [m/min]	Factory setting: - [m/min]
Description:	Display and connector output for the velocity precontrol. For index [0]: Displays the velocity precontrol value after symmetrizing for the force build-up (emulates the closed current control loop). For index [1]: Displays the velocity precontrol value before the switch p1400.10. When the "DSC with spline" function module is activated (r0108.6 = 1, signal source of p1194.0 = 1), then this is the precontrol value generated by the spline. Otherwise, this is the value from the signal source of p1430 (possibly after linear interpolation). The precontrol value generated by the spline is zero if the velocity precontrol for the spline is deactivated (signal source of p1194.4 = 0).		
Index:	[0] = After symmetrization [1] = Before switch p1400.10		
Dependency:	Symmetrizing can be parameterized with p1428 and/or p1429.		

p1433[0...n] SERVO_DBSI_VD	Speed controller reference model natural frequency / n_ctrl RefMod fn		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5030
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.0 [Hz]	Max: 8000.0 [Hz]	Factory setting: 0.0 [Hz]
Description:	Sets the natural frequency of a PT2 element for the reference model of the speed controller.		

Recommendation: The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (actual speed value) are virtually identical when the I component of the speed controller is disabled.

Dependency: Together with p1434 and p1435, the characteristics (in the time domain) of the closed-loop speed control (P) can be emulated.
See also: p1434, p1435

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1433[0...n]	Velocity controller reference model natural frequency / v_ctrl RefMod fn		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5030
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.0 [Hz]	8000.0 [Hz]	0.0 [Hz]
Description:	Sets the natural frequency of a PT2 element for the reference model of the velocity controller.		
Recommendation:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (velocity actual value) are virtually identical when the I component of the velocity controller is disabled.		
Dependency:	Together with p1434 and p1435, the characteristics (in the time domain) of the closed-loop velocity control (P) can be emulated.		
	See also: p1434, p1435		
	Note		
	This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1434[0...n]	Speed controller reference model damping / n_ctrl RefMod D		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5030, 6031
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.000	Max: 5.000	Factory setting: 1.000
Description:	Sets the damping of a PT2 element for the reference model of the speed controller.		
Recommendation:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (actual speed value) are virtually identical when the I component of the speed controller is disabled.		
Dependency:	In conjunction with p1433 and p1435, the characteristics (in time) of the P-controlled speed control loop can be emulated. See also: p1433, p1435		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1434[0...n]	Velocity controller reference model damping / v_ctrl RefMod D		
SERVO_DBSI_VD (Lin)	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.000	Calculated: - Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 5.000	Access level: 3 Function diagram: 5030, 6031 Unit selection: - Expert list: 1 Factory setting: 1.000

Description:	Sets the damping of a PT2 element for the reference model of the velocity controller.
Recommendation:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (velocity actual value) are virtually identical when the I component of the velocity controller is disabled.
Dependency:	Together with p1433 and p1435, the characteristics (in the time domain) of the P-controlled velocity control loop can be emulated. See also: p1433, p1435
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.	

p1435[0...n] SERVO_DBSI_VD	Speed controller reference model dead time / n_ctrRefMod t_dead		
	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5030, 6031
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00	Max: 2.00	Factory setting: 0.00
Description:	Sets the "fractional" dead time for the reference model of the speed controller. This parameter emulates the computing dead time of the proportionally controlled speed control loop. The selected multiplier refers to the speed controller sampling time (dead time= p1435 * p0115[1]).		
Recommendation:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (actual speed value) are virtually identical when the I component of the speed controller is disabled.		
Dependency:	In conjunction with p1433 and p1434, the characteristics (in time) of the P-controlled speed control loop can be emulated. See also: p0115, p1433, p1434		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1435[0...n]	Velocity controller reference model dead time / v_ctrRefMod t_dead		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5030, 6031
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00	Max: 2.00	Factory setting: 0.00
Description:	Sets the "fractional" dead time for the reference model of the velocity controller. This parameter emulates the computing dead time of the proportionally controlled velocity control loop. The selected multiplier refers to the velocity controller sampling time (dead time= p1435 * p0115[1]).		
Recommendation:	The reference model is correctly set when the characteristics of p1439 (reference model output) and p1445 (velocity actual value) are virtually identical when the I component of the velocity controller is disabled.		
Dependency:	Together with p1433 and p1434, the characteristics (in the time domain) of the P-controlled velocity control loop can be emulated. See also: p0115, p1433, p1434		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

r1436	CO: Speed controller reference model speed setpoint output / RefMod n_set outp		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [rpm]	Calculated: - Dyn. index: - Unit group: 3_1 Scaling: p2000 Max: - [rpm]	Access level: 3 Function diagram: 5030, 6031 Unit selection: p0505 Expert list: 1 Factory setting: - [rpm]
Description:	Display and connector output for the speed setpoint at the output of the reference model.		
r1436	CO: Velocity controller reference model velocity_setpoint output / RefMod v_set outp		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [m/min]	Calculated: - Dyn. index: - Unit group: 4_1 Scaling: p2000 Max: - [m/min]	Access level: 3 Function diagram: 5030, 6031 Unit selection: p0505 Expert list: 1 Factory setting: - [m/min]
Description:	Display and connector output for the velocity setpoint at the output of the reference model.		
r1438	CO: Speed controller speed setpoint / n_ctrl n_set		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [rpm]	Calculated: - Dyn. index: - Unit group: 3_1 Scaling: p2000 Max: - [rpm]	Access level: 3 Function diagram: 3001, 5019, 5030, 5042, 5210 Unit selection: p0505 Expert list: 1 Factory setting: - [rpm]
Description:	Display and connector output of the speed setpoint after setpoint limiting for the P component of the speed controller. For U/f operation, the value that is displayed is of no relevance.		
Dependency:	See also: r1439		
	Note In the standard state (the reference model is deactivated), r1438 = r1439.		
r1438	CO: Velocity controller velocity setpoint / v_ctrl v_set		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [m/min]	Calculated: - Dyn. index: - Unit group: 4_1 Scaling: p2000 Max: - [m/min]	Access level: 3 Function diagram: 3001, 5019, 5030, 5042, 5210 Unit selection: p0505 Expert list: 1 Factory setting: - [m/min]
Description:	Displays the velocity setpoint after setpoint limiting for the P component of the velocity controller. For U/f operation, the value that is displayed is of no relevance.		
Dependency:	See also: r1439		
	Note In the standard state (the reference model is deactivated), r1438 = r1439.		

r1439 SERVO_DBSI_VD	Speed setpoint I component / n_set I_comp		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5030, 5040, 6031
	P group: Closed-loop control	Unit group: 3_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2000	Expert list: 1
	Min: - [rpm]	Max: - [rpm]	Factory setting: - [rpm]
Description:	Displays the speed setpoint for the I component of the speed controller (output of the reference model after the setpoint limiting).		
Dependency:	See also: r1438		
	Note In the standard state (the reference model is deactivated), r1438 = r1439.		
r1439 SERVO_DBSI_VD (Lin)	Velocity setpoint I component / v_set I_comp		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5030, 5040, 6031
	P group: Closed-loop control	Unit group: 4_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2000	Expert list: 1
	Min: - [m/min]	Max: - [m/min]	Factory setting: - [m/min]
Description:	Displays the velocity setpoint for the I component of the velocity controller (output of the reference model after the setpoint limiting).		
Dependency:	See also: r1438		
	Note In the standard state (the reference model is deactivated), r1438 = r1439.		
p1441[0...n] SERVO_DBSI_VD	Actual speed smoothing time / n_act t_smth		
	Can be changed: T, U	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 4710, 4715
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 50.00 [ms]	Factory setting: 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the speed actual value.		
Dependency:	See also: r0063		
	Note The speed actual value should be smoothed for encoders with a low pulse number or for resolvers. After this parameter has been changed, we recommend that the speed controller is adapted and/or the speed controller settings checked Kp (p1460) and Tn (p1462). This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1441[0...n]	Actual velocity smoothing time / v_act t_smth		
SERVO_DBSI_VD (Lin)	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [ms]	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 50.00 [ms]	Access level: 3 Function diagram: 4710, 4715 Unit selection: - Expert list: 1 Factory setting: 0.00 [ms]
Description:	Sets the smoothing time constant (PT1) for the velocity actual value.		
Dependency:	See also: r0063		
Note The speed actual value should be smoothed for encoders with a low pulse number or for resolvers. After this parameter has been changed, we recommend that the velocity controller is adapted and/or the velocity controller settings checked Kp (p1460) and Tn (p1462).			
r1454	CO: Speed controller system deviation I component / n_ctrl SysDev Tn		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [rpm]	Calculated: - Dyn. index: - Unit group: 3_1 Scaling: p2000 Max: - [rpm]	Access level: 3 Function diagram: 5040 Unit selection: p0505 Expert list: 1 Factory setting: - [rpm]
Description:	Display and connector output for the system deviation of the I component of the speed controller. When the reference model is inactive (p1433 = 0 Hz), this parameter corresponds to the system deviation of the complete PI controller (r1454 = r0064).		
r1454	CO: Velocity controller system deviation I component / v_ctrl SysDev Tn		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [m/min]	Calculated: - Dyn. index: - Unit group: 4_1 Scaling: p2000 Max: - [m/min]	Access level: 3 Function diagram: 5040 Unit selection: p0505 Expert list: 1 Factory setting: - [m/min]
Description:	Display and connector output for the system deviation of the I component of the velocity controller. When the reference model is inactive (p1433 = 0 Hz), this parameter corresponds to the system deviation of the complete PI controller (r1454 = r0064).		
p1460[0...n]	Speed controller P gain adaptation speed lower / n_ctrl Kp n lower		
SERVO_DBSI_VD	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.0000 [Nms/rad]	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: 17_1 Scaling: - Max: 500000000.0000 [Nms/rad]	Access level: 2 Function diagram: 5040, 5042 Unit selection: p0505 Expert list: 1 Factory setting: 0.3000 [Nms/rad]
Description:	Sets the P gain of the speed controller before the adaptation speed range (0 ... p1464). This value corresponds to the basic setting of the P gain of the speed controller without adaptation (p1461 = 100 %).		
Dependency:	See also: p1461, p1464, p1465		

Note

When automatically calculating the speed controller, only the motor moment of inertia is taken into account (p0341). For higher load moments of inertia (p0342 > 1 or p1498 > 0), you are advised to check the speed controller gain.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1460[0...n]**Velocity controller P gain adaptation velocity lower / v_ctrl Kp n lower**

SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5040, 5042
	P group: Closed-loop control	Unit group: 24_2	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.0000 [Ns/m]	Max: 500000000.0000 [Ns/m]	Factory setting: 10.0000 [Ns/m]
Description:	Sets the P gain of the velocity controller before the adaptation velocity range (0 ... p1464). This value corresponds to the basic setting of the P gain of the velocity controller without adaptation (p1461 = 100 %).		
Dependency:	See also: p1461, p1464, p1465		

Note

When automatically calculating the velocity controller, only the motor inertia is taken into account (p0341). For higher inertias (p0342 > 1 or p1498 > 0), you are advised to check the velocity controller gain.

p1461[0...n]**Speed controller Kp adaptation speed upper scaling / n_ctr Kp n up scal**

SERVO_DBSI_VD	Can be changed: T, U	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5050
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.0 [%]	Max: 200000.0 [%]	Factory setting: 100.0 [%]
Description:	Sets the P gain of the speed controller for the upper adaptation speed range (> p1465). The entry is made referred to the P gain for the lower adaptation speed range of the speed controller (% referred to p1460).		
Dependency:	See also: p1460, p1464, p1465		

Note

When automatically calculating the speed controller, only the motor moment of inertia is taken into account (p0341). For higher load moments of inertia (p0342 > 1 or p1498 > 0), you are advised to check the speed controller gain.
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1461[0...n]**Velocity controller Kp adaptation velocity upper scaling / v_ctr Kp n up scal**

SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5050
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.0 [%]	Max: 200000.0 [%]	Factory setting: 100.0 [%]
Description:	Sets the P gain of the velocity controller for the upper adaptation velocity range (> p1465). The entry is made referred to the P gain for the lower adaptation velocity range of the velocity controller (% referred to p1460).		
Dependency:	See also: p1460, p1464, p1465		

Note

When automatically calculating the velocity controller, only the motor inertia is taken into account (p0341). For higher inertias (p0342 > 1 or p1498 > 0), you are advised to check the velocity controller gain.

p1462[0...n]	Speed controller integral time adaptation speed lower / n_ctrl Tn n lower		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5040, 5042, 6020, 6040
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 100000.00 [ms]	Factory setting: 20.00 [ms]
Description:	Sets the integration time of the speed controller before the adaptation speed range (0 ... p1464). This value corresponds to the basic setting of the integral time of the speed controller without adaptation (p1461 = 100 %).		
Dependency:	See also: p1463, p1464, p1465		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
p1462[0...n]	Velocity contr. integral act. time adaptation velocity lower / v_ctrl Tn n lower		
SERVO_DBSI_VD (Lin)	Can be changed: T, U	Calculated: CALC_MOD_CON	Access level: 2
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5040, 5042
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 100000.00 [ms]	Factory setting: 20.00 [ms]
Description:	Sets the integration time of the velocity controller before the adaptation velocity range (0 ... p1464). This value corresponds to the basic setting of the integral time of the velocity controller without adaptation (p1461 = 100 %).		
Dependency:	See also: p1463, p1464, p1465		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
p1463[0...n]	Speed controller Tn adaptation speed upper scaling / n_ctrl Tn n up scal		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: CALC_MOD_CON	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5050
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.0 [%]	Max: 200000.0 [%]	Factory setting: 100.0 [%]
Description:	Sets the integral time of the speed controller after the adaptation speed range (> p1465). The entry is made referred to the integral time for the lower adaptation speed range of the speed controller (% referred to p1462).		
Dependency:	See also: p1462, p1464, p1465		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1463[0...n]	Velocity controller Tn adaptation velocity upper scaling / v_ctr Tn n up scal		
SERVO_DBSI_VD (Lin)	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.0 [%]	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 200000.0 [%]	Access level: 3 Function diagram: 5050 Unit selection: - Expert list: 1 Factory setting: 100.0 [%]
Description:	Sets the integral time of the velocity controller after the adaptation velocity range (> p1465). The entry is made referred to the integral time for the lower adaptation velocity range of the velocity controller (% referred to p1462).		
Dependency:	See also: p1462, p1464, p1465		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1464[0...n]	Speed controller adaptation speed lower / n_ctrl n lower		
SERVO_DBSI_VD	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [rpm]	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: 3_1 Scaling: - Max: 210000.00 [rpm]	Access level: 3 Function diagram: 5050 Unit selection: p0505 Expert list: 1 Factory setting: 0.00 [rpm]
Description:	Sets the lower adaptation speed of the speed controller. No adaptation is effective below this speed.		
Dependency:	See also: p1460, p1461, p1462, p1463, p1465		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1464[0...n]	Velocity controller adaptation velocity lower / v_ctrl n lower		
SERVO_DBSI_VD (Lin)	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [m/min]	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: 4_1 Scaling: - Max: 1000.00 [m/min]	Access level: 3 Function diagram: 5050 Unit selection: p0505 Expert list: 1 Factory setting: 0.00 [m/min]
Description:	Sets the lower adaptation velocity of the velocity controller. No adaptation is effective below this velocity.		
Dependency:	See also: p1460, p1461, p1462, p1463, p1465		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1465[0...n]	Speed controller adaptation speed upper / n_ctrl n upper		
SERVO_DBSI_VD	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [rpm]	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: 3_1 Scaling: - Max: 210000.00 [rpm]	Access level: 3 Function diagram: 5050 Unit selection: p0505 Expert list: 1 Factory setting: 210000.00 [rpm]
Description:	Sets the upper adaptation speed of the speed controller. No adaptation is effective above this speed. For the proportional gain, p1460 x p1461 is effective. For the integral time, p1462 x p1463 is effective.		
Dependency:	See also: p1460, p1461, p1462, p1463, p1464		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1465[0...n]	Velocity controller adaptation velocity upper / v_ctrl n upper		
SERVO_DBSI_VD (Lin)	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [m/min]	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: 4_1 Scaling: - Max: 1000.00 [m/min]	Access level: 3 Function diagram: 5050 Unit selection: p0505 Expert list: 1 Factory setting: 1000.00 [m/min]
Description:	Sets the upper adaptation velocity of the velocity controller. No adaptation is effective above this velocity. For the proportional gain, p1460 x p1461 is effective. For the integral time, p1462 x p1463 is effective.		
Dependency:	See also: p1460, p1461, p1462, p1463, p1464		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

r1493	CO: Moment of inertia total / M_inert total		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [kgm²]	Calculated: - Dyn. index: - Unit group: 25_1 Scaling: - Max: - [kgm²]	Access level: 3 Function diagram: 5042, 5210 Unit selection: p0100 Expert list: 1 Factory setting: - [kgm²]
Description:	Display and connector output for the parameterized total moment of inertia. The value is calculated as follows: (p0341 * p0342) + p1498 The scaling is not take into account using p1497. When the "moment of inertia estimator" is activated (r0108.10 = 1, p1400.18 = 1) and scaling is deactivated (CI: p1497 = 1), the following applies: The currently estimated value of the moment of inertia estimator is displayed in this parameter.		
Dependency:	See also: p1300		
Note The parameterized total moment of inertia, taking into account p1497, influences the torque precontrol. In encoderless operation or when the torque-speed precontrol with encoder (p1402.4 = 1) is activated, then torque-speed precontrol is activated.			

r1493	CO: Total mass / Total mass		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [kg]	Calculated: - Dyn. index: - Unit group: 27_1 Scaling: - Max: - [kg]	Access level: 3 Function diagram: 5042, 5210 Unit selection: p0100 Expert list: 1 Factory setting: - [kg]
Description:	Display and connector output for the parameterized total mass. The value is calculated as follows: (p0341 * p0342) + p1498 The scaling is not take into account using p1497. When the "moment of inertia estimator" is activated (r0108.10 = 1, p1400.18 = 1) and scaling is deactivated (CI: p1497 = 1), the following applies: The currently estimated value of the moment of inertia estimator is displayed in this parameter.		
Dependency:	See also: p1300		
Note The parameterized total mass, taking into account p1497, influences the force-velocity precontrol. In encoderless operation or when the force-velocity precontrol with encoder (p1402.4 = 1) is activated, then torque-velocity precontrol is activated.			

r1509	CO: Torque setpoint before torque limiting / M_set before M_lim		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [Nm]	Calculated: - Dyn. index: - Unit group: 7_1 Scaling: p2003 Max: - [Nm]	Access level: 3 Function diagram: 5019, 5060, 5610 Unit selection: p0505 Expert list: 1 Factory setting: - [Nm]
Description:	Display and connector output for the total torque setpoint before torque limiting. The value is the sum of the controller output, supplementary torque and where relevant precontrol torque, operation without encoder.		

r1509	CO: Force setpoint before force limiting / F_set before F_lim		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [N]	Calculated: - Dyn. index: - Unit group: 8_1 Scaling: p2003 Max: - [N]	Access level: 3 Function diagram: 5019, 5060, 5610 Unit selection: p0505 Expert list: 1 Factory setting: - [N]
Description:	Display and connector output for the total force setpoint before force limiting. The value is the sum of the controller output, supplementary force and where relevant precontrol force, operation without encoder.		

p1511[0...n]	CI: Supplementary torque 1 / M_suppl 1		
SERVO_DBSI_VD	Can be changed: T Data type: Unsigned32 / FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: -	Calculated: - Dyn. index: CDS, p0170 Unit group: - Scaling: p2003 Max: -	Access level: 8 Function diagram: 5060 Unit selection: - Expert list: 1 Factory setting: 0

Description: Sets the signal source for supplementary torque 1.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1513[0...n]

SERVO_DBSI_VD

CI: Supplementary torque 2 / M_suppl 2

Can be changed: T

Data type: Unsigned32 / FloatingPoint32

P group: Closed-loop control

Not for motor type: REL

Min:

-

Calculated: -

Dyn. index: CDS, p0170

Unit group: -

Scaling: p2003

Max:

-

Access level: 3

Function diagram: 5060

Unit selection: -

Expert list: 1

Factory setting:

0

Description: Sets the signal source for supplementary torque 2.

Note

Supplementary torque 2 can be used for weight equalization, and for example, is included in the manufacturer-specific telegram 136.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1513[0...n]

SERVO_DBSI_VD (Lin)

CI: Supplementary force 2 / F_suppl 2

Can be changed: T

Data type: Unsigned32 / FloatingPoint32

P group: Closed-loop control

Not for motor type: REL

Min:

-

Calculated: -

Dyn. index: CDS, p0170

Unit group: -

Scaling: p2003

Max:

-

Access level: 3

Function diagram: 5060

Unit selection: -

Expert list: 1

Factory setting:

0

Description: Sets the signal source for supplementary force 2.

Note

Supplementary force 2 can be used for weight equalization, and for example, is included in the manufacturer-specific telegram 136.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r1515

SERVO_DBSI_VD

Supplementary torque total / M_suppl total

Can be changed: -

Data type: FloatingPoint32

P group: Closed-loop control

Not for motor type: REL

Min:

- [Nm]

Calculated: -

Dyn. index: -

Unit group: 7_1

Scaling: p2003

Max:

- [Nm]

Access level: 8

Function diagram: 5060

Unit selection: p0505

Expert list: 1

Factory setting:

- [Nm]

Description: Displays the total supplementary torque.

The displayed value is the total of supplementary torque values 1 and 2 (p1511, p1512, p1513, p1514).

r1515	Supplementary force total / F_suppl total		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [N]	Calculated: - Dyn. index: - Unit group: 8_1 Scaling: p2003 Max: - [N]	Access level: 8 Function diagram: 5060 Unit selection: p0505 Expert list: 1 Factory setting: - [N]
Description:	Displays the total supplementary force. The displayed value is the total of supplementary forces 1 and 2 (p1511, p1512, p1513, p1514).		

p1520[0...n]	CO: Torque limit upper/motoring / M_max upper/mot		
SERVO_DBSI_VD	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [Nm]	Calculated: CALC_MOD_LIM_REF Dyn. index: DDS, p0180 Unit group: 7_1 Scaling: p2003 Max: 10000000.00 [Nm]	Access level: 2 Function diagram: 5620, 5630 Unit selection: p0505 Expert list: 1 Factory setting: 0.00 [Nm]
Description:	Sets the fixed upper torque limit or the torque limit when motoring.		
Dependency:	p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating See also: p1521, p1532		
 DANGER For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Negative values when setting the upper torque limit (p1520 < 0) can result in the motor accelerating in an uncontrollable fashion.			
NOTICE A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			

p1520[0...n]	CO: Force limit upper/motoring / F_max upper/mot		
SERVO_DBSI_VD (Lin)	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [N]	Calculated: CALC_MOD_LIM_REF Dyn. index: DDS, p0180 Unit group: 8_1 Scaling: p2003 Max: 10000000.00 [N]	Access level: 2 Function diagram: 5620, 5630 Unit selection: p0505 Expert list: 1 Factory setting: 0.00 [N]
Description:	Sets the fixed upper or force limit when motoring.		
Dependency:	p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating See also: p1521, p1532		
 DANGER For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Negative values when setting the upper torque limit (p1520 < 0) can result in the motor accelerating in an uncontrollable fashion.			
NOTICE A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			

p1521[0...n] SERVO_DBSI_VD	CO: Torque limit lower/regenerative / M_max lower/reg		
	Can be changed: T, U	Calculated: CALC_MOD_LIM_REF	Access level: 2
Description: Dependency:	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5620, 5630
	P group: Closed-loop control	Unit group: 7_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min: -1000000.00 [Nm]	Max: 0.00 [Nm]	Factory setting: 0.00 [Nm]
Sets the fixed lower torque limit or the torque limit when regenerating.			
p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating See also: p1520, p1532			
⚠ DANGER			
For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Positive values when setting the lower torque limit (p1521 > 0) can result in the motor accelerating in an uncontrollable fashion.			
NOTICE			
A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			
p1521[0...n] SERVO_DBSI_VD (Lin)	CO: Force limit lower/regenerative / F_max lower/reg		
	Can be changed: T, U	Calculated: CALC_MOD_LIM_REF	Access level: 2
Description: Dependency:	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5620, 5630
	P group: Closed-loop control	Unit group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min: -1000000.00 [N]	Max: 0.00 [N]	Factory setting: 0.00 [N]
Sets the fixed lower or force limit when regenerating.			
p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating See also: p1520, p1532			
⚠ DANGER			
For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Positive values when setting the lower torque limit (p1521 > 0) can result in the motor accelerating in an uncontrollable fashion.			
NOTICE			
A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			
p1524[0...n] SERVO_DBSI_VD	CO: Torque limit upper/motoring scaling / M_max up/mot scal		
	Can be changed: T, U	Calculated: -	Access level: 3
Description: Dependency:	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5620, 5630
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min: -2000.0 [%]	Max: 2000.0 [%]	Factory setting: 100.0 [%]
Sets the scaling for the upper torque limit or the torque limit when motoring.			
p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating			

NOTICE

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

Note

This parameter can be freely interconnected.

The value has the meaning stated above if it is interconnected from connector input p1528.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1524[0...n]

SERVO_DBSI_VD (Lin)

CO: Force limit upper/motoring scaling / F_max up/mot scal**Can be changed:** T, U**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** DDS, p0180**Function diagram:** 5620, 5630**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** PERCENT**Expert list:** 1**Min:****Max:****Factory setting:**

-2000.0 [%]

2000.0 [%]

100.0 [%]

Description:

Sets the scaling for the upper force limit or the force limit when motoring.

Dependency:

p1400.4 = 0: upper/lower

p1400.4 = 1: motoring / regenerating

NOTICE

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

Note

This parameter can be freely interconnected.

The value has the meaning stated above if it is interconnected from connector input p1528.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1525[0...n]

SERVO_DBSI_VD

CO: Torque limit lower/regenerative scaling / M_max low/reg scal**Can be changed:** T, U**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** DDS, p0180**Function diagram:** 5620, 5630**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** PERCENT**Expert list:** 1**Min:****Max:****Factory setting:**

-2000.0 [%]

2000.0 [%]

100.0 [%]

Description:

Sets the scaling for the lower torque limit or the torque limit when regenerating.

Dependency:

p1400.4 = 0: upper/lower

p1400.4 = 1: motoring / regenerating

NOTICE

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

Note

This parameter can be freely interconnected.

The value has the meaning stated above if it is interconnected from connector input p1528.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1525[0...n]			
SERVO_DBSI_VD (Lin)	CO: Force limit lower/regenerative scaling / F_max low/reg scal		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5620, 5630
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min: -2000.0 [%]	Max: 2000.0 [%]	Factory setting: 100.0 [%]
Description:	Sets the scaling for the lower force limit or the force limit when regenerating.		
Dependency:	p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating		
NOTICE			
A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			
Note			
This parameter can be freely interconnected.			
The value has the meaning stated above if it is interconnected from connector input p1528.			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1528[0...n]			
SERVO_DBSI_VD	CI: Torque limit upper/motoring scaling / M_max up/mot scal		
	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dyn. index: CDS, p0170	Function diagram: 3617, 5609, 5620, 5630
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min: -	Max: -	Factory setting: 1524[0]
Description:	Sets the signal source for the scaling of the upper or motoring torque limit in p1522.		
Dependency:	p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating		
 DANGER			
For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Negative values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.			
NOTICE			
The parameter may be protected as a result of p0922 or p2079 and cannot be changed.			

p1528[0...n]			
SERVO_DBSI_VD (Lin)	CI: Force limit upper/motoring scaling / F_max up/mot scal		
	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dyn. index: CDS, p0170	Function diagram: 3617, 5609, 5620, 5630
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min: -	Max: -	Factory setting: 1524[0]
Description:	Sets the signal source for the scaling of the upper or motoring force limit in p1522.		
Dependency:	p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating		

⚠ DANGER
For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Negative values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.

NOTICE
The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1529[0...n]	CI: Torque limit lower/regenerative scaling / M_max low/reg scal		
SERVO_DBSI_VD	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dyn. index: CDS, p0170	Function diagram: 3617, 5609, 5620, 5630
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min: -	Max: -	Factory setting: 1525[0]
Description:	Sets the signal source for the scaling of the lower torque limit or the regenerative torque limit in p1523.		
Dependency:	p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating		

⚠ DANGER
For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Positive values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.

NOTICE
The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1529[0...n]	CI: Force limit lower/regenerative scaling / F_max low/reg scal		
SERVO_DBSI_VD (Lin)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dyn. index: CDS, p0170	Function diagram: 3617, 5609, 5620, 5630
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: PERCENT	Expert list: 1
	Min: -	Max: -	Factory setting: 1525[0]
Description:	Sets the signal source for the scaling of the lower force limit or the regenerative force limit in p1523.		
Dependency:	p1400.4 = 0: upper/lower p1400.4 = 1: motoring / regenerating		

⚠ DANGER
For p1400.4 = 0 (torque limiting, upper/lower) the following applies: Positive values resulting from the signal source and scaling can cause the motor to accelerate in an uncontrolled manner.

NOTICE
The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1530[0...n] SERVO_DBSI_VD	Power limit motoring / P_max mot Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [kW] Description: Dependency:	Calculated: CALC_MOD_LIM_REF Dyn. index: DDS, p0180 Unit group: 14_5 Scaling: - Max: 100000.00 [kW]	Access level: 2 Function diagram: 5640 Unit selection: p0505 Expert list: 1 Factory setting: 0.00 [kW]
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1530[0...n] SERVO_DBSI_VD (Lin)	Power limit motoring / P_max mot Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.00 [kW] Description: Dependency:	Calculated: CALC_MOD_LIM_REF Dyn. index: DDS, p0180 Unit group: 14_8 Scaling: - Max: 100000.00 [kW]	Access level: 2 Function diagram: 5640 Unit selection: p0505 Expert list: 1 Factory setting: 0.00 [kW]
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1531[0...n] SERVO_DBSI_VD	Power limit regenerative / P_max reg Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: -100000.00 [kW] Description: Dependency:	Calculated: CALC_MOD_LIM_REF Dyn. index: DDS, p0180 Unit group: 14_5 Scaling: - Max: -0.01 [kW]	Access level: 2 Function diagram: 5640 Unit selection: p0505 Expert list: 1 Factory setting: -0.01 [kW]
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1531[0...n] SERVO_DBSI_VD (Lin)	Power limit regenerative / P_max reg		
	Can be changed: T, U	Calculated: CALC_MOD_LIM_REF	Access level: 2
Description: Dependency:	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5640
	P group: Closed-loop control	Unit group: 14_8	Unit selection: p0505
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: -100000.00 [kW]	Max: -0.01 [kW]	Factory setting: -0.01 [kW]
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			
p1532[0...n] SERVO_DBSI_VD	CO: Torque limit offset / M_max offset		
	Can be changed: T, U	Calculated: -	Access level: 3
Description:	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5620, 5630, 5650, 7010, 8012
	P group: Closed-loop control	Unit group: 7_1	Unit selection: p0505
Recommendation: Dependency:	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min: -100000.00 [Nm]	Max: 100000.00 [Nm]	Factory setting: 0.00 [Nm]
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			
p1532[0...n] SERVO_DBSI_VD (Lin)	CO: Force offset, force limit / F_max offset		
	Can be changed: T, U	Calculated: -	Access level: 3
Description: Recommendation: Dependency:	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5620, 5630, 5650, 7010, 8012
	P group: Closed-loop control	Unit group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min: -100000.00 [N]	Max: 100000.00 [N]	Factory setting: 0.00 [N]
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			
⚠ DANGER If the offset is set higher/lower than the lower/upper torque limit, then the unloaded drive can accelerate up to the maximum speed.			
NOTICE A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.			
⚠ DANGER If the offset is set higher/lower than the lower/upper torque limit, then the unloaded drive can accelerate up to the maximum speed.			

NOTICE

A BICO interconnection to a parameter that belongs to a drive data set always acts on the effective data set.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r1538	CO: Upper effective torque limit / M_max upper eff		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [Nm]	Calculated: - Dyn. index: - Unit group: 7_1 Scaling: p2003 Max: - [Nm]	Access level: 8 Function diagram: 5609, 5650 Unit selection: p0505 Expert list: 1 Factory setting: - [Nm]
Description:	Display and connector output for the actual effective upper torque limit.		
	Note The effective upper torque limit is reduced with respect to the selected upper torque limit p1520, if the current limit p0640 is reduced or the rated magnetizing current of the induction motor p0320 is increased. The torque limit p1520 can be re-calculated using p0340 = 1, 3 or 5. This torque limit can be influenced by p0543.		
r1538	CO: Upper force limit effective / F_max upper eff		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [N]	Calculated: - Dyn. index: - Unit group: 8_1 Scaling: p2003 Max: - [N]	Access level: 8 Function diagram: 5609, 5650 Unit selection: p0505 Expert list: 1 Factory setting: - [N]
Description:	Display and connector output for the actual effective upper force limit.		
	Note The effective, upper force limit is reduced with respect to the selected upper force limit p1520 if the current limit p0640 is reduced. The force limit p1520 can be re-calculated using p0340 = 1, 3 or 5.		
r1539	CO: Lower effective torque limit / M_max lower eff		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: - [Nm]	Calculated: - Dyn. index: - Unit group: 7_1 Scaling: p2003 Max: - [Nm]	Access level: 8 Function diagram: 5609, 5650 Unit selection: p0505 Expert list: 1 Factory setting: - [Nm]
Description:	Display and connector output for the actual effective lower torque limit.		
	Note The effective lower torque limit is reduced with respect to the selected lower torque limit p1521, if the current limit p0640 is reduced or the rated magnetizing current of the induction motor p0320 is increased. The following applies for VECTOR: - this is possibly the case for a rotating measurement (see p1960). - additional variable torque limiting is possible (e.g. binector input p1541). The torque limit p1520 can be re-calculated using p0340 = 1, 3 or 5. This torque limit can be influenced by p0543.		

r1539	CO: Lower force limit effective / F_max lower eff		
SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 8
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5609, 5650
	P group: Closed-loop control	Unit group: 8_1	Unit selection: p0505
	Not for motor type: REL	Scaling: p2003	Expert list: 1
	Min:	Max:	Factory setting:
	- [N]	- [N]	- [N]
Description:	Display and connector output for the actual effective lower force limit.		

Note

The effective lower force limit is reduced with respect to the selected lower force limit p1521 if the current limit p0640 is reduced.

The force limit p1520 can be re-calculated using p0340 = 1, 3 or 5.

p1542[0...n]	CI: Travel to fixed stop torque reduction / TfS M_red		
SERVO_DBSI_VD	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dyn. index: CDS, p0170	Function diagram: 5610
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0
Description:	Sets the signal source for the torque reduction when traversing to a fixed stop. This value is converted into a factor and is interconnected to the scaling of the torque limits.		
Dependency:	See also: p1528, p1529, p1544, p1545		

NOTICE

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1542[0...n]	CI: Travel to fixed stop force reduction / TfS F_red		
SERVO_DBSI_VD (Lin)	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / FloatingPoint32	Dyn. index: CDS, p0170	Function diagram: 5610
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0
Description:	Sets the signal source for the force reduction when traversing to a fixed stop. This value is converted into a factor and is interconnected to the scaling of the force limits.		
Dependency:	See also: p1528, p1529, p1544, p1545		

NOTICE

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

p1544	Travel to fixed stop evaluation torque reduction / TfS M_red eval		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 5610
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0 [%]	65535 [%]	100 [%]
Description:	Sets the evaluation for the torque/force reduction when traversing to a fixed stop.		
Dependency:	See also: p1528, p1529, p1542, p1545		

Note

4000 hex (16384 dec) in the MOMRED control word corresponds to a reduction by the percentage specified in this parameter.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1544**Travel to fixed stop evaluation force reduction / TfS F_red eval**

SERVO_DBSI_VD (Lin)

Can be changed: T, U**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 5610**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0 [%]

65535 [%]

100 [%]

Description:

Sets the evaluation for the torque/force reduction when traversing to a fixed stop.

Dependency:

See also: p1528, p1529, p1542, p1545

Note

4000 hex (16384 dec) in the MOMRED control word corresponds to a reduction by the percentage specified in this parameter.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1545[0...n]**Bl: Activates travel to a fixed stop / TfS activation**

SERVO_DBSI_VD

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Unsigned32 / Binary**Dyn. index:** CDS, p0170**Function diagram:** 2520, 3617, 8012**P group:** Commands**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

[0] 0

[1] 0

Description:

Sets the signal source to activate/deactivate the "travel to fixed stop" function

1: Travel to fixed stop is active

0: Travel to fixed stop is inactive

Dependency:

See also: p1542, p1544

NOTICE

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

r1650**Current setpoint torque-generating before filter / Iq_set before filt**

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 5710**P group:** Closed-loop control**Unit group:** 6_2**Unit selection:** p0505**Not for motor type:** REL**Scaling:** p2002**Expert list:** 1**Min:****Max:****Factory setting:**

- [Arms]

- [Arms]

- [Arms]

Description:

Displays the torque generating current setpoint Iqset after the torque limits and the clock cycle interpolation is ahead of the current setpoint filters.

r1650		Current setpoint force-generating before filter / Iq_set before filt			
SERVO_DBSI_VD (Lin)	Can be changed: -		Calculated: -		Access level: 3
	Data type: FloatingPoint32		Dyn. index: -		Function diagram: 5710
	P group: Closed-loop control		Unit group: 6_2		Unit selection: p0505
	Not for motor type: REL		Scaling: p2002		Expert list: 1
	Min:		Max:		Factory setting:
	- [Arms]		- [Arms]		- [Arms]
Description:		Displays the force generating current setpoint Iqset after the force limits and the clock cycle interpolation is ahead of the current setpoint filters.			

r1651		CO: Torque setpoint function generator / M_set FG			
SERVO_DBSI_VD	Can be changed: -		Calculated: -		Access level: 3
	Data type: FloatingPoint32		Dyn. index: -		Function diagram: -
	P group: Closed-loop control		Unit group: 7_1		Unit selection: p0505
	Not for motor type: REL		Scaling: p2003		Expert list: 1
	Min:		Max:		Factory setting:
	- [Nm]		- [Nm]		- [Nm]
Description:		Displays the torque setpoint of the function generator.			

r1651		CO: Force setpoint function generator / F_set FG			
SERVO_DBSI_VD (Lin)	Can be changed: -		Calculated: -		Access level: 3
	Data type: FloatingPoint32		Dyn. index: -		Function diagram: -
	P group: Closed-loop control		Unit group: 8_1		Unit selection: p0505
	Not for motor type: REL		Scaling: p2003		Expert list: 1
	Min:		Max:		Factory setting:
	- [N]		- [N]		- [N]
Description:		Displays the force setpoint of the function generator.			

p1656[0...n]		Activates current setpoint filter / I_set_filt act				
SERVO_DBSI_VD	Can be changed: T, U		Calculated: CALC_MOD_CON		Access level: 3	
	Data type: Unsigned16		Dyn. index: DDS, p0180		Function diagram: 5710	
	P group: Closed-loop control		Unit group: -		Unit selection: -	
	Not for motor type: -		Scaling: -		Expert list: 1	
	Min:		Max:		Factory setting:	
	-		-		0001 bin	
Description:		Setting for activating/deactivating the current setpoint filter.				
Bit array:		Bit	Signal name	1 signal	0 signal	FD
		00	Filter 1	Active	Inactive	-
		01	Filter 2	Active	Inactive	-
		02	Filter 3	Active	Inactive	-
		03	Filter 4	Active	Inactive	-
Dependency:		The individual current setpoint filters are parameterized as of p1657.				

Note						
If not all of the filters are required, then the filters should be used consecutively starting from filter 1.						
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.						

p1657[0...n]	Current setpoint filter 1 type / I_set_filt 1 type		
SERVO_DBSI_VD	Can be changed: T, U Data type: Integer16 P group: Closed-loop control Not for motor type: REL Min: 1	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 2	Access level: 3 Function diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting: 1
Description:	Sets the current setpoint filter 1 as low pass (PT2) or general 2nd-order filter.		
Value:	1: PT2 low pass 2: General 2nd order filter		
Dependency:	The current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
Note For a general 2nd order filter, by inserting the same natural frequency in both the numerator and in the denominator, i.e. bandstop frequency, a bandstop filter is implemented. If the numerator damping of zero is selected, the bandstop frequency is completely suppressed. The denominator damping can be determined from the equation for the 3 dB bandwidth: $f_{3dB} \text{ bandwidth} = 2 * D_{denominator} * f_{bandstop} \text{ frequency}$ This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1658[0...n]	Current setpoint filter 1 denominator natural frequency / I_set_filt1 fn_den		
SERVO_DBSI_VD	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.5 [Hz]	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 16000.0 [Hz]	Access level: 3 Function diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting: 1999.0 [Hz]
Description:	Sets the denominator natural frequency for current setpoint filter 1 (PT2, general filter).		
Dependency:	The current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1659[0...n]	Current setpoint filter 1 denominator damping / I_set_filt 1 D_den		
SERVO_DBSI_VD	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.001	Calculated: CALC_MOD_CON Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 10.000	Access level: 3 Function diagram: 5710, 6710 Unit selection: - Expert list: 1 Factory setting: 0.700
Description:	Sets the denominator damping for current setpoint filter 1.		
Dependency:	The current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1660[0...n] SERVO_DBSI_VD	Current setpoint filter 1 numerator natural frequency / I_set_filt1 fn_num		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5710, 6710
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 1999.0 [Hz]
Description:	Sets the numerator natural frequency for current setpoint filter 1 (general filter).		
Dependency:	The current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1661[0...n] SERVO_DBSI_VD	Current setpoint filter 1 numerator damping / I_set_filt 1 D_num		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5710, 6710
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.000	Max: 10.000	Factory setting: 0.700
Description:	Sets the numerator damping for current setpoint filter 1.		
Dependency:	The current setpoint filter 1 is activated via p1656.0 and parameterized via p1657 ... p1661.		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1662[0...n] SERVO_DBSI_VD	Current setpoint filter 2 type / I_set_filt 2 type		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: DDS, p0180	Function diagram: 5710, 6710
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 1	Max: 2	Factory setting: 1
Description:	Sets current setpoint filter 2 as lowpass filter (PT2) or general 2nd order filter.		
Value:	1: PT2 low pass 2: General 2nd order filter		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		
	Note For a general 2nd order filter, by inserting the same natural frequency in both the numerator and in the denominator, i.e. bandstop frequency, a bandstop filter is implemented. If the numerator damping of zero is selected, the bandstop frequency is completely suppressed. The denominator damping can be determined from the equation for the 3 dB bandwidth: f_3dB bandwidth = 2 * D_denominator * f_bandstop frequency This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1663[0...n] SERVO_DBSI_VD	Current setpoint filter 2 denominator natural frequency / I_set_filt2 fn_den		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5710, 6710
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 1999.0 [Hz]
Description:	Sets the denominator natural frequency for current setpoint filter 2 (PT2, general filter).		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
p1664[0...n] SERVO_DBSI_VD	Current setpoint filter 2 denominator damping / I_set_filt 2 D_den		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5710, 6710
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.001	Max: 10.000	Factory setting: 0.700
Description:	Sets the denominator damping for current setpoint filter 2.		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
p1665[0...n] SERVO_DBSI_VD	Current setpoint filter 2 numerator natural frequency / I_set_filt2 fn_num		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5710, 6710
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.5 [Hz]	Max: 16000.0 [Hz]	Factory setting: 1999.0 [Hz]
Description:	Sets the numerator natural frequency for current setpoint filter 2 (general filter).		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
p1666[0...n] SERVO_DBSI_VD	Current setpoint filter 2 numerator damping / I_set_filt 2 D_num		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: DDS, p0180	Function diagram: 5710, 6710
	P group: Closed-loop control	Unit group: -	Unit selection: -
	Not for motor type: REL	Scaling: -	Expert list: 1
	Min: 0.000	Max: 10.000	Factory setting: 0.700
Description:	Sets the numerator damping for current setpoint filter 2.		
Dependency:	Current setpoint filter 2 is activated via p1656.1 and parameterized via p1662 ... p1666.		

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1667[0...n]

SERVO_DBSI_VD

Current setpoint filter 3 type / I_set_filt 3 type**Can be changed:** T, U**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** DDS, p0180**Function diagram:** 5710**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

1

2

1

Description:

Sets current setpoint filter 3 as lowpass filter (PT2) or general 2nd order filter.

Value:

1: PT2 low pass

2: General 2nd order filter

Dependency:

Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1668[0...n]

SERVO_DBSI_VD

Current setpoint filter 3 denominator natural frequency / I_set_filt3 fn_den**Can be changed:** T, U**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** DDS, p0180**Function diagram:** 5710**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.5 [Hz]

16000.0 [Hz]

1999.0 [Hz]

Description:

Sets the denominator natural frequency for current setpoint filter 3 (PT2, general filter).

Dependency:

Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1669[0...n]

SERVO_DBSI_VD

Current setpoint filter 3 denominator damping / I_set_filt 3 D_den**Can be changed:** T, U**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** DDS, p0180**Function diagram:** 5710**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.001

10.000

0.700

Description:

Sets the denominator damping for current setpoint filter 3.

Dependency:

Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1670[0...n] SERVO_DBSI_VD	Current setpoint filter 3 numerator natural frequency / I_set_filt3 fn_num		
	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.5 [Hz]	Calculated: - Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 16000.0 [Hz]	Access level: 3 Function diagram: 5710 Unit selection: - Expert list: 1 Factory setting: 1999.0 [Hz]
Description:	Sets the numerator natural frequency for current setpoint filter 3 (general filter).		
Dependency:	Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1671[0...n] SERVO_DBSI_VD	Current setpoint filter 3 numerator damping / I_set_filt 3 D_num		
	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.000	Calculated: - Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 10.000	Access level: 3 Function diagram: 5710 Unit selection: - Expert list: 1 Factory setting: 0.700
Description:	Sets the numerator damping for current setpoint filter 3.		
Dependency:	Current setpoint filter 3 is activated via p1656.2 and parameterized via p1667 ... p1671.		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1672[0...n] SERVO_DBSI_VD	Current setpoint filter 4 type / I_set_filt 4 type		
	Can be changed: T, U Data type: Integer16 P group: Closed-loop control Not for motor type: REL Min: 1	Calculated: - Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 2	Access level: 3 Function diagram: 5710 Unit selection: - Expert list: 1 Factory setting: 1
Description:	Sets current setpoint filter 4 as lowpass filter (PT2) or general 2nd order filter.		
Value:	1: PT2 low pass 2: General 2nd order filter		
Dependency:	Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p1673[0...n] SERVO_DBSI_VD	Current setpoint filter 4 denominator natural frequency / I_set_filt4 fn_den		
	Can be changed: T, U Data type: FloatingPoint32 P group: Closed-loop control Not for motor type: REL Min: 0.5 [Hz]	Calculated: - Dyn. index: DDS, p0180 Unit group: - Scaling: - Max: 16000.0 [Hz]	Access level: 3 Function diagram: 5710 Unit selection: - Expert list: 1 Factory setting: 1999.0 [Hz]
Description:	Sets the denominator natural frequency for current setpoint filter 4 (PT2, general filter).		

Dependency: Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1674[0...n] Current setpoint filter 4 denominator damping / I_set_filt 4 D_den

SERVO_DBSI_VD

Can be changed: T, U**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** DDS, p0180**Function diagram:** 5710**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.001

10.000

0.700

Description: Sets the denominator damping for current setpoint filter 4.

Dependency: Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1675[0...n] Current setpoint filter 4 numerator natural frequency / I_set_filt4 fn_den

SERVO_DBSI_VD

Can be changed: T, U**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** DDS, p0180**Function diagram:** 5710**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.5 [Hz]

16000.0 [Hz]

1999.0 [Hz]

Description: Sets the numerator natural frequency for current setpoint filter 4 (general filter).

Dependency: Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1676[0...n] Current setpoint filter 4 numerator damping / I_set_filt 4 D_num

SERVO_DBSI_VD

Can be changed: T, U**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** DDS, p0180**Function diagram:** 5710**P group:** Closed-loop control**Unit group:** -**Unit selection:** -**Not for motor type:** REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.000

10.000

0.700

Description: Sets the numerator damping for current setpoint filter 4.

Dependency: Current setpoint filter 4 is activated via p1656.3 and parameterized via p1672 ... p1676.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p1821[0...n]	Direction of rotation / Dir of rot		
SERVO_DBSI_VD	Can be changed: C2(3) Data type: Integer16	Calculated: - Dyn. index: DDS, p0180	Access level: 3 Function diagram: 4704, 4710, 4711, 4715, 5730, 6730, 6731, 6732
	P group: Motor Not for motor type: - Min: 0	Unit group: - Scaling: - Max: 1	Unit selection: - Expert list: 1 Factory setting: 0
Description:	Setting to change the direction of rotation. If the parameter is changed, it reverses the direction of rotation of the motor and the encoder actual value without changing the setpoint.		
Value:	0: Clockwise 1: Counter-clockwise		
Dependency:	See also: F07434		
NOTICE			
An appropriate fault is output for a drive data set changeover where the direction of rotation changes and the pulses are enabled. After changing parameter p1821, the direction of rotation is not automatically adapted in the safety area. The following parameters can be used to set the direction of rotation for safety monitoring: - p9516.1/p9316.1 "Position actual value sign change" (only for operation with encoder) - p9539/p9339 "SI Motion gearbox direction of rotation reversal" (also in encoderless operation)			
Note			
For operation with the phase sequence U/V/W, the direction of rotation is defined when viewing the face side of the motor output shaft. When changing the direction of rotation, the rotating field direction of the current controller is reversed. The speed actual value (e.g. r0063) is also reversed so that the control sense is kept and internally causing the direction of rotation to be reversed with the same setpoint. Further, the position actual values of the actual encoder are reversed (e.g. r0482[0...2]). This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p1821[0...n]	Direction / Direction		
SERVO_DBSI_VD (Lin)	Can be changed: C2(3) Data type: Integer16	Calculated: - Dyn. index: DDS, p0180	Access level: 3 Function diagram: 4704, 4710, 4711, 4715, 5730, 6730, 6731, 6732
	P group: Motor Not for motor type: - Min: 0	Unit group: - Scaling: - Max: 1	Unit selection: - Expert list: 1 Factory setting: 0
Description:	Setting to change the direction. If the parameter is changed, it reverses the direction of rotation of the motor and the encoder actual value without changing the setpoint.		
Value:	0: Clockwise 1: Counter-clockwise		
Dependency:	See also: F07434		

NOTICE

For a drive data set changeover with differently set direction and pulse enable, an appropriate fault is output. After changing parameter p1821, the direction of rotation is not automatically adapted in the safety area. The following parameters can be used to set the direction of rotation for safety monitoring:

- p9516.1/p9316.1 "Position actual value sign change" (only for operation with encoder)
- p9539/p9339 "SI Motion gearbox direction of rotation reversal" (also in encoderless operation)

Note

For operation with the phase sequence U/V/W, the direction is defined when viewing the face side of the motor output shaft.

When changing the direction, the rotating field direction of the current controller is reversed. The actual velocity (e.g. r0063) is also reversed so that the control sense is kept and internally causing the direction of rotation to be reversed with the same setpoint. Further, the position actual values of the actual encoder are reversed (e.g. r0482[0...2]).

For VECTOR, the following applies:

p1820 can be used to reverse the direction of the motor without reversing the encoder actual value.

p2000

SERVO_DBSI_VD

Reference speed reference frequency / n_ref f_ref**Can be changed:** T**Calculated:** CALC_MOD_ALL**Access level:** 2**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

6.00 [rpm]

210000.00 [rpm]

3000.00 [rpm]

Description:

Sets the reference quantity for speed and frequency.

All speeds or frequencies specified as relative value are referred to this reference quantity.

The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).

Dependency:

See also: p2001, p2002, p2003, r2004

Note

For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1.

If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

Example 1:

The signal of an analog input (e.g. r4055[0]) is connected to a speed setpoint (e.g. p1070[0]). The actual percentage input value is cyclically converted into the absolute speed setpoint using the reference speed (p2000).

Example 2:

The setpoint from PROFIBUS (r2050[1]) is connected to a speed setpoint (e.g. p1070[0]). The actual input value is cyclically converted into a percentage value via the pre-specified scaling 4000 hex. This percentage value is converted to the absolute speed setpoint via reference speed (p2000).

p2000

SERVO_DBSI_VD (Lin)

Reference velocity reference frequency / v_ref f_ref**Can be changed:** T**Calculated:** CALC_MOD_ALL**Access level:** 2**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.60 [m/min]

700.00 [m/min]

120.00 [m/min]

Description:

Sets the reference quantity for velocity and frequency.

All velocities or frequencies specified as relative value are referred to this reference quantity.

The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).

The following applies: Reference frequency (in Hz) = reference velocity (in (m/min) / 60)

Dependency:

See also: p2001, p2002, p2003, r2004

Note

For the automatic calculation ($p0340 = 1$, $p3900 > 0$) an appropriate pre-assignment is only made if the parameter is at the factory setting.

If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

Example 1:

The signal of an analog input (e.g. $r4055[0]$) is connected to a velocity setpoint (e.g. $p1070[0]$). The actual percentage input value is cyclically converted into the absolute velocity setpoint using the reference velocity ($p2000$).

Example 2:

The setpoint from PROFIBUS ($r2050[1]$) is connected to a velocity setpoint (e.g. $p1070[0]$). The actual input value is cyclically converted into a percentage value via the pre-specified scaling 4000 hex. This percentage value is converted to the absolute velocity setpoint via reference velocity ($p2000$).

p2001	Reference voltage / Reference voltage		
SERVO_DBSI_VD	Can be changed: T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Communications	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 10 [Vrms]	Max: 100000 [Vrms]	Factory setting: 1000 [Vrms]
Description:	Sets the reference quantity for voltages. All voltages specified as relative value are referred to this reference quantity. This also applies for direct voltage values (= rms value) like the DC link voltage. The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word). Note: This reference quantity also applies to direct voltage values. It is not interpreted as rms value, but as DC voltage value.		
	Note For the automatic calculation ($p0340 = 1$, $p3900 > 0$) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using $p0573 = 1$. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. For infed units, the parameterized device supply voltage ($p0210$) is pre-assigned as the reference quantity. Example: The actual value of the DC link voltage ($r0070$) is connected to a test socket (e.g. $p0771[0]$). The actual voltage value is cyclically converted into a percentage of the reference voltage ($p2001$) and output according to the parameterized scaling. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p2002	Reference current / I_ref		
SERVO_DBSI_VD	Can be changed: T	Calculated: CALC_MOD_ALL	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Communications	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.10 [Arms]	Max: 100000.00 [Arms]	Factory setting: 100.00 [Arms]
Description:	Sets the reference quantity for currents. All currents specified as relative value are referred to this reference quantity. The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).		

NOTICE

If various DDS are used with different motor data, then the reference quantities remain the same as these are not changed over with the DDS. The resulting conversion factor should be taken into account (e.g. for trace records).

Example:

p2002 = 100 A

Reference quantity 100 A corresponds to 100 %

p0305[0] = 100 A

Rated motor current 100 A for MDS0 in DDS0 --> 100 % corresponds to 100 % of the rated motor current

p0305[1] = 50 A

Rated motor current 50 A for MDS1 in DDS1 --> 100 % corresponds to 200 % of the rated motor current

Note

For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1.

SERVO:

Pre-assigned value for p0338 > 0.001 is p0338, otherwise 2 * p0305.

VECTOR:

Pre-assigned value is p0640.

If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

For infeed units, the rated line current, which is obtained from the rated power and parameterized rated line supply voltage ($p2002 = r0206 / p0210 / 1.73$) is pre-assigned as the reference quantity.

Example:

The actual value of a phase current (r0069[0]) is connected to a test socket (e.g. p0771[0]). The current actual value is cyclically converted into a percentage of the reference current (p2002) and output according to the parameterized scaling.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p2003

SERVO_DBSI_VD

Reference torque / M_ref

Can be changed: T

Data type: FloatingPoint32

P group: Communications

Not for motor type: -

Min:

0.01 [Nm]

Calculated: CALC_MOD_ALL

Dyn. index: -

Unit group: 7_2

Scaling: -

Max:

20000000.00 [Nm]

Access level: 3

Function diagram: -

Unit selection: p0505

Expert list: 1

Factory setting:

1.00 [Nm]

Description:

Sets the reference quantity for torque.

All torques specified as relative value are referred to this reference quantity.

The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).

Note

For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using p0573 = 1.

SERVO:

Pre-assigned value for p0338 and p0334 > 0.001 is p0338 * p0334, otherwise 2 * p0333.

VECTOR:

Pre-assigned value is 2 * p0333.

If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

Example:

The actual value of the total torque (r0079) is connected to a test socket (e.g. p0771[0]). The actual torque is cyclically converted into a percentage of the reference torque (p2003) and output according to the parameterized scaling.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p2003	Reference force / Reference force		
SERVO_DBSI_VD (Lin)	Can be changed: T Data type: FloatingPoint32 P group: Communications Not for motor type: - Min: 0.01 [N]	Calculated: CALC_MOD_ALL Dyn. index: - Unit group: 8_2 Scaling: - Max: 20000000.00 [N]	Access level: 3 Function diagram: - Unit selection: p0505 Expert list: 1 Factory setting: 100.00 [N]
Description:	Sets the reference quantity for forces. All forces specified as relative value are referred to this reference quantity. The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).		
	Note For the automatic calculation (p0340 = 1, p3900 > 0) an appropriate pre-assignment is only made if the parameter is at the factory setting. Pre-assigned value for p0338 and p0334 > 0.001 is p0338 * p0334, otherwise 2 * p0333. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. Example: The actual value of the total force (r0079[0]) is connected to a test socket (e.g. p0771[0]). The actual force is cyclically converted into a percentage of the reference force (p2003) and output according to the parameterized scaling.		
r2004	Reference power / P_ref		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Communications Not for motor type: - Min: - [kW]	Calculated: - Dyn. index: - Unit group: 14_10 Scaling: - Max: - [kW]	Access level: 3 Function diagram: - Unit selection: p0505 Expert list: 1 Factory setting: - [kW]
Description:	Displays the reference quantity for power. All power ratings specified as relative value are referred to this reference quantity. The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).		
Dependency:	This value is calculated as follows: Infeed: Calculated from voltage times current. Closed-loop control: Calculated from torque times speed. See also: p2000, p2001, p2002, p2003		
	Note If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor. The reference power is calculated as follows: - $2 * \pi * \text{reference speed} / 60 * \text{reference torque (motor)}$ - $\text{reference voltage} * \text{reference current} * \text{root}(3) \text{ (infeed)}$		
p2005	Reference angle / Reference angle		
SERVO_DBSI_VD	Can be changed: T Data type: FloatingPoint32 P group: Communications Not for motor type: - Min: 90.00 [°]	Calculated: CALC_MOD_ALL Dyn. index: - Unit group: - Scaling: - Max: 180.00 [°]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 90.00 [°]
Description:	Sets the reference quantity for angle. All angles specified as relative value are referred to this reference quantity. The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).		

Note

For the automatic calculation ($p0340 = 1$, $p3900 > 0$) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using $p0573 = 1$.

If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p2006

SERVO_DBSI_VD

Reference temperature / Ref temp**Can be changed:** T**Calculated:** CALC_MOD_ALL**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Communications**Unit group:** 21_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

50.00 [°C]

300.00 [°C]

100.00 [°C]

Description:

Sets the reference quantity for temperature.

All temperatures specified as relative value are referred to this reference quantity.

The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p2007

SERVO_DBSI_VD

Reference acceleration / a_ref**Can be changed:** T**Calculated:** CALC_MOD_ALL**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**0.01 [rev/s²]500000.00 [rev/s²]0.01 [rev/s²]**Description:**

Sets the reference quantity for acceleration rates.

All acceleration rates specified as relative value are referred to this reference quantity.

The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).

Note

For the automatic calculation ($p0340 = 1$, $p3900 > 0$) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using $p0573 = 1$. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

The reference acceleration is calculated as follows:

Reference speed (p2000) converted from 1/min to 1/s divided by 1 s

--> $p2007 = p2000 \text{ [rpm]} / (60 \text{ [s/min]} * 1 \text{ [s]})$

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p2007

SERVO_DBSI_VD (Lin)

Reference acceleration / a_ref**Can be changed:** T**Calculated:** CALC_MOD_ALL**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Communications**Unit group:** 22_1**Unit selection:** p0505**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**0.01 [m/s²]10000.00 [m/s²]0.01 [m/s²]

Description: Sets the reference quantity for acceleration rates.
 All acceleration rates specified as relative value are referred to this reference quantity.
 The reference quantity corresponds to 100% or 4000 hex (word) or 4000 0000 hex (double word).

Note

For the automatic calculation ($p0340 = 1$, $p3900 > 0$) an appropriate pre-assignment is only made if the parameter is not inhibited from being overwritten using $p0573 = 1$. If a BICO interconnection is established between different physical quantities, then the particular reference quantities are used as internal conversion factor.

The reference acceleration is calculated as follows:

Reference speed ($p2000$) converted from 1/min to 1/s divided by 1 s

--> $p2007 = p2000 \text{ [rpm]} / (60 \text{ [s/min]} * 1 \text{ [s]})$

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p2038**IF1 PROFIdrive STW/ZSW interface mode / PD STW/ZSW IF mode**

SERVO_DBSI_VD

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** -**Function diagram:** -**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

1

1

1

Description: Displays the interface mode of the PROFIdrive control words and status words.

Value: 1: SIMODRIVE 611 universal

Dependency: See also: p0922

NOTICE

The parameter is protected and cannot be changed.

Note

For telegram selection p0922 ($p2079$) = 102, 103, 105, 106, 116, 118, 125, 126, 136, 138, 139, then p2038 is automatically set = 1.

When another telegram is selected, then p2038 is automatically set = 0.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p2038**IF1 PROFIdrive STW/ZSW interface mode / PD STW/ZSW IF mode**

SERVO_DBSI_VD

(EPOS, Pos ctrl, Pos ctrl)

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** -**Function diagram:** -**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

1

1

Description: Displays the interface mode of the PROFIdrive control words and status words.

Value: 0: SINAMICS

1: SIMODRIVE 611 universal

Dependency: See also: p0922

NOTICE

The parameter is protected and cannot be changed.

r2050[0...21]	CO: IF1 PROFIdrive PZD receive word / IF1 PZD recv word				
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Integer16	Dyn. index: -	Function diagram: 2440, 2468		
	P group: Communications	Unit group: -	Unit selection: -		
	Not for motor type: -	Scaling: 4000H	Expert list: 1		
	Min:	Max:	Factory setting:		
	-	-	-		
Description:	Connector output to interconnect PZD (setpoints) with word format received from the fieldbus controller.				
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8 [8] = PZD 9 [9] = PZD 10 [10] = PZD 11 [11] = PZD 12 [12] = PZD 13 [13] = PZD 14 [14] = PZD 15 [15] = PZD 16 [16] = PZD 17 [17] = PZD 18 [18] = PZD 19 [19] = PZD 20 [20] = PZD 21 [21] = PZD 22				
Dependency:	See also: r2060				
	<table><tr><td>NOTICE</td></tr><tr><td>Where there is a multiple interconnection of a connector output, all the connector inputs must either have Integer or FloatingPoint data types. A BICO interconnection for a single PZD can only take place either on r2050 or r2060.</td></tr></table>			NOTICE	Where there is a multiple interconnection of a connector output, all the connector inputs must either have Integer or FloatingPoint data types. A BICO interconnection for a single PZD can only take place either on r2050 or r2060.
NOTICE					
Where there is a multiple interconnection of a connector output, all the connector inputs must either have Integer or FloatingPoint data types. A BICO interconnection for a single PZD can only take place either on r2050 or r2060.					
	<table><tr><td>Note</td></tr><tr><td>IF1: Interface 1</td></tr></table>			Note	IF1: Interface 1
Note					
IF1: Interface 1					

r2053[0...27]	IF1 PROFIdrive diagnostics PZD send word / IF1 diag send word		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: 2450, 2470
	P group: Communications	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Displays the PZD (actual values) with word format sent to the fieldbus controller.		

Index:

- [0] = PZD 1
- [1] = PZD 2
- [2] = PZD 3
- [3] = PZD 4
- [4] = PZD 5
- [5] = PZD 6
- [6] = PZD 7
- [7] = PZD 8
- [8] = PZD 9
- [9] = PZD 10
- [10] = PZD 11
- [11] = PZD 12
- [12] = PZD 13
- [13] = PZD 14
- [14] = PZD 15
- [15] = PZD 16
- [16] = PZD 17
- [17] = PZD 18
- [18] = PZD 19
- [19] = PZD 20
- [20] = PZD 21
- [21] = PZD 22
- [22] = PZD 23
- [23] = PZD 24
- [24] = PZD 25
- [25] = PZD 26
- [26] = PZD 27
- [27] = PZD 28

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Bit 0	ON	OFF	-
	01	Bit 1	ON	OFF	-
	02	Bit 2	ON	OFF	-
	03	Bit 3	ON	OFF	-
	04	Bit 4	ON	OFF	-
	05	Bit 5	ON	OFF	-
	06	Bit 6	ON	OFF	-
	07	Bit 7	ON	OFF	-
	08	Bit 8	ON	OFF	-
	09	Bit 9	ON	OFF	-
	10	Bit 10	ON	OFF	-
	11	Bit 11	ON	OFF	-
	12	Bit 12	ON	OFF	-
	13	Bit 13	ON	OFF	-
	14	Bit 14	ON	OFF	-
	15	Bit 15	ON	OFF	-

Note

IF1: Interface 1

r2060[0...20]	CO: IF1 PROFIdrive PZD receive double word / IF1 PZD recv DW				
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Integer32	Dyn. index: -	Function diagram: 2440, 2468		
	P group: Communications	Unit group: -	Unit selection: -		
	Not for motor type: -	Scaling: 4000H	Expert list: 1		
	Min:	Max:	Factory setting:		
	-	-	-		
Description:	Connector output to interconnect PZD (setpoints) with double word format received from the fieldbus controller.				
Index:	[0] = PZD 1 + 2 [1] = PZD 2 + 3 [2] = PZD 3 + 4 [3] = PZD 4 + 5 [4] = PZD 5 + 6 [5] = PZD 6 + 7 [6] = PZD 7 + 8 [7] = PZD 8 + 9 [8] = PZD 9 + 10 [9] = PZD 10 + 11 [10] = PZD 11 + 12 [11] = PZD 12 + 13 [12] = PZD 13 + 14 [13] = PZD 14 + 15 [14] = PZD 15 + 16 [15] = PZD 16 + 17 [16] = PZD 17 + 18 [17] = PZD 18 + 19 [18] = PZD 19 + 20 [19] = PZD 20 + 21 [20] = PZD 21 + 22				
Dependency:	See also: r2050				
	<table><tr><td>NOTICE</td></tr><tr><td>Where there is a multiple interconnection of a connector output, all the connector inputs must either have Integer or FloatingPoint data types. A BICO interconnection for a single PZD can only take place either on r2050 or r2060. A maximum of 4 indices of the "trace" function can be used.</td></tr></table>			NOTICE	Where there is a multiple interconnection of a connector output, all the connector inputs must either have Integer or FloatingPoint data types. A BICO interconnection for a single PZD can only take place either on r2050 or r2060. A maximum of 4 indices of the "trace" function can be used.
NOTICE					
Where there is a multiple interconnection of a connector output, all the connector inputs must either have Integer or FloatingPoint data types. A BICO interconnection for a single PZD can only take place either on r2050 or r2060. A maximum of 4 indices of the "trace" function can be used.					
	<table><tr><td>Note</td></tr><tr><td>IF1: Interface 1</td></tr></table>			Note	IF1: Interface 1
Note					
IF1: Interface 1					

r2089[0...4]	CO: IF1 send binector-connector converter status word / Bin/con ZSW send		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: 2472
	P group: Communications	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Connector output to interconnect the status words to a PZD send word.		

Index:

- [0] = Status word 1
- [1] = Status word 2
- [2] = Free status word 3
- [3] = Free status word 4
- [4] = Free status word 5

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Bit 0	ON	OFF	-
	01	Bit 1	ON	OFF	-
	02	Bit 2	ON	OFF	-
	03	Bit 3	ON	OFF	-
	04	Bit 4	ON	OFF	-
	05	Bit 5	ON	OFF	-
	06	Bit 6	ON	OFF	-
	07	Bit 7	ON	OFF	-
	08	Bit 8	ON	OFF	-
	09	Bit 9	ON	OFF	-
	10	Bit 10	ON	OFF	-
	11	Bit 11	ON	OFF	-
	12	Bit 12	ON	OFF	-
	13	Bit 13	ON	OFF	-
	14	Bit 14	ON	OFF	-
	15	Bit 15	ON	OFF	-

Note

r2089 together with p2080 to p2084 forms five binector-connector converters.

r2090.0...15 BO: IF1 PROFIdrive PZD1 receive bit-serial / PZD1 recv bitw

SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: 2468
	P group: Communications	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -

Description: Binector output for bit-serial interconnection of PZD1 (normally control word 1) received from the PROFIdrive controller.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Bit 0	ON	OFF	-
	01	Bit 1	ON	OFF	-
	02	Bit 2	ON	OFF	-
	03	Bit 3	ON	OFF	-
	04	Bit 4	ON	OFF	-
	05	Bit 5	ON	OFF	-
	06	Bit 6	ON	OFF	-
	07	Bit 7	ON	OFF	-
	08	Bit 8	ON	OFF	-
	09	Bit 9	ON	OFF	-
	10	Bit 10	ON	OFF	-
	11	Bit 11	ON	OFF	-
	12	Bit 12	ON	OFF	-
	13	Bit 13	ON	OFF	-
	14	Bit 14	ON	OFF	-

15	Bit 15	ON	OFF	-
----	--------	----	-----	---

Note

IF1: Interface 1

r2091.0...15

SERVO_DBSI_VD

BO: IF1 PROFIdrive PZD2 receive bit-serial / PZD2 recv bitw**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 2468**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Binector output for bit-serial interconnection of PZD2 received from the PROFIdrive controller.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Bit 0	ON	OFF	-
01	Bit 1	ON	OFF	-
02	Bit 2	ON	OFF	-
03	Bit 3	ON	OFF	-
04	Bit 4	ON	OFF	-
05	Bit 5	ON	OFF	-
06	Bit 6	ON	OFF	-
07	Bit 7	ON	OFF	-
08	Bit 8	ON	OFF	-
09	Bit 9	ON	OFF	-
10	Bit 10	ON	OFF	-
11	Bit 11	ON	OFF	-
12	Bit 12	ON	OFF	-
13	Bit 13	ON	OFF	-
14	Bit 14	ON	OFF	-
15	Bit 15	ON	OFF	-

Note

IF1: Interface 1

r2092.0...15

SERVO_DBSI_VD

BO: IF1 PROFIdrive PZD3 receive bit-serial / PZD3 recv bitw**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 2468**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Binector output for bit-serial interconnection of PZD3 received from the PROFIdrive controller.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Bit 0	ON	OFF	-
01	Bit 1	ON	OFF	-
02	Bit 2	ON	OFF	-
03	Bit 3	ON	OFF	-
04	Bit 4	ON	OFF	-
05	Bit 5	ON	OFF	-
06	Bit 6	ON	OFF	-

07	Bit 7	ON	OFF	-
08	Bit 8	ON	OFF	-
09	Bit 9	ON	OFF	-
10	Bit 10	ON	OFF	-
11	Bit 11	ON	OFF	-
12	Bit 12	ON	OFF	-
13	Bit 13	ON	OFF	-
14	Bit 14	ON	OFF	-
15	Bit 15	ON	OFF	-

Note

IF1: Interface 1

r2093.0...15

SERVO_DBSI_VD

BO: IF1 PROFIdrive PZD4 receive bit-serial / PZD4 recv bitw**Can be changed:** -**Data type:** Unsigned16**P group:** Communications**Not for motor type:** -**Min:**

-

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** -**Max:**

-

Access level: 3**Function diagram:** 2468**Unit selection:** -**Expert list:** 1**Factory setting:**

-

Description:

Binector output for bit-serial interconnection of PZD4 (normally control word 2) received from the PROFIdrive controller.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Bit 0	ON	OFF	-
01	Bit 1	ON	OFF	-
02	Bit 2	ON	OFF	-
03	Bit 3	ON	OFF	-
04	Bit 4	ON	OFF	-
05	Bit 5	ON	OFF	-
06	Bit 6	ON	OFF	-
07	Bit 7	ON	OFF	-
08	Bit 8	ON	OFF	-
09	Bit 9	ON	OFF	-
10	Bit 10	ON	OFF	-
11	Bit 11	ON	OFF	-
12	Bit 12	ON	OFF	-
13	Bit 13	ON	OFF	-
14	Bit 14	ON	OFF	-
15	Bit 15	ON	OFF	-

Note

IF1: Interface 1

p2103[0...n]

SERVO_DBSI_VD

Bl: 1st acknowledge faults / 1st acknowledge**Can be changed:** T, U**Data type:** Unsigned32 / Binary**P group:** Messages**Not for motor type:** -**Min:**

-

Calculated: -**Dyn. index:** CDS, p0170**Unit group:** -**Scaling:** -**Max:**

-

Access level: 3**Function diagram:** 2441, 2442, 2443, 2447, 2475, 2546, 9220, 9677, 9678**Unit selection:** -**Expert list:** 1**Factory setting:**

0

Description: Sets the first signal source to acknowledge faults.

NOTICE

The parameter may be protected as a result of p0922 or p2079 and cannot be changed.

Note

A fault acknowledgment is triggered with a 0/1 signal.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p2104[0...n]

SERVO_DBSI_VD

BI: 2nd acknowledge faults / 2nd acknowledge**Can be changed:** T, U**Calculated:** -**Access level:** 3**Data type:** Unsigned32 / Binary**Dyn. index:** CDS, p0170**Function diagram:** 2546, 8060**P group:** Messages**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

0

Description: Sets the second signal source to acknowledge faults.

Note

A fault acknowledgment is triggered with a 0/1 signal.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r2109[0...63]A_INF_840_VD,
CU_I_840_VD,
CU_NX_840_VD**Fault time removed in milliseconds / t_flt rem ms****Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** 8050, 8060**P group:** Messages**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

- [ms]

- [ms]

- [ms]

Description: Displays the system runtime in milliseconds when the fault was removed.

Dependency: See also: r0945, r0947, r0948, r0949, r2130, r2136, r3120, r3122

NOTICE

The time comprises r2136 (days) and r2109 (milliseconds).

Note

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

The structure of the fault buffer and the assignment of the indices is shown in r0945.

p2111

SERVO_DBSI_VD

Alarm counter / Alarm counter**Can be changed:** T, U**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 8050, 8065**P group:** Messages**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

65535

0

Description: Number of alarms that have occurred after the last reset.

Dependency: When p2111 is set to 0, the following is initiated:

- all of the alarms of the alarm buffer that have gone [0...7] are transferred into the alarm history [8...63].
- the alarm buffer [0...7] is deleted.

See also: r2122, r2123, r2124, r2125

Note

The parameter is reset to 0 at POWER ON.

r2121

A_INF_840_VD,
SERVO_DBSI_VD

CO: Counter alarm buffer changes / A buff changed

Can be changed: -

Data type: Unsigned16

P group: Messages

Not for motor type: -

Min:

-

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

-

Access level: 3

Function diagram: 8065

Unit selection: -

Expert list: 1

Factory setting:

-

Description:

This counter is incremented every time the alarm buffer changes.

Dependency:

See also: r2122, r2123, r2124, r2125

r2122[0...63]

A_INF_840_VD,
CU_I_840_VD,
CU_NX_840_VD

Alarm code / Alarm code

Can be changed: -

Data type: Unsigned16

P group: Messages

Not for motor type: -

Min:

-

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

-

Access level: 2

Function diagram: 8050, 8065

Unit selection: -

Expert list: 1

Factory setting:

-

Description:

Displays the number of alarms that have occurred.

Dependency:

See also: r2123, r2124, r2125, r2145, r2146, r3121

NOTICE

The properties of the alarm buffer should be taken from the corresponding product documentation.

Note

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

Alarm buffer structure (general principle):

r2122[0], r2124[0], r2123[0], r2125[0] --> alarm 1 (the oldest)

...

r2122[7], r2124[7], r2123[7], r2125[7] --> Alarm 8 (the latest)

When the alarm buffer is full, the alarms that have gone are entered into the alarm history:

r2122[8], r2124[8], r2123[8], r2125[8] --> Alarm 1 (the latest)

...

r2122[63], r2124[63], r2123[63], r2125[63] --> alarm 56 (the oldest)

r2123[0...63]

A_INF_840_VD,
CU_I_840_VD,
CU_NX_840_VD

Alarm time received in milliseconds / t_alarm rcv ms

Can be changed: -

Data type: Unsigned32

P group: Messages

Not for motor type: -

Min:

- [ms]

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

- [ms]

Access level: 3

Function diagram: 8050, 8065

Unit selection: -

Expert list: 1

Factory setting:

- [ms]

Description:

Displays the system runtime in milliseconds when the alarm occurred.

Dependency:

See also: r2122, r2124, r2125, r2145, r2146, r3121

NOTICE

The time comprises r2145 (days) and r2123 (milliseconds).

Note

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

The structure of the alarm buffer and the assignment of the indices is shown in r2122.

r2124[0...63]	Alarm value / Alarm value		
A_INF_840_VD, CU_I_840_VD, CU_NX_840_VD	Can be changed: - Data type: Integer32 P group: Messages Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 8050, 8065 Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays additional information about the active alarm (as integer number).		
Dependency:	See also: r2122, r2123, r2125, r2145, r2146, r3121		
Note The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the alarm buffer and the assignment of the indices is shown in r2122.			

r2125[0...63]	Alarm time removed in milliseconds / t_alarm rem ms		
A_INF_840_VD, CU_I_840_VD, CU_NX_840_VD	Can be changed: - Data type: Unsigned32 P group: Messages Not for motor type: - Min: - [ms]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: - [ms]	Access level: 3 Function diagram: 8050, 8065 Unit selection: - Expert list: 1 Factory setting: - [ms]
Description:	Displays the system runtime in milliseconds when the alarm was cleared.		
Dependency:	See also: r2122, r2123, r2124, r2145, r2146, r3121		
NOTICE The time comprises r2146 (days) and r2125 (milliseconds).			
Note The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the alarm buffer and the assignment of the indices is shown in r2122.			

r2130[0...63]	Fault time received in days / t_fault recv days		
All objects	Can be changed: - Data type: Unsigned16 P group: Messages Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 8060 Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the system runtime in days when the fault occurred.		
Dependency:	See also: r0945, r0947, r0948, r0949, r2109, r2136, r3120, r3122		
NOTICE The time comprises r2130 (days) and r0948 (milliseconds). The time display depends on the selected mode (p3100).			
Note The buffer parameters are cyclically updated in the background (refer to status signal in r2139).			

r2135.0...15	CO/BO: Status word faults/alarms 2 / ZSW fault/alarm 2				
SERVO_DBSI_VD	Can be changed: -		Calculated: -	Access level: 2	
	Data type: Unsigned16		Dyn. index: -	Function diagram: 2548	
	P group: Displays, signals		Unit group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min:		Max:	Factory setting:	
	-		-	-	
Description:	Display and BICO output for the second status word of faults and alarms.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Fault encoder 1	Yes	No	-
	01	Fault encoder 2	Yes	No	-
	02	Fault encoder 3	Yes	No	-
	12	Fault motor overtemperature	Yes	No	8016
	13	Fault power unit thermal overload	Yes	No	8021
	14	Alarm motor overtemperature	Yes	No	8016
	15	Alarm power unit thermal overload	Yes	No	8021

r2136[0...63]	Fault time removed in days / t_{flt} rem days		
A_INF_840_VD, CU_I_840_VD, CU_NX_840_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: 8060
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Displays the system runtime in days when the fault was removed.		
Dependency:	See also: r0945, r0947, r0948, r0949, r2109, r2130, r3120, r3122		
NOTICE			
The time comprises r2136 (days) and r2109 (milliseconds).			
Note			
The buffer parameters are cyclically updated in the background (refer to status signal in r2139).			

r2139.0...15	CO/BO: Status word faults/alarms 1 / ZSW fault/alarm 1				
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 2		
	Data type: Unsigned16	Dyn. index: -	Function diagram: 2548		
	P group: Displays, signals	Unit group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min:	Max:	Factory setting:		
	-	-	-		
Description:	Display and BICO output for status word 1 of faults and alarms.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Being acknowledged	Yes	No	-
	01	Acknowledgment required	Yes	No	-
	03	Fault present	Yes	No	8060
	05	Safety message present	Yes	No	-
	06	Internal message 1 present	Yes	No	-
	07	Alarm present	Yes	No	8065
	08	Internal message 2 present	Yes	No	-
	11	Alarm class bit 0	High	Low	-

12	Alarm class bit 1	High	Low	-
13	Maintenance required	Yes	No	-
14	Maintenance urgently required	Yes	No	-
15	Fault gone/can be acknowledged	Yes	No	-

Note

For bit 03, 05, 07:

These bits are set if at least one fault/alarm occurs. Data is entered into the fault/alarm buffer with delay. For this reason, the fault/alarm buffer should only be read if, after "Fault active" or "Alarm active" occurs, a change is also identified in the buffer (r0944, r9744, r2121).

For bit 06, 08:

These status bits are used for internal diagnostic purposes only.

For bit 12, 11:

These status bits are used for the classification of internal alarm classes and are intended for diagnostic purposes only on certain automation systems with integrated SINAMICS functionality.

r2145[0...63]

A_INF_840_VD,
CU_I_840_VD,
CU_NX_840_VD

Alarm time received in days / t_alarm rcv days

Can be changed: -

Calculated: -

Access level: 3

Data type: Unsigned16

Dyn. index: -

Function diagram: 8065

P group: Messages

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description:

Displays the system runtime in days when the alarm occurred.

Dependency:

See also: r2122, r2123, r2124, r2125, r2146, r3121

NOTICE

The time comprises r2145 (days) and r2123 (milliseconds).

Note

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

r2146[0...63]

A_INF_840_VD,
CU_I_840_VD,
CU_NX_840_VD

Alarm time removed in days / t_alarm rem days

Can be changed: -

Calculated: -

Access level: 3

Data type: Unsigned16

Dyn. index: -

Function diagram: 8065

P group: Messages

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description:

Displays the system runtime in days when the alarm was cleared.

Dependency:

See also: r2122, r2123, r2124, r2125, r2145, r3121

NOTICE

The time comprises r2146 (days) and r2125 (milliseconds).

Note

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

p2161[0...n] SERVO_DBSI_VD	Speed threshold 3 / n_thr val 3				
	Can be changed: T, U		Calculated: CALC_MOD_LIM_REF	Access level: 3	
	Data type: FloatingPoint32		Dyn. index: DDS, p0180	Function diagram: 8010, 8011	
	P group: Messages		Unit group: 3_1	Unit selection: p0505	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min: 0.00 [rpm]		Max: 210000.00 [rpm]	Factory setting: 5.00 [rpm]	
Description:	Sets the speed threshold value for the signal " n_act < speed threshold value 3" (BO: r2199.0).				
Dependency:	See also: r2199				
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.				

p2161[0...n] SERVO_DBSI_VD (Lin)	Velocity threshold value 3 / v_thr val 3				
	Can be changed: T, U		Calculated: CALC_MOD_LIM_REF	Access level: 3	
	Data type: FloatingPoint32		Dyn. index: DDS, p0180	Function diagram: 8010, 8011	
	P group: Messages		Unit group: 4_1	Unit selection: p0505	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min: 0.00 [m/min]		Max: 1000.00 [m/min]	Factory setting: 0.05 [m/min]	
Description:	Sets the velocity threshold value for the signal " v_act < velocity threshold value 3" (BO: r2199.0).				
Dependency:	See also: r2199				
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.				

r2197.1...13 SERVO_DBSI_VD	CO/BO: Status word monitoring 1 / ZSW mon 1				
	Can be changed: -		Calculated: -	Access level: 2	
	Data type: Unsigned16		Dyn. index: -	Function diagram: 2534	
	P group: Messages		Unit group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min: -		Max: -	Factory setting: -	
Description:	Display and BICO output for the first status word of the monitoring functions.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	01	n_act <= speed threshold value 2 p2155	Yes	No	8010
	02	n_act > speed threshold value 2 p2155	Yes	No	8010
	03	n_act >= 0	Yes	No	8011
	06	n_act > n_max	Yes	No	8010
	07	Speed setpoint - actual value deviation in tolerance t_off	Yes	No	8011
	13	n_act > n_max (F07901)	Yes	No	-

Note

For bit 01, 02:
The threshold value is set in p2155 and the hysteresis in p2140.
For bit 03:
The hysteresis is set in p2150.
For bit 06:
The hysteresis is set in p2162.
For bit 07:
The threshold value is set in p2163 and the hysteresis is set in p2164.
For bit 13:
Only for internal Siemens use.

r2197.1...13**CO/BO: Status word monitoring 1 / ZSW mon 1**

SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dyn. index: -	Function diagram: 2534
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-

Description:

Display and BICO output for the first status word of the monitoring functions.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
01	v_act <= velocity threshold value 2 p2155	Yes	No	8010
02	v_act > velocity threshold value 2 p2155	Yes	No	8010
03	v_act >= 0	Yes	No	8011
06	v_act > v_max	Yes	No	8010
07	Velocity setpoint - actual value deviation in tolerance t_off	Yes	No	8011
13	v_act > v_max (F07901)	Yes	No	-

Note

For bit 01, 02:
The threshold value is set in p2155 and the hysteresis in p2140.
For bit 03:
The hysteresis is set in p2150.
For bit 06:
The hysteresis is set in p2162.
For bit 07:
The threshold value is set in p2163 and the hysteresis is set in p2164.
For bit 13:
Only for internal Siemens use.

r2198.4...12**CO/BO: Status word monitoring 2 / ZSW monitor 2**

SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 2
	Data type: Unsigned16	Dyn. index: -	Function diagram: 2536
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-

Description:

Display and BICO output for the second status word of the monitoring functions.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
04	n_set < p2161	Yes	No	8011
05	n_set > 0	Yes	No	8011
06	Motor blocked	Yes	No	8012

10	M_set < torque threshold value 1	Yes	No	8012
11	Load in the alarm range	Yes	No	8013
12	Load in the fault range	Yes	No	8013

Note

For bit 10:

The torque threshold value 1 is set in p2174.

For bit 12:

This bit is reset after the fault cause disappears, even if the fault itself is still present.

r2198.4...12**CO/BO: Status word monitoring 2 / ZSW monitor 2**

SERVO_DBSI_VD (Lin)

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 2536**P group:** Messages**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and BICO output for the second status word of the monitoring functions.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
04	n_set < p2161	Yes	No	8011
05	v_set > 0	Yes	No	8011
06	Motor blocked	Yes	No	8012
10	Force setpoint < force threshold value 1	Yes	No	8012
11	Load in the alarm range	Yes	No	8013
12	Load in the fault range	Yes	No	8013

Note

For bit 10:

The force threshold value 1 is set in p2174.

r2199.0...11**CO/BO: Status word monitoring 3 / ZSW monitor 3**

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 2537**P group:** Messages**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

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-

Description:

Display and BICO output for the third status word of the monitoring functions.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	n_act < speed threshold value 3	Yes	No	8010
01	f or n comparison value reached or exceeded	Yes	No	8010
04	Speed setpoint - actual value deviation in tolerance t_on	Yes	No	8011
05	Ramp-up/ramp-down completed	Yes	No	8011
06	Current below the zero current threshold	Yes	No	8020
11	Torque utilization < torque threshold value 2	Yes	No	8012

Dependency:

See also: F07913

Note

For bit 00:

The speed threshold value 3 is set in p2161.

For bit 01:

The comparison value is set in p2141. We recommend setting the hysteresis (p2142) for canceling the bit to a value lower than that in p2141. Otherwise, the bit is not reset.

For bit 11:

The torque threshold value 2 is set in p2194.

r2199.0...11**CO/BO: Status word monitoring 3 / ZSW monitor 3**

SERVO_DBSI_VD (Lin)

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 2537**P group:** Messages**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and BICO output for the third status word of the monitoring functions.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	v_act < velocity threshold value 3	Yes	No	8010
01	f or v comparison value reached or exceeded	Yes	No	8010
04	Velocity setpoint - actual value deviation in tolerance t_on	Yes	No	8011
05	Ramp-up/ramp-down completed	Yes	No	8011
06	Current below the zero current threshold	Yes	No	8020
11	Force utilization < force threshold value 2	Yes	No	8012

Dependency:

See also: F07913

Note

For bit 00:

The velocity threshold value 3 is set in p2161.

For bit 01:

The comparison value is set in p2141.

For bit 11:

The force threshold value 2 is set in p2194.

p2952[0...n]**Flux characteristic coefficient K01 / Psid_mod K01**

SERVO_DBSI_VD

Can be changed: T, U**Calculated:** -**Access level:** 4**Data type:** FloatingPoint32**Dyn. index:** MDS, p0130**Function diagram:** -**P group:** Motor**Unit group:** -**Unit selection:** -**Not for motor type:** PMSM, SESM, REL**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Sets coefficient K01 of the saturation characteristic for the flux model.

The coefficients (K01, K02, K03) are determined by performing a rotating measurement (p1960). For some Siemens motors, these are also automatically preassigned.

Dependency:

See also: p2953, p2954

Note

For p2952 = 0, the saturation characteristic is deactivated.

p2953[0...n] SERVO_DBSI_VD	Flux characteristic coefficient K02 / Psid_mod K02		
	Can be changed: T, U	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dyn. index: MDS, p0130	Function diagram: -
	P group: Motor	Unit group: -	Unit selection: -
	Not for motor type: PMSM, SESM, REL	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Sets coefficient K02 of the saturation characteristic for the flux model. The coefficients (K01, K02, K03) are determined by performing a rotating measurement (p1960). For some Siemens motors, these are also automatically preassigned.		
Dependency:	See also: p2952, p2954		
	Note For p2952 = 0, the saturation characteristic is deactivated.		
p2954[0...n] SERVO_DBSI_VD	Flux characteristic coefficient K03 / Psid_mod K03		
	Can be changed: T, U	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dyn. index: MDS, p0130	Function diagram: -
	P group: Motor	Unit group: -	Unit selection: -
	Not for motor type: SESM, REL	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Sets coefficient K03 of the saturation characteristic for the flux model. The coefficients (K01, K02, K03) are determined by performing a rotating measurement (p1960). For some Siemens motors, these are also automatically preassigned.		
Dependency:	See also: p2952, p2953		
	Note For p2952 = 0, the saturation characteristic is deactivated.		
p2955[0...n] SERVO_DBSI_VD	Flux characteristic coefficient K04 / Psid_mod K04		
	Can be changed: T, U	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dyn. index: MDS, p0130	Function diagram: -
	P group: Motor	Unit group: -	Unit selection: -
	Not for motor type: ASM, SESM, REL	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Sets the reduction factor of the permanent magnet flux for maximum current (p0323). For some Siemens motors, the coefficient is automatically preassigned.		
	Note For p2954 = 0, the saturation characteristic is deactivated.		
p2980[0...n] SERVO_DBSI_VD	Quadrature axis flux model coefficient K01 / Psiq_mod K01		
	Can be changed: T, U	Calculated: -	Access level: 4
	Data type: FloatingPoint32	Dyn. index: MDS, p0130	Function diagram: -
	P group: Motor	Unit group: -	Unit selection: -
	Not for motor type: ASM, SESM, REL	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -

Description: Definition of the saturation characteristic for the quadrature axis flux model of the synchronous reluctance motor (RESM).
This parameter is coefficient K01.

Note

RESM: reluctance synchronous motor (synchronous reluctance motor)

p2981[0...n] Quadrature axis flux model coefficient K02 / Psiq_mod K02

SERVO_DBSI_VD

Can be changed: T, U

Calculated: -

Access level: 4

Data type: FloatingPoint32

Dyn. index: MDS, p0130

Function diagram: -

P group: Motor

Unit group: -

Unit selection: -

Not for motor type: ASM, SESM, REL

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description: Definition of the saturation characteristic for the quadrature axis flux model of the synchronous reluctance motor (RESM).
This parameter is coefficient K02.

Note

RESM: reluctance synchronous motor (synchronous reluctance motor)

p2982[0...n] Quadrature axis flux model coefficient K03 / Psiq_mod K03

SERVO_DBSI_VD

Can be changed: T, U

Calculated: -

Access level: 4

Data type: FloatingPoint32

Dyn. index: MDS, p0130

Function diagram: -

P group: Motor

Unit group: -

Unit selection: -

Not for motor type: ASM, SESM, REL

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description: Definition of the saturation characteristic for the quadrature axis flux model of the synchronous reluctance motor (RESM).
This parameter is coefficient K03.

Note

RESM: reluctance synchronous motor (synchronous reluctance motor)

p2983[0...n] Quadrature axis flux model coefficient K04 / Psiq_mod K04

SERVO_DBSI_VD

Can be changed: T, U

Calculated: -

Access level: 4

Data type: FloatingPoint32

Dyn. index: MDS, p0130

Function diagram: -

P group: Motor

Unit group: -

Unit selection: -

Not for motor type: ASM, SESM, REL

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description: Definition of the saturation characteristic for the quadrature axis flux model of the synchronous reluctance motor (RESM).
This parameter is coefficient K04.

Note

RESM: reluctance synchronous motor (synchronous reluctance motor)

p3032	MotId: flux characteristic coefficient K01 identified / Psid_mod K01 ident		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Motor identification	Unit group: -	Unit selection: -
	Not for motor type: PMSM, SESM, REL	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Coefficient K01 of the saturation characteristic determined from the motor data identification. This coefficient can be changed after the identification, and accepted in p2952 with p1910/p1960 = -3.		
Dependency:	See also: p2952		

p3033	MotId: flux characteristic coefficient K02 identified / Psid_mod K02 ident		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Motor identification	Unit group: -	Unit selection: -
	Not for motor type: PMSM, SESM, REL	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-100000.0000	100000.0000	0.0000
Description:	Coefficient K02 of the saturation characteristic determined from the motor data identification. This coefficient can be changed after the identification, and accepted in p2953 with p1910/p1960 = -3.		
Dependency:	See also: p2953		

p3034	MotId: flux characteristic coefficient K03 identified / Psid_mod K03 ident		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Motor identification	Unit group: -	Unit selection: -
	Not for motor type: PMSM, SESM, REL	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-100000.0000	100000.0000	0.0000
Description:	Coefficient K03 of the saturation characteristic determined from the motor data identification. This coefficient can be changed after the identification, and accepted in p2954 with p1910/p1960 = -3.		
Dependency:	See also: p2954		

p3035	MotId: quadrature flux model coefficient K01 identified / Psig_mod K01 ident		
SERVO_DBSI_VD	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Motor identification	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.0000	5.0000	0.0000
Description:	Coefficient K01 of the quadrature flux model determined by the motor data identification. This coefficient can be changed after the identification, and accepted in p2954 with p1910/p1960 = -3.		
Dependency:	See also: p2980		

p3036 SERVO_DBSI_VD	MotId: quadrature flux model coefficient K02 identified / Psig_mod K02 ident		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Motor identification	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Coefficient K02 of the quadrature flux model determined by the motor data identification. This coefficient can be changed after the identification, and accepted in p2954 with p1910/p1960 = -3.		
Dependency:	See also: p2981		
p3037 SERVO_DBSI_VD	MotId: quadrature flux model coefficient K03 identified / Psig_mod K03 ident		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Motor identification	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Coefficient K03 of the quadrature flux model determined by the motor data identification. This coefficient can be changed after the identification, and accepted in p2954 with p1910/p1960 = -3.		
Dependency:	See also: p2982		
p3038 SERVO_DBSI_VD	MotId: quadrature flux model coefficient K04 identified / Psig_mod K04 ident		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Motor identification	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Coefficient K04 of the quadrature flux model determined by the motor data identification. This coefficient can be changed after the identification, and accepted in p2954 with p1910/p1960 = -3.		
r3120[0...63] CU_I_840_VD, CU_NX_840_VD, SERVO_DBSI_VD	Component fault / Comp fault		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: 8060
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Displays the component of the fault which has occurred.		
Dependency:	See also: r0945, r0947, r0948, r0949, r2109, r2130, r2136, r3122		
	Note Value = 0: Assignment to a component not possible. The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the fault buffer and the assignment of the indices is shown in r0945.		

r3121[0...63]	Component alarm / Comp alarm																																											
CU_I_840_VD, CU_NX_840_VD, SERVO_DBSI_VD	Can be changed: - Data type: Unsigned32 P group: Messages Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 8065 Unit selection: - Expert list: 1 Factory setting: -																																									
Description:	Displays the component of the alarm which has occurred.																																											
Dependency:	See also: r2122, r2123, r2124, r2125, r2145, r2146																																											
Note Value = 0: Assignment to a component not possible. The buffer parameters are cyclically updated in the background (refer to status signal in r2139). The structure of the alarm buffer and the assignment of the indices is shown in r2122.																																												
r3122[0...63]	Diagnostic attribute fault / Diag_attr fault																																											
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned32 P group: Messages Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: 8060 Unit selection: - Expert list: 1 Factory setting: -																																									
Description:	Displays the diagnostic attribute of the fault which has occurred.																																											
Bit array:	<table><tr><td>Bit</td><td>Signal name</td><td>1 signal</td><td>0 signal</td><td>FD</td></tr><tr><td>00</td><td>Hardware replacement recommended</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>15</td><td>Message has gone</td><td>Yes</td><td>No</td><td>-</td></tr><tr><td>16</td><td>PROFIdrive fault class bit 0</td><td>High</td><td>Low</td><td>-</td></tr><tr><td>17</td><td>PROFIdrive fault class bit 1</td><td>High</td><td>Low</td><td>-</td></tr><tr><td>18</td><td>PROFIdrive fault class bit 2</td><td>High</td><td>Low</td><td>-</td></tr><tr><td>19</td><td>PROFIdrive fault class bit 3</td><td>High</td><td>Low</td><td>-</td></tr><tr><td>20</td><td>PROFIdrive fault class bit 4</td><td>High</td><td>Low</td><td>-</td></tr></table>	Bit	Signal name	1 signal	0 signal	FD	00	Hardware replacement recommended	Yes	No	-	15	Message has gone	Yes	No	-	16	PROFIdrive fault class bit 0	High	Low	-	17	PROFIdrive fault class bit 1	High	Low	-	18	PROFIdrive fault class bit 2	High	Low	-	19	PROFIdrive fault class bit 3	High	Low	-	20	PROFIdrive fault class bit 4	High	Low	-			
Bit	Signal name	1 signal	0 signal	FD																																								
00	Hardware replacement recommended	Yes	No	-																																								
15	Message has gone	Yes	No	-																																								
16	PROFIdrive fault class bit 0	High	Low	-																																								
17	PROFIdrive fault class bit 1	High	Low	-																																								
18	PROFIdrive fault class bit 2	High	Low	-																																								
19	PROFIdrive fault class bit 3	High	Low	-																																								
20	PROFIdrive fault class bit 4	High	Low	-																																								
Dependency:	See also: r0945, r0947, r0948, r0949, r2109, r2130, r2136, r3120																																											

Note

The buffer parameters are cyclically updated in the background (refer to status signal in r2139).

The structure of the fault buffer and the assignment of the indices is shown in r0945.

For bits 20 ... 16:

Bits 20, 19, 18, 17, 16 = 0, 0, 0, 0, 0 --> PROFIdrive message class 0: not assigned

Bits 20, 19, 18, 17, 16 = 0, 0, 0, 0, 1 --> PROFIdrive message class 1: hardware fault/software error

Bits 20, 19, 18, 17, 16 = 0, 0, 0, 1, 0 --> PROFIdrive message class 2: line fault

Bits 20, 19, 18, 17, 16 = 0, 0, 0, 1, 1 --> PROFIdrive message class 3: supply voltage fault

Bits 20, 19, 18, 17, 16 = 0, 0, 1, 0, 0 --> PROFIdrive message class 4: DC link fault

Bits 20, 19, 18, 17, 16 = 0, 0, 1, 0, 1 --> PROFIdrive message class 5: power electronics faulted

Bits 20, 19, 18, 17, 16 = 0, 0, 1, 1, 0 --> PROFIdrive message class 6: overtemperature electronic components

Bits 20, 19, 18, 17, 16 = 0, 0, 1, 1, 1 --> PROFIdrive message class 7: ground fault/phase fault detected

Bits 20, 19, 18, 17, 16 = 0, 1, 0, 0, 0 --> PROFIdrive message class 8: motor overload

Bits 20, 19, 18, 17, 16 = 0, 1, 0, 0, 1 --> PROFIdrive message class 9: communication error to the higher-level control

Bits 20, 19, 18, 17, 16 = 0, 1, 0, 1, 0 --> PROFIdrive message class 10: safe monitoring channel has identified an error

Bits 20, 19, 18, 17, 16 = 0, 1, 0, 1, 1 --> PROFIdrive message class 11: incorrect position actual value/speed actual value or not available

Bits 20, 19, 18, 17, 16 = 0, 1, 1, 0, 0 --> PROFIdrive message class 12: internal (DRIVE-CLiQ) communication error

Bits 20, 19, 18, 17, 16 = 0, 1, 1, 0, 1 --> PROFIdrive message class 13: infeed unit faulted

Bits 20, 19, 18, 17, 16 = 0, 1, 1, 1, 0 --> PROFIdrive message class 14: braking controller/Braking Module faulted

Bits 20, 19, 18, 17, 16 = 0, 1, 1, 1, 1 --> PROFIdrive message class 15: line filter faulted

Bits 20, 19, 18, 17, 16 = 1, 0, 0, 0, 0 --> PROFIdrive message class 16: external measured value/signal state outside the permissible range

Bits 20, 19, 18, 17, 16 = 1, 0, 0, 0, 1 --> PROFIdrive message class 17: application/technology function faulted

Bits 20, 19, 18, 17, 16 = 1, 0, 0, 1, 0 --> PROFIdrive message class 18: error in the parameterization/configuration/ commissioning sequence

Bits 20, 19, 18, 17, 16 = 1, 0, 0, 1, 1 --> PROFIdrive message class 19: general drive fault

Bits 20, 19, 18, 17, 16 = 0, 1, 1, 0, 0 --> PROFIdrive message class 20: auxiliary unit faulted

r3762**AVS/APC load speed controller 1 rate time active / APC n_ctr 1 Tv act**

SERVO_DBSI_VD (APC)

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 5055**P group:** -**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

- [ms]

- [ms]

- [ms]

Description:

Displays the active rate time of the load speed controller 1 for AVS and APC.

The rate time acts on the load acceleration in the branch for filter 2.1 and 2.2.

Note

APC: Advanced Positioning Control

AVS: Active Vibration Suppression

r3988[0...1]**Boot state / Boot_state**

CU_I_840_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** -**Function diagram:** -**P group:** -**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

10800

-

Description:

Index 0:

Displays the boot state.

Index 1:

Displays the partial boot state.

Value:	0:	Not active
	1:	Fatal fault
	10:	Fault
	20:	Reset all parameters
	30:	Drive object modified
	40:	Download using the commissioning tool
	50:	Parameter download using commissioning tool
	90:	Reset Control Unit and delete drive objects
	100:	Start initialization
	101:	Wait for topology input
	110:	Instantiate Control Unit basis
	111:	Insert drive object
	112:	Remove drive object
	113:	Change drive object number
	114:	Change component number
	115:	Parameter download using commissioning tool
	117:	Remove component
	150:	Wait until actual topology determined
	160:	Evaluate topology
	170:	Instantiate Control Unit reset
	180:	Initialization YDB configuration information
	190:	FW update for CU LINK slaves
	200:	First commissioning
	210:	Generate drive packages
	250:	Wait for topology acknowledge
	325:	Wait for input of drive type
	350:	Determine drive type
	360:	Write into topology-dependent parameters
	370:	Wait until p0009 = 0 is set
	380:	Check topology
	550:	Call conversion functions for parameter
	625:	Wait non-cyclic starting DRIVE-CLiQ
	650:	Start cyclic operation
	660:	Evaluate drive commissioning status
	670:	Automatic FW update DRIVE-CLiQ components
	680:	Wait for CU LINK slaves
	690:	Wait non-cyclic starting DRIVE-CLiQ
	700:	Save parameters
	725:	Wait until DRIVE-CLiQ cyclic
	740:	Check the ability to operate
	745:	Start of the time slices
	750:	Interrupt enable
	800:	Initialization finished
	10050:	Wait for synchronization
	10100:	Wait for CU LINK slaves
	10150:	Wait until actual topology determined
	10200:	Evaluate component status
	10250:	Call conversion functions for parameter
	10300:	Preparation cyclic operation

10350: Automatic FW update DRIVE-CLiQ components
10400: Wait for slave properties
10450: Check CX/NX status
10500: Wait until DRIVE-CLiQ cyclic
10550: Carry out warm start
10600: Evaluate, encoder status
10800: Partial boot completed

Index:
[0] = System
[1] = Partial boot

r3988[0...1]
Boot state / Boot_state

CU_NX_840_VD

Can be changed: -

Calculated: -

Access level: 3

Data type: Integer16

Dyn. index: -

Function diagram: -

P group: -

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:
Max:
Factory setting:

0

10800

-

Description:

Index 0:
Displays the boot state.
Index 1:
Displays the partial boot state.

Value:

0: Not active
1: Fatal fault
10: Fault
20: Reset all parameters
30: Drive object modified
40: Download using the commissioning tool
50: Parameter download using commissioning tool
90: Reset Control Unit and delete drive objects
100: Start initialization
101: Wait for topology input
110: Instantiate Control Unit basis
111: Insert drive object
112: Remove drive object
113: Change drive object number
114: Change component number
115: Parameter download using commissioning tool
117: Remove component
150: Wait until actual topology determined
160: Evaluate topology
170: Instantiate Control Unit reset
180: Initialization YDB configuration information
200: First commissioning
210: Generate drive packages
250: Wait for topology acknowledge
325: Wait for input of drive type
350: Determine drive type
360: Write into topology-dependent parameters
370: Wait until p0009 = 0 is set

380:	Check topology
550:	Call conversion functions for parameter
625:	Wait non-cyclic starting DRIVE-CLiQ
650:	Start cyclic operation
660:	Evaluate drive commissioning status
670:	Automatic FW update DRIVE-CLiQ components
680:	Wait for CU LINK slaves
690:	Wait non-cyclic starting DRIVE-CLiQ
700:	Save parameters
725:	Wait until DRIVE-CLiQ cyclic
740:	Check the ability to operate
745:	Start of the time slices
750:	Interrupt enable
800:	Initialization finished
10050:	Wait for synchronization
10100:	Wait for CU LINK slaves
10150:	Wait until actual topology determined
10200:	Evaluate component status
10250:	Call conversion functions for parameter
10300:	Preparation cyclic operation
10350:	Automatic FW update DRIVE-CLiQ components
10400:	Wait for slave properties
10450:	Check CX/NX status
10500:	Wait until DRIVE-CLiQ cyclic
10550:	Carry out warm start
10600:	Evaluate, encoder status
10800:	Partial boot completed

Index:
 [0] = System
 [1] = Partial boot

r3996[0...1]	Parameter write inhibit status / Par_write inhib st		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 1
	Data type: Unsigned8	Dyn. index: -	Function diagram: -
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Displays whether writing to parameters is inhibited. r3996[0] = 0: Parameter write not inhibited. 0 < r3996[0] < 100: Parameter write inhibited. The value shows how the calculations are progressing.		
Index:	[0] = Progress calculations [1] = Cause		
	Note For index [1]: Only for internal Siemens troubleshooting.		

r3998[0...n] First drive commissioning / First drv_com

SERVO_DBSI_VD

Can be changed: -**Data type:** Unsigned16**P group:** -**Not for motor type:** -**Min:**

0

Calculated: -**Dyn. index:** DDS, p0180**Unit group:** -**Scaling:** -**Max:**

65535

Access level: 3**Function diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting:**

-

Description:

Displays whether the drive still has to be commissioned for the first time.

0 = Yes

2 = No

p4618

SERVO_DBSI_VD

Motor Module temperature sensor type / MM temp_sens type**Can be changed:** T, U**Data type:** Integer16**P group:** -**Not for motor type:** -**Min:**

0

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** -**Max:**

6

Access level: 2**Function diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting:**

0

Description:

Sets the type of temperature sensor connected to the Motor Module.

The temperature actual value is displayed via connector output r4619.

Value:

0: No sensor

2: KTY84

5: PT100

6: PT1000

Dependency:

The parameter can only be set if the motor temperature sensor is not connected to the Motor Module (p0600 = 11, X21, X22).

See also: r4619

NOTICE

Users must evaluate the temperature actual value (CO: r4619).

r4619

SERVO_DBSI_VD

CO: Motor Module temperature sensor actual value / MM temp_sens actV**Can be changed:** -**Data type:** FloatingPoint32**P group:** -**Not for motor type:** -**Min:**

- [°C]

Calculated: -**Dyn. index:** -**Unit group:** 21_1**Scaling:** p2006**Max:**

- [°C]

Access level: 2**Function diagram:** -**Unit selection:** p0505**Expert list:** 1**Factory setting:**

- [°C]

Description:

Display and connector output for the temperature sensor connected to the Motor Module

Dependency:

See also: p4618

Note

For r4619 equal to -200.0 °C, the following applies:

This temperature display is not valid (temperature sensor error).

p4622[0...n] SERVO_DBSI_VD	Temperature sensor error lower threshold / Temp_sens low thr		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: MDS, p0130	Function diagram: 8016
	P group: -	Unit group: 21_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -140 [°C]	Max: 20 [°C]	Factory setting: -140 [°C]
Description:	Sets the minimum temperature to identify a temperature sensor error. A temperature sensor error is output below this threshold.		
Dependency:	See also: A07015, F07016		
p4635[0...n] SERVO_DBSI_VD	Encoder scaling factor acceleration / Scale factor accel		
	Can be changed: T	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: EDS, p0140	Function diagram: -
	P group: Encoder	Unit group: 39_1	Unit selection: p0505
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -340.28235E36 [rev/s ²]	Max: 340.28235E36 [rev/s ²]	Factory setting: 0.000 [rev/s ²]
Description:	Sets the scaling factor for acceleration values 1 to 3. The set value specifies the acceleration for sensor value 32767.		
r4636[0...2] SERVO_DBSI_VD	CO: Encoder acceleration value 1 / Acceleration 1		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Encoder	Unit group: 39_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2007	Expert list: 1
	Min: - [rev/s ²]	Max: - [rev/s ²]	Factory setting: - [rev/s ²]
Description:	Displays the first acceleration value, which is transferred via the extended DQ telegram.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		
r4637[0...2] SERVO_DBSI_VD	CO: Encoder acceleration value 2 / Acceleration 2		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Encoder	Unit group: 39_1	Unit selection: p0505
	Not for motor type: -	Scaling: p2007	Expert list: 1
	Min: - [rev/s ²]	Max: - [rev/s ²]	Factory setting: - [rev/s ²]
Description:	Displays the second acceleration value, which is transferred via the extended DQ telegram.		
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3		

r4638[0...2] SERVO_DBSI_VD	CO: Encoder acceleration value 3 / Acceleration 3				
	Can be changed: -	Calculated: -	Access level: 3		
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -		
	P group: Encoder	Unit group: 39_1	Unit selection: p0505		
	Not for motor type: -	Scaling: p2007	Expert list: 1		
	Min:	Max:	Factory setting:		
	- [rev/s²]	- [rev/s²]	- [rev/s²]		
Description:	Displays the third acceleration value, which is transferred via the extended DQ telegram.				
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3				
r4639[0...2] SERVO_DBSI_VD	CO: Encoder supplementary value / Suppl value				
	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Integer16	Dyn. index: -	Function diagram: -		
	P group: Encoder	Unit group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min:	Max:	Factory setting:		
	-	-	-		
Description:	Displays the optional supplementary value, which is transferred via the extended DQ telegram.				
Index:	[0] = Encoder 1 [1] = Encoder 2 [2] = Encoder 3				
p4644 SERVO_DBSI_VD	Simulated motor encoder configuration / Sim mot_enc config				
	Can be changed: T	Calculated: -	Access level: 4		
	Data type: Unsigned16	Dyn. index: -	Function diagram: -		
	P group: -	Unit group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min:	Max:	Factory setting:		
	-	-	0011 bin		
Description:	Sets the configuration for the simulated motor encoder. Using these configuration bits, the response of the encoder status bit "Parking encoder active" (r0481.14) can be set for a pulse inhibit. For bit 00: When the bit is set, after a pulse inhibit, encoder status bit "Parking encoder active" (r0481.14) is temporarily set and then withdrawn. When r0481.14 = 1 is set, the higher-level control system is signaled that the position must be re-referenced. When r0481.14 = 0 is reset, then the higher-level control system can switch the system on again. For bit 01: When the bit is set, a brake is being used (p1215 = 1, 3) and the pulses are canceled for absolute speeds less than the standstill speed (SERVO: p1226, VECTOR: 2 * p1226), then the encoder status bit "Parking encoder active" (r0481.14) is not set. This means that the position is not re-referenced after switching on again. For bit 02: This bit can be used for diagnostics. When the bit is set, encoder status bit "Parking encoder active" (r0481.14) is not set as a result of the pulse inhibit.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	For a pulse inhibit, temporarily set bit r0481.14	Yes	No	-
	01	Bit r0481.14 = 0 for pulse inhibit at standstill with brake	Yes	No	-
	02	For a pulse inhibit, do not set bit r0481.14	Yes	No	-

⚠ DANGER

For bit 02 = 1:

If it cannot be excluded that the motor shaft turns while the pulses are inhibited, then after the pulse enable, the position calculated by the simulated motor encoder can be offset and not detected using encoder status bit "Parking encoder active" (r0481.14). Hazardous operating states can subsequently occur.

Note

Function "Simulated motor encoder" is activated using p0400 = 9010.

For bit 00, 01, 02 = 0, the following applies:

For a pulse inhibit, encoder status bit "Parking encoder active" (r0481.14) is set. The system must be referenced after switching on again.

For bit 02 = 1, the following applies:

This bit has priority over the other bits.

r4950		TEC DO-specific number / TEC DO qty		
SERVO_DBSI_VD		Can be changed: -	Calculated: -	Access level: 4
		Data type: Unsigned16	Dyn. index: -	Function diagram: -
		P group: OEM range	Unit group: -	Unit selection: -
		Not for motor type: -	Scaling: -	Expert list: 1
		Min:	Max:	Factory setting:
		0	32	-
Description:		Number of Technology Extensions installed on this drive object.		
Dependency:		See also: r4951		
		Note DO: Drive Object TEC: Technology Extension		
r4951		TEC DO-specific identifier total length / TEC DO ident tot_l		
SERVO_DBSI_VD		Can be changed: -	Calculated: -	Access level: 4
		Data type: Unsigned16	Dyn. index: -	Function diagram: -
		P group: OEM range	Unit group: -	Unit selection: -
		Not for motor type: -	Scaling: -	Expert list: 1
		Min:	Max:	Factory setting:
		0	288	-
Description:		Displays the total length of the identifier of the Technology Extensions installed on this drive object.		
Dependency:		See also: r4950		
		Note The identifier of a Technology Extension comprises a maximum of 8 characters plus separator. TEC: Technology Extension		
r5386[0...4]		CO: Mot_temp remaining time until fault / t_remain to fault		
SERVO_DBSI_VD		Can be changed: -	Calculated: -	Access level: 3
		Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
		P group: Displays, signals	Unit group: -	Unit selection: -
		Not for motor type: -	Scaling: -	Expert list: 1
		Min:	Max:	Factory setting:
		- [s]	- [s]	- [s]

Description:	Display and connector output for the remaining time until a fault is output due to motor overtemperature. With the motor load remaining the same, the values specify the time remaining until a fault is output. For index [0]: The value indicates the active remaining time until a fault is output. This value is the minimum from indices [1] to [4]; whereby, value -1 is not taken into account. For index [1]: The value indicates the calculated time that remains until a fault is output for the model. For index [2]: The value indicates the time that remains until a fault is output after an alarm has occurred for the model (p0605, p5387). For index [3]: The value indicates the calculated time that remains until a fault is output for the measurement. For index [4]: The value indicates the time that remains until a fault is output after an alarm has occurred for the measurement (p0606).
Index:	[0] = Active time [1] = Thermal time according to the model-based calculation [2] = Timer time after alarm from the model [3] = Timer time after calculation for measurement [4] = Timer time after alarm from the measurement
Note Value = 120 s means that there is no thermal restriction. Value = -1 s means that a prediction is not possible.	

r5387[0...n] SERVO_DBSI_VD	Mot_temp_mod 3 timer / Mod 3 timer		
	Can be changed: -	Calculated: -	Access level: 2
	Data type: FloatingPoint32	Dyn. index: MDS, p0130	Function diagram: 8019
	P group: Displays, signals	Unit group: -	Unit selection: -
	Not for motor type: ASM, SESM, REL	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [s]	- [s]	- [s]
Description:	Displays the timer for monitoring the motor temperature for motor temperature model 3. This timer is started when the temperature alarm threshold (p5390) is exceeded. If the timer has expired and the alarm threshold has, in the meantime, not been fallen below, then fault F07011 is output. If the temperature fault threshold (p5391) is prematurely exceeded before the timer has expired, then fault F07011 is immediately output.		

p7324 A_INF_840_VD (Parallel)	Par_connection VSM voltage asymmetry alarm threshold / VSM volt asym thr		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Modulation	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: PERCENT	Expert list: 1
	Min: 2 [%]	Max: 100 [%]	Factory setting: 10 [%]
Description:	Sets the alarm threshold to detect asymmetry of the VSM voltages in the parallel connection. The deviation between the measured values and average value is evaluated. The reference value for the alarm threshold depends on the device supply voltage and the number of active VSM. The following applies for the reference value: $\text{sqrt}(2) * p0210 * (2 - 1/r7000)$.		
Dependency:	See also: A05066		
	NOTICE		
	For the monitoring function, precisely those Voltage Sensing Modules that are assigned to active power units must be activated.		

Note

For p7324 = 100 %, the monitoring is deactivated.

r7325	CO: Par_connection VSM maximum deviation voltage u1 - u2 / VSM max dev U 12		
A_INF_840_VD (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Displays, signals	Unit group: 5_3	Unit selection: p0505
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min:	Max:	Factory setting:
	- [V]	- [V]	- [V]
Description:	Displays the maximum absolute difference between the measured voltage (u1 - u2) of a VSM, and the corresponding average (mean) value of all VSM (r3661).		
Dependency:	See also: A05066		
	Note The individual VSM voltages are indicated in r7300. The measured average value across all active Voltage Sensing Modules (VSM) is displayed in r3661.		

r7326	CO: Par_connection VSM maximum deviation voltage u2 - u3 / VSM max dev U 23		
A_INF_840_VD (Parallel)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Displays, signals	Unit group: 5_3	Unit selection: p0505
	Not for motor type: -	Scaling: p2001	Expert list: 1
	Min:	Max:	Factory setting:
	- [V]	- [V]	- [V]
Description:	Displays the maximum absolute difference between the measured voltage (u2 - u3) of a VSM, and the corresponding average (mean) value of all VSM (r3662).		
Dependency:	See also: A05066		
	Note The individual VSM voltages are indicated in r7301. The measured average value across all active Voltage Sensing Modules (VSM) is displayed in r3662.		

p8966[0...31]	CI: IF1 PZD extension PZD send word / PZD ext send word		
CU_I_840_VD (PZD ext), CU_NX_840_VD (PZD ext)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer16	Dyn. index: -	Function diagram: -
	P group: Communications	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0
Description:	Selects the PZD (actual values) with word format to be sent to the fieldbus controller. This parameter is only visible when the "PZD extension" function module is activated (p0171.0 = 1).		

Index:	[0] = PZD 1		
	[1] = PZD 2		
	[2] = PZD 3		
	[3] = PZD 4		
	[4] = PZD 5		
	[5] = PZD 6		
	[6] = PZD 7		
	[7] = PZD 8		
	[8] = PZD 9		
	[9] = PZD 10		
	[10] = PZD 11		
	[11] = PZD 12		
	[12] = PZD 13		
	[13] = PZD 14		
	[14] = PZD 15		
	[15] = PZD 16		
	[16] = PZD 17		
	[17] = PZD 18		
	[18] = PZD 19		
	[19] = PZD 20		
	[20] = PZD 21		
	[21] = PZD 22		
	[22] = PZD 23		
	[23] = PZD 24		
	[24] = PZD 25		
	[25] = PZD 26		
	[26] = PZD 27		
	[27] = PZD 28		
	[28] = PZD 29		
	[29] = PZD 30		
	[30] = PZD 31		
	[31] = PZD 32		
Dependency:	The "TRCBUF" Technology Extension license key is required for function module "PZD extension". See also: p8967		
	Note IF1: Interface 1		
p8967[0...30]	CI: IF1 PZD extension PZD send double word / PZD ext send DW		
CU_I_840_VD (PZD ext), CU_NX_840_VD (PZD ext)	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer16	Dyn. index: -	Function diagram: -
	P group: Communications	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: 4000H	Expert list: 1
	Min: -	Max: -	Factory setting: 0
Description:	Selects the PZD (actual values) with double word format to be sent to the fieldbus controller. This parameter is only visible when the "PZD extension" function module is activated (p0171.0 = 1).		

Index:	[0] = PZD 1 + 2
	[1] = PZD 2 + 3
	[2] = PZD 3 + 4
	[3] = PZD 4 + 5
	[4] = PZD 5 + 6
	[5] = PZD 6 + 7
	[6] = PZD 7 + 8
	[7] = PZD 8 + 9
	[8] = PZD 9 + 10
	[9] = PZD 10 + 11
	[10] = PZD 11 + 12
	[11] = PZD 12 + 13
	[12] = PZD 13 + 14
	[13] = PZD 14 + 15
	[14] = PZD 15 + 16
	[15] = PZD 16 + 17
	[16] = PZD 17 + 18
	[17] = PZD 18 + 19
	[18] = PZD 19 + 20
	[19] = PZD 20 + 21
	[20] = PZD 21 + 22
	[21] = PZD 22 + 23
	[22] = PZD 23 + 24
	[23] = PZD 24 + 25
	[24] = PZD 25 + 26
	[25] = PZD 26 + 27
	[26] = PZD 27 + 28
	[27] = PZD 28 + 29
	[28] = PZD 29 + 30
	[29] = PZD 30 + 31
	[30] = PZD 31 + 32

Dependency: The "TRCBUF" Technology Extension license key is required for function module "PZD extension".
See also: p8966

NOTICE

A BICO interconnection for a single PZD can only be realized either with p8966 or p8967.

Note

IF1: Interface 1

r8977[0...31]	IF1 PZD extension diagnostics PZD send word / PZD ext diag word		
CU_I_840_VD (PZD ext), CU_NX_840_VD (PZD ext)	Can be changed: - Data type: Unsigned16 P group: Communications Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the PZD (actual values) with word format sent to the fieldbus controller. This parameter is only visible when the "PZD extension" function module is activated (p0171.0 = 1).		

Index:

[0] = PZD 1
[1] = PZD 2
[2] = PZD 3
[3] = PZD 4
[4] = PZD 5
[5] = PZD 6
[6] = PZD 7
[7] = PZD 8
[8] = PZD 9
[9] = PZD 10
[10] = PZD 11
[11] = PZD 12
[12] = PZD 13
[13] = PZD 14
[14] = PZD 15
[15] = PZD 16
[16] = PZD 17
[17] = PZD 18
[18] = PZD 19
[19] = PZD 20
[20] = PZD 21
[21] = PZD 22
[22] = PZD 23
[23] = PZD 24
[24] = PZD 25
[25] = PZD 26
[26] = PZD 27
[27] = PZD 28
[28] = PZD 29
[29] = PZD 30
[30] = PZD 31
[31] = PZD 32

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Bit 0	ON	OFF	-
	01	Bit 1	ON	OFF	-
	02	Bit 2	ON	OFF	-
	03	Bit 3	ON	OFF	-
	04	Bit 4	ON	OFF	-
	05	Bit 5	ON	OFF	-
	06	Bit 6	ON	OFF	-
	07	Bit 7	ON	OFF	-
	08	Bit 8	ON	OFF	-
	09	Bit 9	ON	OFF	-
	10	Bit 10	ON	OFF	-
	11	Bit 11	ON	OFF	-
	12	Bit 12	ON	OFF	-
	13	Bit 13	ON	OFF	-
	14	Bit 14	ON	OFF	-
	15	Bit 15	ON	OFF	-

Dependency: The "TRCBUF" Technology Extension license key is required for function module "PZD extension".
See also: p8966

IF1: Interface 1

Create MyVirtual Machine simDrive parameters/alarms
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07	Bit 7	ON	OFF	-
08	Bit 8	ON	OFF	-
09	Bit 9	ON	OFF	-
10	Bit 10	ON	OFF	-
11	Bit 11	ON	OFF	-
12	Bit 12	ON	OFF	-
13	Bit 13	ON	OFF	-
14	Bit 14	ON	OFF	-
15	Bit 15	ON	OFF	-
16	Bit 16	ON	OFF	-
17	Bit 17	ON	OFF	-
18	Bit 18	ON	OFF	-
19	Bit 19	ON	OFF	-
20	Bit 20	ON	OFF	-
21	Bit 21	ON	OFF	-
22	Bit 22	ON	OFF	-
23	Bit 23	ON	OFF	-
24	Bit 24	ON	OFF	-
25	Bit 25	ON	OFF	-
26	Bit 26	ON	OFF	-
27	Bit 27	ON	OFF	-
28	Bit 28	ON	OFF	-
29	Bit 29	ON	OFF	-
30	Bit 30	ON	OFF	-
31	Bit 31	ON	OFF	-

Dependency: The "TRCBUF" Technology Extension license key is required for function module "PZD extension".
See also: p8967

NOTICE

A maximum of 4 indices of the "trace" function can be used.

Note

IF1: Interface 1

p9301

SERVO_DBSI_VD

SI Motion enable safety functions (Motor Module) / SI Mtn enable MM

Can be changed: C2(95)

Calculated: -

Access level: 4

Data type: Unsigned32

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

0000 0000 0000 0000 0000
0000 0000 0000 bin

Description: Sets the enable signals for the safe motion monitoring.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Enable SOS/SLS	Enable	Inhibit	-
01	Enable SLP	Enable	Inhibit	-
02	Enable absolute position	Enable	Inhibit	-
03	Enable actual value synchronization	Enable	Inhibit	-
04	Enable SS2ESR	Enable	Inhibit	-
16	Enable SSM hysteresis and filtering	Enable	Inhibit	2823

17	Enable SDI	Enable	Inhibit	2824
18	Enable SS2E	Enable	Inhibit	-
20	Enable SLA	Enable	Inhibit	-
23	Enable deactivation SOS/SLS during an external STOP A	Enable	Inhibit	-
24	Enable transfer SLS limit value via PROFIsafe	Enable	Inhibit	-
25	Enable transfer safe position via PROFIsafe	Enable	Inhibit	-
26	Enable safe gearbox switchover	Enable	Inhibit	-
27	Enable referencing via SCC	Enable	Inhibit	-
28	Enable safe cam	Enable	Inhibit	-
29	Enable synchronous transfer safe position via PROFIsafe	Enable	Inhibit	-

Dependency:

See also: p9501

See also: F01682, F01683

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

A change only becomes effective after a POWER ON.

SCA: Safe Cam

SDI: Safe Direction (safe motion direction)

SLA: Safely-Limited Acceleration

SLP: Safely-Limited Position

SLS: Safely-Limited Speed

SOS: Safe Operating Stop

SP: Safe Position

SS2E: Safe Stop 2 External (Safe Stop 2 with external stop, external STOP D)

SS2ESR: Safe Stop 2 Extended Stop and Retract

SSM: Safe Speed Monitor (safety-relevant feedback signal from the speed monitoring)

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9302

SERVO_DBSI_VD

SI Motion axis type (Motor Module) / SI Mtn AxisType MM**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

1

0

Description:

Sets the axis type (linear axis or rotary axis/spindle).

Value:

0: Linear axis

1: Rotary axis/spindle

Dependency:

See also: p9502

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

For the commissioning tool, after changing over the axis type, the units dependent on the axis type are only updated after a project upload.

A change only becomes effective after a POWER ON.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9305 SI Motion SP modulo value (Motor Module) / SI mtn SP mod MM

SERVO_DBSI_VD

Can be changed: C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0 [°]

737280 [°]

0 [°]

Description:

Sets the modulo value in degrees for rotary axes of the "Safe position" function.

This modulo value is taken into account when safely referencing as well as when transferring the safe position via PROFIsafe when the absolute position is enabled.

The value should be set so that it is precisely at 2^n revolutions, so that when the range that can be represented (± 2048) overflows, this does not cause the position actual value to jump.

The modulo function is deactivated for a value = 0.

Dependency:

See also: F01681

NOTICE

When the "SLP" function is activated, the modulo function must be deactivated as otherwise fault F30681 will be output.

If the absolute position is not enabled, then the parameterized modulo value is not taken into account.

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

SLP: Safely-Limited Position

SP: Safe Position

p9315 SI Motion coarse position value configuration (Motor Module) / SI Mtn s config MM

SERVO_DBSI_VD

Can be changed: C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

0000 0000 0000 0000 0000
0000 0000 0000 bin**Description:**

Sets the encoder configuration for the redundant coarse position value.

The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Incrementer	Yes	No	-
01	Encoder CRC least significant byte first	Yes	No	-
02	Redundant coarse position val. most significant bit left-aligned	Yes	No	-
04	Binary comparison not possible	Yes	No	-
05	Single-channel encoder	Yes	No	-
16	DRIVE-CLiQ encoder	Yes	No	-
17	EnDat 2.2 converter	Yes	No	-

Dependency:

See also: r0474, p9515

p9316	SI Motion encoder configuration, safety functions (Motor Module) / SI Mtn enc_conf MM				
SERVO_DBSI_VD	Can be changed: C2(95)		Calculated: -	Access level: 3	
	Data type: Unsigned16		Dyn. index: -	Function diagram: -	
	P group: Safety Integrated		Unit group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min: -		Max: -	Factory setting: 0000 0000 bin	
Description:	Sets the configuration for the encoder and position actual value. The encoder that is used for the safe motion monitoring functions on the Motor Module must be parameterized in this parameter.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Encoder rotating/linear	Linear	Rotating	-
	01	Position actual value sign change	Yes	No	-
	04	No STOP A after encoder fault for 1 encoder safety	Yes	No	-
Dependency:	See also: p0404, p0410, p9516				
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.					

p9317	SI Motion linear scale grid division (Motor Module) / SI Mtn grid MM				
SERVO_DBSI_VD	Can be changed: C2(95)		Calculated: -	Access level: 3	
	Data type: FloatingPoint32		Dyn. index: -	Function diagram: -	
	P group: Safety Integrated		Unit group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min: 0.00 [nm]		Max: 250000000.00 [nm]	Factory setting: 10000.00 [nm]	
Description:	Sets the grid division for a linear encoder. The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.				
Dependency:	See also: p0407, p9316				
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.					

p9318	SI Motion encoder pulses per revolution (Motor Module) / SI Mtn p/rev MM				
SERVO_DBSI_VD	Can be changed: C2(95)		Calculated: -	Access level: 3	
	Data type: Unsigned32		Dyn. index: -	Function diagram: -	
	P group: Safety Integrated		Unit group: -	Unit selection: -	
	Not for motor type: -		Scaling: -	Expert list: 1	
	Min: 0		Max: 16777215	Factory setting: 2048	
Description:	Sets the number of encoder pulses per revolution for rotary encoders. The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.				
Dependency:	See also: p0408, p9316				
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.					

p9319	SI Motion fine resolution G1_XIST1 (Motor Module) / SI Mtn G1_XIST1 MM		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 2	Max: 18	Factory setting: 11
Description:	Sets the fine resolution for G1_XIST1 in bits. The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.		
Dependency:	See also: p0418 See also: F01670, F01671		
Note For safety functions that have not been enabled (p9301 = 0), the following applies: when booting, p9319 is automatically set the same as p0418. For safety functions that are enabled (p9301 > 0), the following applies: p9319 is checked for agreement with p0418. G1_XIST1: encoder 1 position actual value 1 (PROFIdrive) This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9320	SI Motion spindle pitch (Motor Module) / SI Mtn sp_pitch MM		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.1000 [mm]	Max: 8388.0000 [mm]	Factory setting: 10.0000 [mm]
Description:	Sets the gear ratio between the encoder and load in mm/revolution for a linear axis with rotary encoder. The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.		
Dependency:	See also: p9520		
NOTICE The fourth decimal point can be rounded-off depending on the size of the entered number (from 3 places before the decimal point).			
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9321[0...7]	SI Motion gearbox encoder (motor)/load denom (Motor Module) / SI Mtn grbxDenomMM		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 1	Max: 2147000000	Factory setting: 1
Description:	Sets the denominator for the gearbox between the encoder (or motor in the case of encoderless monitoring functions) and the load. The active gearbox stage can be switched over via PROFIsafe.		

Index:	[0] = Gearbox 1
	[1] = Gearbox 2
	[2] = Gearbox 3
	[3] = Gearbox 4
	[4] = Gearbox 5
	[5] = Gearbox 6
	[6] = Gearbox 7
	[7] = Gearbox 8
Dependency:	See also: p9322

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9322[0...7] SI Motion gearbox encoder (motor)/load numerator (Motor Module) / SI Mtn grbx num MM

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	1	2147000000	1

Description: Sets the numerator for the gearbox between the encoder (or motor in the case of encoderless monitoring functions) and the load.
The active gearbox stage can be switched over via PROFIsafe.

Index:	[0] = Gearbox 1
	[1] = Gearbox 2
	[2] = Gearbox 3
	[3] = Gearbox 4
	[4] = Gearbox 5
	[5] = Gearbox 6
	[6] = Gearbox 7
	[7] = Gearbox 8

Dependency: See also: p9321

Note

In the case of encoderless monitoring functions, the pole pair number must be multiplied by the numerator of the gearbox ratio.

Example:

Gearbox ratio 1:4, pole pair number (r0313) = 2

--> p9321 = 1, p9322 = 8 (4 x 2)

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9323 SI Motion red. coarse position value valid bits (Motor Module) / Valid bits MM

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	16	9

Description: Sets the number of valid bits of the redundant coarse position value.
The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.

Dependency: See also: r0470, p9523

p9324 SI Motion redundant coarse pos. value fine resolution bits (MM) / SI Mtn fine bit MM

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -16	Max: 16	Factory setting: -2

Description: Sets the number of valid bits for the fine resolution of the redundant coarse position value.
The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.

Dependency: See also: r0471, p9524

p9325 SI Motion redundant coarse pos. value relevant bits (MM) / Relevant bits MM

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 16	Factory setting: 16

Description: Sets the number of relevant bits for the redundant coarse position value.
The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.

Dependency: See also: r0472, p9525

p9326 SI Motion encoder assignment (Motor Module) / SI Mtn encoder MM

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 1	Max: 3	Factory setting: 1

Description: Sets the number of the encoder, which is used on the Motor Module for safe motion monitoring functions.

Dependency: For the safe motion monitoring functions, the redundant safety position actual value sensing must be activated in the appropriate encoder data set (p0430.19 = 1).

See also: p0187, p0188, p0189, p9526

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

A change only becomes effective after a POWER ON.

For p9326 = 1, the following applies:

Motor Module uses an encoder for closed-loop speed control, it involves a 1-encoder system.

p9329	SI Motion Gx_XIST1 coarse pos safe most significant bit (MM) / Gx_XIST1 MSB MM		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 31	Factory setting: 14
Description:	Sets the bit number for the safe most significant bit (MSB) of the Gx_XIST1 coarse position. The encoder that is used for the safe motion monitoring functions on the Motor Module/Hydraulic Module must be parameterized in this parameter.		
Dependency:	See also: r0475, p9529		
	Note MSB: Most Significant Bit		
p9339[0...7]	SI Motion gearbox direction of rotation reversal (Motor Module) / SI Mtn grbx rev MM		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 1	Factory setting: 0
Description:	Sets the direction of rotation reversal for the gearbox. 0: No direction of rotation reversal 1: Direction of rotation reversal The active gearbox stage can be switched over via PROFIsafe.		
Index:	[0] = Gearbox 1 [1] = Gearbox 2 [2] = Gearbox 3 [3] = Gearbox 4 [4] = Gearbox 5 [5] = Gearbox 6 [6] = Gearbox 7 [7] = Gearbox 8		
Dependency:	See also: p9321		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
r9398[0...1]	SI Motion actual checksum SI parameters (Motor Module) / SI Mtn act CRC MM		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Displays the checksum for the checked Safety Integrated parameters of the motion monitoring function (actual checksum) on the Motor Module/Hydraulic Module.		
Index:	[0] = Checksum over SI parameters for motion monitoring [1] = Checksum over SI parameters with hardware reference		
Dependency:	See also: p9399		

NoteSI: Safety Integrated

p9399[0...1] SI Motion reference checksum SI parameters (Motor Module) / SI Mtn set CRC MM

SERVO_DBSI_VD

Can be changed: C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0000 hex

FFFF FFFF hex

0000 hex

Description: Sets the checksum for the checked Safety Integrated parameters of the motion monitoring function (reference checksum) on the Motor Module/Hydraulic Module.**Index:**
[0] = Checksum over SI parameters for motion monitoring
[1] = Checksum over SI parameters with hardware reference**Dependency:** See also: r9398

Note

SI: Safety Integrated

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r9481 Number of BICO interconnections / BICO count

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Commands**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 0**Min:****Max:****Factory setting:**

-

-

-

Description: Displays the number of BICO interconnections (signal sinks).**Dependency:** See also: r9482, r9483

NoteThe selected BICO interconnections should be entered into r9482 and r9483.

r9482[0...n] BICO interconnections BI/CI parameters / BICO BI/CI par

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** r9481**Function diagram:** -**P group:** Commands**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 0**Min:****Max:****Factory setting:**

-

-

-

Description: Displays the signal sinks (binector/connector inputs, BI/CI parameters).
The number of BICO interconnections is displayed in r9481.**Dependency:** See also: r9481, r9483

Note

The list is sorted according to signal sources and is structured as follows:

r9842[0]: Interconnection 1 (signal sink, BICO coded), r9843[0]: Interconnection 1 (signal source, BICO coded)

r9842[1]: Interconnection 2 (signal sink, BICO coded), r9843[1]: Interconnection 2 (signal source, BICO coded)

...

r9483[0...n]	BICO interconnections BO/CO parameters / BICO BO/CO par		
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned32 P group: Commands Not for motor type: - Min: -	Calculated: - Dyn. index: r9481 Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: - Unit selection: - Expert list: 0 Factory setting: -
Description:	Displays the signal sources (binector/connector outputs, BO/CO parameters). The number of BICO interconnections is displayed in r9481.		
Dependency:	See also: r9481, r9482		
Note The list is sorted according to signal sources and is structured as follows: r9842[0]: Interconnection 1 (signal sink, BICO coded), r9843[0]: Interconnection 1 (signal source, BICO coded) r9842[1]: Interconnection 2 (signal sink, BICO coded), r9843[1]: Interconnection 2 (signal source, BICO coded) ...			

p9484	BICO interconnections search signal source / BICO s_s srch		
SERVO_DBSI_VD	Can be changed: T, U Data type: Unsigned32 P group: - Not for motor type: - Min: 0	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 4294967295	Access level: 3 Function diagram: - Unit selection: - Expert list: 0 Factory setting: 0
Description:	Sets the signal source (BO/CO parameter, BICO coded) to search in the signal sinks. The question is answered: How often is a connection made to a signal source in the drive object and from which index are these interconnections saved (r9482 and r9483)?		
Dependency:	See also: r9481, r9482, r9483, r9485, r9486		
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

r9485	BICO interconnections signal source search count / BICO s_s srch qty		
SERVO_DBSI_VD	Can be changed: - Data type: Unsigned16 P group: - Not for motor type: - Min: -	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: -	Access level: 3 Function diagram: - Unit selection: - Expert list: 0 Factory setting: -
Description:	Displays the number of BICO interconnections to the signal sink being searched for.		
Dependency:	See also: r9481, r9482, r9483, p9484, r9486		
Note The signal source to be searched is set in p9484 (BICO-coded). The search result is contained in r9482 and r9483 and is specified by the count (r9485) and the first index (r9486).			

r9486	BICO interconnections signal source search first index / BICO s_s srch idx		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 0
	Min:	Max:	Factory setting:
	-	-	-
Description:	Displays the first index of the signal source being searched for.		
Dependency:	See also: r9481, r9482, r9483, p9484, r9485		

Note

The signal source to be searched is set in p9484 (BICO-coded).

The search result is contained in r9482 and r9483 and is specified by the count (r9485) and the first index (r9486).

p9500	SI Motion monitoring clock cycle (Control Unit) / SI Mtn clock CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.50000 [ms]	25.00000 [ms]	12.00000 [ms]
Description:	Sets the monitoring clock cycle for safe motion monitoring.		
Dependency:	See also: p9511 See also: F01652		

Note

A change only becomes effective after a POWER ON.

The monitoring clock cycle must be a multiple of the actual value sensing clock cycle (see the parameter description for p9511).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9501	SI Motion enable safety functions (Control Unit) / SI Mtn enable CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0000 0000 0000 0000 0000 0000 0000 bin

Description: Sets the enable signals for the safe motion monitoring.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Enable SOS/SLS (SBH/SG)	Enable	Inhibit	-
	01	Enable SLP (SE)	Enable	Inhibit	-
	02	Enable absolute position	Enable	Inhibit	-
	03	Enable actual value synchronization	Enable	Inhibit	-
	04	Enable SS2ESR	Enable	Inhibit	-
	16	Enable SSM (n < nx) hysteresis and filtering	Enable	Inhibit	2823
	17	Enable SDI	Enable	Inhibit	2824
	18	Enable SS2E	Enable	Inhibit	-
	20	Enable SLA	Enable	Inhibit	-

23	Enable deactivation SOS/SLS during an external STOP A	Enable	Inhibit	-
24	Enable transfer SLS (SG) limit value via PROFIsafe	Enable	Inhibit	-
25	Enable transfer safe position via PROFIsafe	Enable	Inhibit	-
26	Enable safe gearbox switchover	Enable	Inhibit	-
27	Enable referencing via SCC	Enable	Inhibit	-
28	Enable safe cam	Enable	Inhibit	-
29	Enable synchronous transfer safe position via PROFIsafe	Enable	Inhibit	-

Dependency: See also: F01682, F01683

Note

A change only becomes effective after a POWER ON.

SCA: Safe Cam / SN: Safe software cam

SDI: Safe Direction (safe motion direction)

SLA: Safely-Limited Acceleration

SLS: Safely-Limited Speed / SG: Safely reduced speed

SOS: Safe Operating Stop / SBH: Safe operating stop

SS2ESR: Safe Stop 2 Extended Stop and Retract

SSM: Safe Speed Monitor (safety-relevant feedback signal from the speed monitoring)

p9502

SERVO_DBSI_VD

SI Motion axis type (Control Unit) / SI Mtn ax type CU

Can be changed: C2(95)

Data type: Integer16

P group: Safety Integrated

Not for motor type: -

Min:

0

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

1

Access level: 3

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

0

Description:

Sets the axis type (linear axis or rotary axis/spindle).

Value:

0: Linear axis

1: Rotary axis/spindle

Note

For the commissioning tool, after changing over the axis type, the units dependent on the axis type are only updated after a project upload.

A change only becomes effective after a POWER ON.

p9503

SERVO_DBSI_VD

SI Motion SCA (SN) enable (Control Unit) / SI Mtn SCA enab

Can be changed: C2(95)

Data type: Unsigned32

P group: Safety Integrated

Not for motor type: -

Min:

-

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

-

Access level: 3

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

0000 0000 0000 0000 0000

0000 0000 0000 bin

Description:

Setting to enable the function "Safe Cam" (SCA).

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Enable SCA1 (SN1)	Enable	Inhibit	-
01	Enable SCA2 (SN2)	Enable	Inhibit	-
02	Enable SCA3 (SN3)	Enable	Inhibit	-
03	Enable SCA4 (SN4)	Enable	Inhibit	-
04	Enable SCA5 (SN5)	Enable	Inhibit	-
05	Enable SCA6 (SN6)	Enable	Inhibit	-
06	Enable SCA7 (SN7)	Enable	Inhibit	-

07	Enable SCA8 (SN8)	Enable	Inhibit	-
08	Enable SCA9 (SN9)	Enable	Inhibit	-
09	Enable SCA10 (SN10)	Enable	Inhibit	-
10	Enable SCA11 (SN11)	Enable	Inhibit	-
11	Enable SCA12 (SN12)	Enable	Inhibit	-
12	Enable SCA13 (SN13)	Enable	Inhibit	-
13	Enable SCA14 (SN14)	Enable	Inhibit	-
14	Enable SCA15 (SN15)	Enable	Inhibit	-
15	Enable SCA16 (SN16)	Enable	Inhibit	-
16	Enable SCA17 (SN17)	Enable	Inhibit	-
17	Enable SCA18 (SN18)	Enable	Inhibit	-
18	Enable SCA19 (SN19)	Enable	Inhibit	-
19	Enable SCA20 (SN20)	Enable	Inhibit	-
20	Enable SCA21 (SN21)	Enable	Inhibit	-
21	Enable SCA22 (SN22)	Enable	Inhibit	-
22	Enable SCA23 (SN23)	Enable	Inhibit	-
23	Enable SCA24 (SN24)	Enable	Inhibit	-
24	Enable SCA25 (SN25)	Enable	Inhibit	-
25	Enable SCA26 (SN26)	Enable	Inhibit	-
26	Enable SCA27 (SN27)	Enable	Inhibit	-
27	Enable SCA28 (SN28)	Enable	Inhibit	-
28	Enable SCA29 (SN29)	Enable	Inhibit	-
29	Enable SCA30 (SN30)	Enable	Inhibit	-

Dependency: See also: p9501
See also: F01686

Note

SCA: Safe Cam / SN: Safe software cam

p9505

SERVO_DBSI_VD

SI Motion SP modulo value (Control Unit) / SI mtn SP mod CU

Can be changed: C2(95)	Calculated: -	Access level: 3
Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
P group: Safety Integrated	Unit group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min: 0 [°]	Max: 737280 [°]	Factory setting: 0 [°]

Description: Sets the modulo value in degrees for rotary axes of the "Safe position" function.
This modulo value is taken into account when safely referencing as well as when transferring the safe position via PROFIsafe when the absolute position is enabled.
The value should be set so that it is precisely at 2^n revolutions, so that when the range that can be represented (+/-2048) overflows, this does not cause the position actual value to jump.
The modulo function is deactivated for a value = 0.

Dependency: See also: p9501
See also: F01681

NOTICE

When the "SLP" function is activated, the modulo function must be deactivated as otherwise fault F01681 will be output.
If the absolute position is not enabled, then the parameterized modulo value is not taken into account.

Note

SLP: Safely-Limited Position

SP: Safe Position

p9506**SI Motion function specification (Control Unit) / SI Mtn fct_spc CU**

SERVO_DBSI_VD

Can be changed: C2(95)**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

3

0

Description:

Sets the function specification for the safe motion monitoring.

Value:

0: Safety with encoder and accel_monitoring (SAM) / delay time

1: Safety without encoder with brake ramp (SBR)

2: Safety with encoder with brake ramp (SBR)

3: Safety without encoder with accel_monitoring (SAM) / delay time

Dependency:

See also: C01711

Note

A change only becomes effective after a POWER ON.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9507**SI Motion function specification (Control Unit) / SI Mtn config CU**

SERVO_DBSI_VD

Can be changed: C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

0000 0010 bin

Description:

Sets the function configuration for the safe motion monitoring functions.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Extended message acknowledgment	Yes	No	-
01	Setpoint velocity limit for STOP F	No	Yes	-
02	Actual value sensing encoderless motor type	Synchronous motor	ASM/RESM	-
03	SS1 with OFF3 (brake response)	SS1E external stop	SS1 with OFF3	-
05	Actual value sensing sensorless edge modulation	Yes	No	-
06	Configuration test stop motion monitoring functions	Test automatic	Test manual	-

Dependency:

See also: C01711

Note

For bit 00:

When the function is activated, a safety-relevant acknowledgment (internal event acknowledge) can be performed by selecting/deselecting STO.

For bit 01:

When the function is activated, the active setpoint velocity limit (CO: r9733) is set to zero when STOP F is active.

For bit 02:

This bit defines the type of motor, which the sensorless actual value sensing evaluates.

For bit = 0, the actual velocity is calculated for an induction motor.

For bit = 1, the actual speed is calculated for a synchronous motor. This value depends on the setting in p0300.

Bit = 0 should be set if no motor is defined (p0300 = 0).

For bit 03:

When the bit is activated – when selecting function SS1 or activating a STOP B – an SS1E or a STOP B with stop, which should be externally initiated, is triggered instead of SS1 with a drive-based braking response. As a consequence, brake monitoring (SBR, SAM) is deactivated.

SS1E: Safe Stop 1 external (Safe Stop 1 with external stop)

For bit 05:

This bit defines the type of modulation, which the sensorless actual value sensing evaluates.

For bit = 0, the actual velocity is calculated for space vector modulation.

For bit = 1, the actual velocity is calculated for edge modulation. This value depends on the setting in p1802.

For bit 06:

For the automatic test stop, the test stop can still be initiated via binector input p9705.

The automatic test stop is executed after power up, partial power up or a warm restart.

ASM: Induction motor

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9509

SERVO_DBSI_VD

SI Motion behavior during pulse suppression (Control Unit) / SI Mtn behav IL CU

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: Unsigned32

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

0000 0000 1111 1111 bin

Description:

Sets the behavior of safety functions and their feedback during pulse suppression in encoderless operation.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	SSM during pulse suppression and encoderless	Becomes inactive	Remains active	-
08	SDI during pulse suppression and encoderless	Becomes inactive	Remains active	-

Dependency:

See also: C01711

NOTICE

For bit 00:

If the OFF1 or the OFF3 ramp-down time is too low, or there is an insufficient clearance between the SSM limit speed, and the shutdown speed, then it is possible that the "speed under limit value" signal does not change to 1, because no speed actual value below the SSM limit was able to be identified before pulse cancellation. In this case, the OFF1 or the OFF3 ramp-down time or the clearance between the SSM limit speed and shutdown speed must be increased.

Note

SDI: Safe Direction (safe motion direction)

SSM: Safe Speed Monitor (safety-relevant feedback signal from the speed monitoring)

For bit 00:

For bit = 1 and with the SSM safety function activated, the following applies:

- During pulse suppression, monitoring is switched off and the feedback signal has a 0 signal level.

For bit = 0 and with the SSM safety function activated, the following applies:

- Monitoring continues during pulse suppression. The feedback signal last displayed before pulse suppression is kept and the system goes into the STO state.

For bit 08:

For bit = 1 and with the SDI safety function activated, the following applies:

- During pulse suppression, monitoring is switched off and the status signal indicates inactive.

For bit = 0 and with the SDI safety function activated, the following applies:

- Monitoring continues during pulse suppression. The status signal indicates active and the system goes into the STO state.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9510

SERVO_DBSI_VD

SI Motion isochronous PROFIBUS master / SI Mtn sync master

Can be changed: C2(95)

Data type: Integer16

P group: Safety Integrated

Not for motor type: -

Min:

0

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

1

Access level: 4

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

0

Description:

Setting for isochronous communication between PROFIdrive controller and Control Unit.

The parameter is only relevant, if the safety-relevant motion monitoring functions integrated in the drive have been enabled (p9601.2 = 1).

If a PROFIdrive controller exchanges process data isochronously with the Control Unit, then p9510 must be set to 1. This also applies if the drive itself does not exchange process data isochronously.

Examples for isochronous communication:

- isochronous control for the motion control (e.g. SIMOTION).

- isochronous PROFIsafe master (e.g. SIMATIC S7-400F).

Value:

0: Communication not isochronous

1: Communication isochronous

Dependency:

See also: C01711, A01796

NOTICE

As of firmware version 2.6, the parameter has no effect.

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9511

SERVO_DBSI_VD

SI Motion actual value sensing cycle clock (Control Unit) / SI Mtn act clk CU

Can be changed: C2(95)

Data type: FloatingPoint32

P group: Safety Integrated

Not for motor type: -

Min:

0.00000 [ms]

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

25.00000 [ms]

Access level: 4

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

0.00000 [ms]

Description:	Sets the clock cycle time of the actual value sensing for safe motion monitoring. Setting criteria if the motion monitoring functions are executed with an encoder. - a slower clock cycle time reduces the maximum permissible velocity - however, it ensures a lower load of the Control Unit for safe actual value sensing. - the maximum permissible velocity which, when exceeded, can mean that errors occur during safe actual value sensing, is displayed in r9730. - the isochronous PROFIBUS clock cycle is used as a clock cycle time for actual value sensing with a setting of 0 ms; the setting is 1 ms if isochronous operation is not being used. Setting criteria if the motion monitoring functions are executed without an encoder: - the actual value sensing clock cycle must be set to the same value as the current controller clock cycle (p0115[0]). For SINAMICS S120M, the following applies: Only setting p9511 = 0 or 2 ms is possible (a value of 0 is internally assumed to be 2).
Dependency:	See also: p0115 See also: F01652

Note

The parameter is only active for drive-based motion monitoring functions (p9601.2 = 1).

The monitoring clock cycle from p9500 must be an integer multiple of this parameter.

In the case of motion monitoring functions with encoder, the clock cycle time for actual value sensing must be an integer multiple of the current controller clock cycle and at least 4 times slower than the current controller clock cycle. A factor of at least 8 is recommended.

The clock cycle time of the actual value sensing should not be set to more than 8 ms.

A change only becomes effective after a POWER ON.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9515	SI Motion encoder coarse position value config (Control Unit) / SI Mtn s config CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0000 0000 0000 0000 0000 0000 0000 0000 bin

Description:	Sets the encoder configuration for the redundant coarse position value. The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.
---------------------	--

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Incrementer	Yes	No	-
	01	Encoder CRC least significant byte first	Yes	No	-
	02	Redundant coarse position val. most significant bit left-aligned	Yes	No	-
	04	Binary comparison not possible	Yes	No	-
	05	Single-channel encoder	Yes	No	-
	16	DRIVE-CLiQ encoder	Yes	No	-
	17	EnDat 2.2 converter	Yes	No	-

Dependency:	See also: r0474, p9315
--------------------	------------------------

Note

- after starting the copy function (p9700 = 57 hex), p9515.0...5 are set the same as r0474.

For safety functions that are not enabled (p9501 = 0), the following applies:

- when the system boots, p9515.16 is automatically set the same as p0404.10, p9515.17 the same as p0404.8 & 11.

For safety functions that are enabled (p9501 > 0), the following applies:

- p9515.16 is checked to identify whether it matches with p0404.10, p9515.17 with p0404.8 & 11.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9516		SI Motion encoder configuration safety functions (Control Unit) / SI Mtn enc_conf CU			
SERVO_DBSI_VD	Can be changed: C2(95)		Calculated: -		Access level: 3
	Data type: Unsigned16		Dyn. index: -		Function diagram: -
	P group: Safety Integrated		Unit group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min: -		Max: -		Factory setting: 0000 0000 bin
Description:		Sets the configuration for the motor encoder and position actual value. The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.			
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Motor encoder rotating/linear	Linear	Rotating	-
	01	Position actual value sign change	Yes	No	-
	04	No STOP A after encoder fault for 1 encoder safety	Yes	No	-
Dependency:		See also: p0404, p0410 See also: F01671			

Note

For safety functions that are not enabled (p9501 = 0), the following applies:

- p9516.0 is automatically set the same as p0404.0 when the system boots.

- p9516.1 is automatically set the same as p0410.1 when the system boots.

For safety functions that are enabled (p9501 > 0), the following applies:

- p9516.0 is checked to identify whether it matches p0404.0.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9517		SI Motion linear encoder grid division (Control Unit) / SI Mtn grid CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3	
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -	
	P group: Safety Integrated	Unit group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min: 0.00 [nm]	Max: 250000000.00 [nm]	Factory setting: 10000.00 [nm]	
Description:	Sets the grid division for a linear encoder. The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.			
Dependency:	See also: p0407, p9516 See also: F01671			

Note

For safety functions that have not been enabled (p9501 = 0), the following applies: when booting p9517 is automatically set the same as p0407.

For safety functions that are enabled (p9501 > 0), the following applies: p9517 is checked whether it matches p0407.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9518

SERVO_DBSI_VD

SI Motion encoder pulses per revolution (Control Unit) / SI Mtn puls/rev CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

16777215

2048

Description:

Sets the number of encoder pulses per revolution for rotary encoders.

The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.

Dependency:

See also: p0408, p9516

See also: F01671

Note

For safety functions that have not been enabled (p9501 = 0), the following applies: when booting, p9518 is automatically set the same as p0408.

For safety functions that are enabled (p9501 > 0), the following applies: p9518 is checked whether it matches p0408.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9519

SERVO_DBSI_VD

SI Motion fine resolution G1_XIST1 (Control Unit) / SI Mtn G1_XIST1 CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

2

18

11

Description:

Sets the fine resolution for G1_XIST1 in bits.

The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.

Dependency:

See also: p0418

See also: F01671

Note

G1_XIST1: encoder 1 position actual value 1 (PROFIdrive)

For safety functions that are not enabled (p9501 = 0), the following applies:

- p9519 is automatically set the same as p0418 at run-up.

For safety functions that are enabled (p9501 > 0), the following applies:

- p9519 is checked to see that it matches p0418.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9520			
SERVO_DBSI_VD	SI Motion spindle pitch (Control Unit) / SI Mtn sp_pitch CU		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.1000 [mm]	Max: 8388.0000 [mm]	Factory setting: 10.0000 [mm]
Description:	Sets the gear ratio between the encoder and load in mm/revolution for a linear axis with rotary encoder.		
	NOTICE The fourth decimal point can be rounded-off depending on the size of the entered number (from 3 places before the decimal point).		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p9521[0...7]			
SERVO_DBSI_VD	SI Motion gearbox enc (motor)/load denominator (Control Unit) / SI Mtn grbx den CU		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 1	Max: 2147000000	Factory setting: 1
Description:	Sets the denominator for the gearbox between the encoder (or motor in the case of encoderless monitoring functions) and the load. The active gearbox stage can be switched over via PROFIsafe.		
Index:	[0] = Gearbox 1 [1] = Gearbox 2 [2] = Gearbox 3 [3] = Gearbox 4 [4] = Gearbox 5 [5] = Gearbox 6 [6] = Gearbox 7 [7] = Gearbox 8		
Dependency:	See also: p9522		

p9522[0...7]			
SERVO_DBSI_VD	SI Motion gearbox encoder (motor)/load numerator (Control Unit) / SI Mtn grbx num CU		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 1	Max: 2147000000	Factory setting: 1
Description:	Sets the numerator for the gearbox between the encoder (or motor in the case of encoderless monitoring functions) and the load. The active gearbox stage can be switched over via PROFIsafe.		

Index:

- [0] = Gearbox 1
- [1] = Gearbox 2
- [2] = Gearbox 3
- [3] = Gearbox 4
- [4] = Gearbox 5
- [5] = Gearbox 6
- [6] = Gearbox 7
- [7] = Gearbox 8

Dependency: See also: p9521

Note

In the case of encoderless monitoring functions, the pole pair number must be multiplied by the numerator of the gearbox ratio.

Example:

Gearbox ratio 1:4, pole pair number (r0313) = 2

--> p9521 = 1, p9522 = 8 (4 x 2)

p9523

SERVO_DBSI_VD

SI Motion redundant coarse pos. value valid bits (Control Unit) / Valid bits CU

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: Unsigned16

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0

16

9

Description:

Sets the number of valid bits of the redundant coarse position value.

The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.

Dependency:

See also: r0470, p9323

Note

- after starting the copy function (p9700 = 57 hex), p9523 is set the same as r0470.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9524

SERVO_DBSI_VD

SI Motion Redundant coarse pos. value fine resolution bits (CU) / SI Mtn fine bit CU

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: Integer16

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-16

16

-2

Description:

Sets the number of valid bits for the fine resolution of the redundant coarse position value.

The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.

Dependency:

See also: r0471, p9324

Note

- after starting the copy function (p9700 = 57 hex), p9524 is set the same as r0471.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9525	SI Motion Redundant coarse pos. value relevant bits (CU) / Relevant bits CU		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: Unsigned16 P group: Safety Integrated Not for motor type: - Min: 0	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 16	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 16
Description:	Sets the number of relevant bits for the redundant coarse position value. The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.		
Dependency:	See also: r0472, p9325		
Note For safety functions that are not enabled (p9501 = 0), the following applies: - p9525 is automatically set the same as r0472 when the system boots. For safety functions that are enabled (p9501 > 0), the following applies: - p9525 is checked to see that it matches r0472. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9526	SI Motion encoder assignment second channel / SI Mtn enc chan 2		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: Unsigned32 P group: Safety Integrated Not for motor type: - Min: 1	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 3	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 1
Description:	Sets the number of the encoder that the second channel (control, Motor Module) uses for safe motion monitoring functions.		
Dependency:	For the safe motion monitoring functions, the redundant safety position actual value sensing must be activated in the appropriate encoder data set (p0430.19 = 1). See also: p0187, p0188, p0189		
Note For p9526 = 1, the encoder for the closed-loop speed control is used for the second channel of the motion monitoring functions (1-encoder system). A change only becomes effective after a POWER ON. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

r9527	SI Motion Sensor Module Type second channel / SI Mtn SM chan 2		
SERVO_DBSI_VD	Can be changed: - Data type: Integer16 P group: Safety Integrated Not for motor type: - Min: 0	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 255	Access level: 4 Function diagram: - Unit selection: - Expert list: 1 Factory setting: -
Description:	Displays the Sensor Module type that the second channel uses for the safe motion monitoring functions for sensing actual values in a safety-relevant fashion.		
Value:	0: Sensor Module not assigned 1: Sensor Module with DRT-ASIC 2: DQI encoder 3: EnDat encoder		

255: Sensor Module unknown

NoteSM: Sensor Module

p9529 SI Motion Gx_XIST1 coarse pos. safe most significant bit (CU) / Gx_XIST1 MSB CU

SERVO_DBSI_VD

Can be changed: C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

31

14

Description:

Sets the bit number for the safe most significant bit (MSB) of the Gx_XIST1 coarse position.

The encoder that is used for the safe motion monitoring functions on the Control Unit must be parameterized in this parameter.

Dependency:See also: r0475, p9329

Note

For safety functions that are not enabled (p9501 = 0), the following applies:

- p9529 is automatically set the same as r0475 when the system boots.

For safety functions that are enabled (p9501 > 0), the following applies:

- p9529 is checked to see that it matches r0475.

MSB: Most Significant Bit

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9530 SI Motion standstill tolerance (Control Unit) / SI Mtn standst_tol

SERVO_DBSI_VD

Can be changed: C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.000 [mm]

100.000 [mm]

1.000 [mm]

Description:

Sets the tolerance for the function "Safe Operating Stop" (SOS).

Dependency:See also: C01707

NoteSOS: Safe Operating Stop / SBH: Safe operating stop

p9530 SI Motion standstill tolerance (Control Unit) / SI Mtn standst_tolSERVO_DBSI_VD
(Safety rot)**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.000 [°]

100.000 [°]

1.000 [°]

Description:

Sets the tolerance for the function "Safe Operating Stop" (SOS).

Dependency:See also: C01707

NoteSOS: Safe Operating Stop / SBH: Safe operating stop

p9531[0...3] SERVO_DBSI_VD	SI Motion SLS (SG) limit values (Control Unit) / SI Mtn SLS lim CU		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [mm/min]	Max: 1000000.00 [mm/min]	Factory setting: 2000.00 [mm/min]
Description:	Sets the limit values for the function "Safely-Limited Speed" (SLS).		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLS4		
Dependency:	See also: p9563 See also: C01714		
Note SLS: Safely-Limited Speed / SG: Safely reduced speed			

p9531[0...3] SERVO_DBSI_VD (Safety rot)	SI Motion SLS (SG) limit values (Control Unit) / SI Mtn SLS lim CU		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [rpm]	Max: 1000000.00 [rpm]	Factory setting: 2000.00 [rpm]
Description:	Sets the limit values for the function "Safely-Limited Speed" (SLS).		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLS4		
Dependency:	See also: p9563 See also: C01714		
Note SLS: Safely-Limited Speed / SG: Safely reduced speed			

p9533 SERVO_DBSI_VD	SI Motion SLS setpoint speed limiting (Control Unit) / SI Mtn SLS set_lim		
	Can be changed: T, U	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.000 [%]	Max: 100.000 [%]	Factory setting: 80.000 [%]
Description:	This is an evaluation factor to define the setpoint limit from the selected actual speed limit. The active SLS limit value is evaluated with this factor and is made available as setpoint limit in r9733.		

Dependency: This parameter only has to be parameterized for the motion monitoring functions integrated in the drive (p9601.2 = 1)
 $r9733[0] = p9531[x] \times p9533$ (converted from the load side to the motor side)
 $r9733[1] = -p9531[x] \times p9533$ (converted from the load side to the motor side)
[x] = Selected SLS stage
Conversion factor from the motor side to the load side:
- motor type = rotary and axis type = linear: $p9522 / (p9521 \times p9520)$
- otherwise: $p9522 / p9521$
See also: p9501, p9531, p9601

Note

The active actual speed limit is selected via safety-relevant inputs (SGE).
When selecting SOS or a STOP A ... D, setpoint 0 is specified in r9733.
SLS: Safely-Limited Speed

p9534[0...1]		SI Motion SLP (SE) upper limit values (Control Unit) / SI Mtn SLP up lim	
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2822
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -2147000.000 [mm]	Max: 2147000.000 [mm]	Factory setting: 100000.000 [mm]
Description:	Sets the upper limit for the function "Safely-Limited Position" (SLP).		
Index:	[0] = Limit value SLP1 (SE1) [1] = Limit value SLP2 (SE2)		
Dependency:	See also: p9501, p9535, p9562 See also: C01715		

Note

The following applies to the setting of these limits:
- p9534[x] > p9535[x]
- p9534[x] must lie in the valid traversing range (-737280 ... 737280).
SLP: Safely-Limited Position / SE: Safe software limit switches

p9534[0...1]		SI Motion SLP (SE) upper limit values (Control Unit) / SI Mtn SLP up lim	
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2822
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -2147000.000 [°]	Max: 2147000.000 [°]	Factory setting: 100000.000 [°]
Description:	Sets the upper limit for the function "Safely-Limited Position" (SLP).		
Index:	[0] = Limit value SLP1 (SE1) [1] = Limit value SLP2 (SE2)		
Dependency:	See also: p9501, p9535, p9562 See also: C01715		

Note

The following applies to the setting of these limits:
- p9534[x] > p9535[x]
- p9534[x] must lie in the valid traversing range (-737280 ... 737280).
SLP: Safely-Limited Position / SE: Safe software limit switches

p9535[0...1]	SI Motion SLP (SE) lower limit values (Control Unit) / SI Mtn SLP low lim		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2822
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -2147000.000 [mm]	Max: 2147000.000 [mm]	Factory setting: -100000.000 [mm]
Description:	Sets the lower limit for the function "Safely-Limited Position" (SLP).		
Index:	[0] = Limit value SLP1 (SE1) [1] = Limit value SLP2 (SE2)		
Dependency:	See also: p9501, p9534, p9562 See also: C01715		
Note The following applies to the setting of these limits: - p9534[x] > p9535[x] - p9535[x] must lie in the valid traversing range (-737280 ... 737280). SLP: Safely-Limited Position / SE: Safe software limit switches			

p9535[0...1]	SI Motion SLP (SE) lower limit values (Control Unit) / SI Mtn SLP low lim		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2822
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -2147000.000 [°]	Max: 2147000.000 [°]	Factory setting: -100000.000 [°]
Description:	Sets the lower limit for the function "Safely-Limited Position" (SLP).		
Index:	[0] = Limit value SLP1 (SE1) [1] = Limit value SLP2 (SE2)		
Dependency:	See also: p9501, p9534, p9562 See also: C01715		
Note The following applies to the setting of these limits: - p9534[x] > p9535[x] - p9535[x] must lie in the valid traversing range (-737280 ... 737280). SLP: Safely-Limited Position / SE: Safe software limit switches			

p9536[0...29]	SI Motion SCA (SN) plus cam position (Control Unit) / SI Mtn SCA+		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -2147000.000 [mm]	Max: 2147000.000 [mm]	Factory setting: 10.000 [mm]
Description:	Sets the plus cam position for the function "Safe Cam" (SCA).		

Index:	[0] = Cam position SCA1 (SN1)
	[1] = Cam position SCA2 (SN2)
	[2] = Cam position SCA3 (SN3)
	[3] = Cam position SCA4 (SN4)
	[4] = Cam position SCA5 (SN5)
	[5] = Cam position SCA6 (SN6)
	[6] = Cam position SCA7 (SN7)
	[7] = Cam position SCA8 (SN8)
	[8] = Cam position SCA9 (SN9)
	[9] = Cam position SCA10 (SN10)
	[10] = Cam position SCA11 (SN11)
	[11] = Cam position SCA12 (SN12)
	[12] = Cam position SCA13 (SN13)
	[13] = Cam position SCA14 (SN14)
	[14] = Cam position SCA15 (SN15)
	[15] = Cam position SCA16 (SN16)
	[16] = Cam position SCA17 (SN17)
	[17] = Cam position SCA18 (SN18)
	[18] = Cam position SCA19 (SN19)
	[19] = Cam position SCA20 (SN20)
	[20] = Cam position SCA21 (SN21)
	[21] = Cam position SCA22 (SN22)
	[22] = Cam position SCA23 (SN23)
	[23] = Cam position SCA24 (SN24)
	[24] = Cam position SCA25 (SN25)
	[25] = Cam position SCA26 (SN26)
	[26] = Cam position SCA27 (SN27)
	[27] = Cam position SCA28 (SN28)
	[28] = Cam position SCA29 (SN29)
	[29] = Cam position SCA30 (SN30)

Dependency: See also: p9501, p9503, p9537

Note
SCA: Safe Cam / SN: Safe software cam

p9536[0...29]	SI Motion SCA (SN) plus cam position (Control Unit) / SI Mtn SCA+		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -2147000.000 [°]	Max: 2147000.000 [°]	Factory setting: 10.000 [°]
Description:	Sets the plus cam position for the function "Safe Cam" (SCA).		

Index:	[0] = Cam position SCA1 (SN1)
	[1] = Cam position SCA2 (SN2)
	[2] = Cam position SCA3 (SN3)
	[3] = Cam position SCA4 (SN4)
	[4] = Cam position SCA5 (SN5)
	[5] = Cam position SCA6 (SN6)
	[6] = Cam position SCA7 (SN7)
	[7] = Cam position SCA8 (SN8)
	[8] = Cam position SCA9 (SN9)
	[9] = Cam position SCA10 (SN10)
	[10] = Cam position SCA11 (SN11)
	[11] = Cam position SCA12 (SN12)
	[12] = Cam position SCA13 (SN13)
	[13] = Cam position SCA14 (SN14)
	[14] = Cam position SCA15 (SN15)
	[15] = Cam position SCA16 (SN16)
	[16] = Cam position SCA17 (SN17)
	[17] = Cam position SCA18 (SN18)
	[18] = Cam position SCA19 (SN19)
	[19] = Cam position SCA20 (SN20)
	[20] = Cam position SCA21 (SN21)
	[21] = Cam position SCA22 (SN22)
	[22] = Cam position SCA23 (SN23)
	[23] = Cam position SCA24 (SN24)
	[24] = Cam position SCA25 (SN25)
	[25] = Cam position SCA26 (SN26)
	[26] = Cam position SCA27 (SN27)
	[27] = Cam position SCA28 (SN28)
	[28] = Cam position SCA29 (SN29)
	[29] = Cam position SCA30 (SN30)

Dependency: See also: p9501, p9503, p9537

Note

SCA: Safe Cam / SN: Safe software cam

p9537[0...29]	SI Motion SCA (SN) minus cam position (Control Unit) / SI Mtn SCA-		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-2147000.000 [mm]	2147000.000 [mm]	-10.000 [mm]
Description:	Sets the minus cam position for the function "Safe Cam" (SCA).		

Index:	[0] = Cam position SCA1 (SN1)
	[1] = Cam position SCA2 (SN2)
	[2] = Cam position SCA3 (SN3)
	[3] = Cam position SCA4 (SN4)
	[4] = Cam position SCA5 (SN5)
	[5] = Cam position SCA6 (SN6)
	[6] = Cam position SCA7 (SN7)
	[7] = Cam position SCA8 (SN8)
	[8] = Cam position SCA9 (SN9)
	[9] = Cam position SCA10 (SN10)
	[10] = Cam position SCA11 (SN11)
	[11] = Cam position SCA12 (SN12)
	[12] = Cam position SCA13 (SN13)
	[13] = Cam position SCA14 (SN14)
	[14] = Cam position SCA15 (SN15)
	[15] = Cam position SCA16 (SN16)
	[16] = Cam position SCA17 (SN17)
	[17] = Cam position SCA18 (SN18)
	[18] = Cam position SCA19 (SN19)
	[19] = Cam position SCA20 (SN20)
	[20] = Cam position SCA21 (SN21)
	[21] = Cam position SCA22 (SN22)
	[22] = Cam position SCA23 (SN23)
	[23] = Cam position SCA24 (SN24)
	[24] = Cam position SCA25 (SN25)
	[25] = Cam position SCA26 (SN26)
	[26] = Cam position SCA27 (SN27)
	[27] = Cam position SCA28 (SN28)
	[28] = Cam position SCA29 (SN29)
	[29] = Cam position SCA30 (SN30)

Dependency: See also: p9501, p9503, p9536

Note
SCA: Safe Cam / SN: Safe software cam

p9537[0...29]	SI Motion SCA (SN) minus cam position (Control Unit) / SI Mtn SCA-		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -2147000.000 [°]	Max: 2147000.000 [°]	Factory setting: -10.000 [°]
Description:	Sets the minus cam position for the function "Safe Cam" (SCA).		

Index:	[0] = Cam position SCA1 (SN1)
	[1] = Cam position SCA2 (SN2)
	[2] = Cam position SCA3 (SN3)
	[3] = Cam position SCA4 (SN4)
	[4] = Cam position SCA5 (SN5)
	[5] = Cam position SCA6 (SN6)
	[6] = Cam position SCA7 (SN7)
	[7] = Cam position SCA8 (SN8)
	[8] = Cam position SCA9 (SN9)
	[9] = Cam position SCA10 (SN10)
	[10] = Cam position SCA11 (SN11)
	[11] = Cam position SCA12 (SN12)
	[12] = Cam position SCA13 (SN13)
	[13] = Cam position SCA14 (SN14)
	[14] = Cam position SCA15 (SN15)
	[15] = Cam position SCA16 (SN16)
	[16] = Cam position SCA17 (SN17)
	[17] = Cam position SCA18 (SN18)
	[18] = Cam position SCA19 (SN19)
	[19] = Cam position SCA20 (SN20)
	[20] = Cam position SCA21 (SN21)
	[21] = Cam position SCA22 (SN22)
	[22] = Cam position SCA23 (SN23)
	[23] = Cam position SCA24 (SN24)
	[24] = Cam position SCA25 (SN25)
	[25] = Cam position SCA26 (SN26)
	[26] = Cam position SCA27 (SN27)
	[27] = Cam position SCA28 (SN28)
	[28] = Cam position SCA29 (SN29)
	[29] = Cam position SCA30 (SN30)

Dependency: See also: p9501, p9503, p9536

Note

SCA: Safe Cam / SN: Safe software cam

p9539[0...7] SI Motion gearbox direction of rotation reversal (Control Unit) / SI Mtn grbx rev CU

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	1	0

Description: Sets the direction of rotation reversal for the gearbox.
0: No direction of rotation reversal
1: Direction of rotation reversal
The active gearbox stage can be switched over via PROFIsafe.

Index:

- [0] = Gearbox 1
- [1] = Gearbox 2
- [2] = Gearbox 3
- [3] = Gearbox 4
- [4] = Gearbox 5
- [5] = Gearbox 6
- [6] = Gearbox 7
- [7] = Gearbox 8

Dependency: See also: p9521

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9540

SERVO_DBSI_VD

SI Motion SCA (SN) tolerance (Control Unit) / SI Mtn SCA tol CU

Can be changed: C2(95)	Calculated: -	Access level: 3
Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
P group: Safety Integrated	Unit group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min: 0.0010 [mm]	Max: 10.0000 [mm]	Factory setting: 0.1000 [mm]

Description: Sets the tolerance for the function "Safe Cam" (SCA).
Within this tolerance, both monitoring channels may signal different signal states of the same safe cam.

p9540

SERVO_DBSI_VD
(Safety rot)

SI Motion SCA (SN) tolerance (Control Unit) / SI Mtn SCA tol CU

Can be changed: C2(95)	Calculated: -	Access level: 3
Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
P group: Safety Integrated	Unit group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min: 0.0010 [°]	Max: 10.0000 [°]	Factory setting: 0.1000 [°]

Description: Sets the tolerance for the function "Safe Cam" (SCA).
Within this tolerance, both monitoring channels may signal different signal states of the same safe cam.

p9542

SERVO_DBSI_VD

SI Motion act val comparison tol (cross-check) (Control Unit) / SI Mtn act tol CU

Can be changed: C2(95)	Calculated: -	Access level: 3
Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
P group: Safety Integrated	Unit group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min: 0.0010 [mm]	Max: 360.0000 [mm]	Factory setting: 0.1000 [mm]

Description: Sets the tolerance for the data cross-check of the actual position between the two monitoring channels.
For encoderless motion monitoring functions, the tolerance must be set to a higher value (12 degrees rotary, 1 mm linear).

Dependency: See also: C01711

Note

For a linear axis, the tolerance is internally limited to 10 mm.
For a "linear axis with rotating motor" configuration and factory setting of p9520, p9521 and p9522, the factory setting of p9542 corresponds to a position tolerance of 36 ° on the motor side.
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9542	SI Motion act val comparison tol (cross-check) (Control Unit) / SI Mtn act tol CU		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.0010 [°]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 360.0000 [°]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0.1000 [°]
Description:	Sets the tolerance for the data cross-check of the actual position between the two monitoring channels. For encoderless motion monitoring functions, the tolerance must be set to a higher value (12 degrees rotary, 1 mm linear).		
Dependency:	See also: C01711		

p9543	SI Motion gearbox switching position tolerance factor (CU) / SI Mtn grbx tol CU		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: Integer32 P group: Safety Integrated Not for motor type: - Min: 1	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 1000	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 1
Description:	Sets the factor to increase the tolerance for the data cross-check of the actual position between the two monitoring channels while the gearbox stage is being switched over. This factor is effective when actual value synchronization is activated and when deactivated. Depending on the following tolerance, the following is obtained: - actual value synchronization activated: p9549 * p9543 - actual value synchronization deactivated: p9542 * p9543		
	Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p9544	SI Motion actual value comparison tolerance (referencing) (CU) / SI Mtn ref tol		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.0000 [mm]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 36.0000 [mm]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0.0100 [mm]
Description:	Sets the tolerance for checking the actual values. For an incremental encoder, the actual values are checked after referencing; for an absolute encoder, when switching on.		
Dependency:	See also: C01711		
	Note For linear axes, the maximum value is limited to 1 mm.		

p9544	SI Motion actual value comparison tolerance (referencing) (CU) / SI Mtn ref tol		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.0000 [°]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 36.0000 [°]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0.0100 [°]
Description:	Sets the tolerance for checking the actual values. For an incremental encoder, the actual values are checked after referencing; for an absolute encoder, when switching on.		
Dependency:	See also: C01711		
	Note For linear axes, the maximum value is limited to 1 mm.		
p9545	SI Motion SSM (SGA n < nx) filter time (Control Unit) / SI Mtn SSM fil CU		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.00 [ms]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 500.00 [ms]	Access level: 3 Function diagram: 2823 Unit selection: - Expert list: 1 Factory setting: 0.00 [ms]
Description:	Sets the filter time for the SSM feedback signal to detect standstill (n < nx).		
	Note The filter time is effective only if the function is enabled (p9501.16 = 1). The parameter is included in the data cross-check of the two monitoring channels. The set time is rounded internally to an integer multiple of the monitoring clock cycle. SSM: Safe Speed Monitor (safety-relevant feedback signal from the speed monitoring)		
p9546	SI Motion SSM (SGA n < nx) speed limit (CU) / SI Mtn SSM n_limCU		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.00 [mm/min]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 1000000.00 [mm/min]	Access level: 3 Function diagram: 2823 Unit selection: - Expert list: 1 Factory setting: 20.00 [mm/min]
Description:	Sets the velocity limit for the SSM feedback signal to detect standstill (n < nx). When this limit value is undershot, the signal "SSM feedback signal active" (SGA n < n_x) is set. For p9568 = 0, the value in p9546 is also applicable for SAM/SBR.		
	 CAUTION The following applies for p9506 = 3: The "SAM/SBR" function is deactivated if the selected threshold value is undershot.		
	Note F-DO: Failsafe Digital Output / SGA: Safety-related output SAM: Safe Acceleration Monitor (safe acceleration monitoring) SBR: Safe Brake Ramp (safe brake ramp monitoring) SSM: Safe Speed Monitor (safety-related feedback signal from the speed monitoring) / SGA n < nx: Safety-related output n < nx		

p9546	SI Motion SSM (SGA n < nx) speed limit (CU) / SI Mtn SSM n_limCU		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2823
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [rpm]	Max: 1000000.00 [rpm]	Factory setting: 20.00 [rpm]
Description:	Sets the velocity limit for the SSM feedback signal to detect standstill (n < nx). When this limit value is undershot, the signal "SSM feedback signal active" (SGA n < n_x) is set. For p9568 = 0, the value in p9546 is also applicable for SAM/SBR.		
 CAUTION The following applies for p9506 = 3: The "SAM/SBR" function is deactivated if the selected threshold value is undershot.			
Note F-DO: Failsafe Digital Output / SGA: Safety-related output SAM: Safe Acceleration Monitor (safe acceleration monitoring) SBR: Safe Brake Ramp (safe brake ramp monitoring) SSM: Safe Speed Monitor (safety-related feedback signal from the speed monitoring) / SGA n < nx: Safety-related output n < nx			

p9547	SI Motion SSM (SGA n < nx) velocity hysteresis (CU) / SI Mtn SSM hyst CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2823
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.0010 [mm/min]	Max: 500.0000 [mm/min]	Factory setting: 10.0000 [mm/min]
Description:	Sets the velocity hysteresis for the SSM feedback signal to detect standstill (n < nx).		
Dependency:	See also: C01711		
Note The velocity hysteresis is effective only if the function is enabled (p9501.16 = 1). The parameter is included in the data cross-check of the two monitoring channels. SSM: Safe Speed Monitor (safety-relevant feedback signal from the speed monitoring)			

p9547	SI Motion SSM (SGA n < nx) velocity hysteresis (CU) / SI Mtn SSM hyst CU		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2823
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.0010 [rpm]	Max: 500.0000 [rpm]	Factory setting: 10.0000 [rpm]
Description:	Sets the velocity hysteresis for the SSM feedback signal to detect standstill (n < nx).		
Dependency:	See also: C01711		
Note The velocity hysteresis is effective only if the function is enabled (p9501.16 = 1). The parameter is included in the data cross-check of the two monitoring channels. SSM: Safe Speed Monitor (safety-relevant feedback signal from the speed monitoring)			

p9548	SI Motion SAM actual speed tolerance (Control Unit) / SI Mtn SAM tol CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [mm/min]	Max: 120000.00 [mm/min]	Factory setting: 300.00 [mm/min]
Description:	Sets the velocity tolerance for the "SAM" function.		
Dependency:	See also: C01706		
Note			
SAM: Safe Acceleration Monitor (safe acceleration monitoring)			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9548	SI Motion SAM actual speed tolerance (Control Unit) / SI Mtn SAM tol CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
(Safety rot)	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [rpm]	Max: 120000.00 [rpm]	Factory setting: 300.00 [rpm]
Description:	Sets the velocity tolerance for the "SAM" function.		
Dependency:	See also: C01706		
Note			
SAM: Safe Acceleration Monitor (safe acceleration monitoring)			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9549	SI Motion slip velocity tolerance (Control Unit) / SI Mtn slip tol		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [mm/min]	Max: 6000.00 [mm/min]	Factory setting: 6.00 [mm/min]
Description:	Sets the velocity tolerance that is used for a 2-encoder system in cross-check between the two monitoring channels.		
Dependency:	See also: p9501, p9542		
Note			
If the "actual value synchronization" is not enabled (p9501.3 = 0), then the value parameterized in p9542 is used as tolerance in the data cross-check.			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9549	SI Motion slip velocity tolerance (Control Unit) / SI Mtn slip tol		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.00 [rpm]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 6000.00 [rpm]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 6.00 [rpm]
Description:	Sets the velocity tolerance that is used for a 2-encoder system in cross-check between the two monitoring channels.		
Dependency:	See also: p9501, p9542		
Note If the "actual value synchronization" is not enabled (p9501.3 = 0), then the value parameterized in p9542 is used as tolerance in the data cross-check. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9551	SI Motion SLS(SG) changeover/SOS (SBH) delay time (CU) / SI SLS/SOS t CU		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.00 [ms]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 600000.00 [ms]	Access level: 3 Function diagram: 2819, 2820 Unit selection: - Expert list: 1 Factory setting: 100.00 [ms]
Description:	Sets the delay time for the SLS changeover and for the activation of SOS for the functions "Safely-Limited Speed" (SLS) and "Safe Operating Stop" (SOS). When transitioning from a higher to a lower Safely-Limited Speed level, and when activating Safe Operating Stop (SOS), within this delay time, the "old" speed level remains active. This delay is also applicable when activating SLS from the state "SOS and SLS inactive" and activating SOS from the state "SOS inactive".		
Note The set time is rounded internally to an integer multiple of the monitoring clock cycle. SLS: Safely-Limited Speed / SG: Safely reduced speed SOS: Safe Operating Stop / SBH: Safe operating stop			

p9552	SI Motion transition time STOP C to SOS (SBH) (Control Unit) / SI Mtn t C->SOS CU		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.00 [ms]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 600000.00 [ms]	Access level: 3 Function diagram: 2819 Unit selection: - Expert list: 1 Factory setting: 100.00 [ms]
Description:	Sets the transition time from STOP C to "Safe Operating Stop" (SOS).		
Note The set time is rounded internally to an integer multiple of the monitoring clock cycle. SOS: Safe Operating Stop / SBH: Safe operating stop			
Note This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9553 SI Motion transition time STOP D to SOS (SBH) (Control Unit) / SI Mtn t D->SOS CU

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2819
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [ms]	600000.00 [ms]	100.00 [ms]

Description: Sets the transition time from STOP D to "Safe Operating Stop" (SOS).

Note

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

SOS: Safe Operating Stop / SBH: Safe operating stop

p9554 SI Motion transition time STOP E to SOS (SBH) (Control Unit) / SI Mtn t E->SOS CU

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [ms]	600000.00 [ms]	100.00 [ms]

Description: Sets the transition time from STOP E to "Safe Operating Stop" (SOS).

Note

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

SOS: Safe Operating Stop / SBH: Safe operating stop

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9555 SI Motion transition time STOP F to STOP B (Control Unit) / SI Mtn t F->B CU

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2819
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [ms]	600000.00 [ms]	0.00 [ms]

Description: Sets the transition time from STOP F to STOP B.

Dependency: See also: C01711

Note

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9556 SI Motion STOP A delay time (Control Unit) / SI Mtn IL t_del CU

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2819
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [ms]	3600000.00 [ms]	100.00 [ms]

Description:	Sets the delay time for STOP A after STOP B. In the case of encoderless motion monitoring functions with safe brake ramp monitoring (p9506 = 1) and the OFF3 ramp enabled at the same time (p9507.3 = 0), the parameter has no effect.
Dependency:	See also: p9560 See also: C01701
Note The set time is rounded internally to an integer multiple of the monitoring clock cycle.	

p9557	SI Motion STO test time (Control Unit) / SI Mtn STO t_test		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 10000.00 [ms]	Factory setting: 100.00 [ms]
Description:	Sets the time after which STO must be active when initiating the test stop.		
Dependency:	See also: C01798		
	Note		
	The set time is rounded internally to an integer multiple of the monitoring clock cycle.		
	STO: Safe Torque Off		
	This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p9559	SI Motion forced checking procedure timer (Control Unit) / SI Mtn dyn timer		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [h]	Max: 9000.00 [h]	Factory setting: 8.00 [h]
Description:	Sets the time interval for carrying out the forced checking procedure and testing the safety motion monitoring functions integrated in the drives. Within the parameterized time, the safety functions must have been tested at least once (including deselection of the "STO" function). This monitoring time is reset each time the test is carried out. The signal source to initiate the forced checking procedure is set in p9705.		
Dependency:	See also: p9705 See also: A01697, C01798		
	Note STO: Safe Torque Off This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p9560	SI Motion STO shutdown velocity (Control Unit) / SI Mtn IL v_sh CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [mm/min]	Max: 6000.00 [mm/min]	Factory setting: 0.00 [mm/min]

Description: Sets the shutdown velocity for activating STO.
Below this velocity "standstill" is assumed and for STOP B / SS1, STO is selected.
In the case of encoderless motion monitoring functions, the parameter must be > 0 mm/min (recommended value, 10 mm/min).

Dependency: See also: p9556

Note

The shutdown velocity has no effect for a value = 0.

SS1: Safe Stop 1

STO: Safe Torque Off

p9560

SERVO_DBSI_VD
(Safety rot)

SI Motion pulse suppression shutdown speed (Control Unit) / SI Mtn IL n_sh CU

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0.00 [rpm]

6000.00 [rpm]

0.00 [rpm]

Description: Sets the shutdown speed for the pulse suppression.
Below this speed "standstill" is assumed and for STOP B / SS1, the pulses are suppressed (by changing to STOP A).
In the case of encoderless motion monitoring functions, the parameter must be > 0 rpm (recommended value, 10 rpm).

Dependency: See also: p9556

Note

The shutdown speed has no effect for a value = 0.

SS1: Safe Stop 1

p9562[0...1]

SERVO_DBSI_VD

SI Motion SLP (SE) stop response (Control Unit) / SI Mtn SLP Stop CU

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: Integer16

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0

14

2

Description: Sets the stop response for the "Safely-Limited Position" function (SLP).

Value:

0:	STOP A
1:	STOP B
2:	STOP C
3:	STOP D
4:	STOP E
10:	STOP A with delayed stop response for bus failure
11:	STOP B with delayed stop response for bus failure
12:	STOP C with delayed stop response for bus failure
13:	STOP D with delayed stop response for bus failure
14:	STOP E with delayed stop response for bus failure

Index: [0] = Limit value SLP1 (SE1)

[1] = Limit value SLP2 (SE2)

Dependency: See also: p9534, p9535

Note

In the extended sense, a bus failure should be seen here as a communication error in the control signals of the safety functions (e.g. via PROFIsafe or TM54F).

SLP: Safely-Limited Position / SE: Safe software limit switches

p9563[0...3]	SI Motion SLS (SG)-specific stop response (Control Unit) / SI Mtn SLS stop CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	14	2
Description:	Sets the SLS-specific stop response for the function "Safely-Limited Speed" (SLS). These settings apply to the individual limit values for SLS. An input value of less than 5 signifies personnel protection, from 10 and upwards, machine protection.		
Value:	0: STOP A 1: STOP B 2: STOP C 3: STOP D 4: STOP E 10: STOP A with delayed stop response for bus failure 11: STOP B with delayed stop response for bus failure 12: STOP C with delayed stop response for bus failure 13: STOP D with delayed stop response for bus failure 14: STOP E with delayed stop response for bus failure		
Index:	[0] = Limit value SLS1 [1] = Limit value SLS2 [2] = Limit value SLS3 [3] = Limit value SLS4		
Dependency:	See also: p9531		

NOTICE

In the case of encoderless motion monitoring (p9506/p9306 = 1, 3), only a value of 0 or 1 is permitted.

Note

In the extended sense, a bus failure should be seen here as a communication error in the control signals of the safety functions (e.g. via PROFIsafe or TM54F).

SLS: Safely-Limited Speed / SG: Safely reduced speed

p9564	SI Motion SDI tolerance (Control Unit) / SI Mtn SDI tol CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2824
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.001 [mm]	360.000 [mm]	12.000 [mm]
Description:	Sets the tolerance for the function "Safe motion direction" (SDI). This motion in the monitored direction is still permissible before safety message C01716 is initiated.		
Dependency:	See also: p9565, p9566 See also: C01716		

NoteSDI: Safe Direction (safe motion direction)

p9564SERVO_DBSI_VD
(Safety rot)**SI Motion SDI tolerance (Control Unit) / SI Mtn SDI tol CU****Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 2824**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.001 [°]

360.000 [°]

12.000 [°]

Description:

Sets the tolerance for the function "Safe motion direction" (SDI).

This motion in the monitored direction is still permissible before safety message C01716 is initiated.

Dependency:

See also: p9565, p9566

See also: C01716

NoteSDI: Safe Direction (safe motion direction)

p9565

SERVO_DBSI_VD

SI Motion SDI delay time (Control Unit) / SI Mtn SDI t CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 2824**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.00 [ms]

600000.00 [ms]

100.00 [ms]

Description:

Sets the delay time for the function "Safe motion direction" (SDI).

After selecting the SDI function, then for a maximum of this time, motion in the monitored direction is permissible. This time can therefore be used for braking any motion.

Dependency:

See also: p9564, p9566

See also: C01716

Note

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

SDI: Safe Direction (safe motion direction)

p9566

SERVO_DBSI_VD

SI Motion SDI stop response (Control Unit) / SI Mtn SDI Stop CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** -**Function diagram:** 2824**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

14

1

Description:

Sets the stop response for the "Safe Direction" function (SDI).

This setting applies to both directions of motion.

Value:

0: STOP A

1: STOP B

2: STOP C

3: STOP D

4: STOP E

10: STOP A with delayed stop response for bus failure

11: STOP B with delayed stop response for bus failure

- 12: STOP C with delayed stop response for bus failure
 13: STOP D with delayed stop response for bus failure
 14: STOP E with delayed stop response for bus failure

Dependency: See also: p9564, p9565
 See also: C01716

NOTICE

In the case of encoderless motion monitoring (p9506 = 1), only a value of 0 or 1 is permitted.

Note

In the extended sense, a bus failure should be seen here as a communication error in the control signals of the safety functions (e.g. via PROFIsafe or TM54F).

SDI: Safe Direction (safe motion direction)

p9567 SI Motion switchover velocity to SOS/SLS level (CU) / SI Mtn v_SOS/SG CU

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [mm/min]	1000.00 [mm/min]	0.00 [mm/min]

Description: Sets the switchover velocity to SOS/SLS level.
 Below this velocity limit, for a transition to SOS/SLS level, standstill (zero velocity) is assumed.
 The transition times from STOP C, D, E and the delay time when selecting SOS are canceled if this velocity threshold is fallen below. The wait time from p9569 is started, and after it has expired, SOS is active.
 The transition time from a higher to a lower safety-limited velocity level is canceled when this velocity level is fallen below. The wait time from p9569 is started, and after it has expired, the new SLS level becomes active.
 For a STOP C, this switchover velocity is the only criterion for prematurely activating an SOS. In the other cases mentioned here, it must first have been signaled that the axis had correctly braked.

Dependency: See also: p9501, p9551, p9552, p9553, p9554, p9569

Note

With p9567 = 0, reducing the wait time for the transition to SOS/SLS level is deactivated.

SOS: Safe Operating Stop

SLS: Safely-Limited Speed

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9567 SI Motion switchover velocity to SOS/SLS level (CU) / SI Mtn v_SOS/SG CU

SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [rpm]	1000.00 [rpm]	0.00 [rpm]

Description: Sets the switchover velocity to SOS/SLS level.
 Below this velocity limit, for a transition to SOS/SLS level, standstill (zero velocity) is assumed.
 The transition times from STOP C, D, E and the delay time when selecting SOS are canceled if this velocity threshold is fallen below. The wait time from p9569 is started, and after it has expired, SOS is active.
 The transition time from a higher to a lower safety-limited velocity level is canceled when this velocity level is fallen below. The wait time from p9569 is started, and after it has expired, the new SLS level becomes active.
 For a STOP C, this switchover velocity is the only criterion for prematurely activating an SOS. In the other cases mentioned here, it must first have been signaled that the axis had correctly braked.

Dependency: See also: p9501, p9551, p9552, p9553, p9554, p9569

Note

With p9567 = 0, reducing the wait time for the transition to SOS/SLS level is deactivated.

SOS: Safe Operating Stop

SLS: Safely-Limited Speed

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9568

SERVO_DBSI_VD

SI Motion SAM/SBR velocity limit (Control Unit) / SI Mtn SAM v_limCU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.00 [mm/min]

1000.00 [mm/min]

0.00 [mm/min]

Description:

Sets the velocity limit for the "SAM" and "SBR" functions.

If the drive is being ramped down, but accelerates by the tolerance in p9548/p9348, then the SAM function identifies this and a STOP A is initiated.

The monitoring operates as follows:

- monitoring by SAM is activated for SS1 (or STOP B) and SS2 (or STOP C).
 - the SAM limit value is frozen after the velocity limit in p9568/p9368 is undershot.
 - SAM monitoring is still executed until the transition time to SOS/STO has expired.
-

Note

SAM: Safe Acceleration Monitor (safe acceleration monitoring)

SBR: Safe Brake Ramp (safe brake ramp monitoring)

SSM: Safe Speed Monitor (safety-relevant feedback signal from the speed monitoring)

For p9568 = p9368 = 0, the following applies:

The value in p9546/p9346 (SSM) is applied as the speed limit for SAM/SBR.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9568SERVO_DBSI_VD
(Safety rot)**SI Motion SAM/SBR velocity limit (Control Unit) / SI Mtn SAM v_limCU****Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.00 [rpm]

1000.00 [rpm]

0.00 [rpm]

Description:

Sets the velocity limit for the "SAM" and "SBR" functions.

If the drive is being ramped down, but accelerates by the tolerance in p9548/p9348, then the SAM function identifies this and a STOP A is initiated.

The monitoring operates as follows:

- monitoring by SAM is activated for SS1 (or STOP B) and SS2 (or STOP C).
 - the SAM limit value is frozen after the velocity limit in p9568/p9368 is undershot.
 - SAM monitoring is still executed until the transition time to SOS/STO has expired.
-

Note

SAM: Safe Acceleration Monitor (safe acceleration monitoring)

SBR: Safe Brake Ramp (safe brake ramp monitoring)

SSM: Safe Speed Monitor (safety-relevant feedback signal from the speed monitoring)

For p9568 = p9368 = 0, the following applies:

The value in p9546/p9346 (SSM) is applied as the speed limit for SAM/SBR.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9569	SI Motion transition time to SOS/SLS level after standstill (CU) / SI Mtn t SOS/SG CU		
SERVO_DBSI_VD	Can be changed: C2(95) Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: 0.00 [ms]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 10000.00 [ms]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 100.00 [ms]
Description:	Sets the transition time to SOS/SLS level after standstill. If, at the transition to SOS, standstill is identified (p9567), then SOS is active at the latest after this transition time. If, at the transition from a higher to a lower safely limited velocity level, standstill is identified (p9567), then at the latest after this transition time, the new SLS level becomes active. The transition times from STOP C, D, E and the delay time when selecting SOS / switching over SLS are canceled in this case.		
Dependency:	See also: p9551, p9552, p9553, p9554, p9567		
Note The time set in p9569 is only effective for p9567 > 0. The set time is rounded internally to an integer multiple of the monitoring clock cycle. SOS: Safe Operating Stop SLS: Safely-Limited Speed This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9570	SI Motion acceptance test mode (Control Unit) / SI Mtn acc_mode		
SERVO_DBSI_VD	Can be changed: T, U Data type: Integer16 P group: Safety Integrated Not for motor type: - Min: 0000 hex	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 00AC hex	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0000 hex
Description:	Setting to select and deselect the acceptance test mode.		
Value:	0: [00 hex] Deselect the acceptance test mode 172: [AC hex] Select the acceptance test mode		
Dependency:	See also: p9601 See also: C01799		
Note Acceptance test mode can only be selected if the safe motion monitoring functions are enabled. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9572	SI Motion reference position (Control Unit) / SI mtn ref_pos		
SERVO_DBSI_VD	Can be changed: T, U Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: -737280.000 [mm]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 737280.000 [mm]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0.000 [mm]
Description:	The reference position entered in this parameter, is used as safe absolute position when setting p9573. If errors are identified when performing the plausibility checks, then message C01711 is output with message value 1003		

Note

The unit depends on the selected axis type, linear or rotary axis, in p9502

p9572

SERVO_DBSI_VD
(Safety rot)

SI Motion reference position (Control Unit) / SI mtn ref_pos

Can be changed: T, U

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-737280.000 [°]

737280.000 [°]

0.000 [°]

Description:

The reference position entered in this parameter, is used as safe absolute position when setting p9573.
If errors are identified when performing the plausibility checks, then message C01711 is output with message value 1003

Note

The unit depends on the selected axis type, linear or rotary axis, in p9502

p9573

SERVO_DBSI_VD

SI Motion accept reference position (Control Unit) / SI mtn ref_pos

Can be changed: T, U

Calculated: -

Access level: 3

Data type: Integer16

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0

263

0

Description:

The safe absolute position is rejected or newly set using this parameter.
If errors are identified when performing the plausibility checks, then message C01711 is output with message value 1003

Value:

0: No action
89: Set reference position at standstill
122: Declare reference position invalid
263: Referencing via SCC

Dependency:

See also: p9572

Note

SCC: Safety Control Channel

p9574

SERVO_DBSI_VD

SI Motion safe position scaling (Control Unit) / SI mtn SP scal CU

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: Integer32

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

1

100000

1000

Description:

Sets the scaling factor to transfer the safe position via PROFIsafe in the 16-bit notation.

Dependency:

See also: r9713

Note

The parameter is only effective when PROFIsafe telegram 901 is selected.

By selecting a suitable scaling of the 32 bit position actual value (r9713[0]), it must be ensured that the scaled position actual value is not greater than 16 bit. The scaling is realized by dividing r9713[0] with this scaling factor.

If, during operation, a position actual value is determined that cannot be scaled to 16 bits, then message C0711 is output with value 7001 - along with the STOP F safety stop response.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9576	SI Motion SLA filter time (CU) / SI Mtn SLA filt CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 500.00 [ms]	Factory setting: 0.00 [ms]
Description:	Sets the filter time for the acceleration monitoring with a fine resolution of the acceleration.		

Note

The filter time is only effective if the function is enabled (p9501.20 = 1).

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

The parameter is included in the data cross-check of the two monitoring channels.

SLA: Safely-Limited Acceleration

p9578	SI Motion SLA acceleration limit (CU) / SI Mtn SLA lim_CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [m/s ²]	Max: 1000.00 [m/s ²]	Factory setting: 1.00 [m/s ²]
Description:	Sets the acceleration limit for the "Safely-Limited Acceleration" function (SLA).		
Dependency:	See also: p9579 See also: C01717		

Note

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

SLA: Safely-Limited Acceleration

p9578	SI Motion SLA acceleration limit (CU) / SI Mtn SLA lim_CU		
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [rev/s ²]	Max: 1000.00 [rev/s ²]	Factory setting: 1.00 [rev/s ²]
Description:	Sets the acceleration limit for the "Safely-Limited Acceleration" function (SLA).		
Dependency:	See also: p9579 See also: C01717		

Note

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

SLA: Safely-Limited Acceleration

p9579 SI Motion SLA stop response (Control Unit) / SI Mtn SLA stop CU			
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	14	1
Description:	Sets the stop response for the "Safely-Limited Acceleration" function (SLA).		
Value:	0: STOP A		
	1: STOP B		
	2: STOP C		
	3: STOP D		
	4: STOP E		
	10: STOP A with delayed stop response for bus failure		
	11: STOP B with delayed stop response for bus failure		
	12: STOP C with delayed stop response for bus failure		
	13: STOP D with delayed stop response for bus failure		
	14: STOP E with delayed stop response for bus failure		
Dependency:	See also: p9578		
	See also: C01717		
Note			
The set time is rounded internally to an integer multiple of the monitoring clock cycle.			
SLA: Safely-Limited Acceleration			

p9581 SI Motion brake ramp reference value (Control Unit) / SI Mtn ramp ref CU			
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	600.0000 [mm/min]	1000000.0000 [mm/min]	1500.0000 [mm/min]
Description:	Sets the reference value to define the brake ramp.		
	The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).		
Dependency:	See also: p9582, p9583		
Note			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9581 SI Motion brake ramp reference value (Control Unit) / SI Mtn ramp ref CU			
SERVO_DBSI_VD (Safety rot)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	600.0000 [rpm]	1000000.0000 [rpm]	1500.0000 [rpm]
Description:	Sets the reference value to define the brake ramp.		
	The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).		
Dependency:	See also: p9582, p9583		

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9582

SERVO_DBSI_VD

SI Motion brake ramp delay time (Control Unit) / SI Mtn rp t_del CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

10.00 [ms]

99000.00 [ms]

250.00 [ms]

Description:

Sets the delay time for monitoring the brake ramp.

Monitoring of the brake ramp starts once the delay time has elapsed.

Dependency:

See also: p9581, p9583

Note

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

Internally, the set time is limited downwards to 2 safety monitoring clock cycles (2 * p9500/p9300).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9583

SERVO_DBSI_VD

SI Motion brake ramp monitoring time (Control Unit) / SI Mtn rp t_mon CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.50 [s]

3600.00 [s]

10.00 [s]

Description:

Sets the monitoring time to define the brake ramp.

The rate of rise of the brake ramp depends upon p9581 (reference value) and p9583 (monitoring time).

Dependency:

See also: p9581, p9582

Note

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9585

SERVO_DBSI_VD

SI Motion actual value sensing sensorless fault tolerance (CU) / ActV sl tol CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Integer32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-1

4

-1

Description:

Sets the tolerance of the plausibility monitoring of the current and voltage angle.

A higher value results in a higher degree of ruggedness when reversing at low speeds, as well as in the field weakening range for load steps.

An increase is advantageous, if the current or voltage at the motor become small.

Dependency:

See also: F01681, C01711

NOTICE

Reducing this value can have a negative impact on the actual value sensing and the plausibility check.
When the value is increased, this results in a longer evaluation delay and a higher velocity deviation (r9787).

Note

This parameter is only effective for encoderless actual value sensing (p9506/p9306 = 1, 3).

For synchronous motors, the value 4 must be set.

If value = -1:

- for synchronous motors, the calculation is automatically made with the value 4.

- for induction motors, the calculation is automatically made with a value of 0 (if the code number of the power unit p0201[0] < 14000, otherwise with a value of 2).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9586

SERVO_DBSI_VD

SI Motion actual value sensing sensorless delay time (CU) / ActV sl t_del CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

5.00 [ms]

1000.00 [ms]

100.00 [ms]

Description:

Sets the delay time to evaluate the encoderless actual value sensing after the pulses have been enabled.

The value must be greater than or equal to the motor magnetizing time (p0346).

Dependency:

See also: C01711

**CAUTION**

The safety functionality is only completely guaranteed after this time has expired.

NOTICE

If this value is reduced, this can have a negative impact on the actual value acquisition and plausibility check – and result in Safety message C01711 with the message value 1041 or 1042.

Note

This parameter is only effective for encoderless actual value sensing (p9506/p9306 = 1, 3).

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9587

SERVO_DBSI_VD

SI Motion actual value sensing sensorless filter time (CU) / ActV sl t_filt CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.00 [ms]

100.00 [ms]

25.00 [ms]

Description:

Sets the filter time for smoothing the actual value with sensorless actual value sensing.

NOTICE

A longer filter time results in a longer response time.

Note

This parameter is only effective for encoderless actual value sensing (p9506/p9306 = 1, 3).

The smoothing is realized with a 1st order lowpass filter

For p9587 = minimum value, the filter is deactivated.

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9588		SI Motion actual value sensing sensorless minimum current (CU) / ActV sl I_min CU	
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.00 [%]	1000.00 [%]	10.00 [%]
Description:	Sets the minimum current for encoderless actual value sensing referred to 1 A (i.e. 1 % = 10 mA). - the value must be increased if C01711 has occurred with message value 1042. - the value must be decreased if C01711 has occurred with message value 1041. For synchronous motors, the following condition must be fulfilled: $ p0305 \times p9783 \geq p9588 \times 1.2$		
Recommendation:	If required, the correct value of the motor minimum current should be determined by making the appropriate measurements.		
Dependency:	See also: C01711		
NOTICE			
If this percentage value is reduced excessively, then this can result in a safety message and an inaccurate actual value.			
Note			
This parameter is only effective for encoderless actual value sensing (p9506/p9306 = 1, 3).			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9589		SI Motion act. value sensing sensorless acceleration limit (CU) / ActV sl a_lim CU	
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	10.00 [%]	3300.00 [%]	100.00 [%]
Description:	Sets the acceleration limit to filter velocity fluctuations. If this percentage value is increased, when accelerating, velocity peaks that do not reflect the real velocity characteristic can occur. If this value is decreased, and this dampens the velocity peaks when accelerating. - the value must be increased if C01711 with message value 1043 has occurred. - the value must be lowered if acceleration procedures have led to an excessive Safety actual velocity.		
Recommendation:	The setting of this parameter depends on the motor and closed-loop control, and must be newly determined for each configuration. To do this, a measurement should be performed while the actual value jumps, and the limit in r9785[0] must be set so low using p9589, so that it is exceeded by the value in r9785[1] a maximum of four times per second. The actual value correction filter intervenes at this instant in time. The step is no longer so drastic.		
Dependency:	See also: C01711		

Note

This parameter is only effective for encoderless actual value sensing (p9506/p9306 = 1, 3).

For p9589 = maximum value, the filter is deactivated.

Diagnostics parameter p9784 must be used to correctly set this parameter.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r9590[0...3]

SERVO_DBSI_VD

SI Motion version safety motion monitoring (Control Unit) / SI Mtn version CU**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the Safety Integrated version for the safe monitoring functions.

Index:

[0] = Safety Version (major release)

[1] = Safety Version (minor release)

[2] = Safety Version (baselevel or patch)

[3] = Safety Version (hotfix)

Dependency:

See also: r9770

Note

Example:

r9590[0] = 2, r9590[1] = 60, r9590[2] = 1, r9590[3] = 0 --> SI Motion version V02.60.01.00

p9601

SERVO_DBSI_VD

SI enable functions integrated in the drive (Control Unit) / SI enable fct CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

0000 0000 bin

Description:

Sets the enable signals for the safety functions integrated in the drive and the type of selection on the Control Unit. Not all of the settings listed below will be permissible, depending on the Control Unit and Motor Module or Power Module being used:

0000 hex:
Safety functions integrated in the drive inhibited (no safety function).

0001 hex:
Basic functions are enabled via onboard terminals (permissible for r9771.0 = 1).

0004 hex:
Extended functions via Terminal Module 54F (TM54F) have been enabled (permissible for r9771.5 = 1).

0005 hex:
Extended functions via Terminal Module 54F (TM54F) and the basic functions via onboard terminals have been enabled (permissible for r9771.5 = 1).

0008 hex:
Basic functions are enabled via PROFIsafe (permissible for r9771.6 = 1).

0009 hex:
Basic functions are enabled via PROFIsafe onboard terminals (permissible for r9771.6 = 1).

000C hex:
Extended functions are enabled via PROFIsafe (permissible for r9771.4 = 1).

000D hex:
Extended functions are enabled via PROFIsafe and basic functions via onboard terminals (permissible for r9771.4 = 1).

0024 hex:
Extended functions without selection are enabled (permissible for r9771.16 = 1).

0025 hex:
Extended functions without selection and basic functions via onboard terminals are enabled (permissible for r9771.16 = 1).

0040 hex:
Basic functions are enabled via TM54F

0041 hex:
Basic functions are enabled via TM54F and onboard terminals.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	STO (SH) via terminals (CU) enable	Enable	Inhibit	2810
02	Enable motion monitoring functions integrated in drive (CU)	Enable	Inhibit	-
03	Enable PROFIsafe (CU)	Enable	Inhibit	-
05	Enab motion monit functions integr in drive w/out selection (CU)	Enable	Inhibit	-
06	Basic functions via TM54F	Enable	Inhibit	-

Dependency:

See also: r9771, p9801

Note

A change only becomes effective only after a POWER ON. Exception: Changes to p9601.0 and p9601.7 become effective immediately.

In addition to all of the combinations listed above, using bit 7, the "STO via Power Module terminals" function can be enabled (this is permissible for r9771.19 = 1).

CU: Control Unit

STO: Safe Torque Off / SH: Safe standstill

SS1: Safe Stop 1

SI: Safety Integrated

SMM: Safe Motion Monitoring

F-DI: Failsafe Digital Input

F-DO: Failsafe Digital Output

p9602			
SERVO_DBSI_VD	SI enable Safe Brake Control (Control Unit) / SI enable SBC CU		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: 2814
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 1	Factory setting: 0
Description:	Sets the enable signal for the function "Safe Brake Control" (SBC) on the Control Unit.		
Value:	0: Inhibit SBC 1: Enable SBC		
Dependency:	See also: p9802		
Note			
The "Safe Brake Control" function is not activated until at least one safety monitoring function has been enabled (i.e. p9501 not equal to 0 and/or p9601 not equal to 0).			
It does not make sense to parameterize "no motor holding brake available" and enable "Safe Brake Control" (p1215 = 0, p9602 = p9802 = 1) if there is no motor holding brake.			
The parameterization "motor holding brake the same as sequence control, connection via BICO" and "Safe Brake Control" enabled (p1215 = 3, p9602 = 1, p9802 = 1) is not practical.			
The parameterization "motor holding brake without feedback signals" and "Safe Brake Control" enabled (p1278 = 1, p9602 = 1, p9802 = 1) is not permissible.			
CU: Control Unit			
SBC: Safe Brake Control			
SI: Safety Integrated			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			
p9610			
SERVO_DBSI_VD	SI PROFIsafe address (Control Unit) / SI PROFIsafe CU		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 65534	Factory setting: 0
Description:	Sets the PROFIsafe address for the Control Unit.		
Dependency:	See also: p9810		
Note			
A change only becomes effective after a POWER ON.			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			
p9611			
SERVO_DBSI_VD	SI PROFIsafe telegram selection (Control Unit) / SI Ps telegram CU		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 998	Factory setting: 998
Description:	Sets the PROFIsafe telegram number for the Control Unit.		
Value:	0: No PROFIsafe telegram selected 30: PROFIsafe standard telegram 30, PZD-1/1 31: PROFIsafe standard telegram 31, PZD-2/2		

- 900: PROFIsafe SIEMENS telegram 900, PZD-2/2
 901: PROFIsafe SIEMENS telegram 901, PZD-3/5
 902: PROFIsafe SIEMENS telegram 902, PZD-3/6
 903: PROFIsafe SIEMENS telegram 903, PZD-3/5
 998: Compatibility mode (as for firmware version < 4.5)

Dependency: See also: p9811, p60022

Note

A change only becomes effective after a POWER ON.

For p9601.3 = p9801.3 = 1 (PROFIsafe enabled), the following variants exist when parameterizing PROFIsafe telegram 30:

- p9611 = p9811 = 998 and p60022 = 0
 - p9611 = p9811 = 998 and p60022 = 30
 - p9611 = p9811 = 30 and p60022 = 30
-

p9612

SERVO_DBSI_VD

SI PROFIsafe failure response (Control Unit) / SI Ps fail CU

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: Integer16

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0

1

0

Description: Sets the stop response when PROFIsafe communication fails.

Value:
 0: STOP A
 1: STOP B

Note

For the selected STOP B stop response, in order to ensure that the OFF3 ramp is actually maintained, when just using the Safety Basic Functions, the following must be carefully observed:

- the transition time STOP F to STOP A (p9658, p9858) must be set longer or equal to the SS1 delay time (p9652, p9852).
- if a higher-level control responds to a drive fault by withdrawing the controller enable signals, for faults F01611 and F30611, the message type must be changed to alarm (p2118, p2119).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9620[0...7]

SERVO_DBSI_VD

BI: SI signal source for STO (SH)/SBC/SS1 (Control Unit) / SI s_s STO/SS1 CU

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: Unsigned32 / Binary

Dyn. index: -

Function diagram: 2810

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

0

Description: Sets the signal source for the following functions on the Control Unit:

STO: Safe Torque Off / SH: Safe standstill

SBC: Safe Brake Control

SS1: Safe Stop 1 (time monitored)

Dependency: See also: p9601

Note

The following signal sources are permitted:

- fixed zero (standard setting).
- digital inputs DI 0 ... 7, 16, 17, 20, 21 on the Control Unit 320-2 (CU320-2).
- digital inputs DI 0 ... 3, 16, 17 on the Controller Extensions (CX32-2, NX10.3, NX15.3).
- digital inputs DI 0 ... 3, 16 on the Control Unit 310-2 (CU310-2).

It is not permitted to establish an interconnection to a digital input in the simulation mode.

For a parallel circuit configuration of n power units, the following applies:

p9620[0] = Signal source for power unit 1

...

p9620[n-1] = signal source for power unit n

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9650

SERVO_DBSI_VD

SI SGE changeover discrepancy time (Control Unit) / SI SGE chgov t CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 2810**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.00 [ms]

2000.00 [ms]

500.00 [ms]

Description:

Sets the discrepancy time to change over the safety-related inputs (SGE) on the Control Unit.

An SGE changeover is not effective simultaneously due to the different runtimes in the two monitoring channels. After an SGE changeover, dynamic data is not subject to a data cross-check during this discrepancy time.

Dependency:

See also: p9850

Note

For a data cross-check between p9650 and p9850, a difference of one Safety monitoring clock cycle is tolerated.

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

SGE: Safety-related input (e.g. STO terminals)

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9651

SERVO_DBSI_VD

SI STO/SBC/SS1 debounce time (Control Unit) / SI STO t_debou CU**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0.00 [ms]

100.00 [ms]

0.00 [ms]

Description:

Sets the debounce time for the failsafe digital inputs used to control STO/SBC/SS1.

NOTICE

To filter noise pulses or test impulses from F-DOs, there is the following dependency on the parameter p0799[0]:

- if p0799[0] is less than 1 ms, then p9651 = 1 ms or a multiple integer of 1 ms.

- if p0799[0] is greater or equal to 1 ms, then p9651 must = p0799[0] - or must be a multiple integer of p0799[0].

Note

The debounce time is rounded to whole milliseconds. It specifies the maximum duration of a fault pulse at the failsafe digital inputs with no reaction/influence on the selection or deselection of the Safety Basic Functions.

Example:

Debounce time = 1 ms: Fault pulses of 1 ms are filtered; only pulses longer than 2 ms are processed.

Debounce time = 3 ms: Fault pulses of 3 ms are filtered; only pulses longer than 4 ms are processed.

The set debounce time impacts the response time of the safety function.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9652	SI Safe Stop 1 delay time / SI Stop 1 t_{del}		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [s]	Max: 300.00 [s]	Factory setting: 0.00 [s]
Description:	Sets the delay time of the pulse suppression for the function "Safe Stop 1" (SS1) on the Control Unit to brake along the OFF3 down ramp (p1135).		
Recommendation:	In order that the drive can completely ramp-down along the OFF3 ramp and a motor holding brake that is possibly available can close, then the delay time should be set as follows: Motor holding brake parameterized: delay time $\geq p1135 + p1228 + p1217$ Motor holding brake not parameterized: delay time $\geq p1135 + p1228$		
Dependency:	See also: p1135, p9852		
	Note Pulse cancellation after failure of PROFIsafe communication is delayed by this time if "STOP B" is set (p9612 = 1). For a data cross-check between p9652 and p9852, a difference of one safety monitoring clock cycle is tolerated. The set time is rounded internally to an integer multiple of the monitoring clock cycle. SS1: Safe Stop 1		

p9653	SI Safe Stop 1 drive-based braking response / SI SS1 drv resp		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 1	Factory setting: 0
Description:	Sets the drive-based braking response for the "Safe Stop 1" (SS1) function.		
Value:	0: SS1 with OFF3 1: SS1E external stop		
	Note SS1: Safe Stop 1 SS1E: Safe Stop 1 external (Safe Stop 1 with external stop) SS1E requires the externally initiated stop in order to be in conformance with stop Category 1. With this parameter, a switchover is made from SS1 to SS1E, and the drive-based braking response of function SS1 (time controlled) of the Basic Functions is deactivated. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

p9658	SI transition time STOP F to STOP A (Control Unit) / SI STOP F->A CU		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2802
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 30000.00 [ms]	Factory setting: 0.00 [ms]
Description:	Sets the transition period from STOP F to STOP A on the Control Unit.		
Dependency:	See also: r9795, p9858 See also: F01611		
Note			
For a data cross-check between p9658 and p9858, a difference of one safety monitoring clock cycle is tolerated. The set time is rounded internally to an integer multiple of the monitoring clock cycle.			
If a higher-level control responds to a drive fault by withdrawing the controller enable signals, for faults F01611 and F30611, the message type must be changed to alarm (p2118, p2119). As a consequence, the drive can still be braked in a controlled fashion during this delay time.			
STOP F: Defect in a monitoring channel (error in the data cross-check)			
STOP A: STO as a result of a fault detected by Safety Integrated			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9659	SI forced checking procedure timer / SI FCP Timer		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2810
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [h]	Max: 9000.00 [h]	Factory setting: 8.00 [h]
Description:	Sets the time interval for carrying out the forced checking procedure and testing the Safety switch-off signal paths. Within the parameterized time, STO must have been deselected at least once. The monitoring time is reset each time that STO is deselected.		
Dependency:	See also: A01699		
Note			
STO: Safe Torque Off / SH: Safe standstill			
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

r9660	SI forced checking procedure remaining time / SI FCP remain		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: - [h]	Max: - [h]	Factory setting: - [h]
Description:	Displays the time remaining before dynamization and testing of the safety switch-off signal paths (forced checking procedure).		
Dependency:	See also: A01699		

p9700	SI Motion copy function / SI Mtn copy fct		
SERVO_DBSI_VD	Can be changed: T, U Data type: Integer16 P group: Safety Integrated Not for motor type: - Min: 0000 hex	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 00D0 hex	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0000 hex
Description:	Setting to start the required copy function. After starting, the appropriate parameters are copied from the Control Unit to the Motor Module. Once copying is complete, the parameter is automatically reset to zero.		
Value:	0: [00 hex] Copy function ended 29: [1D hex] Start copy function node identifier 46: [2E hex] start copy function encoder parameters 87: [57 hex] Start copy function SI parameters 208: [D0 hex] Start copy function SI basic parameters		
Note For value = 57 hex, 2E hex and D0 hex: The value can only be set if the safety commissioning mode is set and the Safety Integrated password was entered. For value = D0 hex: The following parameters are copied after starting the copy function: p9601 --> p9801, p9602 --> p9802, p9610 --> 9810, p9611 --> 9811, p9621 --> 9821, p9622 --> 9822, p9650 --> p9850, p9651 --> p9851, p9652 --> p9852, p9658 --> p9858, p9697 --> p9897 This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9701	Acknowledge SI motion data change / Ackn SI Mtn dat		
SERVO_DBSI_VD	Can be changed: T, U Data type: Integer16 P group: Safety Integrated Not for motor type: - Min: 0000 hex	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: 00EC hex	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0000 hex
Description:	Setting to transfer the reference checksums from the associated actual checksums after changes (SI parameters, hardware). After transferring the reference checksums, parameters are automatically reset to zero.		
Value:	0: [00 hex] Data unchanged 172: [AC hex] Acknowledge data change complete 220: [DC hex] Acknowledge SI basic parameter change 236: [EC hex] Acknowledge hardware CRC		
Dependency:	See also: r9398, p9399, r9728, p9729, r9798, p9799, r9898, p9899		
Note For value = AC and DC hex: These values can only be set if the safety commissioning mode is set and the Safety Integrated password was entered. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

r9703.0...31

SERVO_DBSI_VD

CO/BO: SI Motion SCA status signal (Control Unit) / SI Mtn SCA stat CU**Can be changed:** -**Data type:** Unsigned32**P group:** Safety Integrated**Not for motor type:** -**Min:**

-

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** -**Max:**

-

Access level: 3**Function diagram:** 2840, 2905**Unit selection:** -**Expert list:** 1**Factory setting:**

-

Description:

Display and BICO output for the status signals of the SCA function in monitoring channel 1.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Position at safe output cam 1	Yes	No	-
01	Position at safe output cam 2	Yes	No	-
02	Position at safe output cam 3	Yes	No	-
03	Position at safe output cam 4	Yes	No	-
04	Position at safe output cam 5	Yes	No	-
05	Position at safe output cam 6	Yes	No	-
06	Position at safe output cam 7	Yes	No	-
07	Position at safe output cam 8	Yes	No	-
08	Position at safe output cam 9	Yes	No	-
09	Position at safe output cam 10	Yes	No	-
10	Position at safe output cam 11	Yes	No	-
11	Position at safe output cam 12	Yes	No	-
12	Position at safe output cam 13	Yes	No	-
13	Position at safe output cam 14	Yes	No	-
14	Position at safe output cam 15	Yes	No	-
15	Position at safe output cam 16	Yes	No	-
16	Position at safe output cam 17	Yes	No	-
17	Position at safe output cam 18	Yes	No	-
18	Position at safe output cam 19	Yes	No	-
19	Position at safe output cam 20	Yes	No	-
20	Position at safe output cam 21	Yes	No	-
21	Position at safe output cam 22	Yes	No	-
22	Position at safe output cam 23	Yes	No	-
23	Position at safe output cam 24	Yes	No	-
24	Position at safe output cam 25	Yes	No	-
25	Position at safe output cam 26	Yes	No	-
26	Position at safe output cam 27	Yes	No	-
27	Position at safe output cam 28	Yes	No	-
28	Position at safe output cam 29	Yes	No	-
29	Position at safe output cam 30	Yes	No	-
30	Function "Safe cam" active	Yes	No	-
31	Function "Safe cam" valid	Yes	No	-

Note

SCA: Safe Cam

This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero.

p9705			
SERVO_DBSI_VD	BI: SI Motion test stop signal source / SI Mtn test stop		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Unsigned32 / Binary	Dyn. index: -	Function diagram: 2837
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	0
Description:	Sets the signal source for the test stop of the safe motion monitoring functions.		
	NOTICE		
	Before setting the signal source in p9705 it must be ensured that the signal source is at a logical 0. If, in the Safety commissioning mode, the signal source in p9705 is set - and it already has a logical 1 - then a test stop is immediately initiated and the messages C01711/C30711 are output with message value 1005.		
	Note		
	It is not permissible to use TM54F inputs to start the test stop. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
r9707[0...2]			
SERVO_DBSI_VD	CO: SI Motion diagnostics encoder position actual value GX_XIST1 / SI Mtn XIST1		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Display:		
	Index0: of the encoder actual value GX_Xist1,		
	Index1: of the encoder actual value GX_Xist1 in the clock cycle, from which the subsequently transferred reference position was received.		
	Index2: the difference between index1 and index0 while waiting for the reference position to be transferred.		
	Index1 and index2 are only relevant for safety monitoring functions requiring an encoder with absolute reference when the "Referencing via SCC" function is enabled (p9501 Bit27 = 1)		
Index:	[0] = Encoder actual value Xist1 on CU		
	[1] = Encoder actual value Xist1 latched for referencing		
	[2] = Xist1 latched - reference position difference		
	Note		
	The parameter is only available for Safety Integrated with encoder		
r9708[0...5]			
SERVO_DBSI_VD	SI Motion diagnostics safe position / SI mtn safe pos		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2822, 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [mm]	- [mm]	- [mm]
Description:	Displays the actual load-side actual values of both monitoring channels and their difference.		

Index:	[0] = Load-side actual value on the CU [1] = Load-side actual value on the second channel [2] = Load-side actual value difference CU - second channel [3] = Load-side max. actual value difference CU - second channel [4] = Load-side actual value as safe position via PROFIsafe [5] = Load-side additional actual value difference CU - second channel
Dependency:	See also: r9713

Note

For index [0]:

The display of the load-side position actual value on the Control Unit is updated in the monitoring clock cycle.

For index [1]:

The display of the load-side position actual value on the second channel is updated in the KDV clock cycle (r9724) and delayed by one KDV clock cycle.

For index [2]:

The difference between the load-side position actual value on the Control Unit and load-side position actual value in the second channel is updated in the KDV clock cycle (r9724) and delayed by one KDV clock cycle.

For index [3]:

The maximum difference between the load-side position actual value on the Control Unit and the load-side position actual value on the second channel.

For index [4]:

Displays the load-side position actual value when enabling the function "Safe position via PROFIsafe".

The value is an average value from the values in index 0 and 1.

When the function is not enabled, the content corresponds to the value in index 0.

For index [5]:

The display of the maximum additional difference between the load-side position actual value on the Control Unit, and the load-side position actual value in the second channel, which can occur as a result of the actual value sensing delay in the EnDat 2.2 converter.

Input in p9542: p9708[3] + p9708[5], after performing the measurement for the mechanical tolerance by performing a test run, where, after completion, the maximum tolerance that has occurred is displayed in p9708[3].

KDV: Data cross-check

r9708[0...5]

SERVO_DBSI_VD
(Safety rot)

SI Motion diagnostics safe position / SI mtn safe pos

Can be changed: -

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dyn. index: -

Function diagram: 2822, 2836

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

- [°]

- [°]

- [°]

Description:

Displays the actual load-side actual values of both monitoring channels and their difference.

Index:

[0] = Load-side actual value on the CU
[1] = Load-side actual value on the second channel
[2] = Load-side actual value difference CU - second channel
[3] = Load-side max. actual value difference CU - second channel
[4] = Load-side actual value as safe position via PROFIsafe
[5] = Load-side additional actual value difference CU - second channel

Dependency:

See also: r9713

Note

For index [0]:

The display of the load-side position actual value on the Control Unit is updated in the monitoring clock cycle.

For index [1]:

The display of the load-side position actual value on the second channel is updated in the KDV clock cycle (r9724) and delayed by one KDV clock cycle.

For index [2]:

The difference between the load-side position actual value on the Control Unit and load-side position actual value in the second channel is updated in the KDV clock cycle (r9724) and delayed by one KDV clock cycle.

For index [3]:

The maximum difference between the load-side position actual value on the Control Unit and the load-side position actual value on the second channel.

For index [4]:

Displays the load-side position actual value when enabling the function "Safe position via PROFIsafe".

The value is an average value from the values in index 0 and 1.

When the function is not enabled, the content corresponds to the value in index 0.

For index [5]:

The display of the maximum additional difference between the load-side position actual value on the Control Unit, and the load-side position actual value in the second channel, which can occur as a result of the actual value sensing delay in the EnDat 2.2 converter.

Input in p9542: p9708[3] + p9708[5], after performing the measurement for the mechanical tolerance by performing a test run, where, after completion, the maximum tolerance that has occurred is displayed in p9708[3].

KDV: Data cross-check

r9713[0...5]	CO: SI Motion diagnostics position actual value load side / SI Mtn s_act load		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Display and connector output for the actual load-side actual values of both monitoring channels and their difference.		
Index:	[0] = Load-side actual value on the CU [1] = Load-side actual value on the second channel [2] = Load-side actual value difference CU - second channel [3] = Load-side max. actual value difference CU - second channel [4] = Load-side actual value as safe position via PROFIsafe [5] = Load-side additional actual value difference CU - second channel		
Dependency:	See also: r9708		

Note

Regarding the units, this parameter should be interpreted as follows:

- linear axis: μm

- rotary axis: mdegrees

The value of this parameter is displayed in r9708 with units (mm or degrees).

The display is updated in the safety monitoring clock cycle.

For index [0]:

The display of the load-side position actual value on the Control Unit is updated in the monitoring clock cycle.

For index [1]:

The display of the load-side position actual value on the second channel is updated in the KDV clock cycle (r9724) and delayed by one KDV clock cycle.

For index [2]:

The difference between the load-side position actual value on the Control Unit and load-side position actual value in the second channel is updated in the KDV clock cycle (r9724) and delayed by one KDV clock cycle.

For index [3]:

The maximum difference between the load-side position actual value on the Control Unit and the load-side position actual value on the second channel.

For index [4]:

Displays the load-side position actual value when enabling the function "Safe position via PROFIsafe".

The value is an average value from the values in index 0 and 1.

For a 16-bit notation, the value is influenced using the scaling factor (p9574/p9374).

When the function is not enabled, the content corresponds to the value in index 0.

For index [5]:

The display of the maximum additional difference between the load-side position actual value on the Control Unit, and the load-side position actual value in the second channel, which can occur as a result of the actual value sensing delay in the EnDat 2.2 converter.

Input in p9542: r9713[3] + r9713[5], after performing the measurement for the mechanical tolerance by performing a test run, where, after completion, the maximum tolerance that has occurred is displayed in r9713[3].

KDV: Data cross-check

r9714[0...3]

SERVO_DBSI_VD

CO: SI motion diagnostics velocity / SI Mtn diag v

Can be changed: -

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dyn. index: -

Function diagram: -

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

- [mm/min]

- [mm/min]

- [mm/min]

Description:

Displays the velocity actual values for the motion monitoring functions on the Control Unit.

Index:

[0] = Load-side velocity actual value on the Control Unit

[1] = Actual SAM/SBR speed limit on the Control Unit

[2] = Actual SLS speed limit on the Control Unit

[3] = Actual SLA speed limit on the Control Unit

Dependency:

See also: r9732

NOTICE

For index [2]:

This SLS velocity limit can, as a result of conversion into the internal monitoring format, deviate from the specified SLS velocity limit (see r9732).

Note

The display is updated in the safety monitoring clock cycle.

For linear axes, the following unit applies: millimeters per minute

For rotary axes, the following unit applies: revolutions per minute

r9714[0...3]	CO: SI motion diagnostics velocity / SI Mtn diag v		
SERVO_DBSI_VD (Safety rot)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [rpm]	- [rpm]	- [rpm]
Description:	Displays the velocity actual values for the motion monitoring functions on the Control Unit.		
Index:	[0] = Load-side velocity actual value on the Control Unit [1] = Actual SAM/SBR speed limit on the Control Unit [2] = Actual SLS speed limit on the Control Unit [3] = Actual SLA speed limit on the Control Unit		
Dependency:	See also: r9732		
NOTICE			
For index [2]: This SLS velocity limit can, as a result of conversion into the internal monitoring format, deviate from the specified SLS velocity limit (see r9732).			
Note			
The display is updated in the safety monitoring clock cycle. For linear axes, the following unit applies: millimeters per minute For rotary axes, the following unit applies: revolutions per minute			

r9720.0...29	CO/BO: SI Motion control signals integrated in the drive / SI Mtn integ STW				
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3		
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2840, 2905		
	P group: Safety Integrated	Unit group: -	Unit selection: -		
	Not for motor type: -	Scaling: -	Expert list: 1		
	Min:	Max:	Factory setting:		
	-	-	-		
Description:	Control signals for safety-relevant motion monitoring functions integrated in the drive.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Deselect STO	Yes	No	-
	01	Deselect SS1	Yes	No	-
	02	Deselect SS2	Yes	No	-
	03	Deselect SOS	Yes	No	-
	04	Deselect SLS	Yes	No	-
	06	Deselect SLP	Yes	No	2822
	07	Acknowledgment	Signal edge active	No	-
	08	Deselect SLA	Yes	No	2838
	09	Select SLS bit 0	Set	Not set	-
	10	Select SLS bit 1	Set	Not set	-
	12	Deselect SDI positive	Yes	No	2824
	13	Deselect SDI negative	Yes	No	2824
	19	Select SLP position range	SLP2	SLP1	2822
	23	Deselect SCA	Yes	No	-
	24	Select gearbox bit 0	Set	Not set	-
	25	Select gearbox bit 1	Set	Not set	-
	26	Select gearbox bit 2	Set	Not set	-
	27	Gear change	Set	Not set	-

28	Deselect SS2E	Yes	No	-
29	Deselect SS2ESR	Yes	No	-

Note

This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero.

r9721.0...15

SERVO_DBSI_VD

CO/BO: SI Motion status signals (Control Unit) / SI Mtn stat_sig CU

Can be changed: -	Calculated: -	Access level: 3
Data type: Unsigned32	Dyn. index: -	Function diagram: -
P group: Safety Integrated	Unit group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min:	Max:	Factory setting:
-	-	-

Description:

Display and BICO output for the status signals of the safe motion monitoring functions on monitoring channel 1.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	SOS or SLS active	Yes	No	-
01	SOS active	Yes	No	-
02	Pulse enable	Deleted	Enabled	-
03	Active SLS stage bit 0	Set	Not set	-
04	Active SLS stage bit 1	Set	Not set	-
05	Velocity below limit value n_x	Yes	No	-
06	SLP active	Yes	No	-
07	Safely referenced	Yes	No	-
08	SDI positive active	Yes	No	-
09	SDI negative active	Yes	No	-
10	SLP active position area	SLP2	SLP1	-
11	SLA active	Yes	No	-
12	STOP A or STOP B or STO or SS1 active	Yes	No	2819
13	STOP C or SS2 active	Yes	No	2819
14	STOP D or SS2E active	Yes	No	2819
15	STOP E active	Yes	No	-

Note

This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero.

r9722.0...31

SERVO_DBSI_VD

CO/BO: SI Motion drive-integrated status signals (Control Unit) / SI Mtn int stat CU

Can be changed: -	Calculated: -	Access level: 3
Data type: Unsigned32	Dyn. index: -	Function diagram: 2840, 2905
P group: Safety Integrated	Unit group: -	Unit selection: -
Not for motor type: -	Scaling: -	Expert list: 1
Min:	Max:	Factory setting:
-	-	-

Description:

Status signal for safety-relevant motion monitoring functions integrated in the drive on monitoring channel 1.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	STO or safe pulse suppression active	Yes	No	-
01	SS1 active	Yes	No	-
02	SS2 active	Yes	No	-
03	SOS active	Yes	No	-
04	SLS active	Yes	No	-

06	SLP active	Yes	No	2822
07	Internal event	No	Yes	-
08	SLA active	Yes	No	2838
09	Active SLS stage bit 0	Set	Not set	-
10	Active SLS stage bit 1	Set	Not set	-
11	SOS selected	Yes	No	-
12	SDI positive active	Yes	No	2824
13	SDI negative active	Yes	No	2824
15	SSM (speed below limit value)	Yes	No	2823
19	SLP active position area	SLP2	SLP1	2822
22	SP valid	Yes	No	-
23	Safely referenced	Yes	No	-
27	SS2ESR active	Yes	No	-
28	SS2E active	Yes	No	-
30	SLP limit upper maintained	Yes	No	2822
31	SLP limit lower maintained	Yes	No	2822

NOTICE

For bit 07:

The signal state behaves in an opposite way to the PROFIsafe Standard.

Note

This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero.

For bit 07:

An internal event is displayed if a STOP A ... F is active.

r9722.0...31	CO/BO: SI Motion drive-integrated status signals (Control Unit) / SI Mtn int stat CU				
SERVO_DBSI_VD (Lin)	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned32		Dyn. index: -		Function diagram: 2840, 2905
	P group: Safety Integrated		Unit group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min:		Max:		Factory setting:
	-		-		-
Description:	Status signal for safety-relevant motion monitoring functions integrated in the drive on monitoring channel 1.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	STO or safe pulse suppression active	Yes	No	-
	01	SS1 active	Yes	No	-
	02	SS2 active	Yes	No	-
	03	SOS active	Yes	No	-
	04	SLS active	Yes	No	-
	06	SLP active	Yes	No	2822
	07	Internal event	No	Yes	-
	08	SLA active	Yes	No	2838
	09	Active SLS stage bit 0	Set	Not set	-
	10	Active SLS stage bit 1	Set	Not set	-
	11	SOS selected	Yes	No	-
	12	SDI positive active	Yes	No	2824
	13	SDI negative active	Yes	No	2824
	15	SSM (velocity below limit value)	Yes	No	2823
	19	SLP active position area	SLP2	SLP1	2822

22	SP valid	Yes	No	-
23	Safely referenced	Yes	No	-
27	SS2ESR active	Yes	No	-
28	SS2E active	Yes	No	-
30	SLP limit upper maintained	Yes	No	2822
31	SLP limit lower maintained	Yes	No	2822

NOTICE

For bit 07:

The signal state behaves in an opposite way to the PROFIsafe Standard.

Note

This parameter is only supplied with actual values in the case of Safety Integrated Extended Functions. For Safety Integrated Basic Functions (SBC, SS1, STO), the value is equal to zero.

For bit 07:

An internal event is displayed if a STOP A ... F is active.

r9723.0...17

SERVO_DBSI_VD

CO/BO: SI Motion diagnostic signals integrated in the drive / SI Mtn integ diag**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the diagnostic signals for safety-relevant motion monitoring functions integrated in the drive.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Forced checking procedure required	Yes	No	-
01	STOP F and then STOP B active	Yes	No	2819
02	Communication failure delay time active	Yes	No	-
03	Actual value sensing supplies valid value	Yes	No	2821
04	Encoderless act val sensing acc to technique for U/f control	Yes	No	-
09	Safe pulse suppression active	Yes	No	-
12	Test stop active	Yes	No	-
16	SAM/SBR active	Yes	No	2820
17	Position referenced	Yes	No	2821

Note

For bit 00:

A required dynamization is also displayed via alarm A01679.

For bit 01:

This bit can be used, to execute a drive-based or control-based ESR.

For bit 02:

This bit is set if communication fails and the delay time of the stop response is running.

For bit 04:

When sensing the velocity without encoder, a distinction is made between the closed-loop speed controlled and open-loop speed controlled (U/f) modes.

For bit 09:

Safe pulse cancellation is a state that can only occur for the combination of velocity sensing without encoder (p9506) and drive-integrated motion monitoring functions without selection (p9601.5). In this state, internally an STO is initiated, which can be withdrawn again using an OFF1 enable.

For bit 12:

Test stop active, is also displayed using safety message C01798.

ESR: Extended Stop and Retract

SAM: Safe Acceleration Monitor (safe acceleration monitoring)

SBR: Safe Brake Ramp (safe brake ramp monitoring)

p9726

SERVO_DBSI_VD

SI Motion user agreement selection/deselection / SI Mtn UserAgr sel

Can be changed: T, U

Data type: Integer16

P group: Safety Integrated

Not for motor type: -

Min:

0000 hex

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

00AC hex

Access level: 3

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

0000 hex

Description:

Setting to select and deselect the user agreement.

Value:

0: [00 hex] Deselect user agreement

172: [AC hex] Select user agreement

Dependency:

See also: r9727

r9727

SERVO_DBSI_VD

SI Motion user agreement inside the drive / SI Mtn UserAgr int

Can be changed: -

Data type: Integer16

P group: Safety Integrated

Not for motor type: -

Min:

-

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

-

Access level: 3

Function diagram: 2822

Unit selection: -

Expert list: 1

Factory setting:

-

Description:

Displays the internal state of the user agreement.

Value = 0: User agreement is not set.

Value = AC hex: User agreement is set.

Dependency:

See also: p9726

r9728[0...2]

SERVO_DBSI_VD

SI Motion actual checksum SI parameters / SI Mtn act CRC

Can be changed: -

Data type: Unsigned32

P group: Safety Integrated

Not for motor type: -

Min:

-

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

-

Access level: 3

Function diagram: -

Unit selection: -

Expert list: 1

Factory setting:

-

Description:	Displays the checksum over the checked Safety Integrated parameters of the motion monitoring functions (actual checksum).
Index:	[0] = Checksum over SI parameters for motion monitoring [1] = Checksum over SI parameters for actual values [2] = Checksum over SI parameters for hardware
Dependency:	See also: p9729 See also: F01680

p9729[0...2] SI Motion reference checksum SI parameters / SI Mtn ref CRC

SERVO_DBSI_VD	Can be changed: C2(95) Data type: Unsigned32 P group: Safety Integrated Not for motor type: - Min: 0000 hex	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: FFFF FFFF hex	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: 0000 hex
Description:	Sets the checksum using the checksum-tested Safety Integrated parameters for motion monitoring functions (reference checksum).		
Index:	[0] = Checksum over SI parameters for motion monitoring [1] = Checksum over SI parameters for actual values [2] = Checksum over SI parameters for hardware		
Dependency:	See also: r9728 See also: F01680		

Note

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r9730 SI Motion Safe maximum velocity / SI mtn safe v_max

SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: - [mm/min]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: - [mm/min]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: - [mm/min]
Description:	Displays the safe maximum velocity (on the load side) that is permissible for the safe motion monitoring functions as a result of the actual value sensing. This parameter indicates up to which load velocity the safe encoder actual values (redundant encoder coarse position) can still be correctly detected as a result of the particular encoder parameterization. This parameter is only of significance for enabled safety with encoder (otherwise "0").		

Note

If the value displayed is exceeded, message C01711 is output indicating relevant subsequent faults.

r9730 SI Motion Safe maximum velocity / SI mtn safe v_max

SERVO_DBSI_VD (Safety rot)	Can be changed: - Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: - [rpm]	Calculated: - Dyn. index: - Unit group: - Scaling: - Max: - [rpm]	Access level: 3 Function diagram: - Unit selection: - Expert list: 1 Factory setting: - [rpm]
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Description: Displays the safe maximum velocity (on the load side) that is permissible for the safe motion monitoring functions as a result of the actual value sensing.
 This parameter indicates up to which load velocity the safe encoder actual values (redundant encoder coarse position) can still be correctly detected as a result of the particular encoder parameterization.
 This parameter is only of significance for enabled safety with encoder (otherwise "0").

Note

If the value displayed is exceeded, message C01711 is output indicating relevant subsequent faults.

r9731	SI Motion safe position accuracy / SI Mtn pos_accur		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [mm]	- [mm]	- [mm]
Description:	Displays the safe position accuracy (load side). As a result of the actual value sensing for safe motion monitoring functions, this accuracy can be achieved as the maximum. In the case of a 2-encoder system, the accuracy of the poorer encoder - as a result of the number of encoder pulses - is displayed here.		
	Note The parameter is only of significance for enabled safety with encoder (otherwise "0").		

r9731	SI Motion safe position accuracy / SI Mtn pos_accur		
SERVO_DBSI_VD (Safety rot)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [°]	- [°]	- [°]
Description:	Displays the safe position accuracy (load side). As a result of the actual value sensing for safe motion monitoring functions, this accuracy can be achieved as the maximum. In the case of a 2-encoder system, the accuracy of the poorer encoder - as a result of the number of encoder pulses - is displayed here.		
	Note The parameter is only of significance for enabled safety with encoder (otherwise "0").		

r9732[0...1]	SI Motion velocity resolution / SI Mtn v_res		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [mm/min]	- [mm/min]	- [mm/min]

Description:	Displays the velocity resolution for safety-relevant motion monitoring functions. For index [0]: Displays the safe velocity resolution (load side). Setpoints for velocity limits or parameter changes for velocities below this threshold have no effect. For index [1]: Displays the safe velocity accuracy based on the safe encoder accuracy
Index:	[0] = Actual velocity resolution [1] = Minimum velocity resolution
Note For index [0]: This parameter does not provide any information about the actual accuracy of the velocity sensing. This depends on the type of actual value sensing, the gear factors as well as the quality of the encoder being used. Conversion of: (internal fixed value/ Tsi) to mm/min (linear) or rpm (rotary) with Tsi = p9500 (SI Motion monitoring clock cycle). Example: For Tsi = 12 ms, r9732[0] = 5 mm/min (linear) or 1/72 rpm (rotary) is obtained. For index [1]: - for a 2-encoder system with non safety-capable encoders, this means the poorer value for both encoders. Index[1] takes into account the coarse resolution of the encoder only Internal calculation, which also incorporates the factor for the motor-load side conversion, the gearbox ratio and the safety monitoring clock cycle. Result returns mm/min (linear) or rpm (rotary). - for safety without encoder, index 1 is not relevant, and is always the value of zero.	

r9732[0...1] SERVO_DBSI_VD (Safety rot)	SI Motion velocity resolution / SI Mtn v_res		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [rpm]	- [rpm]	- [rpm]
Description:	Displays the velocity resolution for safety-relevant motion monitoring functions. For index [0]: Displays the safe velocity resolution (load side). Setpoints for velocity limits or parameter changes for velocities below this threshold have no effect. For index [1]: Displays the safe velocity accuracy based on the safe encoder accuracy		
Index:	[0] = Actual velocity resolution [1] = Minimum velocity resolution		
Note For index [0]: This parameter does not provide any information about the actual accuracy of the velocity sensing. This depends on the type of actual value sensing, the gear factors as well as the quality of the encoder being used. Conversion of: (internal fixed value/ Tsi) to mm/min (linear) or rpm (rotary) with Tsi = p9500 (SI Motion monitoring clock cycle). Example: For Tsi = 12 ms, r9732[0] = 5 mm/min (linear) or 1/72 rpm (rotary) is obtained. For index [1]: - for a 2-encoder system with non safety-capable encoders, this means the poorer value for both encoders. Index[1] takes into account the coarse resolution of the encoder only Internal calculation, which also incorporates the factor for the motor-load side conversion, the gearbox ratio and the safety monitoring clock cycle. Result returns mm/min (linear) or rpm (rotary). - for safety without encoder, index 1 is not relevant, and is always the value of zero.			

r9733[0...2]	CO: SI Motion setpoint speed limit effective / SI Mtn set_lim		
SERVO_DBSI_VD	Can be changed: - Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: - [rpm]	Calculated: - Dyn. index: - Unit group: 3_1 Scaling: p2000 Max: - [rpm]	Access level: 3 Function diagram: 2820, 2824, 3630 Unit selection: p0505 Expert list: 1 Factory setting: - [rpm]
Description:	Displays the necessary setpoint speed limit as a result of the selected motion monitoring functions. Contrary to the parameterization of the SI limit values, this parameter specifies the motor-side limit value and not the load-side limit value.		
Recommendation:	For the ramp-function generator, by appropriately interconnecting the speed limits p1051 and p1052 with r9733[0, 1], a drive-based setpoint velocity limiting can be realized. - CI: p1051 = r9733[0] - CI: p1052 = r9733[1] Additional limiting can also be activated using connector input p1085 and p1088.		
Index:	[0] = Setpoint limiting positive [1] = Setpoint limiting negative [2] = Setpoint limit absolute		
Dependency:	For SLS: r9733[0] = p9531[x] x p9533 (converted from the load side to the motor side) For SDI negative: r9733[0] = 0 For SLS: r9733[1] = - p9531[x] x p9533 (converted from the load side to the motor side) For SDI positive: r9733[1] = 0 [x] = Selected SLS stage Conversion factor from the motor side to the load side: - motor type = rotary and axis type = linear: p9522 / (p9521 x p9520) - otherwise: p9522 / p9521 See also: p9531, p9533		
NOTICE			
If p1051 = r9733[0] is interconnected, p1052 = r9733[1] must also be interconnected and vice versa. If only the absolute value of the setpoint velocity limiting is required, r9733[2] must be interconnected.			
Note			
The unit changeover between linear and rotary axis is not implemented via the safety changeover (p9502) but by the linear motor changeover. If the "SLS" or "SDI" function is not selected, r9733[0] shows p1082 and r9733[1] shows -p1082. The display in r9733 can be delayed by up to one Safety monitoring clock cycle as compared to the display in r9719/r9720 and r9721/r9722. When selecting SOS or a STOP A ... D, setpoint 0 is specified in r9733.			

r9733[0...2]	CO: SI Motion setpoint speed limit effective / SI Mtn set_lim		
SERVO_DBSI_VD (Lin)	Can be changed: - Data type: FloatingPoint32 P group: Safety Integrated Not for motor type: - Min: - [m/min]	Calculated: - Dyn. index: - Unit group: 4_1 Scaling: p2000 Max: - [m/min]	Access level: 3 Function diagram: 2820, 2824, 3630 Unit selection: p0505 Expert list: 1 Factory setting: - [m/min]
Description:	Displays the necessary setpoint speed limit as a result of the selected motion monitoring functions. Contrary to the parameterization of the SI limit values, this parameter specifies the motor-side limit value and not the load-side limit value.		

Recommendation: For the ramp-function generator, by appropriately interconnecting the speed limits p1051 and p1052 with r9733[0, 1], a drive-based setpoint velocity limiting can be realized.

- Cl: p1051 = r9733[0]

- Cl: p1052 = r9733[1]

Additional limiting can also be activated using connector input p1085 and p1088.

Index: [0] = Setpoint limiting positive

[1] = Setpoint limiting negative

[2] = Setpoint limit absolute

Dependency: For SLS: r9733[0] = p9531[x] x p9533 (converted from the load side to the motor side)

For SDI negative: r9733[0] = 0

For SLS: r9733[1] = - p9531[x] x p9533 (converted from the load side to the motor side)

For SDI positive: r9733[1] = 0

[x] = Selected SLS stage

Conversion factor from the motor side to the load side:

- motor type = rotary and axis type = linear: p9522 / (p9521 x p9520)

- otherwise: p9522 / p9521

See also: p9531, p9533

NOTICE

If p1051 = r9733[0] is interconnected, p1052 = r9733[1] must also be interconnected and vice versa.

If only the absolute value of the setpoint velocity limiting is required, r9733[2] must be interconnected.
--

Note

The unit changeover between linear and rotary axis is not implemented via the safety changeover (p9502) but by the linear motor changeover.

If the "SLS" or "SDI" function is not selected, r9733[0] shows p1082 and r9733[1] shows -p1082.

The display in r9733 can be delayed by up to one Safety monitoring clock cycle as compared to the display in r9719/r9720 and r9721/r9722.

When selecting SOS or a STOP A ... D, setpoint 0 is specified in r9733.

r9734.0...15

SERVO_DBSI_VD

CO/BO: SI Safety Information Channel status word S_ZSW1B / SIC S_ZSW1B**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and BICO output for status word S_ZSW1B of the Safety Information Channel.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	STO active	Yes	No	-
01	SS1 active	Yes	No	-
02	SS2 active	Yes	No	-
03	SOS active	Yes	No	-
04	SLS active	Yes	No	-
05	SOS selected	Yes	No	-
06	SLS selected	Yes	No	-
07	Internal event	Yes	No	-
08	SLA selected	Yes	No	-
09	Select SLS bit0	Yes	No	-
10	Select SLS bit1	Yes	No	-
12	SDI positive selected	Yes	No	-
13	SDI negative selected	Yes	No	-

14	ESR retract requested	Yes	No	-
15	Safety message present	Yes	No	-

Note

SIC: Safety Information Channel

For bit 07:

An internal event is displayed if a STOP A ... F is active.

p9740**SI Motion user agreement selection/deselection MM / SI mtn UserAgr MM**

SERVO_DBSI_VD

Can be changed: T, U**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0000 hex

00AC hex

0000 hex

Description:

Setting to select and deselect the user agreement on the Motor Module/Hydraulic Module.

Value:

0: [00 hex] Deselect user agreement

172: [AC hex] Select user agreement

Dependency:

See also: r9741

r9741**SI Motion user agreement inside the drive MM / SI Mtn UserAgr int**

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Integer16**Dyn. index:** -**Function diagram:** 2822**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the internal state of the user agreement.

Value = 0: User agreement is not set.

Value = AC hex: User agreement is set.

Dependency:

See also: p9740

r9743.4...15**CO/BO: SI Safety Information Channel status word S_ZSW2B / SIC S_ZSW2B**

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and BICO output for status word S_ZSW2B of the Safety Information Channel.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
04	SLP selected position area	SLP2	SLP1	-
07	SLP selected and user agreement set	Yes	No	-
08	SDI positive selected	Yes	No	-
09	SDI negative selected	Yes	No	-
12	Test stop active	Yes	No	-
13	Test stop required	Yes	No	-
14	Reference position required	Yes	No	-
15	Reference trigger command identified or reference position valid	Yes	No	-

NoteSIC: Safety Information Channel

r9744	SI message buffer changes counter / SI msg_buffer chg		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Displays the changes of the safety message buffer. This counter is incremented every time that the safety message buffer changes.		
Recommendation:	This is used to check whether the safety message buffer has been read out consistently.		
Dependency:	See also: r9747, r9748, r9749, r9754		

r9745[0...63]	SI components / SI comp		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Displays the component of the safety message that has occurred.		
	Note Value = 0: Assignment to a component not possible.		

r9747[0...63]	SI message code / SI msg_code		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Displays the numbers of safety messages that have occurred.		
Dependency:	See also: r9744, r9748, r9749, r9754		

Note

The messages type "safety message" (Cxxxxx) are entered in the message fault buffer.

Message buffer structure (principle):

r9747[0], r9748[0], r9749[0], r9753[0], r9754[0], r9755[0], r9756[0] --> Actual message case, safety message 1

...

r9747[7], r9748[7], r9749[7], r9753[7], r9754[7], r9755[7], r9756[7] --> Actual message case, safety message 8

r9747[8], r9748[8], r9749[8], r9753[8], r9754[8], r9755[8], r9756[8] --> 1st acknowledged message case, safety message 1

...

r9747[15], r9748[15], r9749[15], r9753[15], r9754[15], r9755[15], r9756[15] --> 1st acknowledged message case, safety message 8

...

r9747[56], r9748[56], r9749[56], r9753[56], r9754[56], r9755[56], r9756[56] --> 7th acknowledged message case, safety message 1

...

r9747[63], r9748[63], r9749[63], r9753[63], r9754[63], r9755[63], r9756[63] --> 7th acknowledged message case, safety message 8

r9748[0...63]	SI message time received in milliseconds / SI t_msg rcv ms		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: -
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [ms]	- [ms]	- [ms]
Description:	Displays the relative system runtime in milliseconds when the safety message occurred.		
Dependency:	See also: r9744, r9747, r9749, r9754		
r9749[0...63]	SI message value / SI msg_value		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer32	Dyn. index: -	Function diagram: -
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Displays the additional information about the safety message that occurred (as integer number).		
Dependency:	See also: r9744, r9747, r9748, r9754		
r9754[0...63]	SI message time received in days / SI t_msg rcv days		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Messages	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-
Description:	Displays the relative system runtime in days when the safety message occurred.		
Dependency:	See also: r9744, r9747, r9748, r9749		

p9761 SERVO_DBSI_VD	SI password input / SI password inp		
	Can be changed: C1, T	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2800
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0000 hex	Max: FFFF FFFF hex	Factory setting: 0000 hex
Description:	Enters the Safety Integrated password.		
Dependency:	See also: F01659		
	Note It is not possible to change Safety Integrated parameters until the Safety Integrated password has been entered. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		
r9765 SERVO_DBSI_VD	SI Motion forced check procedure remaining time (Control Unit) / SI Mtn dyn remain		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: - [h]	Max: - [h]	Factory setting: - [h]
Description:	Displays the time remaining until the next dynamization and testing of the safety motion monitoring functions integrated in the drives. The signal source to initiate the forced checking procedure is parameterized in p9705.		
Dependency:	See also: p9705 See also: C01798		
r9768[0...7] SERVO_DBSI_VD	SI PROFIsafe receive control words (Control Unit) / SI Ps PZD recv CU		
	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: -
Description:	Displays the received PROFIsafe telegram on the Control Unit.		
Index:	[0] = PZD 1 [1] = PZD 2 [2] = PZD 3 [3] = PZD 4 [4] = PZD 5 [5] = PZD 6 [6] = PZD 7 [7] = PZD 8		
Dependency:	See also: r9769		
	Note The PROFIsafe trailer at the end of the telegram is also displayed (2 words).		

r9769[0...7]				
SERVO_DBSI_VD				
SI PROFIsafe send status words (Control Unit) / SI Ps PZD send CU				
	Can be changed: -		Calculated: -	
	Data type: Unsigned16		Dyn. index: -	
	P group: Safety Integrated		Unit group: -	
	Not for motor type: -		Scaling: -	
	Min:		Max:	
				Access level: 3
				Function diagram: -
				Unit selection: -
				Expert list: 1
				Factory setting:
				-
Description: Displays the PROFIsafe telegram to be sent on the Control Unit.				
Index:				
[0] = PZD 1				
[1] = PZD 2				
[2] = PZD 3				
[3] = PZD 4				
[4] = PZD 5				
[5] = PZD 6				
[6] = PZD 7				
[7] = PZD 8				
Dependency: See also: r9768				
Note				
The PROFIsafe trailer at the end of the telegram is also displayed (2 words).				

r9770[0...3]				
SERVO_DBSI_VD				
SI version drive-integrated safety function (Control Unit) / SI version CU				
	Can be changed: -		Calculated: -	
	Data type: Unsigned16		Dyn. index: -	
	P group: Safety Integrated		Unit group: -	
	Not for motor type: -		Scaling: -	
	Min:		Max:	
				Access level: 3
				Function diagram: 2802
				Unit selection: -
				Expert list: 1
				Factory setting:
				-
Description: Displays the Safety Integrated version for the drive-integrated safety functions on the Control Unit.				
Index:				
[0] = Safety Version (major release)				
[1] = Safety Version (minor release)				
[2] = Safety Version (baselevel or patch)				
[3] = Safety Version (hotfix)				
Note				
Example:				
r9770[0] = 2, r9770[1] = 60, r9770[2] = 1, r9770[3] = 0 --> Safety version V02.60.01.00				

r9771				
SERVO_DBSI_VD				
SI common functions (Control Unit) / SI common fct CU				
	Can be changed: -		Calculated: -	
	Data type: Unsigned32		Dyn. index: -	
	P group: Safety Integrated		Unit group: -	
	Not for motor type: -		Scaling: -	
	Min:		Max:	
				Access level: 3
				Function diagram: 2804
				Unit selection: -
				Expert list: 1
				Factory setting:
				-
Description: Displays the supported Safety Integrated monitoring functions supported on both monitoring channels. The Control Unit determines this display.				
Bit array:	Bit	Signal name	1 signal	0 signal
	00	STO supported via terminals	Yes	No
	01	SBC supported	Yes	No
	02	Extended Functions supported (p9501 > 0)	Yes	No
				FD
				2804
				2804
				2804

03	SS1 supported	Yes	No	2804
04	Extended Functions PROFIsafe supported	Yes	No	-
05	Extended Functions integrated in drive supported (p9601.2 = 1)	Yes	No	-
06	Basic Functions PROFIsafe supported	Yes	No	-
07	Extended Functions encoderless supported	Yes	No	-
08	Safe Brake Adapter supported	Yes	No	-
09	Basic Functions PROFIsafe for parallel connection supported	Yes	No	-
10	Extended Functions integrated in drive for parallel connection	Yes	No	-
11	Extended Functions SDI supported	Yes	No	-
12	Extended Functions SSM encoderless supported	Yes	No	-
13	ESR delay of the pulse suppression	Yes	No	-
14	SBC for parallel connection supported	Yes	No	-
15	SLS limit SP supported via PROFIsafe	Yes	No	-
16	Safety functions without selection, SLP, SS1E supported	Yes	No	-
17	Safe gearbox stage switchover ref supported via SCC	Yes	No	-
18	Controlling Basic Functions with TM54F	Yes	No	-
20	STOP B for PROFIsafe failure supported	Yes	No	-
21	SBR with encoder and SS2E supported	Yes	No	-
22	SCA, deactivation SOS/SLS during an external STOP A	Yes	No	-
23	Synchronous safe position and SLA via PROFIsafe supported	Yes	No	-
24	SLA filtering and fine resolution, SS2ESR supported	Yes	No	-

Note

CU: Control Unit

ESR: Extended Stop and Retract

SBC: Safe Brake Control

SBR: Safe Brake Ramp (safe brake ramp monitoring)

SCA: Safe Cam

SCC: Safety Control Channel

SDI: Safe Direction (safe motion direction)

SI: Safety Integrated

SLA: Safely-Limited Acceleration

SLP: Safely-Limited Position

SLS: Safely-Limited Speed / SG: Safely reduced speed

SOS: Safe Operating Stop / SBH: Safe operating stop

SP: Safe Position

SS1: Safe Stop 1

SS1E: Safe Stop 1 External (Safe Stop 1 with external stop)

SS2E: Safe Stop 2 External (Safe Stop 2 with external stop, external STOP D)

SS2ESR: Safe Stop 2 Extended Stop and Retract

SSM: Safe Speed Monitor (safety-related feedback signal from the speed monitoring) / SGA n < nx: Safety-related output n < nx

STO: Safe Torque Off / SH: Safe standstill

For bit 16:

SS1E is supported for Safety Extended Functions.

r9772.0...23

SERVO_DBSI_VD

CO/BO: SI status (Control Unit) / SI status CU

Can be changed: -

Data type: Unsigned32

P group: Safety Integrated

Not for motor type: -

Min:

-

Calculated: -

Dyn. index: -

Unit group: -

Scaling: -

Max:

-

Access level: 2

Function diagram: 2804

Unit selection: -

Expert list: 1

Factory setting:

-

Description: Display and BICO output for the Safety Integrated status on the Control Unit.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	STO or safe pulse cancellation on CU selected	Yes	No	2810
	01	STO or safe pulse cancellation on CU active	Yes	No	2810
	02	SS1 delay time active on CU	Yes	No	2810
	04	SBC requested	Yes	No	2814
	05	SS1 on CU selected (Basic Functions)	Yes	No	-
	06	SS1 on CU active (Basic Functions)	Yes	No	-
	07	STO terminal state on CU (Basic Functions)	High	Low	-
	09	STOP A cannot be acknowledged active	Yes	No	2802
	10	STOP A active	Yes	No	2802
	15	STOP F active	Yes	No	2802
	16	STO cause: Safety comm. mode	Yes	No	-
	17	STO cause selection via terminal (Basic Functions)	Yes	No	-
	18	STO cause selection via Safe Motion Monitoring (SMM)	Yes	No	-
	19	STO cause actual value missing or safe pulse cancellation	Yes	No	-
	20	STO cause selection PROFIsafe/TM54F (Basic Functions)	Yes	No	-
	21	STO cause selection on other monitoring channel	Yes	No	-
	22	SS1 cause selection terminal (Basic Functions)	Yes	No	-
	23	SS1 cause selection PROFIsafe/TM54F (Basic Functions)	Yes	No	-

Dependency: See also: r9872

Note

For bit 00:

When STO or "Safe pulse cancellation" is selected, the cause is displayed in bits 16 ... 21.

For bit 01:

- For p9772.1 = 1 and p9772.19 = 0, an STO from the Safety Basic functions is active.

- For p9772.1 = 1 and p9772.19 = 1, safe pulse cancellation is active, if safety functions without selection are activated via p9601.2/p9801.2 = 1 and p9601.5/p9801.5 = 1.

Note:

If p9601.0 = 1 and p9601.2 = 1 and p9801.5 = 1 then for bit 0 and 1, the STO function applies.

For bit 05:

When SS1 is selected, the cause is displayed in bits 22 and 23.

For bit 18:

When the bit is set, STO is selected via PROFIsafe or Terminal Module 54F (TM54F).

For bit 19:

With SMM encoderless no actual value sensing is possible on account of OFF2.

With SMM with encoder no actual value sensing is possible on account of parking.

For Safety functions without selection, safe pulse cancellation to selected (p9772.19 = 1).

SMM: Safe Motion Monitoring

For bit 22 and 23:

These bits show via which path the SS1 has been triggered, i.e. what has started the SS1 delay time.

If the SS1 delay time is not started (e.g. because an STO is triggered at the same time), neither of the two bits is set.

r9773.0...31

SERVO_DBSI_VD

CO/BO: SI status (Control Unit + Motor Module) / SI status CU+MM

Can be changed: -

Calculated: -

Access level: 2

Data type: Unsigned32

Dyn. index: -

Function diagram: 2804

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description: Display and BICO output of the Safety Integrated status on the drive (Control Unit + Motor Module).

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	STO selected in drive	Yes	No	2804
	01	STO active in drive	Yes	No	2804
	02	SS1 delay time active in the drive	Yes	No	2804
	04	SBC requested	Yes	No	2804
	05	SS1 selected in the drive (Basic Functions)	Yes	No	-
	06	SS1 active in the drive (Basic Functions)	Yes	No	-
	31	Test stop required for STO	Yes	No	2810

Note

This status is formed from the AND operation of the relevant status of the two monitoring channels.

r9774.0...31

SERVO_DBSI_VD

CO/BO: SI status (group STO) / SI stat group STO**Can be changed:** -**Calculated:** -**Access level:** 2**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** 2804**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Display and BICO output for the Safety Integrated status of the group to which this drive belongs.
These signals are an AND logic operation of the individual status signals of the drives included in this group.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	STO selected in group	Yes	No	2804
01	STO active in group	Yes	No	2804
02	SS1 delay time active in group	Yes	No	-
04	SBC requested in group	Yes	No	2804
05	SS1 selected in group (Basic Functions)	Yes	No	-
06	SS1 active in group (Basic Functions)	Yes	No	-
31	Switch-off signal paths of the group must be tested	Yes	No	2804

Dependency:

See also: p9620, r9773

NOTICE

If a drive belonging to a group is deactivated via p0105, then the signals in r9774 can no longer be correctly displayed (Remedy: Before deactivating, remove this drive from the group).

Note

A group is formed by appropriately grouping the terminals for the function "Safe Torque Off" (STO).
The status of a group of n drives is, for drives 1 to n - 1 displayed with a delay of one monitoring clock cycle; this is a system-related effect.

r9776.0...3

SERVO_DBSI_VD

BO: SI diagnostics / SI diag**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

The parameter is used for diagnostics.

Bit array:

Bit	Signal name	1 signal	0 signal	FD
00	Safety parameter changed POWER ON required	Yes	No	-
01	Safety functions enabled	Yes	No	-
02	Safety component replaced and data save required	Yes	No	-

03	Safety component replaced and acknowledge/save required	Yes	No	-
----	---	-----	----	---

Note

For bit 00 = 1:

At least one Safety parameter has been changed that will only take effect after a POWER ON.

For bit 01 = 1:

Safety functions (basic functions or extended functions) have been enabled and are active.

For bit 02 = 1:

A safety-relevant component has been replaced. Data save required (p0977 = 1 or p0971 = 1 or "copy RAM to ROM").

For bit 03 = 1:

A safety-relevant component has been replaced. Acknowledge (p9702 = 29) and save (p0977 = 1 or p0971 = 1 or "Copy RAM to ROM") required.

r9780**SI monitoring clock cycle (Control Unit) / SI monitor_clk CU**

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 2802**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

- [ms]

- [ms]

- [ms]

Description:

Displays the clock cycle time for the Safety Integrated Basic Functions on the Control Unit.

Dependency:

See also: p0115

Note

Information regarding the relationship between monitoring clock cycle and response times can be found in the following references:

- SINAMICS S120 Function Manual Safety Integrated

- technical documentation for the particular product

r9795**SI diagnostics STOP F (Control Unit) / SI diag STOP F CU**

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 2**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** 2802**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the number of the cross-checked data item which has caused STOP F on the Control Unit.

Dependency:

See also: r9895

See also: F01611

Note

A complete list of numbers for cross-checked data items appears in fault F01611.

r9798**SI actual checksum SI parameters (Control Unit) / SI act_checksum CU**

SERVO_DBSI_VD

Can be changed: -**Calculated:** -**Access level:** 3**Data type:** Unsigned32**Dyn. index:** -**Function diagram:** 2800**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the checksum over the checked Safety Integrated parameters on the Control Unit (actual checksum).

Dependency:

See also: p9799, r9898

p9799	SI reference checksum SI parameters (Control Unit) / SI set_checksum CU			
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3	
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2800	
	P group: Safety Integrated	Unit group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min:	Max:	Factory setting:	
	0000 hex	FFFF FFFF hex	0000 hex	
Description:	Sets the checksum for the checked Safety Integrated parameters on the Control Unit (reference checksum).			
Dependency:	See also: r9798, p9899			
	Note			
	This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			
p9801	SI enable functions integrated in the drive (Motor Module) / SI enable fct MM			
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3	
	Data type: Unsigned16	Dyn. index: -	Function diagram: -	
	P group: Safety Integrated	Unit group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min:	Max:	Factory setting:	
	-	-	0000 0000 bin	
Description:	Sets the enable signals for the safety functions integrated in the drive and the type of selection on the Motor Module. Not all of the settings listed below will be permissible, depending on the Control Unit and Motor Module or Power Module being used: 0000 hex: Safety functions integrated in the drive inhibited (no safety function). 0001 hex: Basic functions are enabled via onboard terminals (permissible for r9871.0 = 1). 0004 hex: Extended functions via Terminal Module 54F (TM54F) have been enabled (permissible for r9871.5 = 1). 0005 hex: Extended functions via Terminal Module 54F (TM54F) and the basic functions via onboard terminals have been enabled (permissible for r9871.5 = 1). 0008 hex: Basic functions are enabled via PROFIsafe (permissible for r9871.6 = 1). 0009 hex: Basic functions are enabled via PROFIsafe onboard terminals (permissible for r9871.6 = 1). 000C hex: Extended functions are enabled via PROFIsafe (permissible for r9871.4 = 1). 000D hex: Extended functions are enabled via PROFIsafe and basic functions via onboard terminals (permissible for r9871.4 = 1). 0024 hex: Extended functions without selection are enabled (permissible for r9871.16 = 1). 0025 hex: Extended functions without selection and basic functions via onboard terminals are enabled (permissible for r9871.16 = 1). 0040 hex: Basic functions are enabled via TM54F 0041 hex: Basic functions are enabled via TM54F and onboard terminals.			
Bit array:	Bit	Signal name	1 signal	0 signal
	00	STO (SH) via terminals (MM) enable	Enable	Inhibit
				FD
				2810

02	Enable motion monitoring functions integrated in drive (MM)	Enable	Inhibit	-
03	Enable PROFIsafe (MM)	Enable	Inhibit	-
05	Enab motion monit functions integr in drive w/out selection (MM)	Enable	Inhibit	-
06	Basic functions via TM54F	Enable	Inhibit	-

Dependency: See also: p9601

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

A change only becomes effective after a POWER ON. Exception: Changes to p9801.0 become effective immediately.

MM: Motor Module

SI: Safety Integrated

SMM: Safe Motion Monitoring

STO: Safe Torque Off / SH: Safe standstill

SS1: Safe Stop 1

F-DI: Failsafe Digital Input

F-DO: Failsafe Digital Output

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9802

SERVO_DBSI_VD

SI enable Safe Brake Control (Motor Module) / SI enable SBC MM

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: Integer32

Dyn. index: -

Function diagram: 2814

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0

1

0

Description: Sets the enable signal for the "Safe Brake Control" function (SBC) on the Motor Module.

0: Inhibit SBC

1: Enable SBC

Dependency: See also: p9602

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

The "Safe Brake Control" function is not activated until at least one safety monitoring function has been enabled (i.e. p9501 not equal to 0 and/or p9801 not equal to 0).

It does not make sense to parameterize "no motor holding brake available" and enable "Safe Brake Control" (p1215 = 0, p9602 = p9802 = 1) if there is no motor holding brake.

The parameterization "motor holding brake the same as sequence control, connection via BICO" and "Safe Brake Control" enabled (p1215 = 3, p9602 = 1, p9802 = 1) is not practical.

The parameterization "motor holding brake without feedback signals" and "Safe Brake Control" enabled (p1278 = 1, p9602 = 1, p9802 = 1) is not permissible.

MM: Motor Module

SBC: Safe Brake Control

SI: Safety Integrated

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9810

SERVO_DBSI_VD

SI PROFIsafe address (Motor Module) / SI Ps address MM**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

65534

0

Description:

Sets the PROFIsafe address of the Motor Module/Hydraulic module.

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

A change only becomes effective after a POWER ON.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9811

SERVO_DBSI_VD

SI PROFIsafe telegram selection (Motor Module) / SI Ps telegram MM**Can be changed:** C2(95)**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

998

998

Description:

Sets the PROFIsafe telegram number for the Motor Module/Hydraulic Module.

Value:

0: No PROFIsafe telegram selected

30: PROFIsafe standard telegram 30, PZD-1/1

31: PROFIsafe standard telegram 31, PZD-2/2

900: PROFIsafe SIEMENS telegram 900, PZD-2/2

901: PROFIsafe SIEMENS telegram 901, PZD-3/5

902: PROFIsafe SIEMENS telegram 902, PZD-3/6

903: PROFIsafe SIEMENS telegram 903, PZD-3/5

998: Compatibility mode (as for firmware version < 4.5)

Dependency:

See also: p9611, p60022

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

A change only becomes effective after a POWER ON.

For p9601.3 = p9801.3 = 1 (PROFIsafe enabled), the following variants exist when parameterizing PROFIsafe telegram 30:

- p9611 = p9811 = 998 and p60022 = 0
- p9611 = p9811 = 998 and p60022 = 30
- p9611 = p9811 = 30 and p60022 = 30

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9850	SI SGE changeover discrepancy time (Motor Module) / SI SGE chgov t MM		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2810
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [μs]	Max: 2000000.00 [μs]	Factory setting: 500000.00 [μs]
Description:	Sets the discrepancy time to change over the safety-related inputs (SGE) on the Motor Module/Hydraulic Module. An SGE changeover is not effective simultaneously due to the different runtimes in the two monitoring channels. After an SGE changeover, dynamic data is not subject to a data cross-check during this discrepancy time.		
Dependency:	See also: p9650		
NOTICE			
This parameter is overwritten by the copy function of the safety functions integrated in the drive.			
Note			
For a data cross-check between p9650 and p9850, a difference of one Safety monitoring clock cycle is tolerated. Rounding effects can occur in the last decimal place of the parameterized time. The set time is rounded internally to an integer multiple of the monitoring clock cycle. SGE: Safety-related input (e.g. STO terminals) This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9851	SI STO/SBC/SS1 debounce time (Motor Module) / SI STO t_debou MM		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [μs]	Max: 100000.00 [μs]	Factory setting: 0.00 [μs]
Description:	Sets the debounce time for the EP terminal of the Motor Module.		
NOTICE			
This parameter is overwritten by the copy function of the safety functions integrated in the drive.			
Note			
Rounding effects can occur in the last decimal place of the parameterized time. The debounce time is rounded to whole milliseconds. It specifies the maximum duration of a fault pulse at the failsafe digital inputs with no reaction/influence on the selection or deselection of the Safety Basic Functions. Example: Debounce time = 1 ms: Fault pulses of 1 ms are filtered; only pulses longer than 2 ms are processed. Debounce time = 3 ms: Fault pulses of 3 ms are filtered; only pulses longer than 4 ms are processed. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p9852	SI Safe Stop 1 delay time (Motor Module) / SI Stop 1 t_del MM		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.00 [ms]	Max: 300000.00 [ms]	Factory setting: 0.00 [ms]
Description:	Sets the delay time of the pulse suppression for the function "Safe Stop 1" (SS1) on the Motor Module to brake along the OFF3 down ramp (p1135).		

Recommendation: In order that the drive can completely ramp-down along the OFF3 ramp and a motor holding brake that is possibly available can close, then the delay time should be set as follows:

Motor holding brake parameterized: delay time $\geq p1135 + p1228 + p1217$

Motor holding brake not parameterized: delay time $\geq p1135 + p1228$

Dependency: See also: p1135, p9652

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

Pulse cancellation after failure of PROFIsafe communication is delayed by this time if "STOP B" is set (p9812 = 1).

For a data cross-check between p9652 and p9852, a difference of one safety monitoring clock cycle is tolerated.

Rounding effects can occur in the last decimal place of the parameterized time.

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

SS1: Safe Stop 1

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p9858

SERVO_DBSI_VD

SI transition time STOP F to STOP A (Motor Module) / SI STOP F->A MM

Can be changed: C2(95)

Calculated: -

Access level: 3

Data type: FloatingPoint32

Dyn. index: -

Function diagram: 2802

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

0.00 [μs]

30000000.00 [μs]

0.00 [μs]

Description:

Sets the transition period from STOP F to STOP A on the Motor Module/Hydraulic Module.

Dependency:

See also: p9658, r9895

See also: F30611

NOTICE

This parameter is overwritten by the copy function of the safety functions integrated in the drive.

Note

For a data cross-check between p9658 and p9858, a difference of one safety monitoring clock cycle is tolerated.

Rounding effects can occur in the last decimal place of the parameterized time.

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

If a higher-level control responds to a drive fault by withdrawing the controller enable signals, for faults F01611 and F30611, the message type must be changed to alarm (p2118, p2119). As a consequence, the drive can still be braked in a controlled fashion during this delay time.

STOP F: Defect in a monitoring channel (error in the data cross-check)

STOP A: STO as a result of a fault detected by Safety Integrated

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r9872.0...24

SERVO_DBSI_VD

CO/BO: SI status list (Motor Module) / SI status MM

Can be changed: -

Calculated: -

Access level: 2

Data type: Unsigned32

Dyn. index: -

Function diagram: 2804

P group: Safety Integrated

Unit group: -

Unit selection: -

Not for motor type: -

Scaling: -

Expert list: 1

Min:

Max:

Factory setting:

-

-

-

Description:

Display and BICO output for the Safety Integrated status on the Motor Module/Hydraulic Module.

Bit array:

Bit **Signal name**

1 signal

0 signal

FD

00 STO selected on MM

Yes

No

2810

01	STO on MM act	Yes	No	2810
02	SS1 delay time active on MM	Yes	No	2810
03	Safe Brake Adapter feedback signal	Yes	No	2814
04	SBC requested	Yes	No	2814
05	SS1 selected on MM (Basic Functions)	Yes	No	-
06	SS1 active on MM (Basic Functions)	Yes	No	-
07	STO terminal state on MM (Basic Functions)	High	Low	-
09	STOP A cannot be acknowledged active	Yes	No	2802
10	STOP A active	Yes	No	2802
15	STOP F active	Yes	No	2802
16	STO cause: Safety comm. mode	Yes	No	-
17	STO cause selection via terminal (Basic Functions)	Yes	No	-
18	STO cause: selection via SMM	Yes	No	-
20	STO cause selection PROFIsafe/TM54F (Basic Functions)	Yes	No	-
21	STO cause selection on other monitoring channel	Yes	No	-
22	SS1 cause selection terminal (Basic Functions)	Yes	No	-
23	SS1 cause selection PROFIsafe/TM54F (Basic Functions)	Yes	No	-
24	Slave MM ready for communication	Yes	No	-

Dependency:

See also: r9772

NOTICE

If communication between both monitoring channels is interrupted (e.g. by switching off the power unit), this display parameter is no longer updated. The last transferred status of the Motor Module/Hydraulic Module is displayed.

Note

For bit 00:

When STO is selected, the cause is displayed in bits 16 ... 21.

For bit 05:

When SS1 is selected, the cause is displayed in bits 22 and 23.

For bit 18:

When the bit is set, STO is selected via PROFIsafe or Terminal Module 54F (TM54F).

SMM: Safe Motion Monitoring

For bit 22, 23:

These bits show via which path the SS1 was triggered, i.e. what has started the SS1 delay time.

If the SS1 delay time is not started (e.g. because an STO is triggered at the same time), neither of the two bits is set.

For bit 24:

Only for a parallel connection and active motion monitoring functions.

r9881[0...11]

SERVO_DBSI_VD

SI Motion Sensor Module Node Identifier second channel / SI Mtn SM Ident**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** Unsigned8**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

-

Description:

Displays the Node Identifier of the Sensor Module that the second channel uses for the motion monitoring functions.

r9895	SI diagnostics STOP F (Motor Module) / SI diag STOP F MM			
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 2	
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2802	
	P group: Safety Integrated	Unit group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min:	Max:	Factory setting:	
	-	-	-	
Description:	Displays the number of the cross-checked data item which caused STOP F on the Motor Module/Hydraulic Module.			
Dependency:	See also: r9795 See also: F30611			
Note				
The complete list of numbers for data cross-check is listed in Fault F30611.				
r9898	SI actual checksum SI parameters (Motor Module) / SI act_checksum MM			
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3	
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2800	
	P group: Safety Integrated	Unit group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min:	Max:	Factory setting:	
	-	-	-	
Description:	Displays the checksum for the checked Safety Integrated parameters on the Motor Module/Hydraulic Module (actual checksum).			
Dependency:	See also: r9798, p9899			
p9899	SI reference checksum SI parameters (Motor Module) / SI set_checksum MM			
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3	
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2800	
	P group: Safety Integrated	Unit group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min:	Max:	Factory setting:	
	0000 hex	FFFF FFFF hex	0000 hex	
Description:	Sets the checksum for the checked Safety Integrated parameters on the Motor Module/Hydraulic Module (reference checksum).			
Dependency:	See also: p9799, r9898			
Note				
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.				
p10201	SI Motion SBT enable / SBT enable			
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3	
	Data type: Unsigned32	Dyn. index: -	Function diagram: -	
	P group: Safety Integrated	Unit group: -	Unit selection: -	
	Not for motor type: -	Scaling: -	Expert list: 1	
	Min:	Max:	Factory setting:	
	-	-	0000 bin	
Description:	Sets the enable for the safe brake test.			
Bit array:	Bit	Signal name	1 signal	0 signal
	00	Enable safe brake test	Yes	No
				FD
				-

Note

SBT: Safe Brake Test

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10202[0...1]

SERVO_DBSI_VD

SI Motion SBT brake selection / SBT brake sel**Can be changed:** C2(95)**Data type:** Integer16**P group:** Safety Integrated**Not for motor type:** -**Min:**

0

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** -**Max:**

2

Access level: 3**Function diagram:** -**Unit selection:** -**Expert list:** 1**Factory setting:**

0

Description:

Selects the brakes to be tested.

Value:

0: Inhibit

1: Test motor holding brake

2: Test external brake

Index:

[0] = Brake 1

[1] = Brake 2

Dependency:

See also: p10203, p10235

See also: A01785

Note

It is not possible to test two motor holding brakes. An appropriate message is output for an incorrect parameterization. The brake to be tested is selected using p10230[2] or p10235.2.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10203

SERVO_DBSI_VD

SI Motion SBT control selection / SBT control sel**Can be changed:** C2(95)**Data type:** Integer16**P group:** Safety Integrated**Not for motor type:** -**Min:**

0

Calculated: -**Dyn. index:** -**Unit group:** -**Scaling:** -**Max:**

2

Access level: 3**Function diagram:** 2837**Unit selection:** -**Expert list:** 1**Factory setting:**

0

Description:

Selects the control for the safe brake test.

Value:

0: SBT via SCC (p10235)

1: SBT via BICO (p10230)

2: SBT for test stop selection (p9705/p10250.8)

Dependency:

See also: p9705, p10235, p10250

Note

SCC: Safety Control Channel

For a value = 2, the following applies:

Brake 1 with sequence 1 (p10210[0], p10211[0], p10212[0], p10218) is tested. Brake 1 must be configured as motor holding brake (p10202[0] = 1).

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10204 SERVO_DBSI_VD	SI Motion SBT motor type / SBT motor type		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0	Max: 1	Factory setting: 0
Description:	Selects the motor type for the safe brake test.		
Value:	0: Rotating 1: Linear		
Dependency:	See also: F01787		
Note For safety functions that are not enabled (p9501 = 0), the following applies: - p10204 is automatically set the same as r0108.12 when the system boots. When the safe brake test is enabled (10201.0 = 1), the following applies: - p10204 is checked when the system boots to see that it matches r0108.12. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p10208[0...1] SERVO_DBSI_VD	SI Motion SBT test torque ramp time / SBT M_test t_ramp		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 20 [ms]	Max: 10000 [ms]	Factory setting: 1000 [ms]
Description:	Sets the time, during which the test torque is ramped up against the closed brake. The test torque is then ramped down after the safe brake test.		
Index:	[0] = Brake 1 [1] = Brake 2		
Note The set time is rounded internally to an integer multiple of the monitoring clock cycle. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p10208[0...1] SERVO_DBSI_VD (Lin)	SI Motion SBT test force ramp time / SBT F_test t_ramp		
	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 20 [ms]	Max: 10000 [ms]	Factory setting: 1000 [ms]
Description:	Sets the time, during which the test force is ramped up against the closed brake. The test force is then ramped down after the safe brake test.		
Index:	[0] = Brake 1 [1] = Brake 2		
Note The set time is rounded internally to an integer multiple of the monitoring clock cycle. This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p10209[0...1]	SI Motion SBT brake holding torque / SBT brake M_stop		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.05 [Nm]	Max: 60000.00 [Nm]	Factory setting: 10.00 [Nm]
Description:	Sets the effective holding torque on the motor side of the brake to be tested.		
Index:	[0] = Brake 1 [1] = Brake 2		
Dependency:	The holding torque of an external brake should be converted to the motor side. Conversion factor: - motor type = rotary and axis type = linear: p9522 / (p9521 x p9520) - otherwise: p9522 / p9521 Further, the efficiency of the mechanical system should be taken into account. See also: p10210, p10220		
Note The test torque effective for the brake test can be set for each sequence using a factor (p10210, p10220). This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.			

p10209[0...1]	SI Motion SBT brake holding force / SBT brake F_stop		
SERVO_DBSI_VD (Lin)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 1.00 [N]	Max: 100000.00 [N]	Factory setting: 10.00 [N]
Description:	Sets the holding force of the brake to be tested.		
Index:	[0] = Brake 1 [1] = Brake 2		
Dependency:	See also: p10210, p10220		
Note The effective test force can be set for each sequence using a factor (p10210, p10220).			

p10210[0...1]	SI Motion SBT test torque factor sequence 1 / SBT M_test fact 1		
SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.30	Max: 1.00	Factory setting: 1.00
Description:	Sets the factor for the test torque of sequence 1 for the safe brake test. The factor is referred to the holding torque of the brake (p10209).		
Index:	[0] = Brake 1 [1] = Brake 2		
Dependency:	See also: p10209, p10235		

Note

The test sequence is selected using p10230[4] or p10235.4.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10210[0...1] SI Motion SBT test force factor sequence 1 / SBT F_test fact 1

SERVO_DBSI_VD (Lin)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.30	Max: 1.00	Factory setting: 1.00
Description:	Sets the factor for the test force of sequence 1 for the safe brake test. The factor is referred to the holding force of the brake (p10209).		
Index:	[0] = Brake 1 [1] = Brake 2		
Dependency:	See also: p10209, p10235		

Note

The test sequence is selected using p10230[4] or p10235.4.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10211[0...1] SI Motion SBT test duration sequence 1 / SBT t_test seq 1

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 20 [ms]	Max: 10000 [ms]	Factory setting: 1000 [ms]
Description:	Sets the test duration for sequence 1 for the safe brake test. The test torque is available for this time at the closed brake.		
Index:	[0] = Brake 1 [1] = Brake 2		
Dependency:	See also: p10235		

Note

The test sequence is selected using p10230[4] or p10235.4.

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10211[0...1] SI Motion SBT test duration sequence 1 / SBT t_test seq 1

SERVO_DBSI_VD (Lin)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 20 [ms]	Max: 10000 [ms]	Factory setting: 1000 [ms]
Description:	Sets the test duration for sequence 1 for the safe brake test. The test force is available for this time at the closed brake.		

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10235

Note

The test sequence is selected using p10230[4] or p10235.4.

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10212[0...1] SI Motion SBT position tolerance sequence 1 / SBT pos_tol seq 1

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.001 [mm]	360.000 [mm]	1.000 [mm]

Description: Sets the tolerated position deviation for sequence 1 for the safe brake test.

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10235

Note

The test sequence is selected using p10230[4] or p10235.4.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10212[0...1] SI Motion SBT position tolerance sequence 1 / SBT pos_tol seq 1

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
(Safety rot)	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.001 [°]	360.000 [°]	1.000 [°]

Description: Sets the tolerated position deviation for sequence 1 for the safe brake test.

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10235

Note

The test sequence is selected using p10230[4] or p10235.4.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10220[0...1] SI Motion SBT test torque factor sequence 2 / SBT M_test fact 2

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.30	1.00	1.00

Description: Sets the factor for the test torque of sequence 2 for the safe brake test.

The factor is referred to the holding torque of the brake (p10209).

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10209, p10235

Note

The test sequence is selected using p10230[4] or p10235.4.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10220[0...1] SI Motion SBT test force factor sequence 2 / SBT F_test fact 2

SERVO_DBSI_VD (Lin)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 0.30	Max: 1.00	Factory setting: 1.00

Description: Sets the factor for the test force of sequence 2 for the safe brake test.
The factor is referred to the holding force of the brake (p10209).

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10209, p10235

Note

The test sequence is selected using p10230[4] or p10235.4.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10221[0...1] SI Motion SBT test duration sequence 2 / SBT t_test seq 2

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 20 [ms]	Max: 10000 [ms]	Factory setting: 1000 [ms]

Description: Sets the test duration for sequence 2 for the safe brake test.
The test torque is available for this time at the closed brake.

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10235

Note

The test sequence is selected using p10230[4] or p10235.4.

The set time is rounded internally to an integer multiple of the monitoring clock cycle.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10221[0...1] SI Motion SBT test duration sequence 2 / SBT t_test seq 2

SERVO_DBSI_VD (Lin)	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: 20 [ms]	Max: 10000 [ms]	Factory setting: 1000 [ms]

Description: Sets the test duration for sequence 2 for the safe brake test.
The test force is available for this time at the closed brake.

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10235

Note

The test sequence is selected using p10230[4] or p10235.4.
The set time is rounded internally to an integer multiple of the monitoring clock cycle.
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10222[0...1] SI Motion SBT position tolerance sequence 2 / SBT pos_tol seq 2

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.001 [mm]	360.000 [mm]	1.000 [mm]

Description: Sets the tolerated position deviation for sequence 2 for the safe brake test.

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10235

Note

The test sequence is selected using p10230[4] or p10235.4.
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

p10222[0...1] SI Motion SBT position tolerance sequence 2 / SBT pos_tol seq 2

SERVO_DBSI_VD	Can be changed: C2(95)	Calculated: -	Access level: 3
(Safety rot)	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0.001 [°]	360.000 [°]	1.000 [°]

Description: Sets the tolerated position deviation for sequence 2 for the safe brake test.

Index: [0] = Brake 1
[1] = Brake 2

Dependency: See also: p10235

Note

The test sequence is selected using p10230[4] or p10235.4.
This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r10231 SI Motion SBT control word diagnostics / SBT STW diag

SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2836, 2837
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-

Description: Displays the diagnostic bits for the control word of the safe brake test

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Select brake test	Yes	No	-
	01	Start brake test	Yes	No	-
	02	Select brake	Brake 2	Brake 1	-
	03	Select test torque sign	Negative	Positive	-
	04	Select test sequence	Test sequence 2	Test sequence 1	-
	05	External brake status	Closed	Open	-

Dependency: See also: p10203

Note

The bits indicate the actual control signals of the control set in p10203.

r10231 SI Motion SBT control word diagnostics / SBT STW diag

SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2836, 2837
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-

Description: Displays the diagnostic bits for the control word of the safe brake test

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Select brake test	Yes	No	-
	01	Start brake test	Yes	No	-
	02	Select brake	Brake 2	Brake 1	-
	03	Select test force sign	Negative	Positive	-
	04	Select test sequence	Test sequence 2	Test sequence 1	-
	05	External brake status	Closed	Open	-

Dependency: See also: p10203

Note

The bits indicate the actual control signals of the control set in p10203.

r10234.0...15 CO/BO: SI Safety Information Channel status word S_ZSW3B / SIC S_ZSW3B

SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Unsigned32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	-	-	-

Description: Display and BICO output for status word S_ZSW3B of the Safety Information Channel.

Bit array:	Bit	Signal name	1 signal	0 signal	FD
	00	Brake test selected	Yes	No	-
	01	Setpoint input drive/external	Drive	External	-
	02	Active brake	Brake 2	Brake 1	-
	03	Brake test active	Yes	No	-
	04	Brake test result	Successful	Erroneous/not	-
	05	Brake test completed	Yes	No	-
	06	External brake request	Close	Open	-
	07	Actual load sign	Negative	Positive	-

11	SS2E active	Yes	No	-
12	SS2ESR active	Yes	No	-
14	Acceptance test SLP (SE) active	Yes	No	-
15	Acceptance test mode selected	Yes	No	-

Note

SIC: Safety Information Channel

SLP: Safely-Limited Position / SE: Safe software limit switches

SS2E: Safe Stop 2 External (Safe Stop 2 with external stop, external STOP D)

SS2ESR: Safe Stop 2 Extended Stop and Retract

For bits 05, 04:

For r10234.4 = 0 signal, it is possible to make a distinction as to whether the brake test was executed with error - or has still not been executed - using bit 5.

Bit 5/4 = 0/0: The brake test has still not been executed since the last warm restart or POWER ON.

Bit 5/4 = 1/0: The last brake test that was executed had an error.

p10235

SERVO_DBSI_VD

CI: SI Safety Control Channel control word S_STW3B / SCC S_STW3B**Can be changed:** T**Calculated:** -**Access level:** 3**Data type:** Unsigned32 / Integer16**Dyn. index:** -**Function diagram:** 2837**P group:** -**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

-

-

0

Description:

Sets the signal source for control word S_STW3B of the Safety Control Channel.

Dependency:

This parameter is used as control word for the safe brake test only for "SBT via SCC" (p10203 = 0).

See also: p10203

Note

SBT: Safe Brake Test

SCC: Safety Control Channel

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

r10240

SERVO_DBSI_VD

SI Motion SBT test torque diagnostics / SBT M_test diag**Can be changed:** -**Calculated:** -**Access level:** 3**Data type:** FloatingPoint32**Dyn. index:** -**Function diagram:** 2836**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

- [Nm]

- [Nm]

- [Nm]

Description:

Displays the effective maximum test torque on the motor side for a safe brake test.

Dependency:

The test torque for an external brake should be converted to the load side.

Conversion factor:

- motor type = rotary and axis type = linear: (p9521 x p9520) / p9522

- otherwise: p9521 / p9522

Further, the efficiency of the mechanical system should be taken into account.

See also: p10210, p10220

Note

The value remains displayed until the start of the next test sequence.

r10240	SI Motion SBT test force diagnostics / SBT F_test diag		
SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [N]	- [N]	- [N]
Description:	Displays the maximum test force for a safe brake test.		
Dependency:	See also: p10210, p10220		

Note

The value remains displayed until the start of the next test sequence.

r10241	SI Motion SBT load torque diagnostics / SBT M_load diag		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [Nm]	- [Nm]	- [Nm]
Description:	Displays the load torque for a safe brake test. When initializing the brake test, this load torque is available at the drive.		

Note

The value remains displayed until the brake test is deselected.

r10241	SI Motion SBT load force diagnostics / SBT F_load diag		
SERVO_DBSI_VD (Lin)	Can be changed: -	Calculated: -	Access level: 3
	Data type: FloatingPoint32	Dyn. index: -	Function diagram: 2836
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	- [N]	- [N]	- [N]
Description:	Displays the load force for a safe brake test. When initializing the brake test, this load force is available at the drive.		

Note

The value remains displayed until the brake test is deselected.

r10242	SI Motion SBT state diagnostics / SBT state diag		
SERVO_DBSI_VD	Can be changed: -	Calculated: -	Access level: 3
	Data type: Integer16	Dyn. index: -	Function diagram: -
	P group: Safety Integrated	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min:	Max:	Factory setting:
	0	16	-
Description:	Displays the actual state of the safe brake test.		
Value:	0: Brake test inactive, wait for SBT selection		
	1: Setpoint input drive		
	2: Determining the load		
	3: Brake test is initialized, wait for start of test sequence		

- 4: Start test sequence
- 5: Closing the brake, establishing the test torque
- 6: Brake test active, wait for test duration sequence
- 7: Reduce test torque
- 8: Wait for the brake to open
- 9: Brake test successfully completed, wait for start deselection
- 10: Change to brake test initialized - fault acknowledgment
- 11: Brake test canceled, torque is reduced
- 12: Brake test canceled, wait for brake to open
- 13: Brake test ended with error, wait for acknowledgment
- 14: Brake opening timer elapsed
- 15: Error when initializing the brake test, wait for acknowledgment
- 16: Change to brake test inactive, acknowledgment active

p10250	CI: SI Safety Control Channel control word S_STW1B / SCC S_STW1B		
SERVO_DBSI_VD	Can be changed: T	Calculated: -	Access level: 3
	Data type: Unsigned32 / Integer16	Dyn. index: -	Function diagram: -
	P group: -	Unit group: -	Unit selection: -
	Not for motor type: -	Scaling: -	Expert list: 1
	Min: -	Max: -	Factory setting: 0
Description:	Sets the signal source for control word S_STW1B of the Safety Control Channel.		
Dependency:	See also: p10203, r10251		
	Note SCC: Safety Control Channel This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.		

r10251.8...13	CO/BO: SI Safety Control Channel control word S_STW1B diagnostics / SCC S_STW1B diag				
SERVO_DBSI_VD	Can be changed: -		Calculated: -		Access level: 3
	Data type: Unsigned32		Dyn. index: -		Function diagram: -
	P group: -		Unit group: -		Unit selection: -
	Not for motor type: -		Scaling: -		Expert list: 1
	Min:		Max:		Factory setting:
	-		-		-
Description:	Display and BICO output for the diagnostics of control word S_STW1B of the Safety Control Channel.				
Bit array:	Bit	Signal name	1 signal	0 signal	FD
	08	Extended Functions test stop selection	Selected	Not selected	2837
	09	Extended Functions referencing trigger	Selected	Not selected	-
	10	Extended Functions referencing reset	Selected	Not selected	-
	12	Extended Functions premature SOS after STOP D	Selected	Not selected	-
	13	Close brake from control	Selected	Not selected	-
Dependency:	See also: p10250				
Note					
SCC: Safety Control Channel					
For bit 13:					
The following BICO interconnection is required for brake control via SCC:					
BI: p0858 = r10251.13					

p60022 PROFIsafe telegram selection / Ps tel_sel

SERVO_DBSI_VD

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** -**P group:** Safety Integrated**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

0

903

0

Description:

Sets the telegram number for PROFIsafe.

Value:

0: No PROFIsafe telegram selected

902: PROFIsafe SIEMENS telegram 902, PZD-3/6

903: PROFIsafe SIEMENS telegram 903, PZD-3/5

Dependency:

See also: p9611, p9811

Note

For p9601.3 = p9801.3 = 1 (PROFIsafe enabled), the following variants exist when parameterizing PROFIsafe telegram 30:

- p9611 = p9811 = 998 and p60022 = 0

- p9611 = p9811 = 998 and p60022 = 30

- p9611 = p9811 = 30 and p60022 = 30

p60122 IF1 PROFIdrive SIC/SCC telegram selection / IF1 SIC/SCC tel

SERVO_DBSI_VD

Can be changed: T**Calculated:** -**Access level:** 3**Data type:** Unsigned16**Dyn. index:** -**Function diagram:** 2423**P group:** Communications**Unit group:** -**Unit selection:** -**Not for motor type:** -**Scaling:** -**Expert list:** 1**Min:****Max:****Factory setting:**

701

999

999

Description:

Sets the telegram for the Safety Information Channel (SIC) / Safety Control Channel (SCC).

The SIC/SCC telegram p60122 is attached directly to the PZD telegram p0922/p2079.

Value:

701: Supplementary telegram 701, PZD-2/5

999: No telegram

Dependency:

For p8864 equal to 999, then p60122 is locked.

Note

The clearance to the PZD telegram can be increased using p2070/p2071.

After changing p0922/p2079 or p2070/p2071, then p60122 must be set again.

The telegram interconnections can only be changed if p60122 and p0922 are both set to 999.

This parameter cannot be changed in the Create MyVirtualMachine and Run MyVirtualMachine commissioning screen forms. This guarantees the functionality of the simplified drive model.

Explanation of the alarms

SINAMICS alarms

The faults and states detected by the individual components of the drive system are indicated by alarms. These SINAMICS alarms are categorized into faults and alarms.

The differences between faults and alarms are as follows:

Fault	What happens when a fault occurs? • The appropriate fault reaction is initiated. • Status signal ZSW1.3 is set. • The fault is entered in the fault buffer. How are faults eliminated? • Removing the cause of the fault • Acknowledging the fault
Alarm	What happens when an alarm occurs? • Status signal ZSW1.7 is set. • The alarm is entered into the alarm buffer. How are alarms eliminated? • Alarms acknowledge themselves. When the cause has been removed, the warnings reset themselves automatically.

Structure of SINAMICS alarm descriptions

The descriptions of the SINAMICS alarms have the following layout:

<Alarm No.> <Location data> <Alarm text>

Message value:

Drive object:

Reaction:

Acknowledgment:

Cause:

Help:

Each alarm is uniquely identified using the <Alarm no.> and the <Alarm text>.

The <Location data> is optional display information. Location data can include:

- Axis name and drive number
- OR -
- Bus and slave number of the PROFINET or PROFIBUS DP component affected

The place holder <Location data> is kept in the alarm description for this optional information.

The description of the SINAMICS alarms is classified according to the following categories:

- **Message value**
The information provided under the message value tells you about the composition of the fault/alarm value.
Example:
Message value: Component number: %1, cause: %2
This fault value or alarm value contains information about the component number and cause. The entries %1 and %2 are place holders, which are filled appropriately in online operation with the commissioning software.
- **Drive object**
For each alarm (fault/alarm) it is specified in which drive object this message can be found. A message can belong to either one, several or all drive objects.
- **Response**
Specifies the default reaction in the event of a fault.
- **Acknowledgment**
Specifies the default method of acknowledging the fault/alarm after the cause has been removed.
- **Cause**
Describes the possible causes. For the cause of the alarm, the fault/alarm value is essentially prepared in text form.
- **Remedy**
Describes the possible remedies.

A detailed description of the fault responses and acknowledgment can be found in Section: System reactions for SINAMICS alarms (Page 285)

References

Further information regarding the SINAMICS messages can be obtained in the following manual: SINAMICS S120/S150 List Manual.

Syntax:

The notation of the messages can be derived as follows:

Messages initiated from the SINAMICS drive are always displayed in SINUMERIK Operate with a "2" followed by a 5-digit number.

In the SINAMICS S120/S150 List Manual, the messages begin with a letter, followed by a 5-digit number:

- "F" (Fault)
- "A" (Alarm)
- "N" (No Report)
- "C" (Safety message)

If you require further information on the alarms, you can find a description in the SINAMICS List Manual by replacing the first number "2" of the message number with the letter "F" or "A".

Example:

You can find the description of alarm **207841** in the SINAMICS S120/S150 List Manual under fault **F07841**.

You can find the description of alarm **201797** in the SINAMICS S120/S150 List Manual under safety_message **C01797**.

Synchronization of time and date

Note**Synchronization of time**

The SINAMICS drives have no real-time clock. The time of day and date of the SINAMICS clock are synchronized in a 10-second cycle with the SINUMERIK real-time clock.

The result is that after a change of the date and/or the clock of the SINUMERIK real-time clock up to 10 seconds expire until these changes have been synchronized with the SINAMICS drives.

If in this up period of up to 10 seconds SINAMICS messages occur (number range 200000 - 299999), this SINAMICS messages still receive the outdated date/time stamp. The SINUMERIK alarms (alarm numbers < 200000 and > 300000) triggered as a result of the SINAMICS messages, on the other hand, already receive the new date/ time stamp.

See also

System reactions for SINAMICS alarms (Page 285)

System reactions for SINAMICS alarms

Fault reactions

The fault reactions used and their meanings are described below:

Designation	NONE
Response	None
Description	No reaction when a fault occurs.

Designation	OFF1
Response	Brake along the ramp-function generator deceleration ramp followed by pulse disable.
Description	Closed loop speed control (p1300 = 20, 21) - The immediate specification of $n_{set} = 0$ at the ramp generator deceleration ramp (p1121) causes the drive to be braked. - When zero speed is detected, the motor holding brake (if parameterized) is closed (p1215). The pulses are suppressed when the brake closing time (p1217) expires. The standstill is detected when the speed actual value falls below the speed threshold (p1226) or when the monitoring time (p1227) started when speed setpoint $\leq$ speed threshold (p1226) has expired. Closed loop torque control (p1300 = 23) - The following applies to closed-loop torque control: Reaction as for OFF2. - When the system switches to torque control with p1501, the following applies: There is no separate braking reaction. If the actual speed value falls below the speed threshold (p1226) or the timer stage (p1227) has expired, the motor holding brake (if parameterized) is closed. The pulses are suppressed when the brake closing time (p1217) expires.

Designation	OFF1_DELAYED
Response	As for OFF1, however delayed.
Description	Faults with this fault reaction only become effective after the delay time on p3136 has expired. The remaining time up to OFF1 is displayed in r3137.

Designation	OFF2
Response	Internal/external pulse disable
Description	Closed-loop speed and torque control - Instantaneous pulse suppression, the drive "coasts" to a standstill. - The motor holding brake (if parameterized) is closed immediately. - Switch-on inhibit is activated.

Designation	OFF3
Response	Brake along the OFF3 deceleration ramp followed by pulse disable
Description	Closed loop speed control (p1300 = 20, 21) - n_set = 0 is input immediately to brake the drive along the OFF3 deceleration ramp (p1135). - When zero speed is detected, the motor holding brake (if parameterized) is closed. The pulses are suppressed when the closing time of the holding brake (p1217) expires. Standstill is detected when the speed actual value falls below the speed threshold (p1226) or when the monitoring time (p1227) started when speed setpoint ≤ speed threshold (p1226) has expired. - Switch-on inhibit is activated. Closed loop torque control (p1300 = 23) - Switchover to speed-controlled operation and other reactions as described for speed-controlled operation.

Designation	STOP2
Response	n_set = 0
Description	- n_set = 0 is input immediately to brake the drive along the OFF3 deceleration ramp (p1135). - The drive remains in closed-loop speed control.

Designation	IASC/DCBRK
Response	-
Description	- For synchronous motors, the following applies: If a fault occurs with this fault reaction, an internal armature short-circuit is triggered. The conditions for p1231 = 4 must be observed. - For induction motors, the following applies: If a fault occurs with this fault reaction, DC braking is triggered. DC braking must have been commissioned (p1232, p1233, p1234).

Designation	ENCODER
Response	Internal/external pulse disable (p0491)
Description	The fault reaction ENCODER is applied as a function of the setting in p0491. Factory setting: p0491 = 0 → Encoder fault causes OFF2

Acknowledging faults

Specifies the default method of acknowledging faults after the cause has been removed.

Designation	POWER ON
Description	The fault is acknowledged by a POWER ON process (switch drive unit off and on again). Note: If the fault cause has still not been resolved, then the fault is immediately displayed again after booting.

Designation	IMMEDIATELY
Description	Faults can be acknowledged on one drive object or on all drive objects as follows: - Acknowledge by setting parameter: - p3981 = 0 → 1 - Acknowledge via binector inputs: - p2103 BI: 1. Acknowledge faults - p2104 BI: 2. Acknowledge faults - p2105 BI: 3. Acknowledge faults - Acknowledge all faults - p2102 BI: Acknowledge all faults Note - These faults can also be acknowledged by a POWER ON. - If the cause of the fault has not been removed the fault is not cleared after acknowledgement. - Safety Integrated faults The "Safe Torque Off" (STO) function must be deselected before these faults are acknowledged.

Designation	PULSE DISABLE
Description	The fault can only be acknowledged with a pulse disable (r0899.11 = 0). The same options are available for acknowledging as described under acknowledgment with IMMEDIATE.

Reference to SINAMICS parameters

In some fields, reference is made to a SINAMICS parameter for the alarms.

The parameter number consists the letter "p" or "r" as prefix, followed by a 4-digit number (xxxx), and optionally an index, e.g. p0918[0...3].

References

A detailed description of the SINAMICS parameters is provided in the:
SINAMICS S120/S150 List Manual

simDrive alarms

'All objects: A_INF_840_VD, CU_I_840_VD, CU_NX_840_VD, SERVO_DBSI_VD

F01681	SI Motion P1 (CU): Incorrect parameter value
Drive object:	SERVO_DBSI_VD
Message class:	Error in the parameterization / configuration / commissioning procedure (18)
Message value:	Parameter: %1, supplementary information: %2
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The parameter cannot be parameterized with this value. Note: This message does not result in a safety stop response. Fault value (r0949, interpret decimal): yyyyxxx dec: yyyy = supplementary information, xxxx = parameter yyyy = 0: No additional information available. xxxx = 9578: SLA is enabled (p9501.20 = 1). Acceleration limit is too low (p9578). The acceleration resolution is no longer sufficient (r9790). The minimum limit is x-times the acceleration resolution: - firmware version 5.2 and higher: $10 * r9790[1]$
Remedy:	Correct parameter (if required, also on another monitoring channel, p9801). For xxxx = 9578: - observe the information in r9790. Note: SCA: Safe Cam SCSE: Single Channel Safety Encoder (single-channel encoder) SS2E: Safe Stop 2 External (Safe Stop 2 with external stop, external STOP D) SS2ESR: Safe Stop 2 Extended Stop and Retract SLA: Safely-Limited Acceleration

A01697	SI Motion: Test stop for motion monitoring functions required
Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	-
Reaction:	NONE
Acknowledge:	NONE

Cause:	The time set in p9559 for the forced checking procedure (test stop) for the safe motion monitoring functions has been exceeded. A new forced checking procedure is required. After the next time the forced checking procedure is selected, the message is withdrawn and the monitoring time is reset. Note: - this message does not result in a safety stop response. - As the switch-off signal paths are not automatically checked during booting, an alarm is always issued once booting is complete. - the test must be performed within a defined, maximum time interval (p9559, maximum of 9000 hours) in order to comply with the requirements as laid down in the standards for timely fault detection and the conditions to calculate the failure rates of safety functions (PFH value). Operation beyond this maximum time period is permissible if it can be ensured that the forced checking procedure is performed before persons enter the hazardous area and who are depending on the safety functions correctly functioning. See also: p9559 (SI Motion forced checking procedure timer (Control Unit)), r9765 (SI Motion forced check procedure remaining time (Control Unit))
Remedy:	Carry out the forced checking procedure of the safety motion monitoring functions. The signal source to select the forced checking procedure is set via binector input p9705. Notice: It is not permissible to use TM54F inputs to select the forced checking procedure. Note: SI: Safety Integrated See also: p9705 (SI Motion test stop signal source)

A01699 SI P1 (CU): Test stop for STO required

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	-
Reaction:	NONE
Acknowledge:	NONE
Cause:	The time set in p9659 for the forced checking procedure (test stop) for the "STO" function has been exceeded. A new forced checking procedure is required. After the next time the "STO" function is deselected, the message is withdrawn and the monitoring time is reset. Note: - this message does not result in a safety stop response. - the test must be performed within a defined, maximum time interval (p9659) in order to comply with the requirements as laid down in the standards for timely fault detection and the conditions to calculate the failure rates of safety functions (PFH value). Operation beyond this maximum time period is permissible if it can be ensured that the forced checking procedure is performed before persons enter the hazardous area and who are depending on the safety functions correctly functioning. See also: p9659 (SI forced checking procedure timer), r9660 (SI forced checking procedure remaining time)
Remedy:	Select STO and then deselect again. Note: CU: Control Unit SI: Safety Integrated STO: Safe Torque Off / SH: Safe standstill

C01700 SI Motion P1 (CU): STOP A initiated

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	-
Reaction:	OFF2
Acknowledge:	IMMEDIATELY

Cause:	The drive is stopped via a STOP A (STO via the safety switch-off signal path of the Control Unit). Possible causes: - stop request from the second monitoring channel. - STO not active after parameterized time (p9557) after test stop selection. - subsequent response to the message C01706 "SI Motion CU: SAM/SBR limit exceeded". - subsequent response to the message C01714 "SI Motion CU: Safely-Limited Speed exceeded". - subsequent response to the message C01701 "SI Motion CU: STOP B initiated". - subsequent response to the message C01715 "SI Motion CU: Safely-Limited Position exceeded". - subsequent response to the message C01716 "SI Motion CU: tolerance for safe motion direction exceeded".
Remedy:	- remove the cause of the fault on the second monitoring channel. - carry out a diagnostics routine for message C01706. - carry out a diagnostics routine for message C01714. - carry out a diagnostics routine for message C01701. - carry out a diagnostics routine for message C01715. - carry out a diagnostics routine for message C01716. - check the value in p9557 (where available), increase the value if necessary, and carry out a POWER ON - check the switch-off signal path of the Control Unit (check DRIVE-CLiQ communication if it has been implemented) - replace the Motor Module, Power Module or Hydraulic Module. - replace Control Unit. This message can be acknowledged without a POWER ON as follows (safe acknowledgment): - Terminal Module 54F (TM54F). - onboard F-DI (only CU310-2). - PROFIsafe. - machine control panel. Note: SAM: Safe Acceleration Monitor (safe acceleration monitoring) SBR: Safe Brake Ramp (safe brake ramp monitoring) SI: Safety Integrated

C01701	SI Motion P1 (CU): STOP B initiated
Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	-
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The drive is stopped via a STOP B (braking along the OFF3 deceleration ramp). As a result of this fault, after the time parameterized in p9556 has expired, or the speed threshold parameterized in p9560 has been undershot, message C01700 "STOP A initiated" is output. Possible causes: - subsequent response to the message C01714 "SI Motion CU: Safely-Limited Speed exceeded". - subsequent response to the message C01711 "SI Motion CU: Defect in a monitoring channel". - subsequent response to the message C01707 "SI Motion CU: tolerance for safe operating stop exceeded". - subsequent response to the message C01715 "SI Motion CU: Safely-Limited Position exceeded". - subsequent response to the message C01716 "SI Motion CU: tolerance for safe motion direction exceeded".

Remedy:

- carry out a diagnostics routine for message C01714.
- carry out a diagnostics routine for message C01711.
- carry out a diagnostics routine for message C01707.
- carry out a diagnostics routine for message C01715.
- carry out a diagnostics routine for message C01716.

This message can be acknowledged without a POWER ON as follows (safe acknowledgment):

- Terminal Module 54F (TM54F).
- onboard F-DI (only CU310-2).
- PROFIsafe.
- machine control panel.

Note:
SI: Safety Integrated

C01707 SI Motion P1 (CU): Tolerance for safe operating stop exceeded

Drive object: SERVO_DBSI_VD

Message class: Safety monitoring channel has identified an error (10)

Message value: -

Reaction: NONE

Acknowledge: IMMEDIATELY

Cause: The actual position has distanced itself further from the target position than the standstill tolerance.
The drive is shut down by the message C01701 "SI Motion: STOP B initiated".

Remedy:

- check whether safety faults are present and if required carry out the appropriate diagnostic routines for the particular faults.
- check whether the standstill tolerance matches the accuracy and control dynamic performance of the axis.
- carry out a POWER ON.

This message can be acknowledged without a POWER ON as follows (safe acknowledgment):

- Terminal Module 54F (TM54F).
- onboard F-DI (only CU310-2).
- PROFIsafe.
- machine control panel.

Note:
SI: Safety Integrated
SOS: Safe Operating Stop / SBH: Safe operating stop
See also: p9530 (SI Motion standstill tolerance (Control Unit))

C01708 SI Motion P1 (CU): STOP C initiated

Drive object: SERVO_DBSI_VD

Message class: Safety monitoring channel has identified an error (10)

Message value: -

Reaction: STOP2

Acknowledge: IMMEDIATELY

Cause: The drive is stopped via a STOP C (braking along the OFF3 deceleration ramp).
"Safe Operating Stop" (SOS) is activated after the parameterized time has expired.
Possible causes:

- subsequent response to the message C01714 "SI Motion CU: Safely-Limited Speed exceeded".
- subsequent response to the message C01715 "SI Motion CU: Safely-Limited Position exceeded".
- subsequent response to the message C01716 "SI Motion CU: tolerance for safe motion direction exceeded".

See also: p9552 (SI Motion transition time STOP C to SOS (SBH) (Control Unit))

Remedy:

- remove the cause of the fault at the control.
- carry out a diagnostics routine for message C01714/C01715/C01716.

This message can be acknowledged without a POWER ON as follows (safe acknowledgment):

- Terminal Module 54F (TM54F).
- onboard F-DI (only CU310-2).
- PROFIsafe.
- machine control panel.

Note:

SI: Safety Integrated

SOS: Safe Operating Stop / SBH: Safe operating stop

C01709 SI Motion P1 (CU): STOP D initiated

Drive object: SERVO_DBSI_VD

Message class: Safety monitoring channel has identified an error (10)

Message value: -

Reaction: NONE

Acknowledge: IMMEDIATELY

Cause: The drive is stopped via a STOP D (braking along the path).
 "Safe Operating Stop" (SOS) is activated after the parameterized time has expired.
 Possible causes:

- subsequent response to the message C01714 "SI Motion CU: Safely-Limited Speed exceeded".
- subsequent response to the message C01715 "SI Motion CU: Safely-Limited Position exceeded".
- subsequent response to the message C01716 "SI Motion CU: tolerance for safe motion direction exceeded".

See also: p9553 (SI Motion transition time STOP D to SOS (SBH) (Control Unit))

Remedy:

- remove the cause of the fault at the control.
- carry out a diagnostics routine for message C01714/C01715/C01716.

This message can be acknowledged without a POWER ON as follows (safe acknowledgment):

- Terminal Module 54F (TM54F).
- onboard F-DI (only CU310-2).
- PROFIsafe.
- machine control panel.

Note:

SI: Safety Integrated

SOS: Safe Operating Stop / SBH: Safe operating stop

C01711 SI Motion P1 (CU): Defect in a monitoring channel

Drive object: SERVO_DBSI_VD

Message class: Safety monitoring channel has identified an error (10)

Message value: %1

Reaction: NONE

Acknowledge: IMMEDIATELY

Cause:	When cross-comparing the two monitoring channels, the drive detected a difference between the input data or results of the monitoring functions and initiated a STOP F. One of the monitoring functions no longer reliably functions - i.e. safe operation is no longer possible. If at least one monitoring function is active, then after the parameterized timer has expired, the message C01701 "SI Motion: STOP B initiated" is output. The message value that resulted in a STOP F is displayed in r9725. If the drive is operated together with a SINUMERIK, the message values are described in message 27001 of SINUMERIK, with the exception of the following message values, which can only occur in SINAMICS: The following described message values involve the data cross-check between the two monitoring channels (safety functions integrated in the drive). The message values may also occur in the following cases if the cause that is explicitly mentioned does not apply: - cycle times not uniformly parameterized (p9500/p9300 and p9511/p9311) - differently parameterized axis types (p9502/p9302). - excessively fast cycle times (p9500/p9300, p9511/p9311). - For message values 3, 44 ... 57, 232 and 1-encoder system, differently set encoder parameters. - For message values 3, 44 ... 57, 232 and 2-encoder system, encoder parameters that have not been correctly set. - incorrect synchronization. Message value (r9749, interpret decimal): 0 to 999: Number of the cross-compared data that resulted in this fault. Message values that are not subsequently listed are only for internal Siemens troubleshooting. 3: The position actual value differential (r9713[0/1]) between the two monitoring channels is greater than the tolerance in p9542. 1005: - for safe motion monitoring functions without encoder: pulses already suppressed for test stop selection. - for safe motion monitoring functions with encoder: STO already active for test stop selection. See also: p9555 (SI Motion transition time STOP F to STOP B (Control Unit))
Remedy:	The following generally applies: The monitoring clock cycles in both channels and the axis types should be checked for equality and the same setting applied if necessary. If the error continues to be identified, increasing the monitoring clock cycles may resolve it. For message value = 3: Commissioning phase: - check the encoder parameters, and if required, correct (p9516/p9316, p9517/p9317, p9518/p9318, p9520/p9320, p9521/p9321, p9522/p9322, p9526/p9326). In operation: - check the mechanical design and the encoder signals. - if closed-loop control with edge modulation is parameterized (p1802[x] = 9): parameterize edge modulation for actual value sensing without encoder (p9507.5 = p9307.5 = 1). For message value = 1005: - for safe motion monitoring functions without encoder: check the conditions for pulse enable. - for safe motion monitoring functions with encoder: check the conditions for STO deselection. Note: For a Power Module, the test stop should always be performed when the pulses are enabled (independent of whether with encoder or without encoder). See also: p9500 (SI Motion monitoring clock cycle (Control Unit))

C01714 SI Motion P1 (CU): Safely-Limited Speed exceeded

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	%1
Reaction:	NONE
Acknowledge:	IMMEDIATELY

Cause:	The drive has moved faster than that specified by the velocity limit value (p9531). The drive is stopped as a result of the configured stop response (p9563). Message value (r9749, interpret decimal): 100: SLS1 exceeded. 200: SLS2 exceeded. 300: SLS3 exceeded. 400: SLS4 exceeded.
Remedy:	- check the traversing/motion program in the control. - check limits for SLS and if required adapt accordingly (p9531). This message can be acknowledged without a POWER ON as follows (safe acknowledgment): - Terminal Module 54F (TM54F). - onboard F-DI (only CU310-2). - PROFIsafe. - machine control panel. Note: SI: Safety Integrated SLS: Safely-Limited Speed / SG: Safely reduced speed See also: p9531 (SI Motion SLS (SG) limit values (Control Unit)), p9563 (SI Motion SLS (SG)-specific stop response (Control Unit))

C01715 SI Motion P1 (CU): Safely-Limited Position exceeded

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	%1
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The axis has moved past a parameterized position that is monitored by the "SLP" function. Message value (r9749, interpret decimal): 10: SLP1 violated. 20: SLP2 violated.
Remedy:	- check the traversing/motion program in the control. - check the limits for "SLP" function and if required, adapt (p9534, p9535). This message can be acknowledged without a POWER ON as follows (safe acknowledgment): Prerequisite: - deselect "SLP" function and retract the axis into the permitted position range. Carry out a safe acknowledgment using one of the following options: - Terminal Module 54F (TM54F). - onboard F-DI (only CU310-2). - PROFIsafe. - machine control panel. Note: SI: Safety Integrated SLP: Safely-Limited Position / SE: Safe software limit switches See also: p9534 (SI Motion SLP (SE) upper limit values (Control Unit)), p9535 (SI Motion SLP (SE) lower limit values (Control Unit))

C01716 SI Motion P1 (CU): Tolerance for safe motion direction exceeded

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	%1
Reaction:	NONE

Acknowledge:	IMMEDIATELY
Cause:	The tolerance for the "safe motion direction" function was exceeded. The drive is stopped as a result of the configured stop response (p9566). Message value (r9749, interpret decimal): 0: Tolerance for the "safe motion direction positive" function exceeded. 1: Tolerance for the "safe motion direction negative" function exceeded.
Remedy:	- check the traversing/motion program in the control. - check the tolerance for "SDI" function and if required, adapt (p9564). This message can be acknowledged without a POWER ON as follows (safe acknowledgment): Prerequisite: - deselect the "SDI" function and if required select again. Carry out a safe acknowledgment using one of the following options: - Terminal Module 54F (TM54F). - onboard F-DI (only CU310-2). - PROFIsafe. - machine control panel. Note: SDI: Safe Direction (safe motion direction) SI: Safety Integrated See also: p9564 (SI Motion SDI tolerance (Control Unit)), p9565 (SI Motion SDI delay time (Control Unit)), p9566 (SI Motion SDI stop response (Control Unit))

C01717 SI Motion P1 (CU): SLA limit exceeded

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	%1
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The acceleration limit for the "Safely-Limited Acceleration" function was exceeded. The drive is stopped as a result of the configured stop response (p9579). Message value (r9749, interpret decimal): 1: The monitoring of the finely resolved acceleration and possibly filtered acceleration has violated the acceleration limit.
Remedy:	- check the traversing/motion program in the control. - check the acceleration limit for the "SLA" function and if required, adapt (p9578). - carry out a safe acknowledgment. For message value = 1: Analyze the causes using r9789[0], r9789[1] and r9789[2]. Note: SI: Safety Integrated SLA: Safely-Limited Acceleration See also: p9578 (SI Motion SLA acceleration limit (CU)), p9579 (SI Motion SLA stop response (Control Unit))

A01780 SBT When selected, the brake is closed

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	Following brakes are closed: %1 bin
Reaction:	NONE
Acknowledge:	NONE

Cause: When selecting the brake test or starting the brake test, not all of the brakes were open.
Alarm value (r2124, interpret binary):
Bit 0 = 1:
The internal brake is closed.
Bit 1 = 1:
The external brake is closed (p10230.5, p10235.5, p10202).
Note:
The alarm is also issued, if a brake has not been configured in p10202.
SBT: Safe Brake Test
See also: p10202 (SI Motion SBT brake selection), p10235 (SI Safety Control Channel control word S_STW3B)

Remedy: Open all brakes and reselect the brake test (p10230.0, p10235.0).

A01782 SBT brake test incorrect control

Drive object: SERVO_DBSI_VD
Message class: Safety monitoring channel has identified an error (10)
Message value: Fault cause: %1 bin
Reaction: NONE
Acknowledge: NONE

Cause: The brake test was canceled as a result of incorrect control.
Alarm value (r2124, interpret binary):
Bit 0:
The safe brake test was canceled by resetting the brake test selection.
Bit 1:
The safe brake test was canceled by resetting the brake test start.
Note:
SBT: Safe Brake Test
See also: p10202 (SI Motion SBT brake selection)

Remedy:

- check parameterization of the brake test (p10202).
- check as to whether alarm A01785 is present, and if so, evaluate.
- carry out a safe acknowledgment.
- if required, restart the brake test.

A01784 SBT brake test canceled with fault

Drive object: SERVO_DBSI_VD
Message class: Safety monitoring channel has identified an error (10)
Message value: Fault cause: %1
Reaction: NONE
Acknowledge: NONE

Cause:	The safe brake test was canceled as a result of a fault. Alarm value (r2124, interpret binary): Bit 17 = 1: fault in the brake test sequence (cause, see bits 0 ... 10). Bit 22 = 1: internal software error. Bit 24 = 1: the tested internal brake was opened while the brake test was active. Bit 25 = 1: the tested external brake was opened while the brake test was active. Cause for alarm value bit 17: Bit 0 = 1: operation when selecting the brake test not enabled (r0899.2 = 0). Bit 4 = 1: A fault has occurred with stop response (e.g. OFF1, OFF2 or OFF3) - or the pulse enable was withdrawn (e.g. STO selected or operation no longer enabled). Bit 5 = 1: when selecting the brake test the axis speed setpoint is too high. Bit 6 = 1: the actual speed (r0063) of the axis is too high (e.g. brake does not hold during the brake test). Note: SBT: Safe Brake Test
Remedy:	- remove the fault cause. - carry out a safe acknowledgment. - if required, restart the brake test. For bit 17 = 1 with bit 6 = 1 or bit 23 = 1: If the brake closing time of the motor holding brake (p1217) has been set too low, then at the start of the brake test, the brake is closed too late. The brake closing time should be adapted (p1217).

C01797 SI Motion P1 (CU): Axis not safely referenced

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	%1
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The standstill position saved before switching off does not match the actual position determined at switch-on. Message value (r9749, interpret decimal): 1: Axis not safely referenced. 2: User agreement missing.
Remedy:	If safe automatic referencing is not possible the user must issue a user agreement for the new position using the softkey. This mean that this position is then designated as safety-relevant. Note: SI: Safety Integrated

C01798 SI Motion P1 (CU): Test stop for motion monitoring functions running

Drive object:	SERVO_DBSI_VD
Message class:	Safety monitoring channel has identified an error (10)
Message value:	-
Reaction:	NONE
Acknowledge:	IMMEDIATELY
Cause:	The forced checking procedure (test stop) for the safe motion monitoring functions is currently in progress.
Remedy:	Not necessary. The message is automatically withdrawn when the test stop has been completed. Note: SI: Safety Integrated

F07841 Drive: Infeed operation withdrawn

Drive object:	SERVO_DBSI_VD
Message class:	Infeed faulted (13)

Message value:	-
Reaction:	OFF2
Acknowledge:	IMMEDIATELY
Cause:	The signal "infeed operation" was withdrawn in operation. - interconnection of the binector input for the signal "infeed operation" is either incorrect or missing (p0864). - the enable signals of the infeed were disabled. - due to a fault, the infeed withdraws the signal "infeed operation".
Remedy:	- check the interconnection of the binector input for the "infeed operation" signal (p0864). - check the enable signals of the infeed and if required, enable. - remove and acknowledge an infeed fault. Note: If this drive is intended to back up the DC link regeneratively, then the fault response must be parameterized for NONE, OFF1 or OFF3. so that the drive can continue to operate even after the infeed fails.

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